

WORDPRESS DEPLOYMENT IN AWS

Task 1: Add VPC Components & Use Multiple Regions:

Objective:

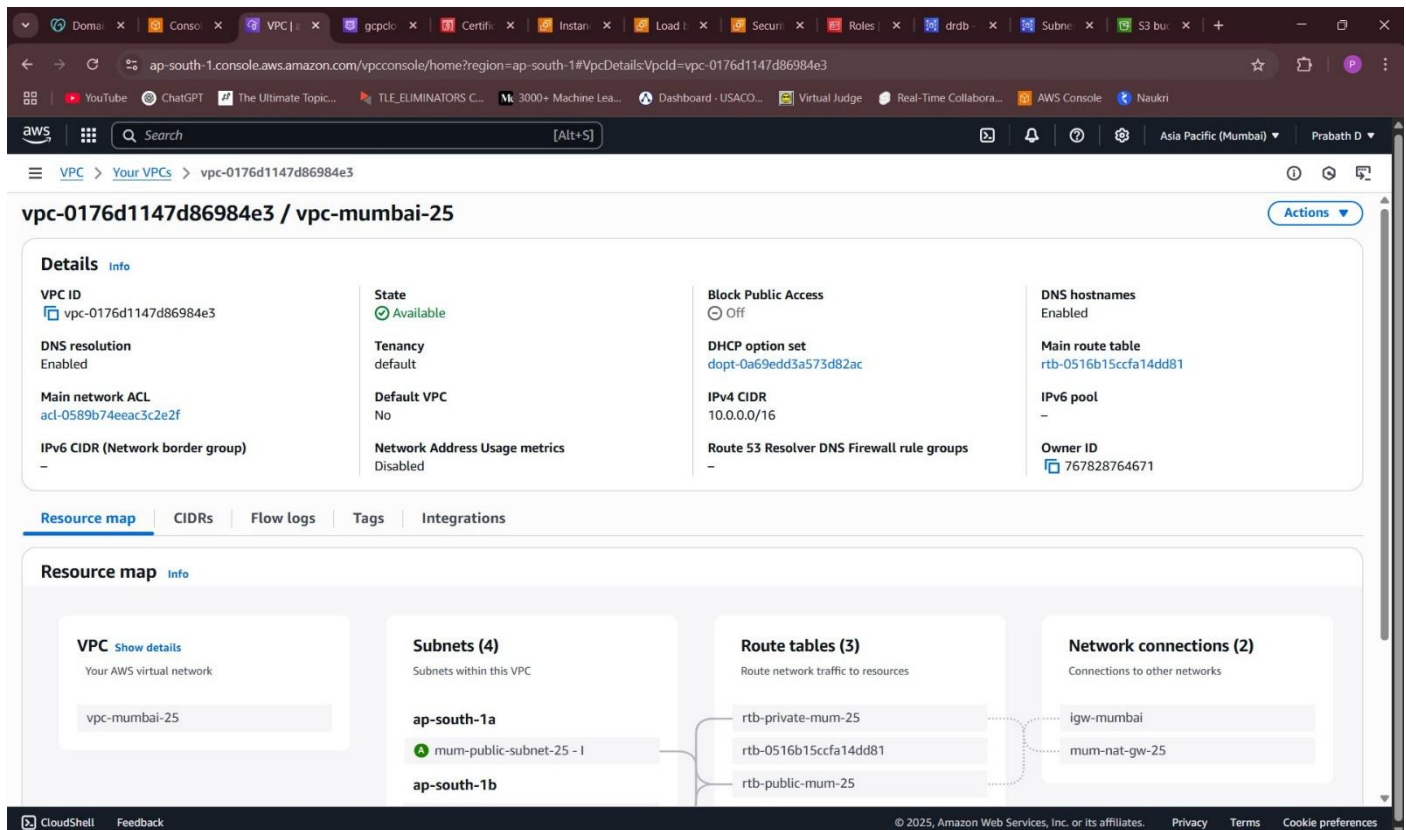
Set up a custom VPC in multiple AWS regions (Mumbai & Ohio) with complete networking (subnets, route tables, NAT, IGW), configure EC2 & RDS, IAM role, set up S3 bucket implement SSL certificate request and DNS linking using ACM + Route 53, connected to GoDaddy domain.

Mumbai Region:

Steps:

i.VPC Creation:

- Click your VPC, then click create VPC then give a name as “vpc-mumbai-25”
- Then set the IP address 10.0.0.0/16, then set both DNS hostname and DNS resolution as enabled, then click create.



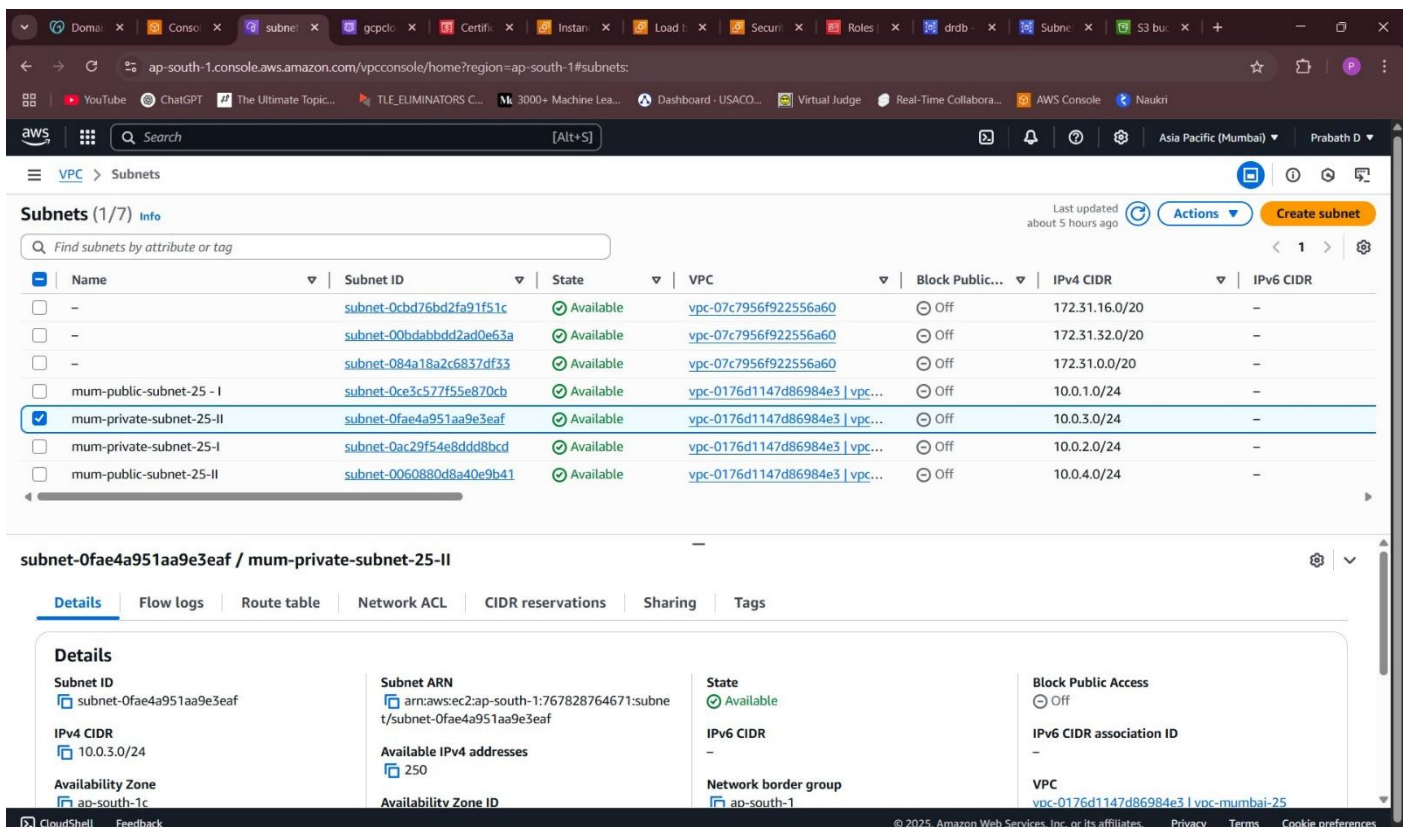
The screenshot displays the AWS Management Console interface for a VPC named "vpc-mumbai-25" in the Asia Pacific (Mumbai) region. The VPC ID is vpc-0176d1147d86984e3. The state is "Available". The DNS resolution is "Enabled". The main network ACL is "acl-0589b74eeac3c2e2f". The IPv4 CIDR is "10.0.0.0/16". The DHCP option set is "dopt-0a69edd3a573d82ac". The route table is "rtb-0516b15ccfa14dd81". The owner ID is "767828764671". The resource map shows the VPC connected to four subnets (ap-south-1a, ap-south-1b, and two others), three route tables (rtb-private-mum-25, rtb-0516b15ccfa14dd81, and rtb-public-mum-25), and two network connections (igw-mumbai and mum-nat-gw-25).

ii. Subnet Setup:

- Go to subnets, click create subnet after that assign a name for the subnets, in this task we need two subnets for public and private.
- Then enter the below details particularly for public and private subnet, by assigning the IP Addresses, then click create subnet
- Public Subnet 1: For EC2 + NAT
- Public Subnet 2: For EC2
- Private Subnet 1: For RDS Prod
- Private Subnet 2: For RDS DR

CIDR Blocks (Mumbai):

- 10.0.1.0/24 (Public subnet 1)
- 10.0.2.0/24 (Public subnet 2)
- 10.0.3.0/24 (Private subnet 1)
- 10.0.4.0/24 (Private subnet 2)



The screenshot shows the AWS Management Console interface for the Asia Pacific (Mumbai) region. The 'Subnets' page is active, displaying a list of subnets. The subnet 'mum-private-subnet-25-ii' is selected, and its details are shown below the list.

Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR	IPv6 CIDR
-	subnet-0cbd76bd2fa91f51c	Available	vpc-07c7956f922556a60	Off	172.31.16.0/20	-
-	subnet-00bdabbdd2ad0e63a	Available	vpc-07c7956f922556a60	Off	172.31.32.0/20	-
-	subnet-084a18a2c6837df33	Available	vpc-07c7956f922556a60	Off	172.31.0.0/20	-
mum-public-subnet-25 - I	subnet-0ce3c577f55e870cb	Available	vpc-0176d1147d86984e3 vpc...	Off	10.0.1.0/24	-
mum-private-subnet-25-ii	subnet-0fae4a951aa9e3eaf	Available	vpc-0176d1147d86984e3 vpc...	Off	10.0.3.0/24	-
mum-private-subnet-25-i	subnet-0ac29f54e8ddd8bcd	Available	vpc-0176d1147d86984e3 vpc...	Off	10.0.2.0/24	-
mum-public-subnet-25-II	subnet-0060880d8a40e9b41	Available	vpc-0176d1147d86984e3 vpc...	Off	10.0.4.0/24	-

subnet-0fae4a951aa9e3eaf / mum-private-subnet-25-ii

Details

Subnet ID subnet-0fae4a951aa9e3eaf	Subnet ARN arn:aws:ec2:ap-south-1:767828764671:subnet/subnet-0fae4a951aa9e3eaf	State Available	Block Public Access Off
IPv4 CIDR 10.0.3.0/24	Available IPv4 addresses 250	IPv6 CIDR -	IPv6 CIDR association ID -
Availability Zone ap-south-1c	Availability Zone ID ap-south-1	Network border group ap-south-1	VPC vpc-0176d1147d86984e3 vpc-mumbai-25

ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#SubnetDetails:subnetId=subnet-0ce3c577f5e870cb

Search [Alt+S]

Asia Pacific (Mumbai) Prabath D

VPC > Subnets > subnet-0ce3c577f5e870cb

subnet-0ce3c577f5e870cb / mum-public-subnet-25 - I

Actions

Details

Subnet ID

subnet-0ce3c577f5e870cb

IPv4 CIDR

10.0.1.0/24

Availability Zone

ap-south-1a

Route table

rtb-01d7b447432b5b441 | rtb-public-mum-25

Auto-assign IPv6 address

No

IPv4 CIDR reservations

-

Resource name DNS A record

Disabled

Subnet ARN

arn:aws:ec2:ap-south-1:767828764671:subnet/subnet-0ce3c577f5e870cb

Available IPv4 addresses

250

Availability Zone ID

aps1-az1

Network ACL

-

Auto-assign customer-owned IPv4 address

No

IPv6 CIDR reservations

-

Resource name DNS AAAA record

Disabled

State

Available

IPv6 CIDR

-

Network border group

ap-south-1

Default subnet

No

Customer-owned IPv4 pool

-

IPv6-only

No

DNS64

Disabled

Block Public Access

Off

IPv6 CIDR association ID

-

VPC

vpc-0176d1147d86984e3 | vpc-mumbai-25

Auto-assign public IPv4 address

No

Outpost ID

-

Hostname type

IP name

Owner

767828764671

Flow logs

Route table

Network ACL

CIDR reservations

Sharing

Tags

Flow logs

Search

Actions

Create flow log

< 1 >

Name

Flow log ID

Filter

Destination type

Destination name

IAM role ARN

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ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#SubnetDetails:subnetId=subnet-0ac29f54e8ddd8bcd

Search [Alt+S]

Asia Pacific (Mumbai) Prabath D

VPC > Subnets > subnet-0ac29f54e8ddd8bcd

subnet-0ac29f54e8ddd8bcd / mum-private-subnet-25-I

Actions

Details

Subnet ID

subnet-0ac29f54e8ddd8bcd

IPv4 CIDR

10.0.2.0/24

Availability Zone

ap-south-1b

Route table

rtb-0aaee2fc7e3c115ad | rtb-private-mum-25

Auto-assign IPv6 address

No

IPv4 CIDR reservations

-

Resource name DNS A record

Disabled

Subnet ARN

arn:aws:ec2:ap-south-1:767828764671:subnet/subnet-0ac29f54e8ddd8bcd

Available IPv4 addresses

250

Availability Zone ID

aps1-az3

Network ACL

-

Auto-assign customer-owned IPv4 address

No

IPv6 CIDR reservations

-

Resource name DNS AAAA record

Disabled

State

Available

IPv6 CIDR

-

Network border group

ap-south-1

Default subnet

No

Customer-owned IPv4 pool

-

IPv6-only

No

DNS64

Disabled

Block Public Access

Off

IPv6 CIDR association ID

-

VPC

vpc-0176d1147d86984e3 | vpc-mumbai-25

Auto-assign public IPv4 address

No

Outpost ID

-

Hostname type

IP name

Owner

767828764671

Flow logs

Route table

Network ACL

CIDR reservations

Sharing

Tags

Flow logs

Search

Actions

Create flow log

< 1 >

Name

Flow log ID

Filter

Destination type

Destination name

IAM role ARN

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The screenshot displays the AWS Management Console interface for a subnet. The breadcrumb navigation shows 'VPC > Subnets > subnet-0060880d8a40e9b41'. The main heading is 'subnet-0060880d8a40e9b41 / mum-public-subnet-25-II'. Below this, there are several tabs: 'Details', 'Flow logs', 'Route table', 'Network ACL', 'CIDR reservations', 'Sharing', and 'Tags'. The 'Details' tab is active, showing various attributes of the subnet in a grid layout. The 'Flow logs' tab is also visible at the bottom, showing a search bar and a table with columns for Name, Flow log ID, Filter, Destination type, Destination name, and IAM role ARN.

Details			
Subnet ID subnet-0060880d8a40e9b41	Subnet ARN arn:aws:ec2:ap-south-1:767828764671:subnet/subnet-0060880d8a40e9b41	State Available	Block Public Access Off
IPv4 CIDR 10.0.4.0/24	Available IPv4 addresses 251	IPv6 CIDR -	IPv6 CIDR association ID -
Availability Zone ap-south-1b	Availability Zone ID aps1-az3	Network border group ap-south-1	VPC vpc-0176d1147d86984e3 vpc-mumbai-25
Route table rtb-01d7b447432b5b441 rtb-public-mum-25	Network ACL -	Default subnet No	Auto-assign public IPv4 address No
Auto-assign IPv6 address No	Auto-assign customer-owned IPv4 address No	Customer-owned IPv4 pool -	Outpost ID -
IPv4 CIDR reservations -	IPv6 CIDR reservations -	IPv6-only No	Hostname type IP name
Resource name DNS A record Disabled	Resource name DNS AAAA record Disabled	DNS64 Disabled	Owner 767828764671

iii. Internet Gateway + NAT Gateway:

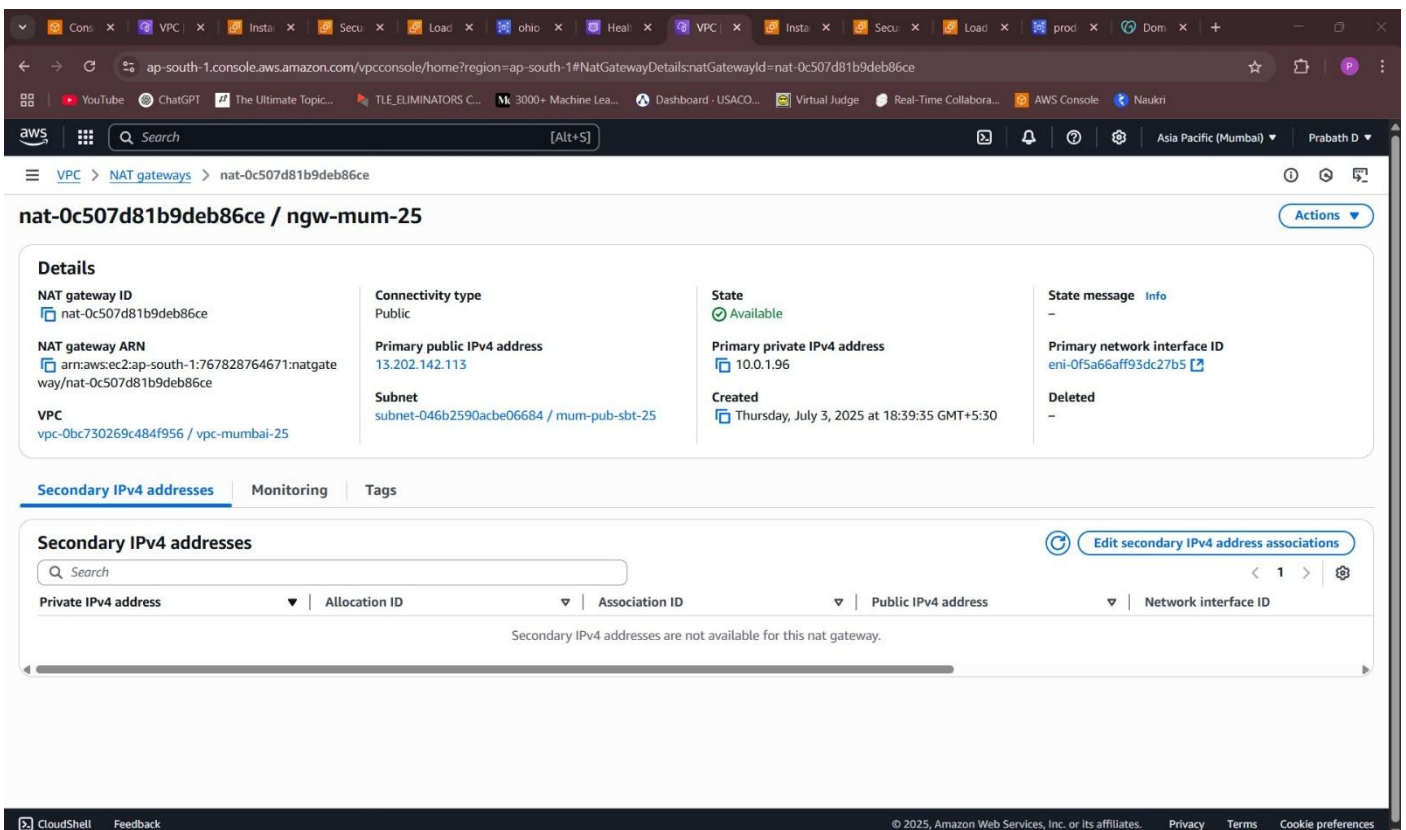
- Create and attach IGW in Mumbai VPC.

The screenshot displays the AWS Management Console interface for an Internet Gateway. The breadcrumb navigation shows 'VPC > Internet gateways > igw-0867acd1a59c8cf48'. The main heading is 'igw-0867acd1a59c8cf48 / igw-mumbai'. Below this, there are several tabs: 'Details', 'Tags', and 'Manage tags'. The 'Details' tab is active, showing various attributes of the Internet Gateway in a grid layout. The 'Tags' tab is also visible at the bottom, showing a search bar and a table with columns for Key and Value.

Details			
Internet gateway ID igw-0867acd1a59c8cf48	State Attached	VPC ID vpc-0176d1147d86984e3 vpc-mumbai-25	Owner 767828764671

Tags	
Key	Value
Name	igw-mumbai

- Create NAT Gateway in Public Subnet (with Elastic IP)



iv. Route Tables (Mumbai):

- Then after creating both Internet Gateway and NAT Gateway, go to route tables, click create, then assign names for both public and private route tables, then associate both the tables by clicking routes then attach the network
- Public Route Table → 0.0.0.0/0 → IGW.
- Then click subnet associations in public route table, then click edit association after that click both public subnet 1 and public subnet 2.
- Private Route Table → 0.0.0.0/0 → NAT.
- Then click subnet associations in private route table, then click edit association after that click both private subnet 1 and private subnet 2.

ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#RouteTableDetails:RouteTableId=rtb-01d7b447432b5b441

Search

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VPCRoute tablesrtb-01d7b447432b5b441

rtb-01d7b447432b5b441 / rtb-public-mum-25

Actions

DetailsInfo

Route table ID
rtb-01d7b447432b5b441

VPC
vpc-0176d1147d86984e3 | vpc-mumbai-25

Main
No

Owner ID
767828764671

Explicit subnet associations
2 subnets

Edge associations
-

Routes

Subnet associations

Edge associations

Route propagation

Tags

Explicit subnet associations (2)

Edit subnet associations

Find subnet association

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
mum-public-subnet-25 - I	subnet-0ce3c577f55e870cb	10.0.1.0/24	-
mum-public-subnet-25-II	subnet-0060880d8a40e9b41	10.0.4.0/24	-

Subnets without explicit associations (0)

Edit subnet associations

Find subnet association

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
------	-----------	-----------	-----------

No subnets without explicit associations
All your subnets are associated with a route table.

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ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#RouteTableDetails:RouteTableId=rtb-01d7b447432b5b441

Search

[Alt+S]

Asia Pacific (Mumbai)Prabath D

VPCRoute tablesrtb-01d7b447432b5b441

rtb-01d7b447432b5b441 / rtb-public-mum-25

Actions

DetailsInfo

Route table ID
rtb-01d7b447432b5b441

VPC
vpc-0176d1147d86984e3 | vpc-mumbai-25

Main
No

Owner ID
767828764671

Explicit subnet associations
2 subnets

Edge associations
-

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (2)

Edit routes

Filter routes

Destination	Target	Status	Propagated
0.0.0.0/0	igw-0867acd1a59c8cf48	Active	No
10.0.0.0/16	local	Active	No

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Domain xConso xVPC | xgcpcl xCertifi xInstan xLoad xSecuri xRoles xdrdb xSubne xS3 bu x+

ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#RouteTableDetails:RouteTableId=rtb-0aaee2fc7e3c115ad

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VPC > Route tables > rtb-0aaee2fc7e3c115ad

Actions

DetailsInfo

Route table ID

rtb-0aaee2fc7e3c115ad

VPC

vpc-0176d1147d86984e3 | vpc-mumbai-25

Main

No

Owner ID

767828764671

Explicit subnet associations

2 subnets

Edge associations

-

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (2)

BothEdit routes

Filter routes

< 1 >

Destination

Target

Status

Propagated

0.0.0.0/0

nat-0f80ac4cfb0ae2260

Active

No

10.0.0/16

local

Active

No

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Domain xConso xVPC | xgcpcl xCertifi xInstan xLoad xSecuri xRoles xdrdb xSubne xS3 bu x+

ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#RouteTableDetails:RouteTableId=rtb-0aaee2fc7e3c115ad

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aws

Search[Alt+S]

Asia Pacific (Mumbai)Prabath D

VPC > Route tables > rtb-0aaee2fc7e3c115ad

Actions

DetailsInfo

Route table ID

rtb-0aaee2fc7e3c115ad

VPC

vpc-0176d1147d86984e3 | vpc-mumbai-25

Main

No

Owner ID

767828764671

Explicit subnet associations

2 subnets

Edge associations

-

Routes

Subnet associations

Edge associations

Route propagation

Tags

Explicit subnet associations (2)

Edit subnet associations

Find subnet association

< 1 >

Name

Subnet ID

IPv4 CIDR

IPv6 CIDR

mum-private-subnet-25-II

subnet-0fae4a951aa9e3eaf

10.0.3.0/24

-

mum-private-subnet-25-I

subnet-0ac29f54e8ddd8bcd

10.0.2.0/24

-

Subnets without explicit associations (0)

Edit subnet associations

The following subnets have not been explicitly associated with any route tables and are therefore associated with the main route table:

Find subnet association

< 1 >

Name

Subnet ID

IPv4 CIDR

IPv6 CIDR

No subnets without explicit associations

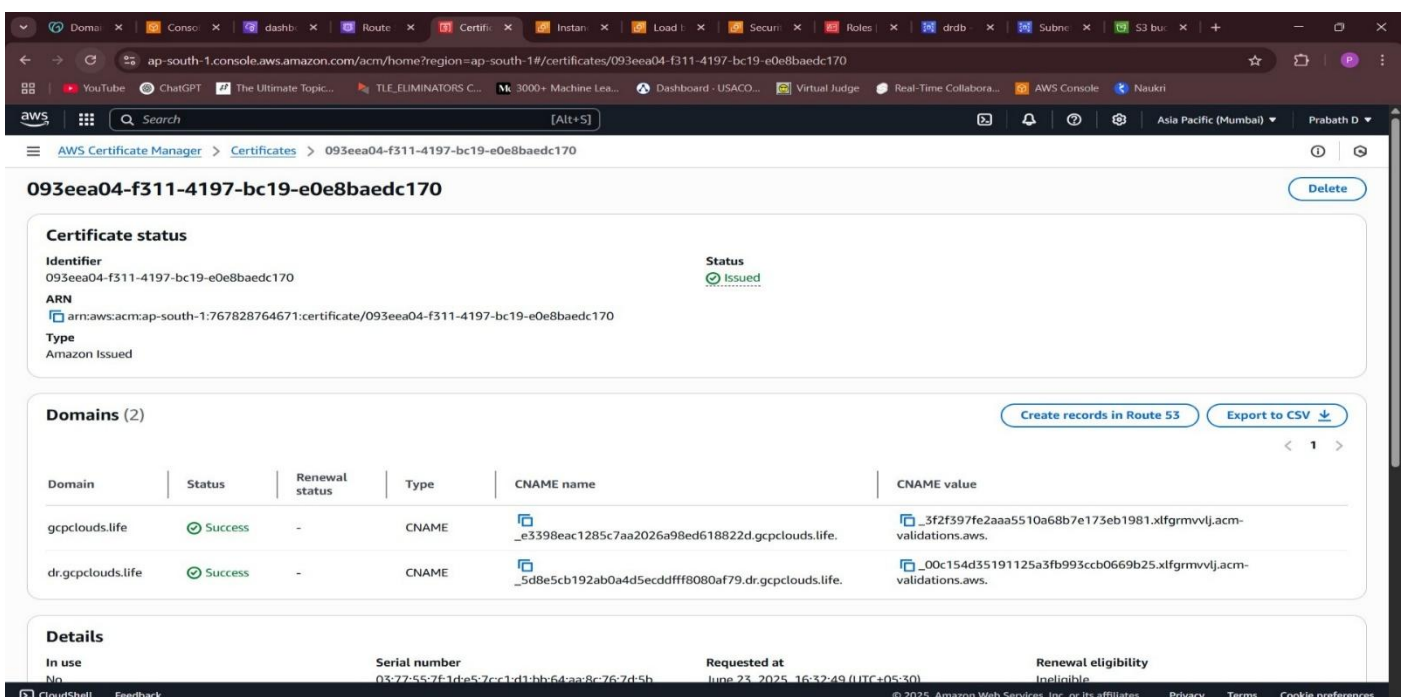
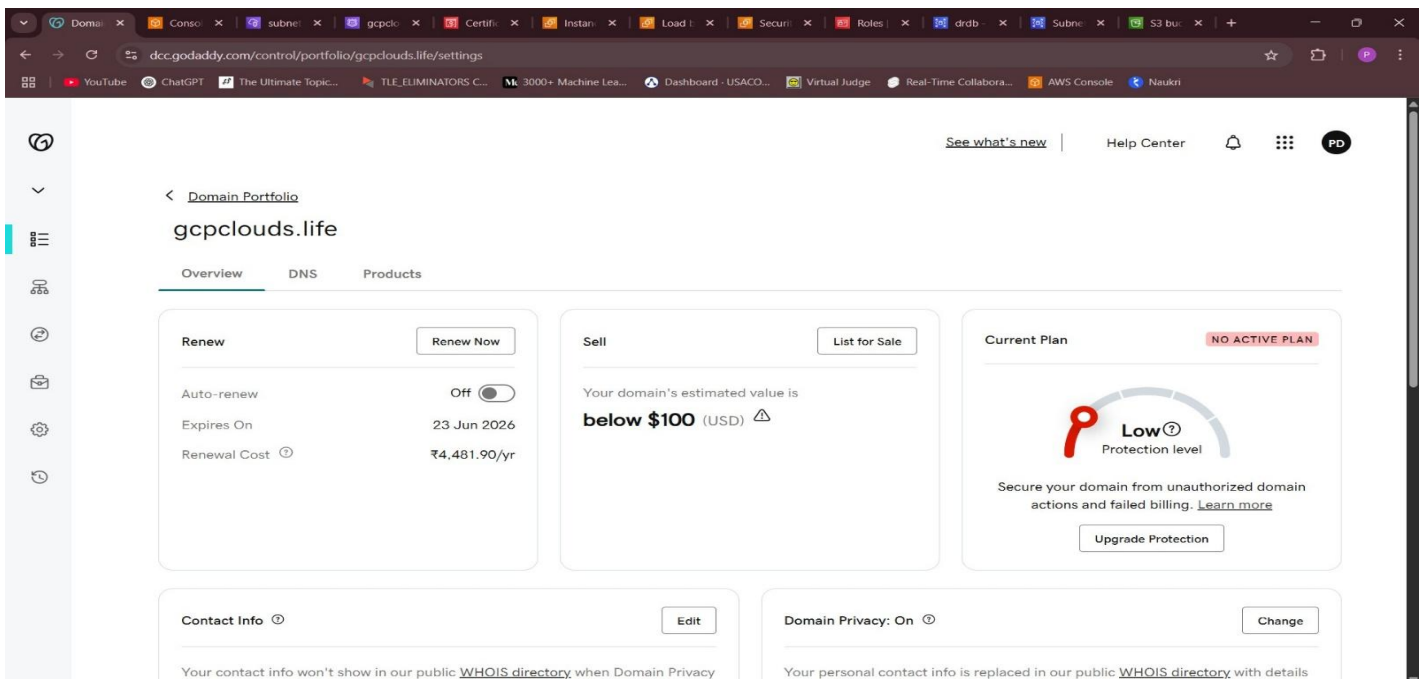
All your subnets are associated with a route table.

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v. AWS Certificate Manager Creation:

- Request public certificate as per the registered domain.
- Go to ACM click request, then click public then name it with gcpclouds.life, then click add another certificate in that name it as 'dr.gcpclouds.life'.
- Then click create, after the creation records click create records in Route53, then after selecting both SSL certificate that click create.
- Then refresh the Route53 after the creation of hosted zone, it would be appearing with CNAME and CVALUE.
- Purchased a domain in GoDaddy named with 'gcpclouds.life'.



vi. AWS Route53 setup & link to GoDaddy domain DNS:

- Create hosted zone.
- Map the nameservers under GoDaddy domain.
- Create records set from ACM to R53.

Public **gcpclouds.life** Info [Delete zone](#) [Test record](#) [Configure query logging](#)

► Hosted zone details [Edit hosted zone](#)

Records (6) DNSSEC signing Hosted zone tags (0)

Records (6) Info [Delete record](#) [Import zone file](#) [Create record](#)

Automatic mode is the current search behavior optimized for best filter results. [To change modes go to settings.](#)

Filter records by property or value Type Routing p... Alias < 1 > ⚙

☐	Record name	Type	Routin...	Differ...	Alias	Value/Route traffic to	TTL (seconds)
☐	gcpclouds.life	A	Failover	Primary	Yes	dualstack.prodlib-723851817...	-
☐	gcpclouds.life	A	Failover	Secondary	Yes	dualstack.dr1b-154749572.a...	-
☐	gcpclouds.life	NS	Simple	-	No	ns-1211.awsdns-23.org. ns-154.awsdns-19.com. ns-1554.awsdns-02.co.uk. ns-744.awsdns-29.net.	172800
☐	gcpclouds.life	SOA	Simple	-	No	ns-1211.awsdns-23.org. aws...	900
☐	_e3398eac1285c7aa2026a98ed618822d.gcpclouds.life	CNAME	Simple	-	No	_3f2f397fe2aaa5510a68b7e...	300
☐	_5d8e5cb192ab0a4d5ecdff8080af79.dr.gcpclouds.life	CNAME	Simple	-	No	_00c154d35191125a3fb993...	300

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DNS Records Forwarding **Nameservers** Premium DNS Hostnames DS Records

Nameservers determine where your DNS is hosted and where you add, edit or delete your DNS records.

Using custom nameservers [Change Nameservers](#)

Nameservers ⓘ

ns-1211.awsdns-23.org

ns-154.awsdns-19.com

ns-1554.awsdns-02.co.uk

ns-744.awsdns-29.net

vii. AWS RDS Setup:

- Create Two RDS
 - I. New Security Group
 - II. New Subnet Group
 - III. PRODUCTION DB (SECURITY GROUP MAKE AS ALL TRAFFIC)
 - IV. DR DB (SECURITY GROUP MAKE AS ALL TRAFFIC)

The screenshot displays the AWS Management Console for the 'awsproject-subnet' subnet group. A green notification bar at the top states 'Successfully created awsproject-subnet. View subnet group'. The main content area is titled 'awsproject-subnet' and includes the following details:

- Subnet group details:**
 - VPC ID: vpc-0176d1147d86984e3
 - ARN: arn:aws:rds:ap-south-1:767828764671:subgrp:awsproject-subnet
 - Supported network types: IPv4
 - Description: awsproject-subnet
- Subnets (2):**

Availability zone	Subnet name	Subnet ID	CIDR block
ap-south-1c	mum-private-subnet-25-II	subnet-0fae4a951aa9e3eaf	10.0.3.0/24
ap-south-1b	mum-private-subnet-25-I	subnet-0ac29f54e8ddd8bcd	10.0.2.0/24
- Tags (0):** A 'Manage tags' button is available.

The screenshot displays the AWS Management Console for the 'proddb' database instance. The left sidebar shows the 'Aurora and RDS' navigation menu. The main content area is titled 'proddb' and includes the following details:

- DB identifier:** prodadb
- Status:** Available
- Role:** Instance
- Engine:** MySQL Community
- Recommendations:**
- CPU:** 3.76%
- Class:** db.t3.micro
- Current activity:** 0 Connections
- Region & AZ:** ap-south-1c

The 'Connectivity & security' tab is selected, showing the following details:

- Endpoint & port:**
 - Endpoint: prodadb.cngac84c2in0.ap-south-1.rds.amazonaws.com
 - Port: 3306
- Networking:**
 - Availability Zone: ap-south-1c
 - VPC: vpc-mumbai-25 (vpc-0176d1147d86984e3)
 - Subnet group: awsproject-subnet
 - Subnets: subnet-0ac29f54e8ddd8bcd, subnet-0fae4a951aa9e3eaf
 - Network type: IPv4
- Security:**
 - VPC security groups: awsprojectsg (sg-067a443aeced89167) - Active
 - Publicly accessible: No
 - Certificate authority: rds-ca-rsa2048-g1
 - Certificate authority date: May 20, 2061, 00:10 (UTC+05:30)
 - DB instance certificate expiration date: June 26, 2026, 16:07 (UTC+05:30)

Aurora and RDS

Dashboard

Databases

Query editor

Performance insights

Snapshots

Exports in Amazon S3

Automated backups

Reserved instances

Proxies

Subnet groups

Parameter groups

Option groups

Custom engine versions

Zero-ETL integrations

New

Events

Event subscriptions

Recommendations

0

Certificate update

No proxies

You don't have any proxies.

Create proxy

Security group rules (3)

Filter by Security group rules

< 1 >

Security group	Type	Rule
awsprojectsg (sg-067a443aecef89167)	CIDR/IP - Inbound	0.0.0.0/0
awsprojectsg (sg-067a443aecef89167)	CIDR/IP - Inbound	0.0.0.0/0
awsprojectsg (sg-067a443aecef89167)	CIDR/IP - Outbound	0.0.0.0/0

Replication (1)

Filter by Replication

< 1 >

DB identifier	Role	Region & AZ	Replication source	Replication state	Lag
proddb	Instance	ap-south-1c	-	-	-

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Aurora and RDS

Dashboard

Databases

Query editor

Performance insights

Snapshots

Exports in Amazon S3

Automated backups

Reserved instances

Proxies

Subnet groups

Parameter groups

Option groups

Custom engine versions

Zero-ETL integrations

New

Events

Event subscriptions

Recommendations

0

Certificate update

DB identifier

drdb

CPU

2.90%

Status

Available

Class

db.t3.micro

Role

Instance

Current activity

0 Connections

Engine

MySQL Community

Region & AZ

ap-south-1b

Recommendations

Connectivity & security

Monitoring

Logs & events

Configuration

Zero-ETL integrations

Maintenance & backups

Data

Connectivity & security

Endpoint & port

Endpoint

[drdb.cngac84c2in0.ap-south-1.rds.amazonaws.com](#)

Port

3306

Networking

Availability Zone

ap-south-1b

VPC

[vpc-mumbai-25 \(vpc-0176d1147d86984e3\)](#)

Subnet group

[awsproject-subnet](#)

Subnets

[subnet-0ac29f54e8ddd8bcd](#)
[subnet-0fae4a951aa9e3eaf](#)

Network type

IPv4

Security

VPC security groups

[awsprojectsg \(sg-067a443aecef89167\)](#)

Active

Publicly accessible

No

Certificate authority

[Info](#)
[rds-ca-rsa2048-g1](#)

Certificate authority date

May 20, 2061, 00:10 (UTC+05:30)

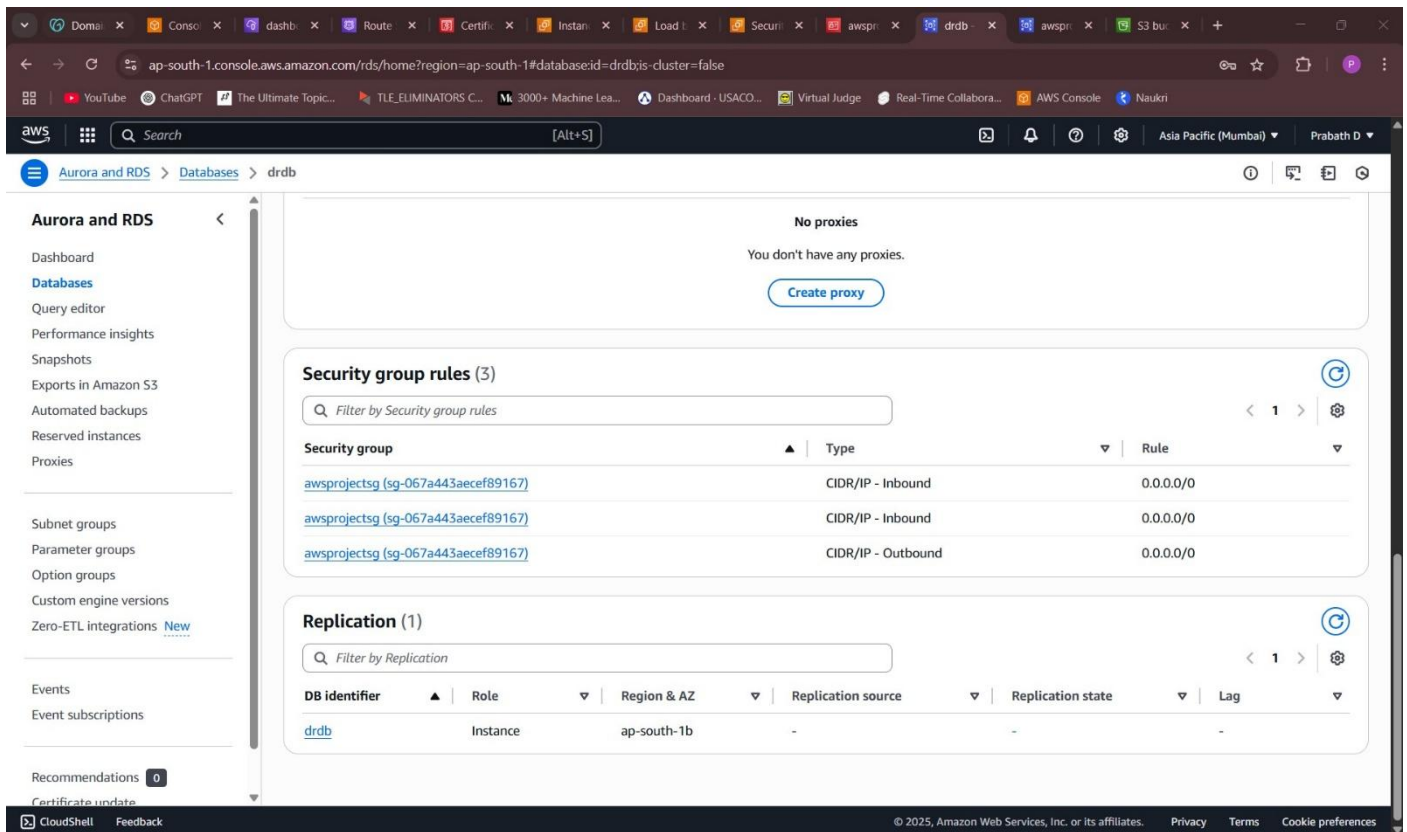
DB instance certificate expiration date

June 26, 2026, 15:59 (UTC+05:30)

CloudShell

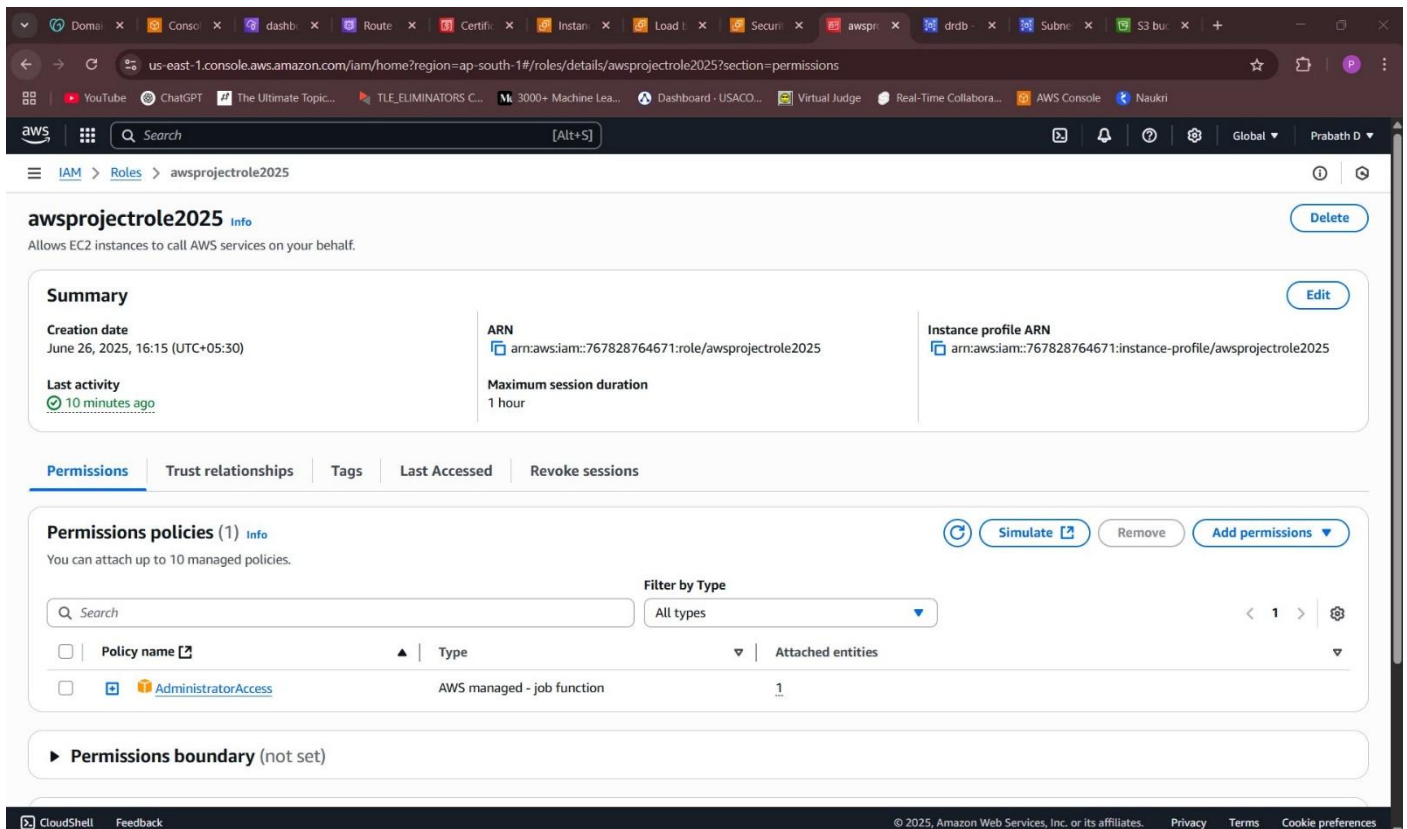
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viii. AWS IAM Role:

- Create an IAM role with full access.



ix. AWS EC2 Creation:

- Create two EC2 with IAM roles enables & add user data under advanced section
- Two instances need to setup for Mumbai region and adding the details in advanced section.

User data:

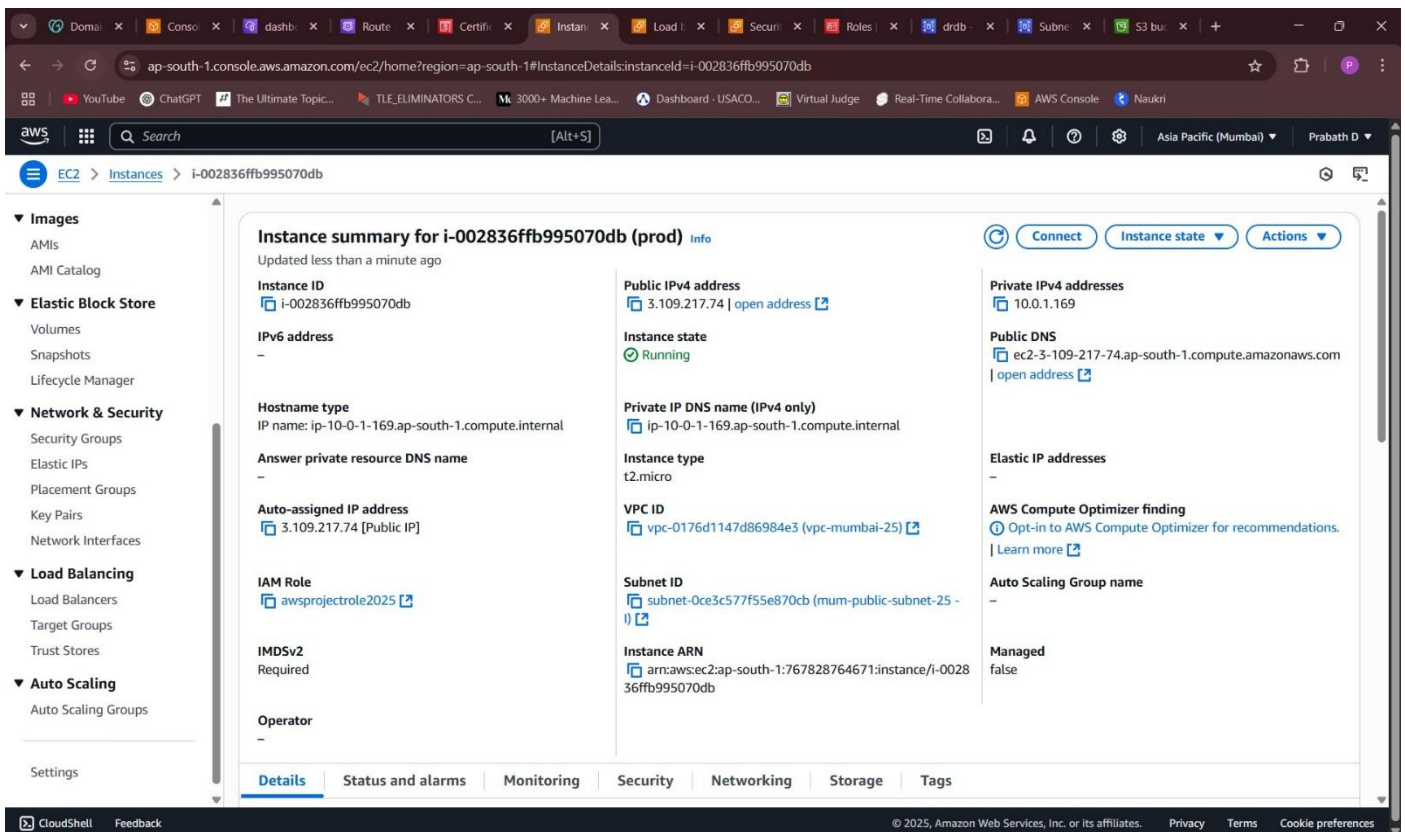
```
#!/bin/bash
yum install httpd php-mysql -y
amazon-linux-extras install -y php7.3
cd /var/www/html
echo "healthy" > healthy.html
wget https://wordpress.org/latest.tar.gz
tar -xzf latest.tar.gz
cp -r wordpress/* /var/www/html/
rm -rf wordpress
rm -rf latest.tar.gz
chmod -R 755 wp-content
chown -R apache:apache wp-content
wget https://s3.amazonaws.com/bucketforwordpresslab-donotdelete/htaccess.txt
mv htaccess.txt .htaccess
chkconfig httpd on
service httpd start
```

- This is the data need to be added in both EC2 instances.

The screenshot displays the AWS Management Console interface for an EC2 instance. The left sidebar shows navigation options like Images, Elastic Block Store, Network & Security, Load Balancing, and Auto Scaling. The main content area shows the 'Instance summary for i-08fc18592dcdbf44a (dr)'. Key details include:

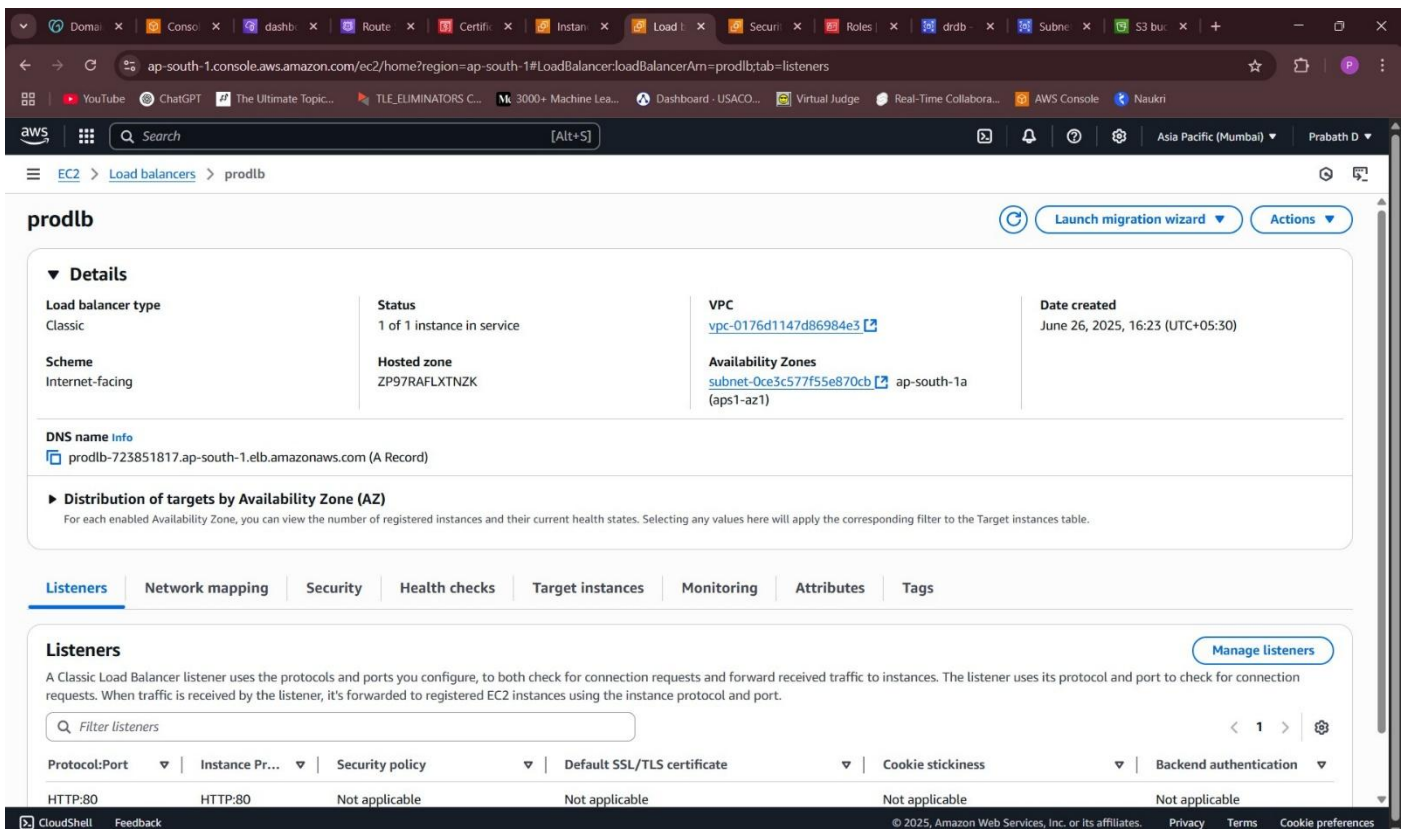
- Instance ID:** i-08fc18592dcdbf44a
- Public IPv4 address:** 3.110.46.203
- Instance state:** Running
- Private IP DNS name (IPv4 only):** ip-10-0-4-163.ap-south-1.compute.internal
- Instance type:** t2.micro
- VPC ID:** vpc-0176d1147d86984e3 (vpc-mumbai-25)
- Subnet ID:** subnet-0060880d8a40e9b41 (mum-public-subnet-25-ii)
- Instance ARN:** arn:aws:ec2:ap-south-1:767828764671:instance/i-08fc18592dcdbf44a
- IAM Role:** awsprojectrole2025
- IMDSv2:** Required
- Operator:** -
- Public DNS:** ec2-3-110-46-203.ap-south-1.compute.amazonaws.com
- Elastic IP addresses:** -
- AWS Compute Optimizer finding:** Opt-in to AWS Compute Optimizer for recommendations.
- Auto Scaling Group name:** -
- Managed:** false

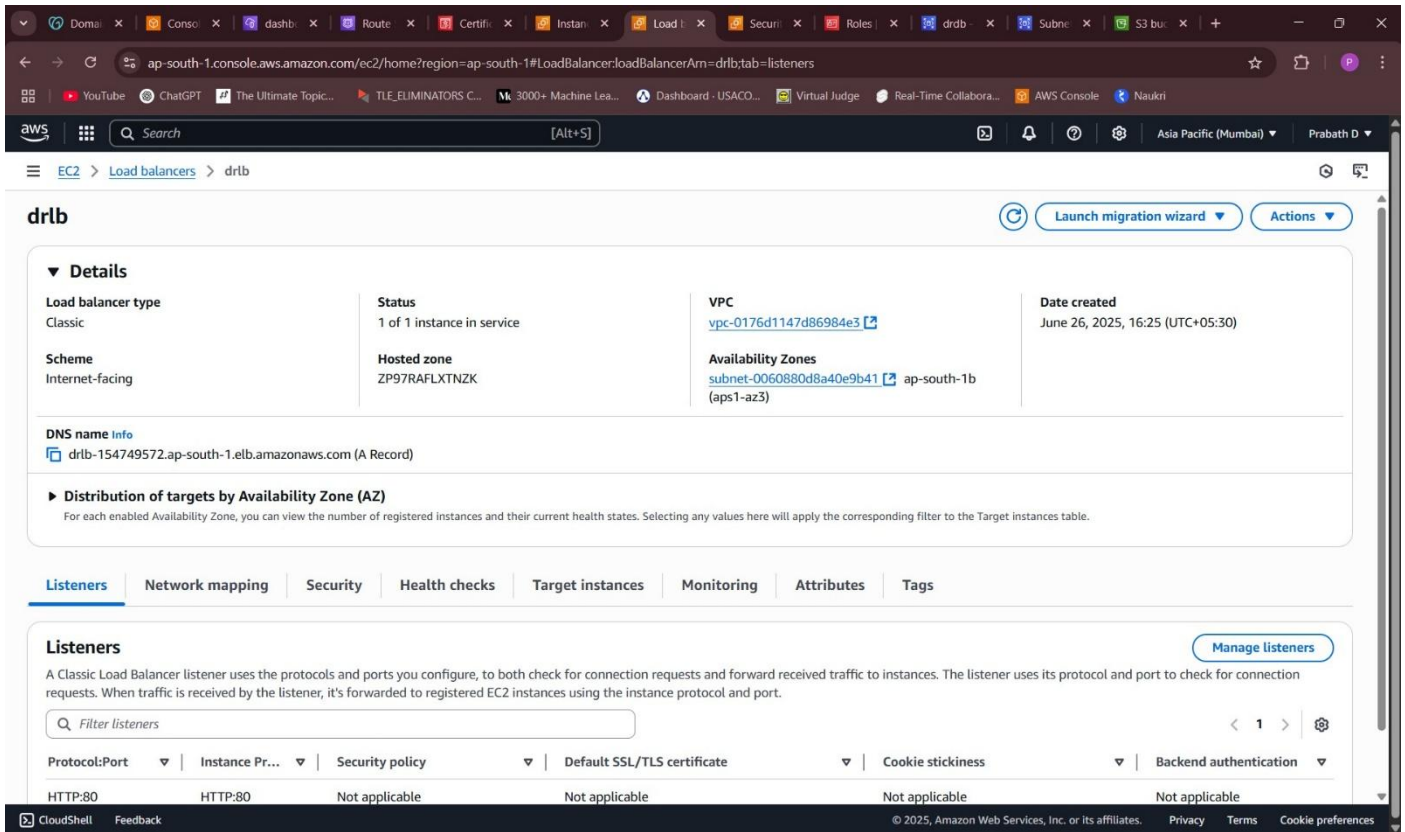
The bottom of the console shows tabs for Details, Status and alarms, Monitoring, Security, Networking, Storage, and Tags.



x. AWS ELB Creation & Setup:

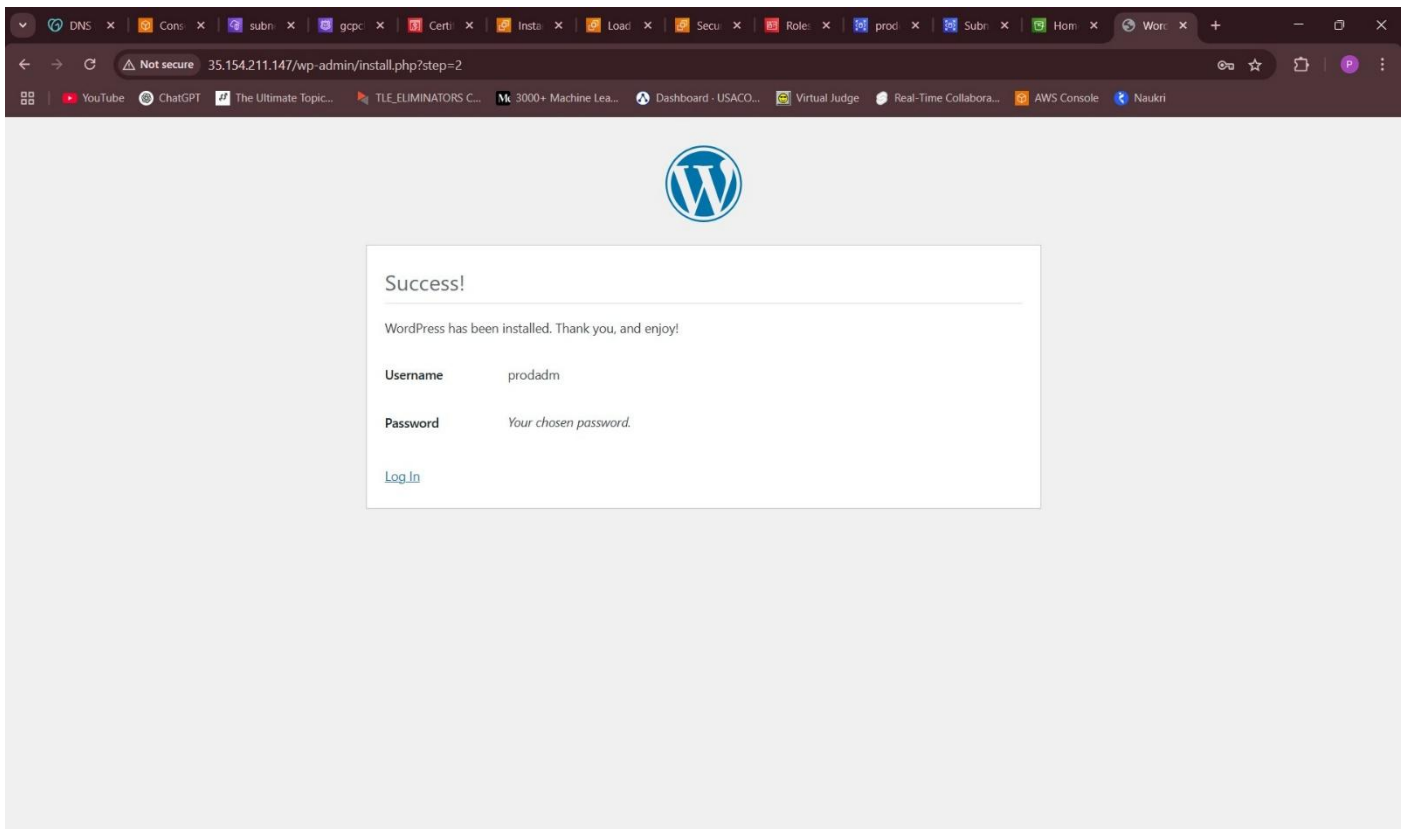
- Create two Classic ELB and map appropriate EC2 to it.

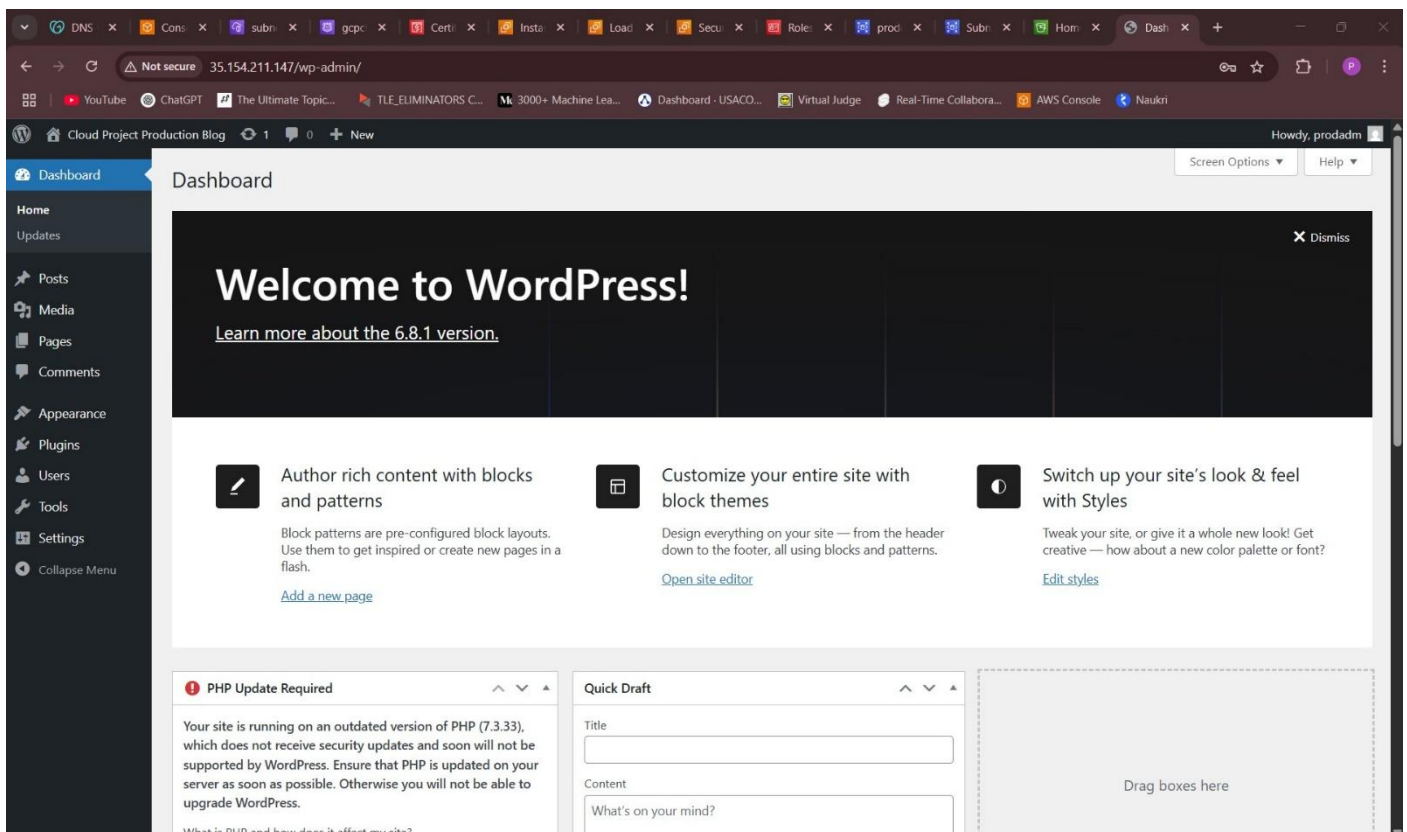
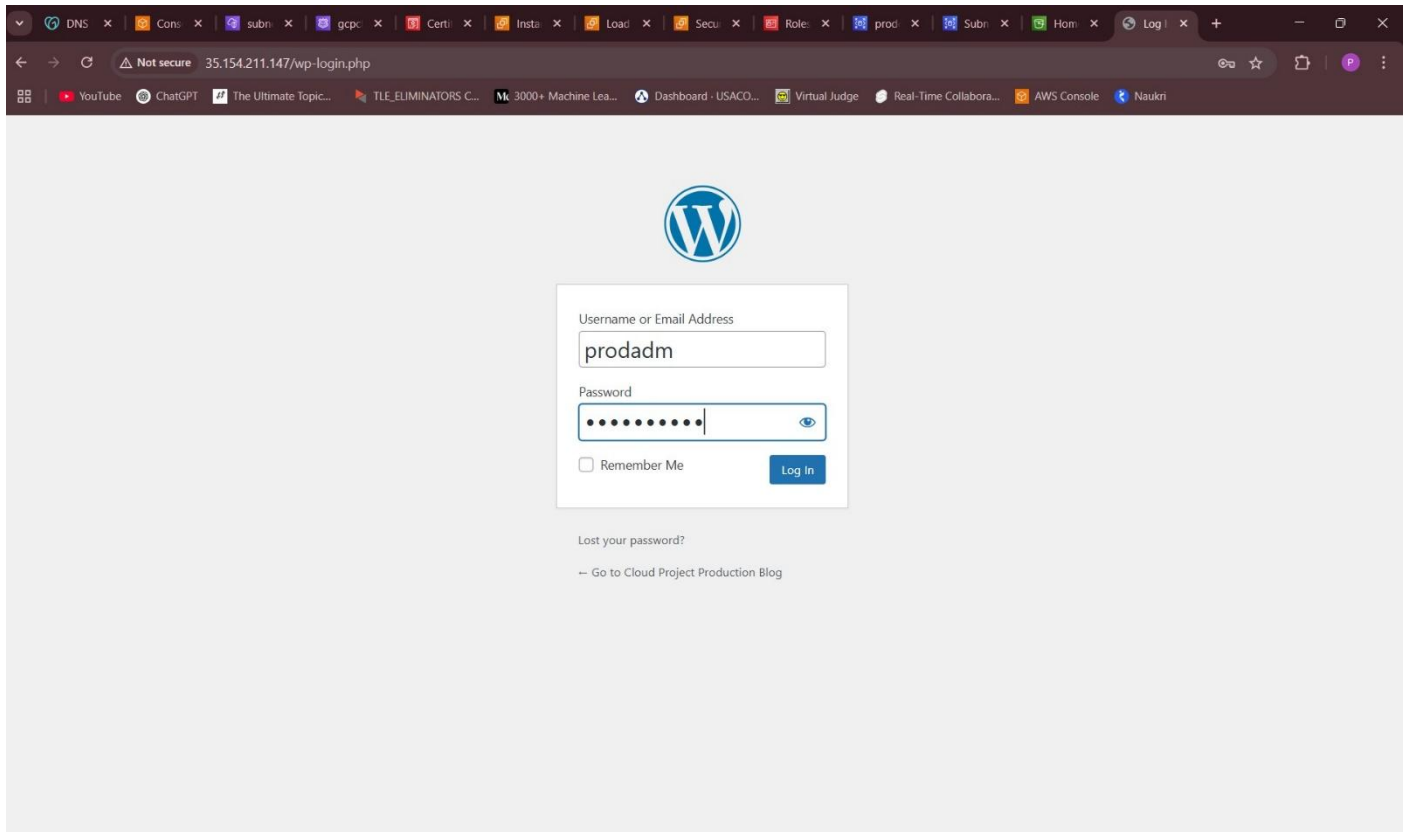


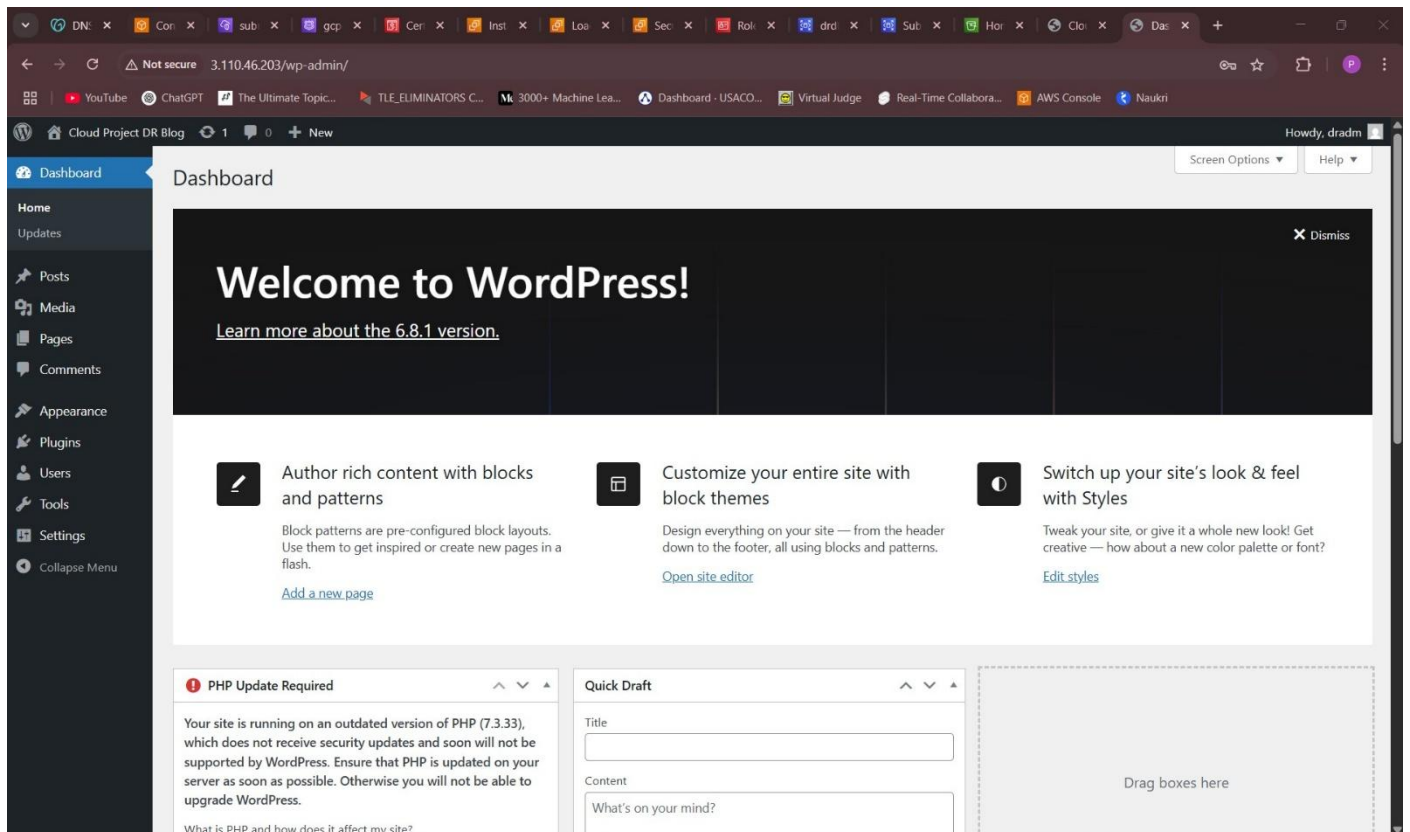
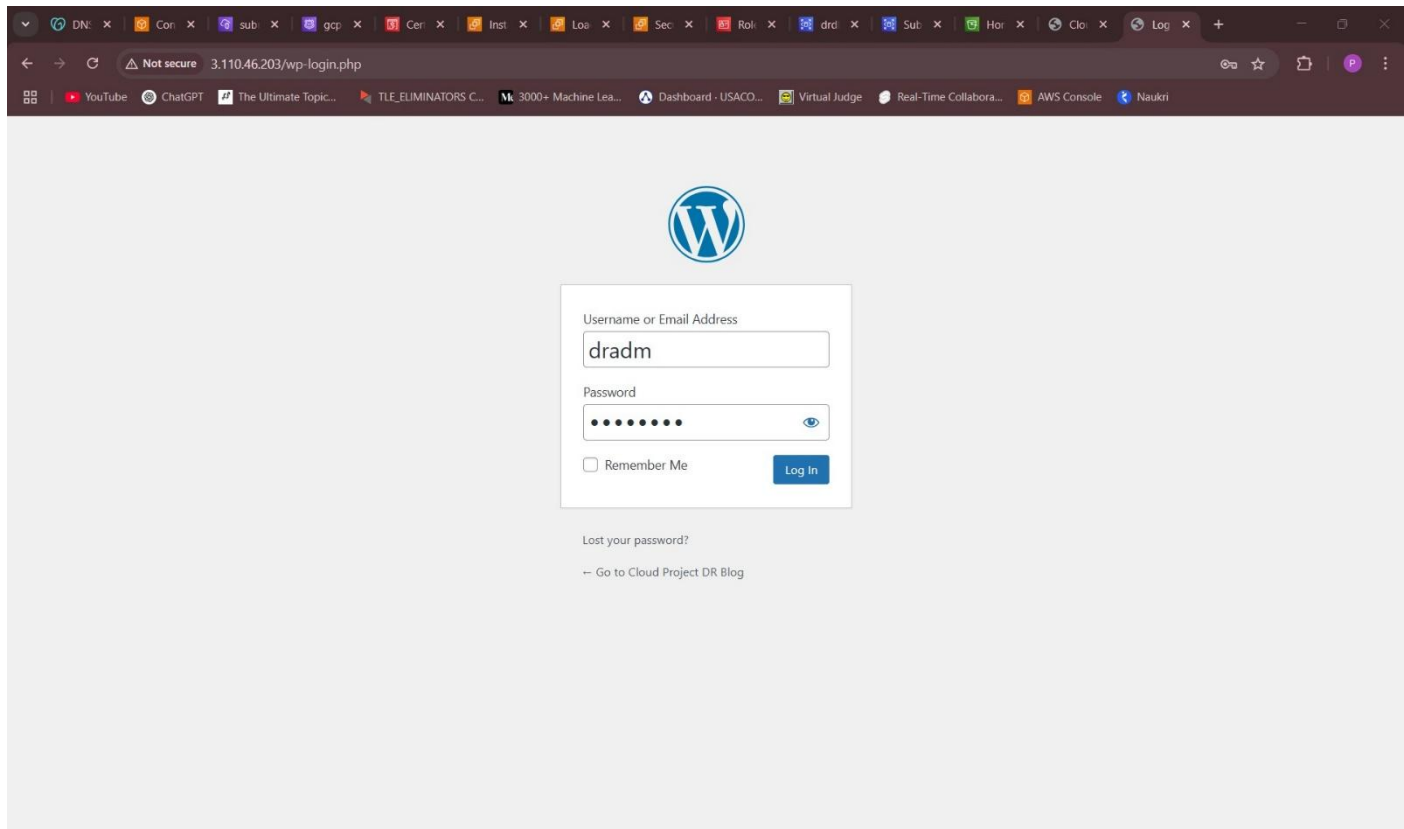


xi. Add the DB details under WordPress Application for both DR & Production:

- Create and edit the wp-config.php file in both the EC2s.

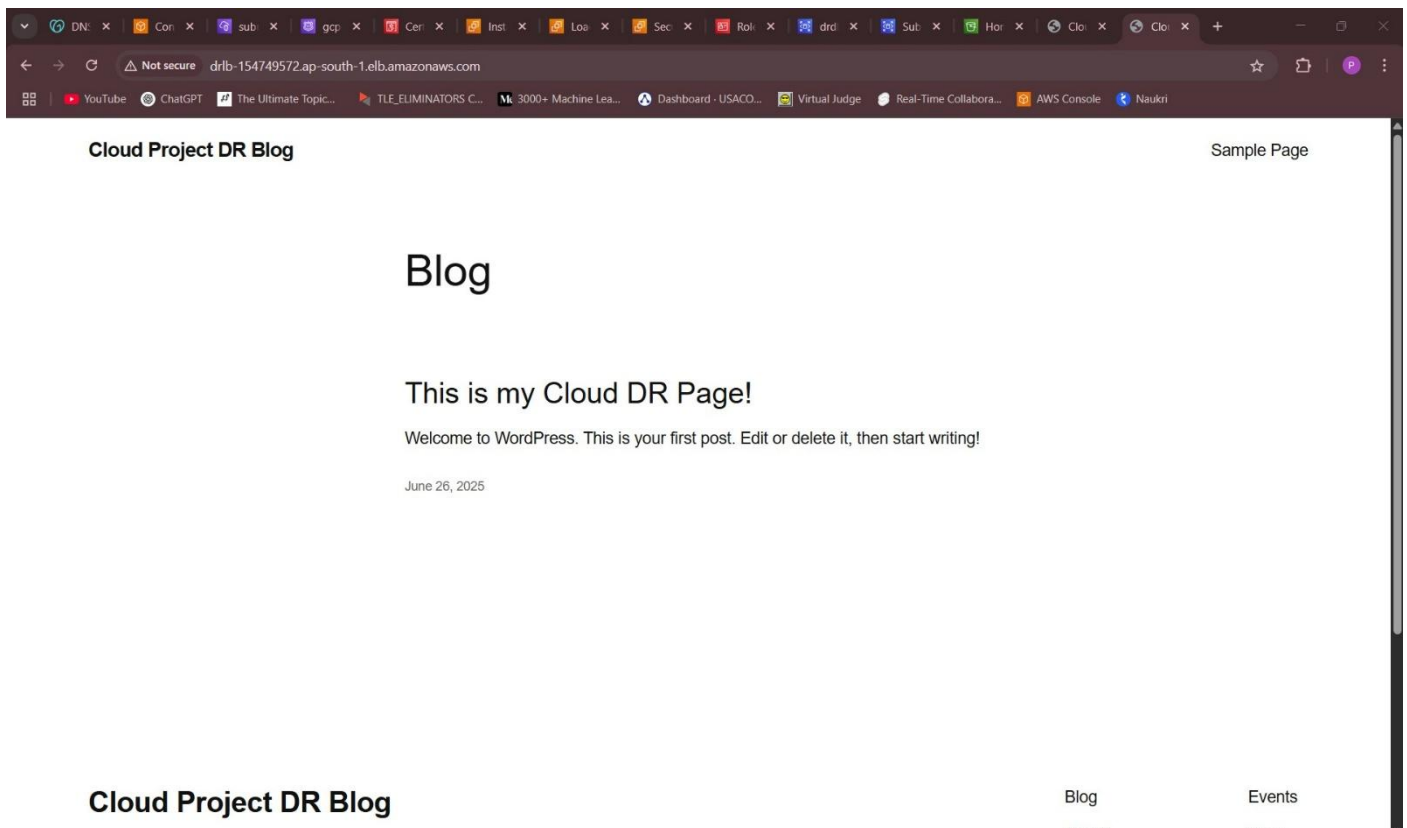
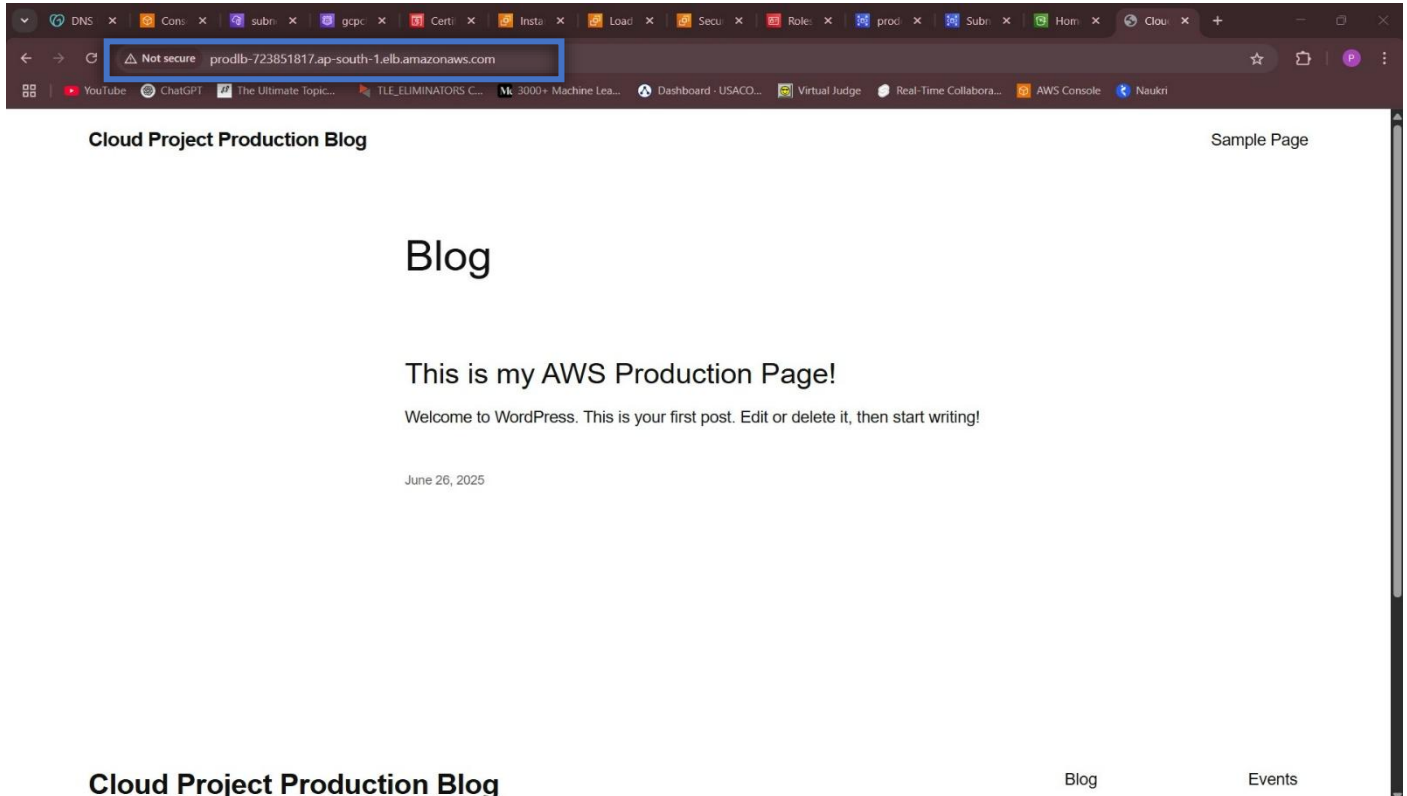






xii. Launch the WordPress page & post it with appropriate DR & Production texts to get difference in the application page:

- Now, with the help of DNS name of the prod1b and dr1b or IP Address of any one of EC2, paste it in the browser and load the page.



xiii. AWS R53 & ELB Sync:

- Refresh the R53
- Go to primary health check. Click create health check, then give a name then by mapping the created production ELB by giving the path in '/' as index.html.
- Create Record set and map appropriate ELB under alias by adding health checks in a particular region.

The screenshot shows the AWS Route 53 console for a health check named 'primary-health-check'. The configuration details are as follows:

Property	Value
ID	1192c1b6-c952-48b6-8249-196287627d4d
URL	http://prod1b-723851817.ap-south-1.elb.amazonaws.com:80/
Specified endpoint by	Domain name
State	Enabled
Status	Healthy
Inverted	No

Below the configuration, there are tabs for Metrics, Alarms, Health checkers, and Tags. The Metrics tab is active, showing a line graph for 'Health check status' and a bar chart for 'Health checkers that report the endpoint healthy (percent)'.

The screenshot shows the AWS Route 53 console for a hosted zone named 'gpcclouds.life'. The 'Records (6)' tab is active, displaying a table of DNS records:

Record name	Type	Routing	Difference	Alias	Value/Route traffic to	TTL (seconds)
gpcclouds.life	A	Failover	Primary	Yes	dualstack.prod1b-723851817...	-
gpcclouds.life	A	Failover	Secondary	Yes	dualstack.dr1b-154749572.a...	-
gpcclouds.life	NS	Simple	-	No	ns-1211.awsdns-23.org, ns-154.awsdns-19.com, ns-1554.awsdns-02.co.uk, ns-744.awsdns-29.net	172800
gpcclouds.life	SOA	Simple	-	No	ns-1211.awsdns-23.org. aws...	900
_e3398eac1285c7aa2026a98ed618822d.gpcclouds.life	CNAME	Simple	-	No	_3f2f397fe2aaa5510a68b7e...	300
_5d8e5cb192ab0a4d5ecddfff8080af79.dr.gpcclouds.life	CNAME	Simple	-	No	_00c154d35191125a3fb993...	300

xiv. AWS S3 Bucket Creation:

- Go to S3 click create bucket, then create two S3 buckets for Production and DR.
- Set the crontab job's in the EC2 in small intervals to copy the content from production to S3 & S3 to DR.
- Add the media sets for production and DR.
- The media sets are:
 - wordpress-mediaasset-project
 - wordpress-code-project
- The content that must be given in server instance,

Production:

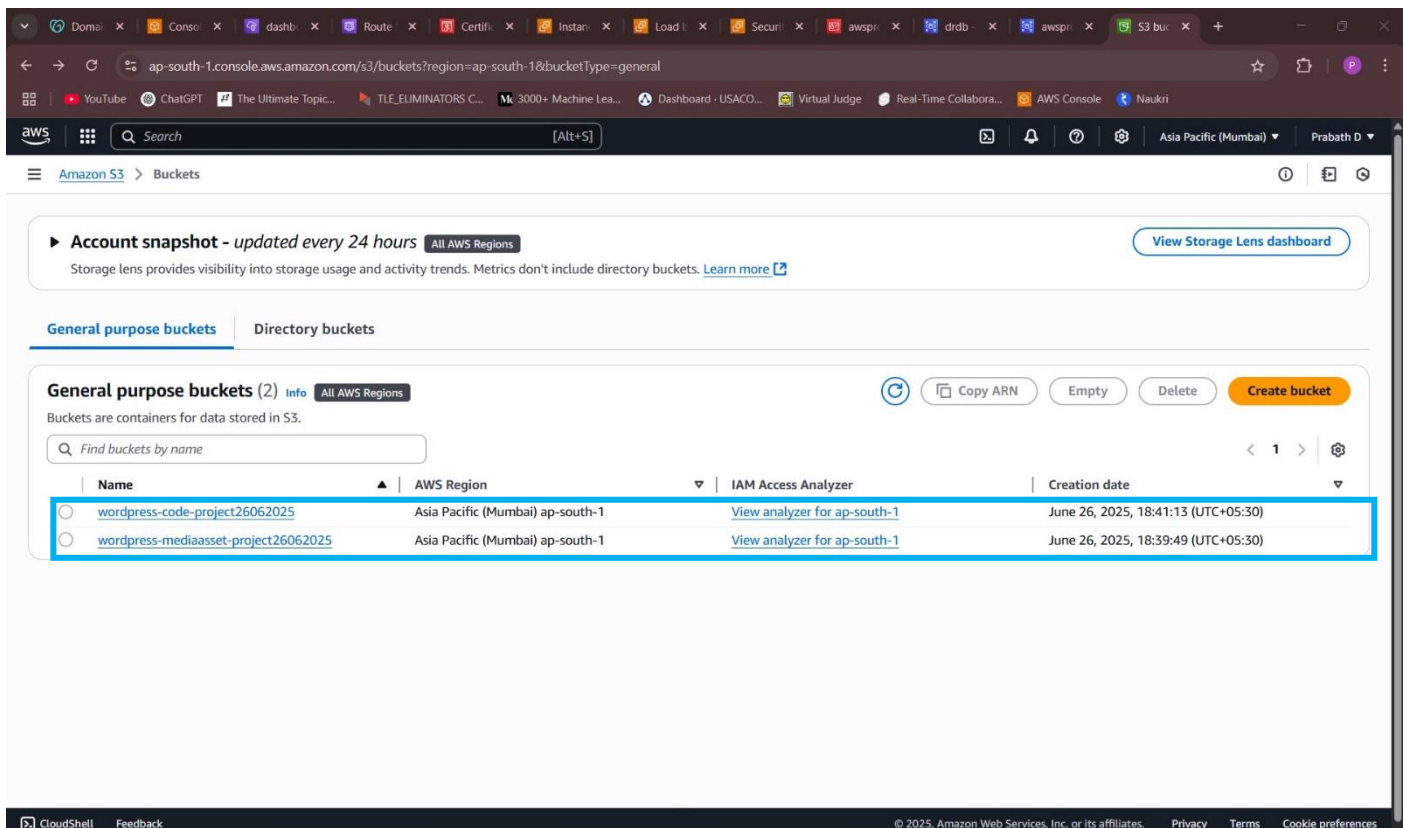
```
* / 2 * * * * aws s3 sync --delete /var/www/html/wp-content/uploads s3://wordpress-mediaasset-project
```

```
* / 2 * * * * aws s3 sync --delete /var/www/html/ s3://wordpress-code-project
```

DR:

```
* / 2 * * * * aws s3 sync --delete s3://wordpress-mediaasset-project /var/www/html/wp-content/uploads
```

```
* / 2 * * * * aws s3 sync --delete s3://wordpress-code-project /var/www/html/
```



```

PuTTY (inactive)
login as: ec2-user
Authenticating with public key "imported-openssh-key"
Last login: Thu Jun 26 11:35:14 2025 from 106.222.221.106

#
~\  #####
~~~\  #####\
~~~\  #####|
~~~\  \#/
~~~\  V~'  '--->
~~~\  .
~~~\  /
~~~\  /m/'

Amazon Linux 2
AL2 End of Life is 2026-06-30.

A newer version of Amazon Linux is available!
Amazon Linux 2023, GA and supported until 2028-03-15.
https://aws.amazon.com/linux/amazon-linux-2023/

[ec2-user@ip-10-0-1-169 ~]$ sudo -i
[root@ip-10-0-1-169 ~]# crontab -e
no crontab for root - using an empty one
crontab: installing new crontab
[root@ip-10-0-1-169 ~]# crontab -l
*/2 * * * * aws s3 sync --delete /var/www/html/wp-content/uploads s3://wordpress-mediaasset-project26062025
*/2 * * * * aws s3 sync --delete /var/www/html/ s3://wordpress-code-project26062025

[root@ip-10-0-1-169 ~]#
login as: ec2-userrec2

```

Crontab(Prod)

```

PuTTY (inactive)
[ec2-user@ip-10-0-4-163 ~]$ cd /var/www/html
[root@ip-10-0-4-163 html]# ls
healthy.html      wp-admin          wp-cron.php       wp-mail.php
index.php         wp-blog-header.php wp-includes       wp-settings.php
license.txt       wp-comments-post.php wp-links-opml.php wp-signup.php
readme.html      wp-config-sample.php wp-load.php       wp-trackback.php
wp-activate.php  wp-content        wp-login.php      xmlrpc.php
[root@ip-10-0-4-163 html]# vi wp-config.php
[root@ip-10-0-4-163 html]#
login as: ec2-user
Authenticating with public key "imported-openssh-key"
Last login: Thu Jun 26 11:11:15 2025 from 106.222.221.106

#
~\  #####
~~~\  #####\
~~~\  #####|
~~~\  \#/
~~~\  V~'  '--->
~~~\  .
~~~\  /
~~~\  /m/'

Amazon Linux 2
AL2 End of Life is 2026-06-30.

A newer version of Amazon Linux is available!
Amazon Linux 2023, GA and supported until 2028-03-15.
https://aws.amazon.com/linux/amazon-linux-2023/

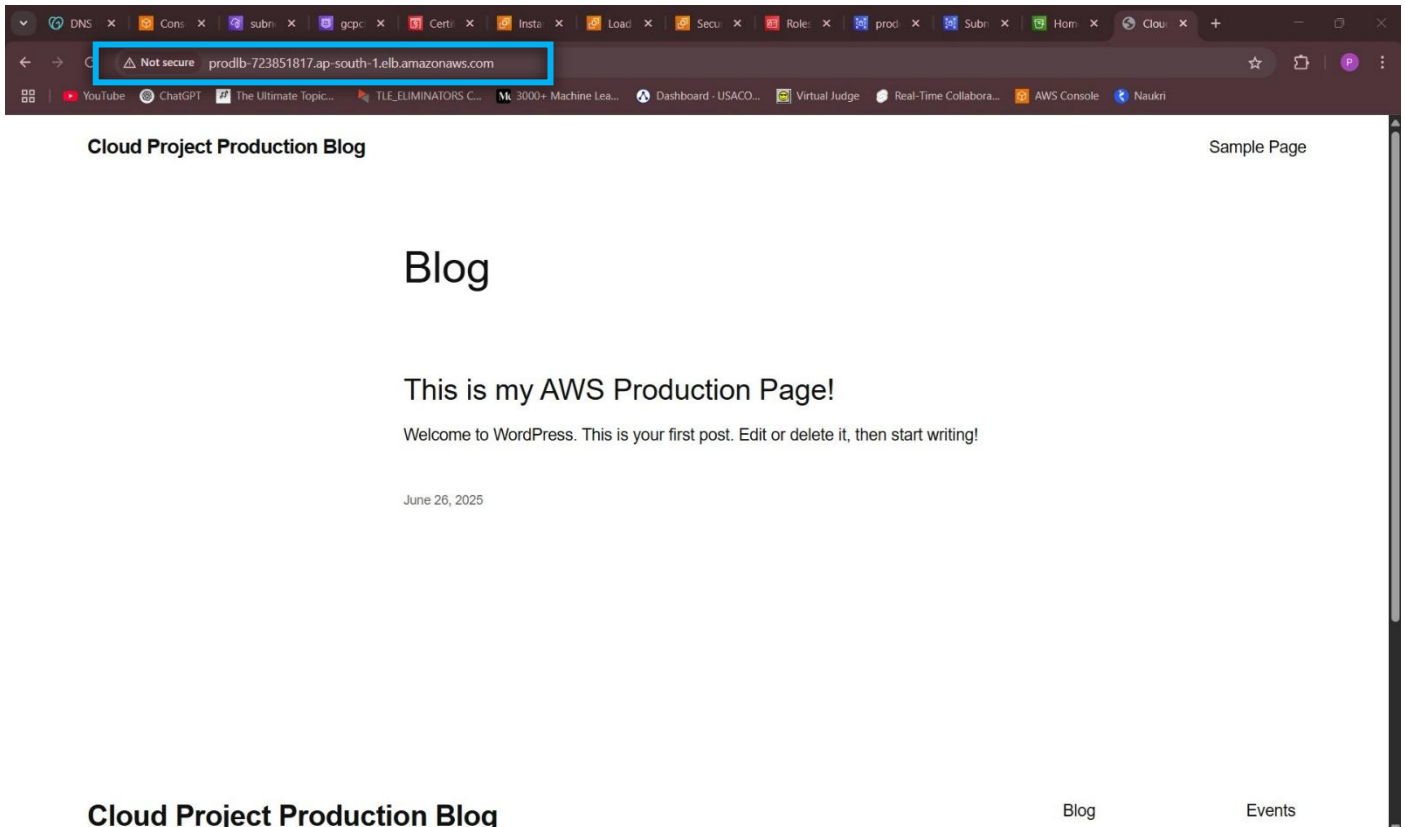
[ec2-user@ip-10-0-4-163 ~]$ sudo -i
[root@ip-10-0-4-163 ~]# crontab -e
no crontab for root - using an empty one
crontab: installing new crontab
[root@ip-10-0-4-163 ~]# crontab -l
*/2 * * * * aws s3 sync --delete s3://wordpress-mediaasset-project26062025 /var/www/html/wp-content/uploads
*/2 * * * * aws s3 sync --delete s3://wordpress-code-project26062025 /var/www/html/

[root@ip-10-0-4-163 ~]#
login as:

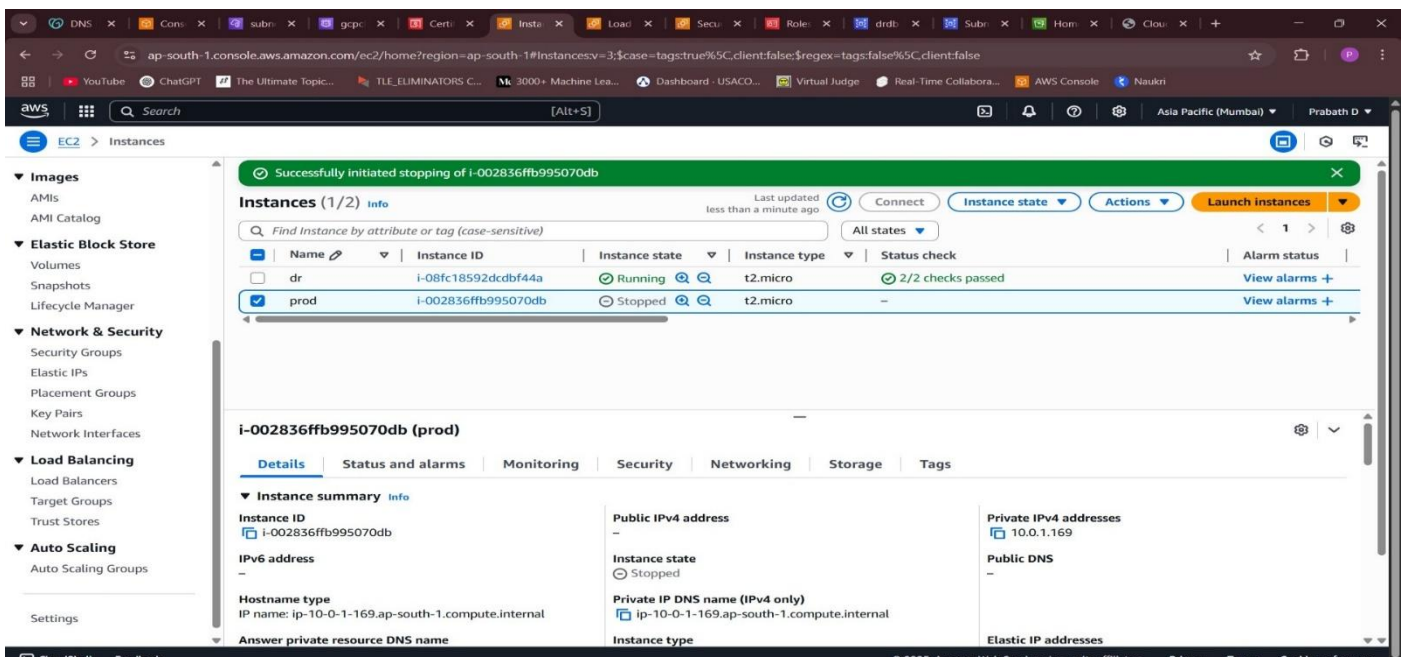
```

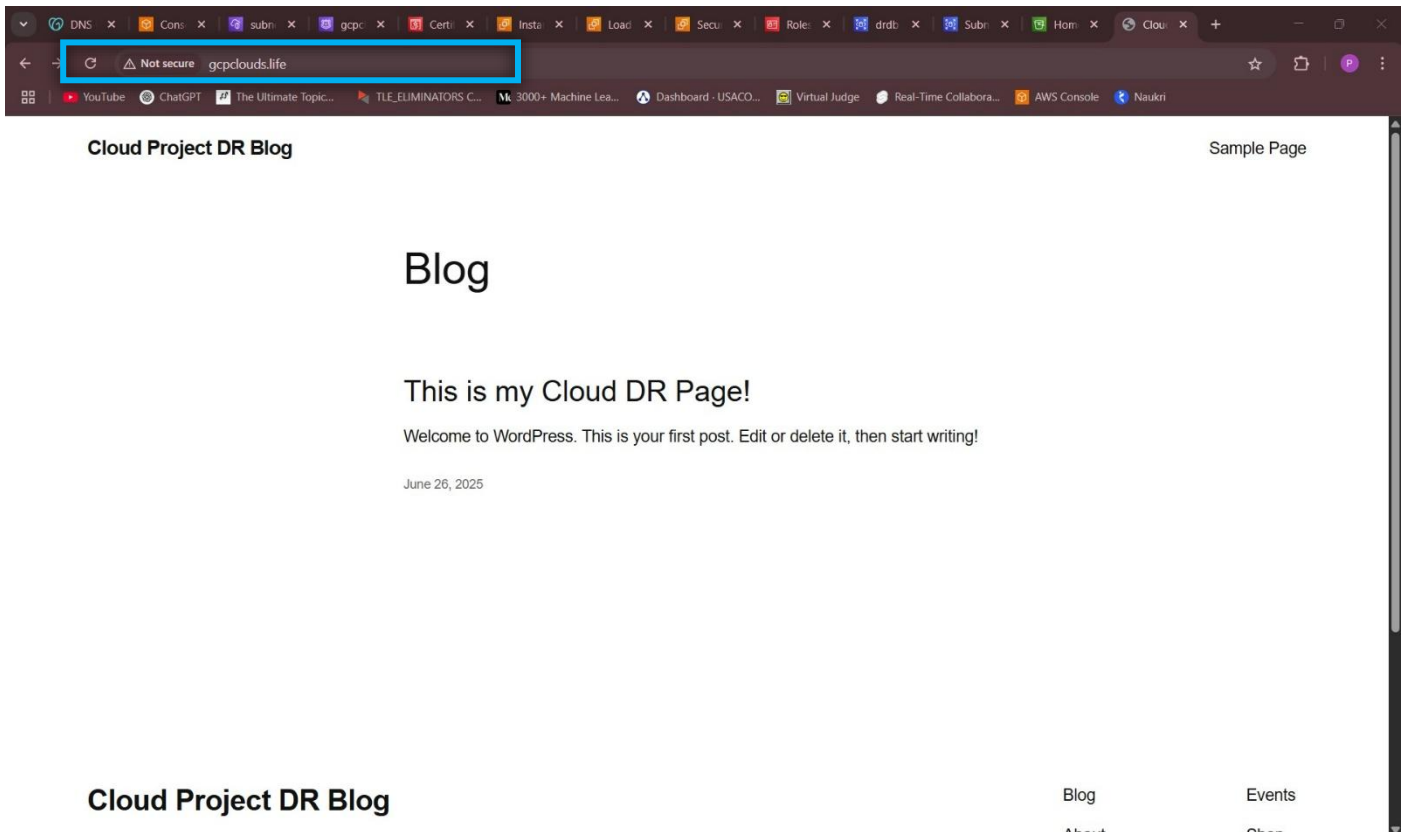
Crontab(DR)

- Now, wait for few minutes and then refresh the page.
- In this task, specifically the failover routing must be applied to view the Production page and DR page, if one availability zone is failed it redirects to the other zone, because of the custom VPC that is used in this.
- The production blog that is hosted in the web browser in the form of DNS name.



- Now, go to EC2 instances and stop the Production instance for some time, after 2-3 mins, the health check in the route53 becomes unhealthy
- With that, by refreshing the page production DNS name that is hosted in the web browser it will redirect to the DR DNS name.
- By giving the name as gcpcLOUDS.life it will show the DR Blog.





- Once, again if you start the instance of Production instance the web page will get refresh and shows the Production Blog
- The setup for deploying the wordpress page in one region is completed. Now, we must do the same for another region.

Ohio Region:

In this region, the implementation will slightly differ with the configuration. Other than this, the procedure to configure all the things is same.

Steps:

i. VPC Creation:

- Go to your VPC.
- Click your VPC, then click create VPC then give a name as “vpc-ohio”
- Then set the IP address 10.1.0.0/16, then set both DNS hostname and DNS resolution as enabled, then click create.

The screenshot displays the AWS Management Console interface for a VPC named **vpc-01be684b1008a4d02** in the **us-east-2** region, specifically in the **Ohio** Availability Zone. The console shows various configuration details and a resource map.

VPC Details:

- VPC ID:** vpc-01be684b1008a4d02
- State:** Available
- DNS resolution:** Enabled
- Main network ACL:** acl-0a19cdb12d7ea9c60
- IPv6 CIDR:** -
- Tenancy:** default
- Default VPC:** No
- Network Address Usage metrics:** Disabled
- Block Public Access:** Off
- DHCP option set:** dopt-047742a9f79b1e4b2
- IPv4 CIDR:** 10.1.0.0/16
- Route 53 Resolver DNS Firewall rule groups:** -
- DNS hostnames:** Disabled
- Main route table:** rtb-0d180f1dff8c24a3a
- IPv6 pool:** -
- Owner ID:** 767828764671

Resource map:

- VPC:** vpc-ohio
- Subnets (4):**
 - us-east-2a: ohio-pub-subnet-1
 - us-east-2b: ohio-pub-subnet-1
- Route tables (3):**
 - ohio-pvt-rtb
 - rtb-0d180f1dff8c24a3a
 - ohio-pub-rtb
- Network connections (2):**
 - igw-ohio
 - ngw-ohio

ii. Subnet Setup:

- Go to subnets, click create subnet after that assign a name for the subnets, in this task we need two subnets for public and private.
- Then enter the below details particularly for public and private subnet, by assigning the IP Addresses, then click create subnet
- Public Subnet 1: For EC2 + NAT
- Public Subnet 2: For EC2
- Private Subnet 1: For RDS Prod
- Private Subnet 2: For RDS DR

CIDR Blocks (Mumbai):

- 10.1.1.0/24 (Public subnet 1)
- 10.1.2.0/24 (Public subnet 2)
- 10.1.3.0/24 (Private subnet 1)
- 10.1.4.0/24 (Private subnet 2)

Domain P...Console H...Certificate...VPC | us-e...gcpclouds...Health ch...Instances...Load bal...drdb - Dai...Roles | IAM...+

us-east-2.console.aws.amazon.com/vpconsole/home?region=us-east-2#SubnetDetails:subnetId=subnet-0a7b1cbc0b0b8a1dd

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United States (Ohio)Prabath D

VPCSubnets subnet-0a7b1cbc0b0b8a1dd

subnet-0a7b1cbc0b0b8a1dd / ohio-pub-subnet-2

Actions

Details

Subnet ID

subnet-0a7b1cbc0b0b8a1dd

IPv4 CIDR

10.1.2.0/24

Availability Zone

us-east-2b

Network ACL

acl-0a19cdb12d7ea9c60

Auto-assign customer-owned IPv4 address

No

IPv6 CIDR reservations

-

Resource name DNS AAAA record

Disabled

Subnet ARN

arn:aws:ec2:us-east-2:767828764671:subnet/subnet-0a7b1cbc0b0b8a1dd

Available IPv4 addresses

251

Availability Zone ID

use2-az2

Default subnet

No

Customer-owned IPv4 pool

-

IPv6-only

No

DNS64

Disabled

State

Available

IPv6 CIDR

-

VPC

vpc-01be684b1008a4d02 | vpc-ohio

Auto-assign public IPv4 address

No

Outpost ID

-

Hostname type

IP name

Owner

767828764671

Block Public Access

Off

IPv6 CIDR association ID

-

Route table

rtb-0eeb04e2b5394309e | ohio-pub-rtb

Auto-assign IPv6 address

No

IPv4 CIDR reservations

-

Resource name DNS A record

Disabled

Flow logs

Route table

Network ACL

CIDR reservations

Sharing

Tags

Flow logs

Search

Actions

Create flow log

1

NameFlow log IDFilterDestination typeDestination nameIAM role ARN

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Domain P...Console H...Certificate...VPC | us-e...gcpclouds...Health ch...Instances...Load bal...drdb - Dai...Roles | IAM...+

us-east-2.console.aws.amazon.com/vpconsole/home?region=us-east-2#SubnetDetails:subnetId=subnet-0c88515d0f65bcb3b

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VPCSubnets subnet-0c88515d0f65bcb3b

subnet-0c88515d0f65bcb3b / ohio-pvt-subnet-1

Actions

Details

Subnet ID

subnet-0c88515d0f65bcb3b

IPv4 CIDR

10.1.3.0/24

Availability Zone

us-east-2b

Network ACL

acl-0a19cdb12d7ea9c60

Auto-assign customer-owned IPv4 address

No

IPv6 CIDR reservations

-

Resource name DNS AAAA record

Disabled

Subnet ARN

arn:aws:ec2:us-east-2:767828764671:subnet/subnet-0c88515d0f65bcb3b

Available IPv4 addresses

251

Availability Zone ID

use2-az2

Default subnet

No

Customer-owned IPv4 pool

-

IPv6-only

No

DNS64

Disabled

State

Available

IPv6 CIDR

-

VPC

vpc-01be684b1008a4d02 | vpc-ohio

Auto-assign public IPv4 address

No

Outpost ID

-

Hostname type

IP name

Owner

767828764671

Block Public Access

Off

IPv6 CIDR association ID

-

Route table

rtb-04329ce48d68a40f4 | ohio-pvt-rtb

Auto-assign IPv6 address

No

IPv4 CIDR reservations

-

Resource name DNS A record

Disabled

Flow logs

Route table

Network ACL

CIDR reservations

Sharing

Tags

Flow logs

Search

Actions

Create flow log

1

NameFlow log IDFilterDestination typeDestination nameIAM role ARN

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us-east-2.console.aws.amazon.com/vpcconsole/home?region=us-east-2#SubnetDetails:subnetId=subnet-0f0fc95f0c8c6ab38

subnets > subnet-0f0fc95f0c8c6ab38

subnet-0f0fc95f0c8c6ab38 / ohio-pvt-subnet-2

Details

Subnet ID
subnet-0f0fc95f0c8c6ab38

IPv4 CIDR
10.1.4.0/24

Availability Zone
us-east-2c

Network ACL
acl-0a19cdb12d7ea9c60

Auto-assign customer-owned IPv4 address
No

IPv6 CIDR reservations
-

Resource name DNS AAAA record
Disabled

Subnet ARN
arn:aws:ec2:us-east-2:767828764671:subnet/subnet-0f0fc95f0c8c6ab38

Available IPv4 addresses
251

Availability Zone ID
use2-az3

Default subnet
No

Customer-owned IPv4 pool
-

IPv6-only
No

DNS64
Disabled

State
Available

IPv6 CIDR
-

VPC
vpc-01be684b1008a4d02 | vpc-ohio

Auto-assign public IPv4 address
No

Outpost ID
-

Hostname type
IP name

Owner
767828764671

Block Public Access
Off**IPv6 CIDR association ID**
-**Route table**
rtb-04329ce48d68a40f4 | ohio-pvt-rtb**Auto-assign IPv6 address**
No**IPv4 CIDR reservations**
-**Resource name DNS A record**
Disabled**Flow logs** | **Route table** | **Network ACL** | **CIDR reservations** | **Sharing** | **Tags****Flow logs**

Search

Name

Flow log ID

Filter

Destination type

Destination name

IAM role ARN

< 1 >

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iii. Internet Gateway + NAT Gateway:

- Create and attach IGW in Ohio VPC.

us-east-2.console.aws.amazon.com/vpcconsole/home?region=us-east-2#InternetGateway:internetGatewayId=igw-009219a6d1df517e7

vpc > Internet gateways > igw-009219a6d1df517e7

igw-009219a6d1df517e7 / igw-ohio

Details Info

Internet gateway ID
igw-009219a6d1df517e7

State
Attached

VPC ID
vpc-01be684b1008a4d02 | vpc-ohio

Owner
767828764671

Tags

Search tags

Key

Value

Name igw-ohio

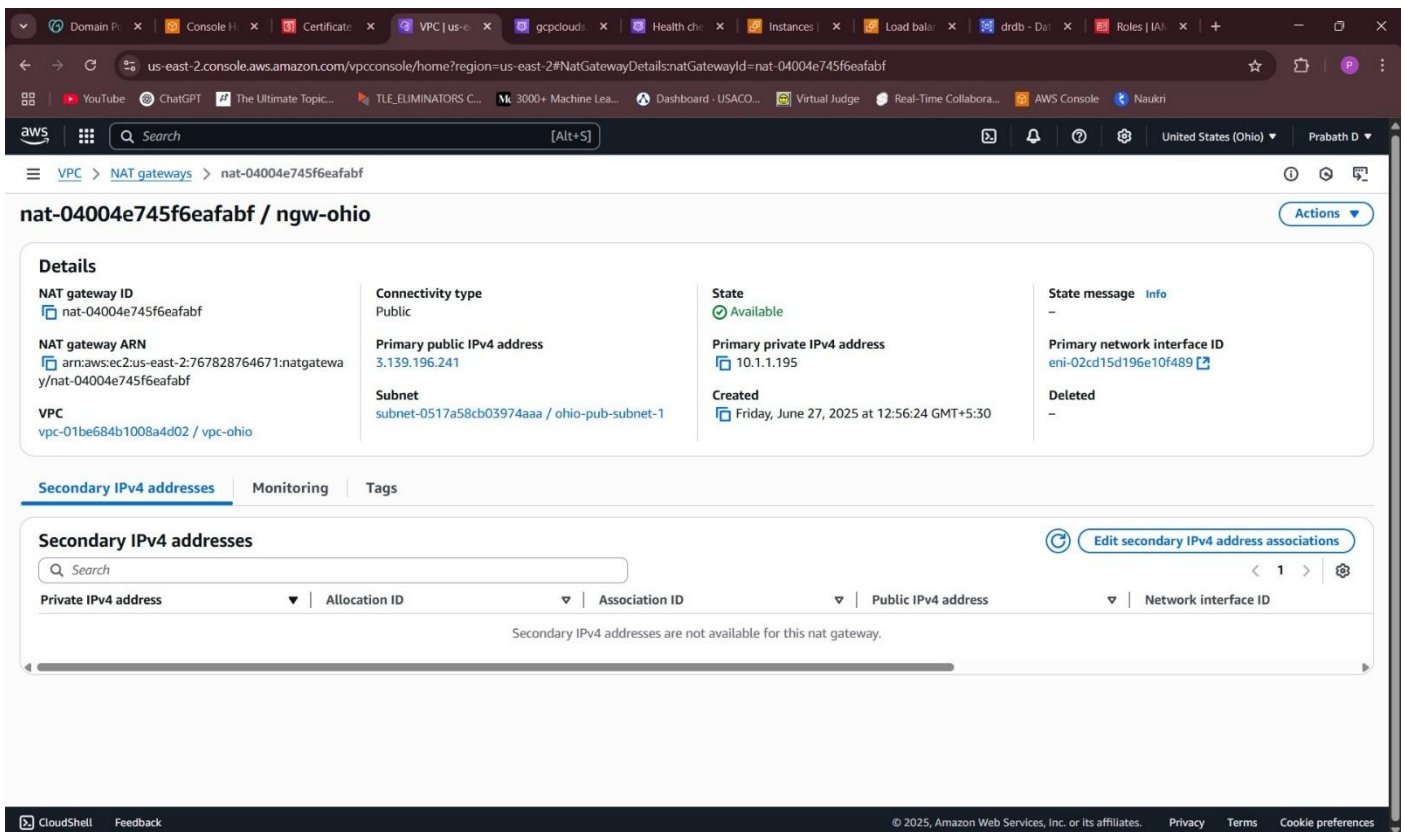
Manage tags

< 1 >

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- Create NAT Gateway in Public Subnet (with Elastic IP).



nat-04004e745f6eafabf / ngw-ohio

Details

NAT gateway ID nat-04004e745f6eafabf	Connectivity type Public	State Available	State message info -
NAT gateway ARN arn:aws:ec2:us-east-2:767828764671:natgateway/nat-04004e745f6eafabf	Primary public IPv4 address 3.139.196.241	Primary private IPv4 address 10.1.1.195	Primary network interface ID eni-02cd15d196e10f489
VPC vpc-01be684b1008a4d02 / vpc-ohio	Subnet subnet-0517a58cb03974aaa / ohio-pub-subnet-1	Created Friday, June 27, 2025 at 12:56:24 GMT+5:30	Deleted -

Secondary IPv4 addresses

Search

Private IPv4 address | Allocation ID | Association ID | Public IPv4 address | Network interface ID

Secondary IPv4 addresses are not available for this nat gateway.

iv. Route Tables (Mumbai):

- Then after creating both Internet Gateway and NAT Gateway, go to route tables, click create, then assign names for both public and private route tables, then associate both the tables by clicking routes then attach the network
- Public Route Table → 0.0.0.0/0 → IGW.
- Then click subnet associations in public route table, then click edit association after that click both public subnet 1 and public subnet 2.
- Private Route Table → 0.0.0.0/0 → NAT.
- Then click subnet associations in private route table, then click edit association after that click both private subnet 1 and private subnet 2.

Domain P x Console H x Certificate x VPC | us-e x gcpclouds x Health ch x Instances x Load bal x drdb - Dai x Roles | IA x +

us-east-2.console.aws.amazon.com/vpconsole/home?region=us-east-2#RouteTableDetails:RouteTableId=rtb-0eeb04e2b5394309e

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United States (Ohio) ▾ Prabath D ▾

VPC > Route tables > rtb-0eeb04e2b5394309e

rtb-0eeb04e2b5394309e / ohio-pub-rtb

Actions ▾

Details Info

Route table ID
rtb-0eeb04e2b5394309e

VPC
vpc-01be684b1008a4d02 | vpc-ohio

Main
No

Owner ID
767828764671

Explicit subnet associations
2 subnets

Edge associations
-

Routes Subnet associations Edge associations Route propagation Tags

Routes (2)

Filter routes

Both ▾ Edit routes

< 1 > ⚙

Destination	Target	Status	Propagated
0.0.0.0/0	igw-009219a6d1df517e7	Active	No
10.1.0.0/16	local	Active	No

Domain P x Console H x Certificate x VPC | us-e x gcpclouds x Health ch x Instances x Load bal x drdb - Dai x Roles | IA x +

us-east-2.console.aws.amazon.com/vpconsole/home?region=us-east-2#RouteTableDetails:RouteTableId=rtb-0eeb04e2b5394309e

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aws Search [Alt+S]

United States (Ohio) ▾ Prabath D ▾

VPC > Route tables > rtb-0eeb04e2b5394309e

rtb-0eeb04e2b5394309e / ohio-pub-rtb

Actions ▾

Details Info

Route table ID
rtb-0eeb04e2b5394309e

VPC
vpc-01be684b1008a4d02 | vpc-ohio

Main
No

Owner ID
767828764671

Explicit subnet associations
2 subnets

Edge associations
-

Routes Subnet associations Edge associations Route propagation Tags

Explicit subnet associations (2)

Find subnet association

Edit subnet associations

< 1 > ⚙

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
ohio-pub-subnet-1	subnet-0517a58cb03974aaa	10.1.1.0/24	-
ohio-pub-subnet-2	subnet-0a7b1cbc0b0b8a1dd	10.1.2.0/24	-

Subnets without explicit associations (0)

The following subnets have not been explicitly associated with any route tables and are therefore associated with the main route table:

Find subnet association

Edit subnet associations

< 1 > ⚙

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
------	-----------	-----------	-----------

No subnets without explicit associations
All your subnets are associated with a route table.

Domain P...Console H...Certificate...VPC | us-e...gcpcloud...Health ch...Instances...Load bal...drdb - Da...Roles | IA...+

us-east-2.console.aws.amazon.com/vpconsole/home?region=us-east-2#RouteTableDetails:RouteTableId=rtb-04329ce48d68a40f4

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VPCRoute tablesrtb-04329ce48d68a40f4

rtb-04329ce48d68a40f4 / ohio-pvt-rtb

DetailsInfo

Route table ID
rtb-04329ce48d68a40f4

VPC
vpc-01be684b1008a4d02 | vpc-ohio

Main
No

Owner ID
767828764671

Explicit subnet associations
2 subnets

Edge associations
-

RoutesSubnet associationsEdge associationsRoute propagationTags

Routes (2)

Filter routes

Destination	Target	Status	Propagated
0.0.0.0/0	nat-04004e745f6eafabf	Active	No
10.1.0.0/16	local	Active	No

CloudShellFeedback

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Domain P...Console H...Certificate...VPC | us-e...gcpcloud...Health ch...Instances...Load bal...drdb - Da...Roles | IA...+

us-east-2.console.aws.amazon.com/vpconsole/home?region=us-east-2#RouteTableDetails:RouteTableId=rtb-04329ce48d68a40f4

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United States (Ohio)Prabath D

VPCRoute tablesrtb-04329ce48d68a40f4

rtb-04329ce48d68a40f4 / ohio-pvt-rtb

DetailsInfo

Route table ID
rtb-04329ce48d68a40f4

VPC
vpc-01be684b1008a4d02 | vpc-ohio

Main
No

Owner ID
767828764671

Explicit subnet associations
2 subnets

Edge associations
-

RoutesSubnet associationsEdge associationsRoute propagationTags

Explicit subnet associations (2)

Find subnet association

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
ohio-pvt-subnet-1	subnet-0c88515d0f65bcb3b	10.1.3.0/24	-
ohio-pvt-subnet-2	subnet-0f0fc95f0c8c6ab38	10.1.4.0/24	-

Subnets without explicit associations (0)

Find subnet association

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
------	-----------	-----------	-----------

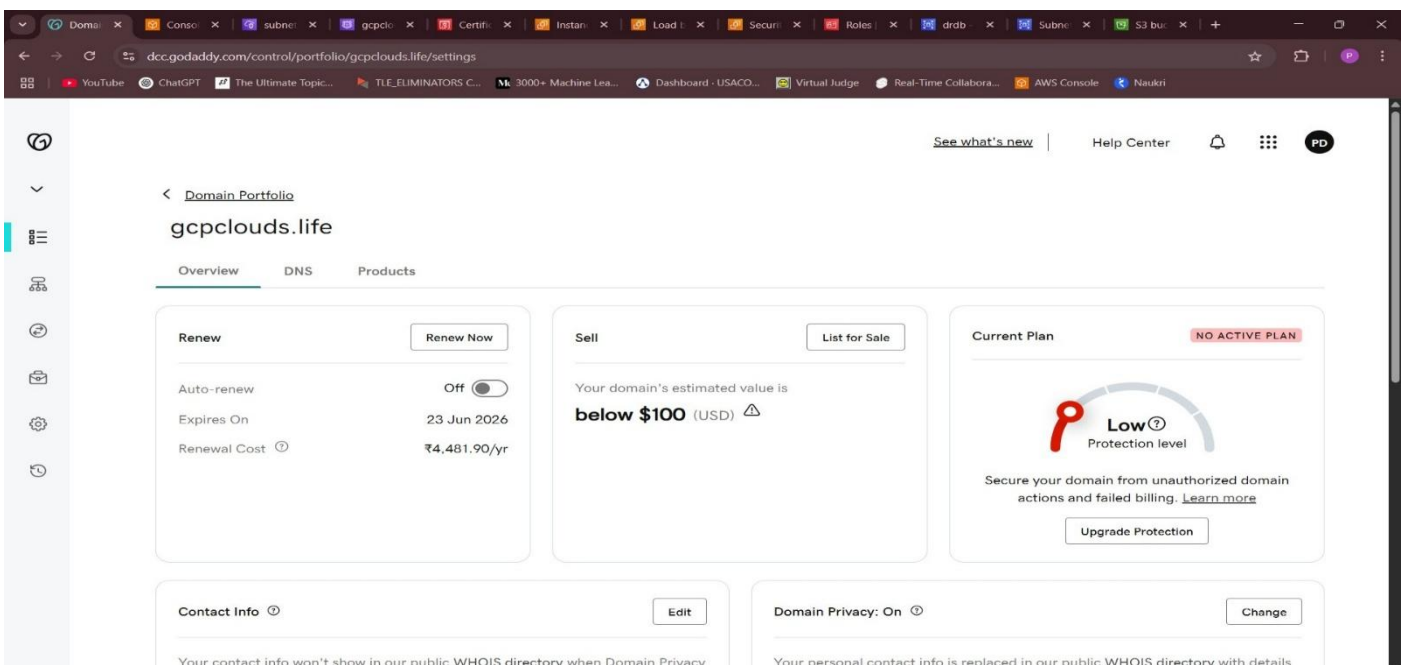
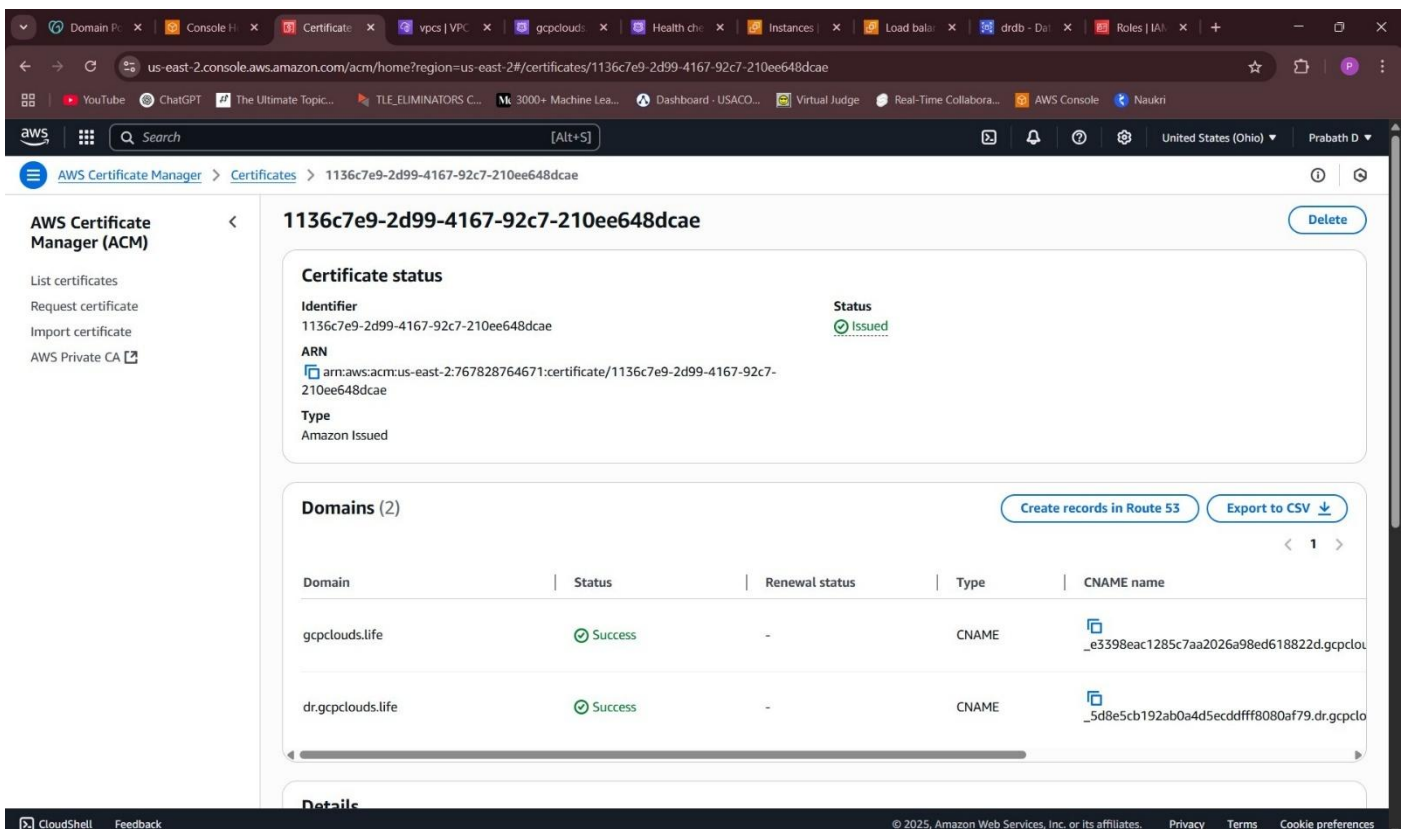
No subnets without explicit associations
All your subnets are associated with a route table.

CloudShellFeedback

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v. AWS Certificate Manager Creation:

- Request public certificate as per the registered domain.
- Go to ACM click request, then click public then name it with gcpclouds.life, then click add another certificate in that name it as 'dr.gcpclouds.life'.
- Then click create, after the creation records click create records in Route53, then after selecting both SSL certificate that click create.
- Then refresh the Route53 after the creation of hosted zone, it would be appearing with CNAME and CVALUE.
- Purchased a domain in GoDaddy named with 'gcpclouds.life'.



vi. AWS Route53 setup & link to GoDaddy domain DNS:

- Create hosted zone.
- Map the nameservers under GoDaddy domain.
- Create records set from ACM to R53.

The screenshot shows the AWS Route53 console for the hosted zone 'gcpclouds.life'. The 'Records (6)' tab is selected, displaying a table of DNS records. The record for 'gcpclouds.life' with Type 'NS' and Value 'ns-1350.awsdns-40.org, ns-475.awsdns-59.com, ns-953.awsdns-55.net, ns-1715.awsdns-22.co.uk' is highlighted with a blue box.

Record name	Type	Routing	Difference	Alias	Value/Route traffic to	TTL (s)	Health check ID	Evaluation
gcpclouds.life	A	Failover	Secondary	Yes	dualstack.dri1b-1371757702.us-east-2...	-	-	Yes
gcpclouds.life	A	Failover	Primary	Yes	dualstack.prod1b-491463309.us-east-...	-	e3b55fed-1f8f-4f86-91e5-7e58f716781b	Yes
gcpclouds.life	NS	Simple	-	No	ns-1350.awsdns-40.org, ns-475.awsdns-59.com, ns-953.awsdns-55.net, ns-1715.awsdns-22.co.uk	172800	-	-
gcpclouds.life	SOA	Simple	-	No	ns-1350.awsdns-40.org, awsdns-host...	900	-	-
_e3398eac12...	CNAME	Simple	-	No	_3f2f397fe2aaa5510a68b7e173eb19...	300	-	-
_5d8e5cb19...	CNAME	Simple	-	No	_00c154d35191125a3fb993ccb0669b...	300	-	-

The screenshot shows the GoDaddy Nameservers settings page. The 'Nameservers' tab is selected, and the 'Using custom nameservers' option is chosen. The custom nameservers listed are: ns-1350.awsdns-40.org, ns-475.awsdns-59.com, ns-953.awsdns-55.net, and ns-1715.awsdns-22.co.uk.

Nameservers determine where your DNS is hosted and where you add, edit or delete your DNS records.

Using custom nameservers Change Nameservers

Nameservers ⓘ

ns-1350.awsdns-40.org

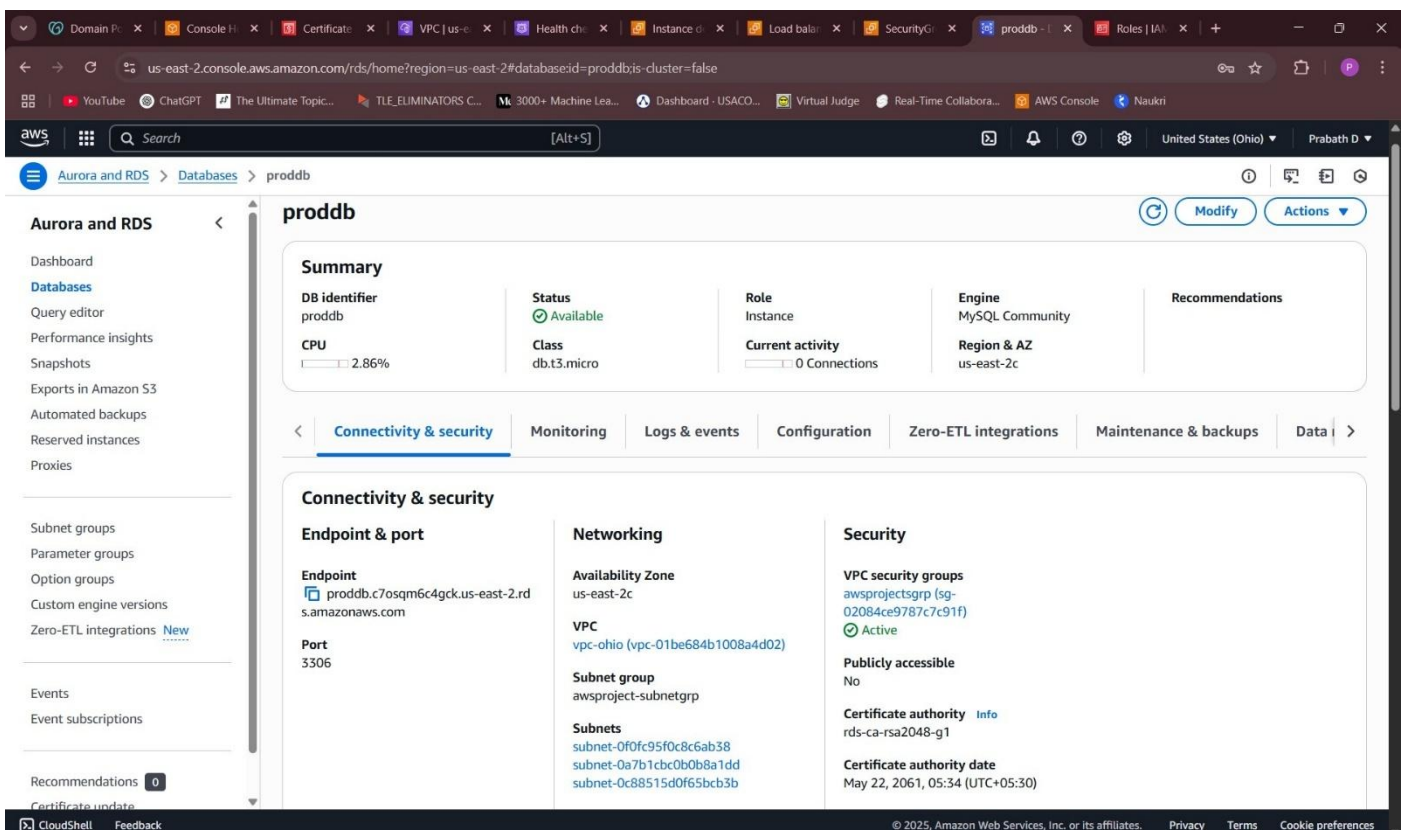
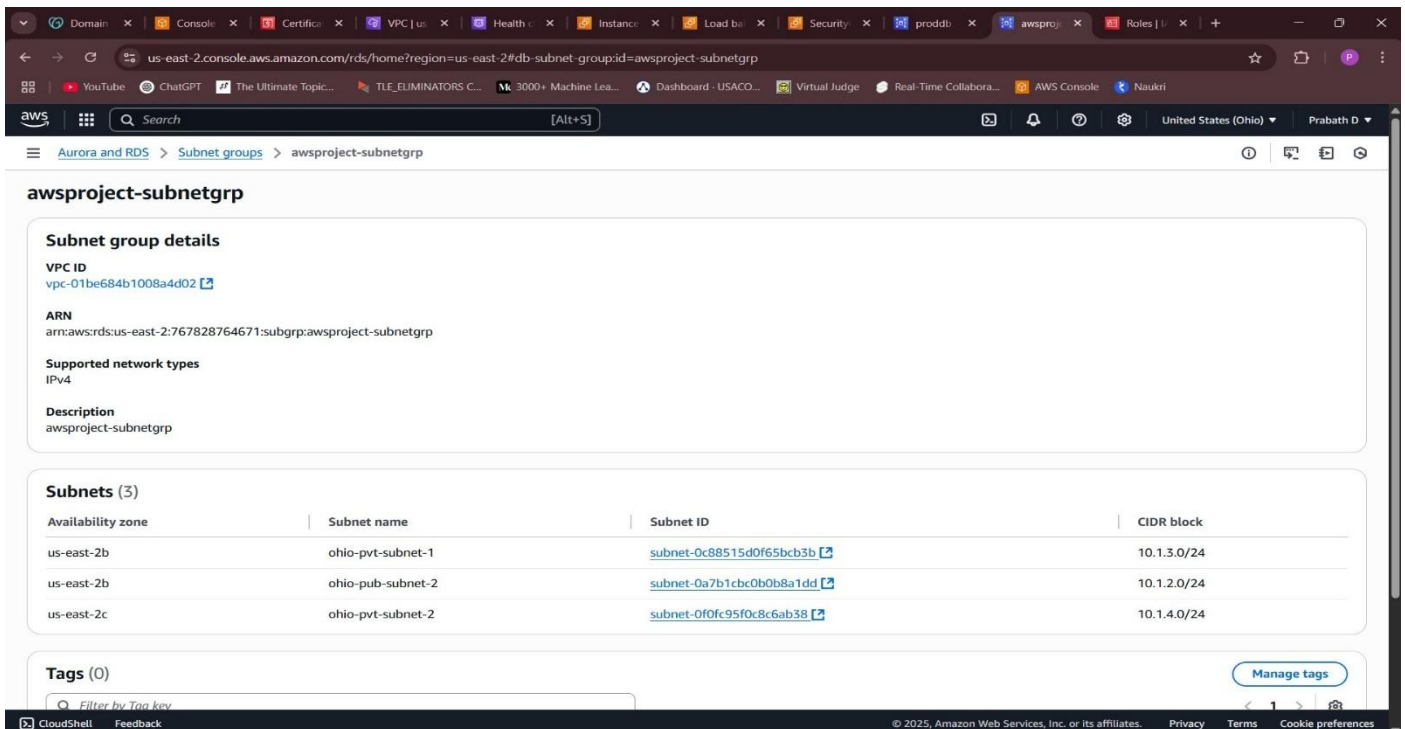
ns-475.awsdns-59.com

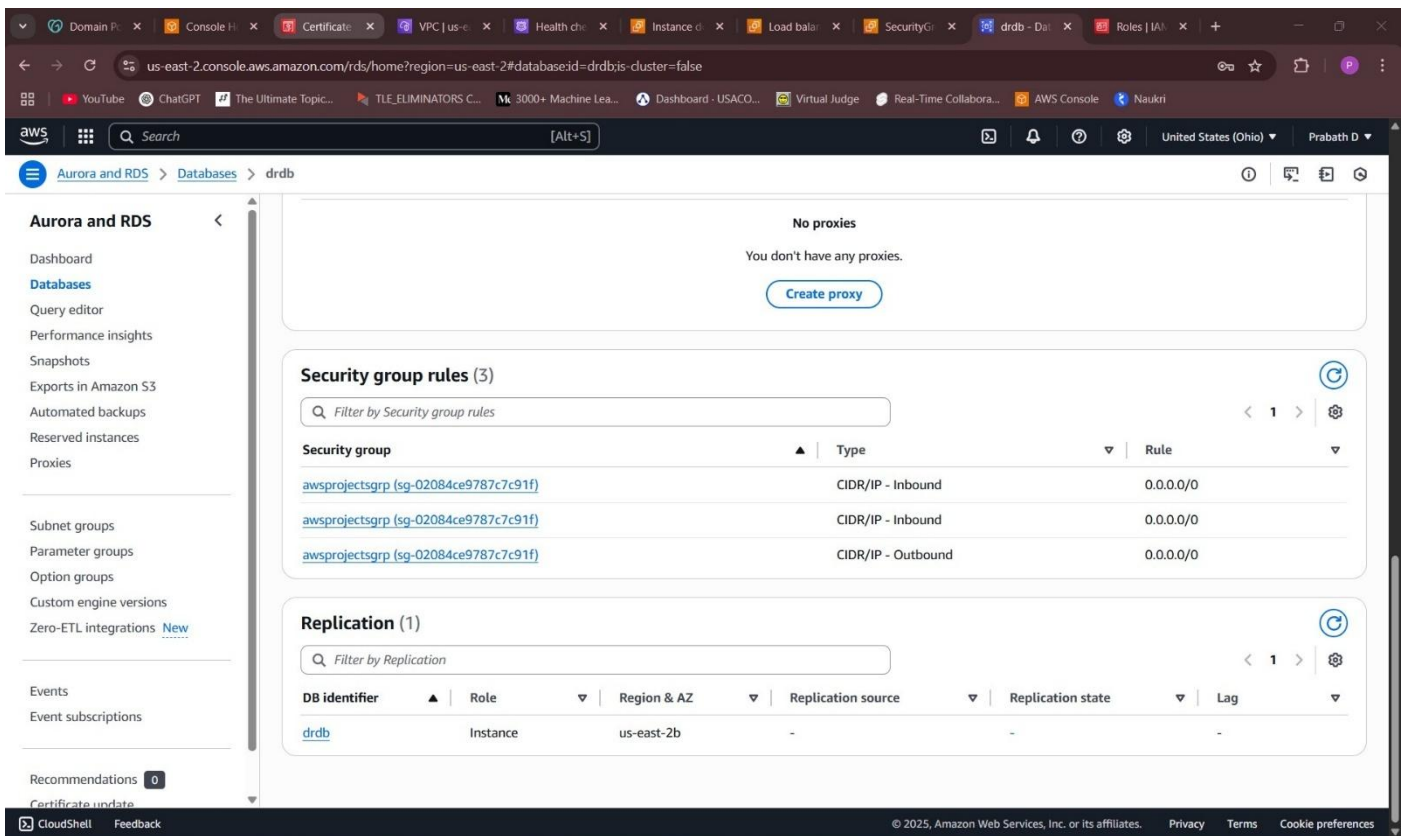
ns-953.awsdns-55.net

ns-1715.awsdns-22.co.uk

vii. AWS RDS Setup:

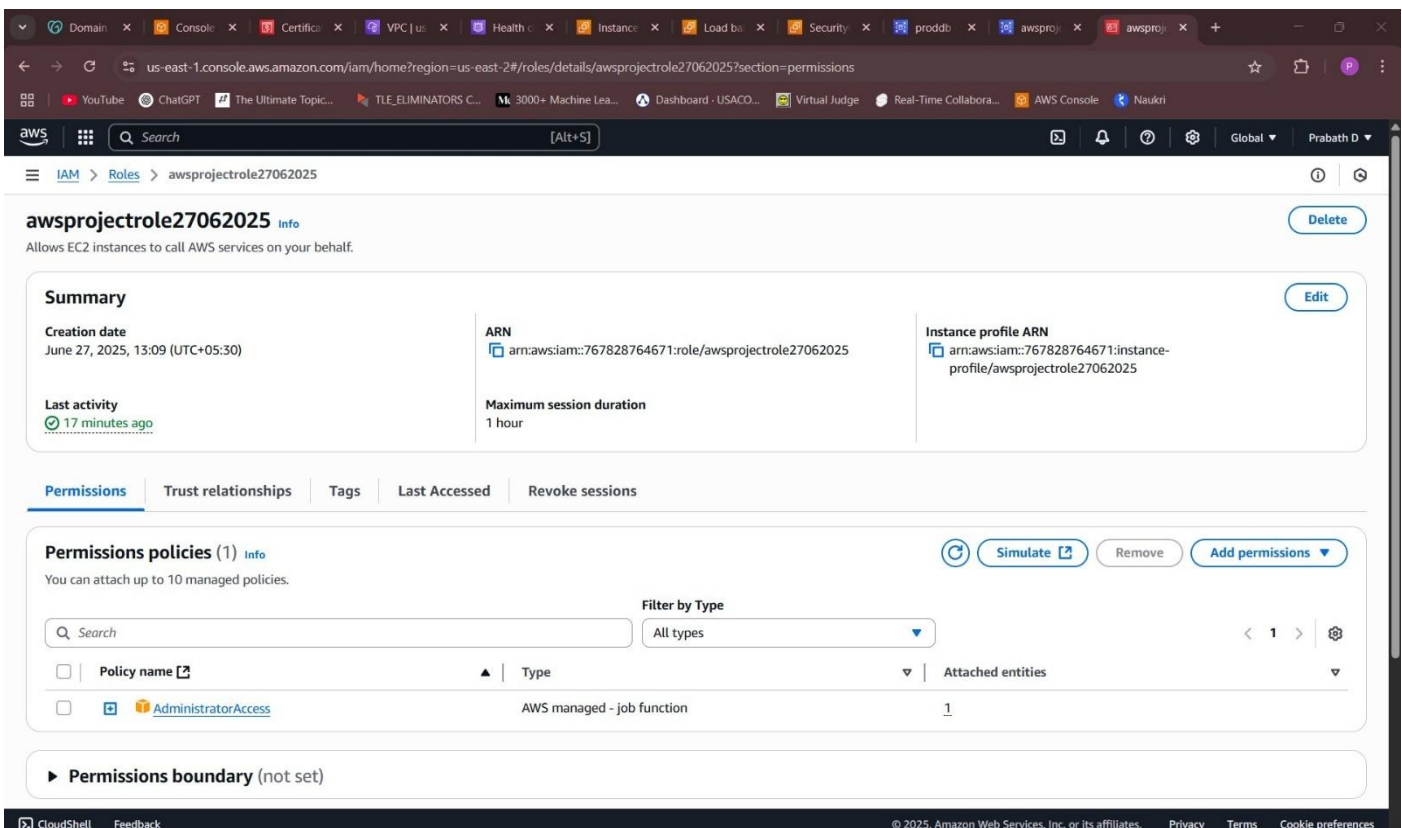
- Create Two RDS
 - i. New Security Group
 - ii. New Subnet Group
 - iii. PRODUCTION DB (SECURITY GROUP MAKE AS ALL TRAFFIC)
 - iv. DR DB (SECURITY GROUP MAKE AS ALL TRAFFIC)





viii. AWS IAM Role:

- Create an IAM role with full access.



ix. AWS EC2 Creation:

- Create two EC2 with IAM roles enables & add user data under advanced section
- Two instances need to setup for Mumbai region and adding the details in advanced section.

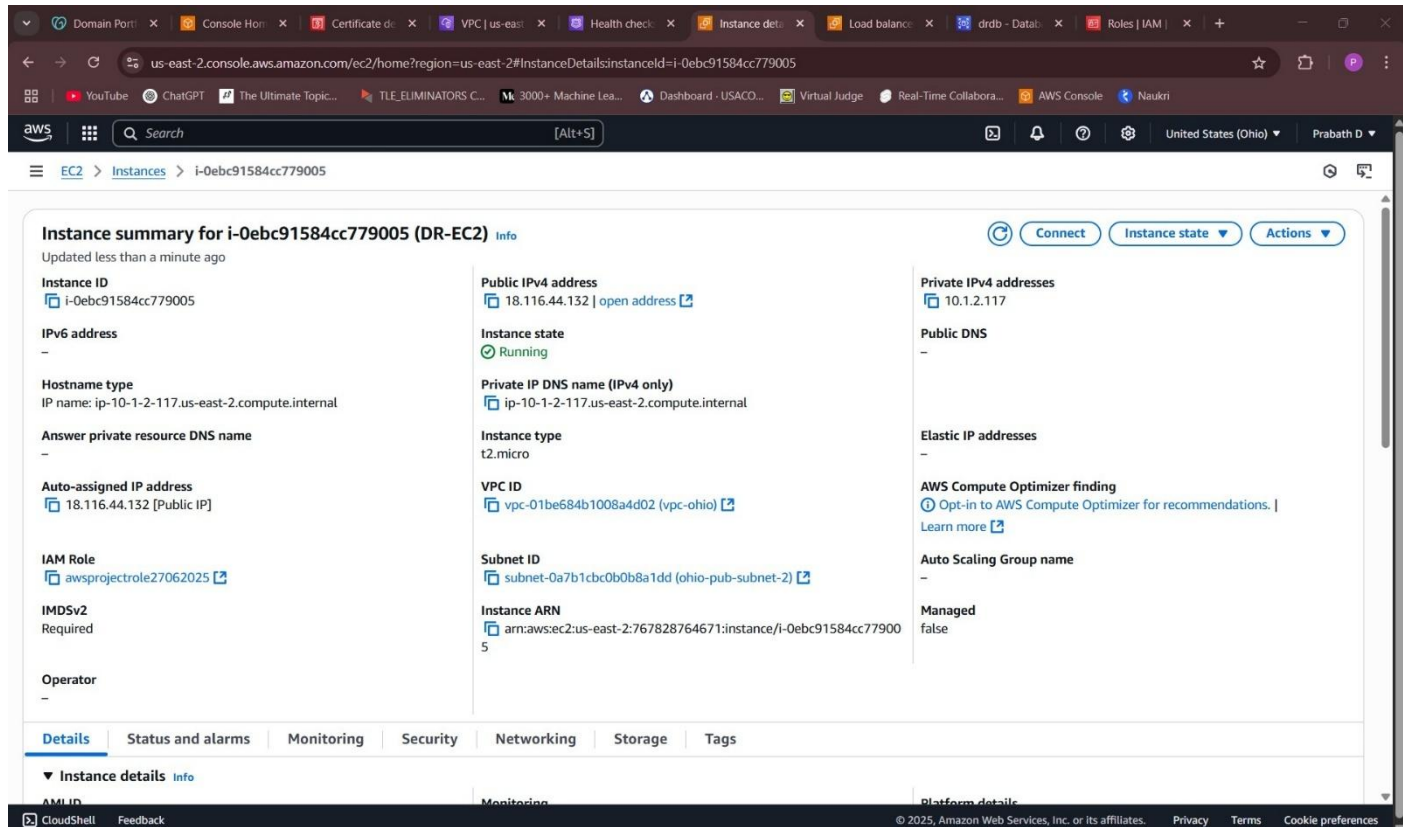
User data:

```
#!/bin/bash
yum install httpd php-mysql -y
amazon-linux-extras install -y php7.3
cd /var/www/html
echo "healthy" > healthy.html
wget https://wordpress.org/latest.tar.gz
tar -xzf latest.tar.gz
cp -r wordpress/* /var/www/html/
rm -rf wordpress
rm -rf latest.tar.gz
chmod -R 755 wp-content
chown -R apache:apache wp-content
wget https://s3.amazonaws.com/bucketforwordpresslab-donotdelete/htaccess.txt
mv htaccess.txt .htaccess
chkconfig httpd on
service httpd start
```

- This is the data need to be added in both EC2 instances.

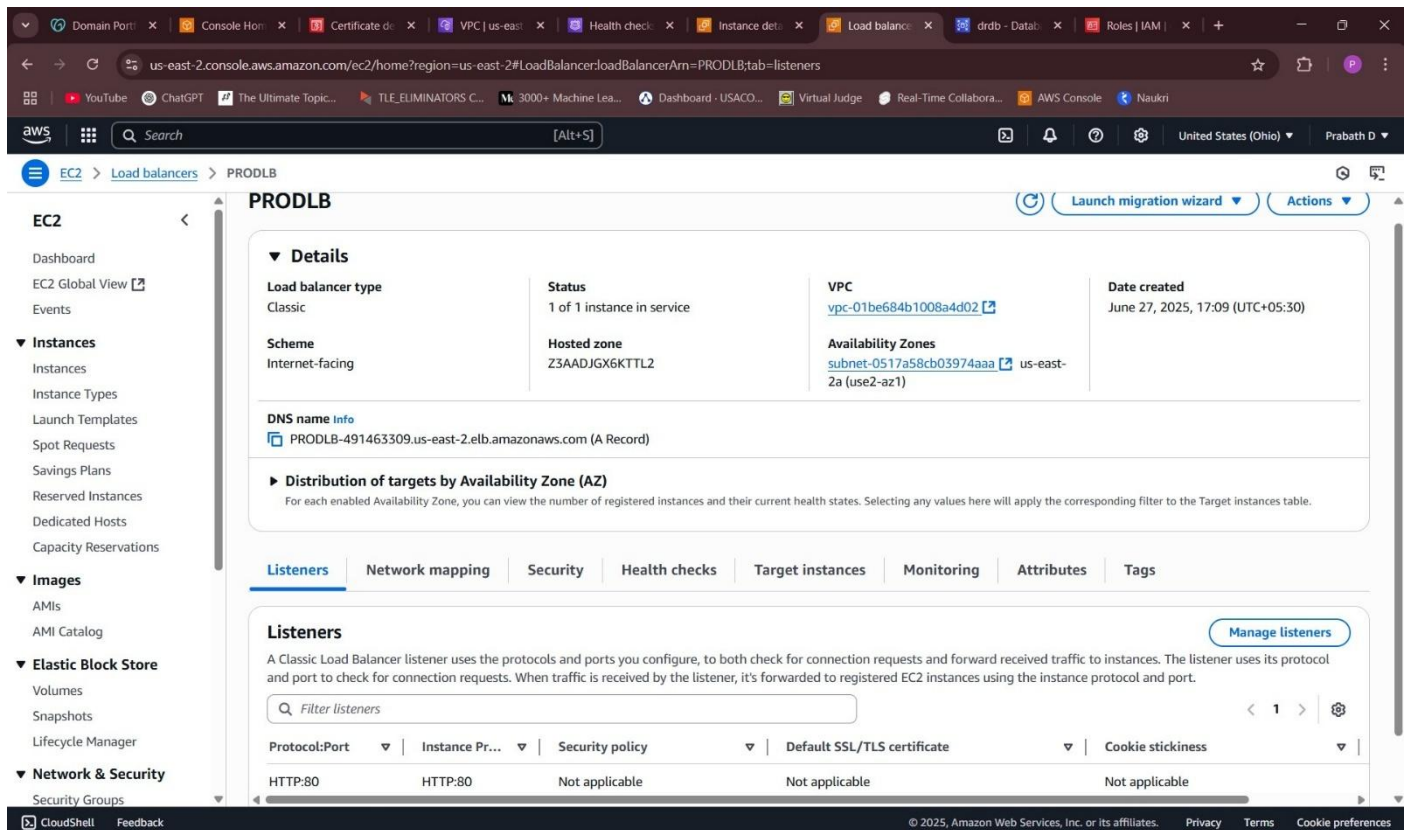
The screenshot displays the AWS Management Console interface for an EC2 instance. The browser address bar shows the URL: `us-east-2.console.aws.amazon.com/ec2/home?region=us-east-2#InstanceDetails:instanceId=i-0de70d6ce0c7738af`. The console header includes the AWS logo, a search bar, and navigation links for various services like VPC, IAM, and Load Balancing. The main content area is titled "Instance summary for i-0de70d6ce0c7738af (PROD-EC2)" and includes tabs for "Details", "Status and alarms", "Monitoring", "Security", "Networking", "Storage", and "Tags". The "Details" tab is active, showing a table of instance information. The instance is in a "Running" state, using a "t2.micro" instance type, and is associated with the "awsprojectrole27062025" IAM role. The public IPv4 address is 3.141.28.126, and the private IPv4 address is 10.1.1.79. The instance is part of a VPC named "vpc-01be684b1008a4d02" and is associated with the subnet "subnet-0517a58cb03974aaa". The instance ARN is "arn:aws:ec2:us-east-2:767828764671:instance/i-0de70d6ce0c7738af".

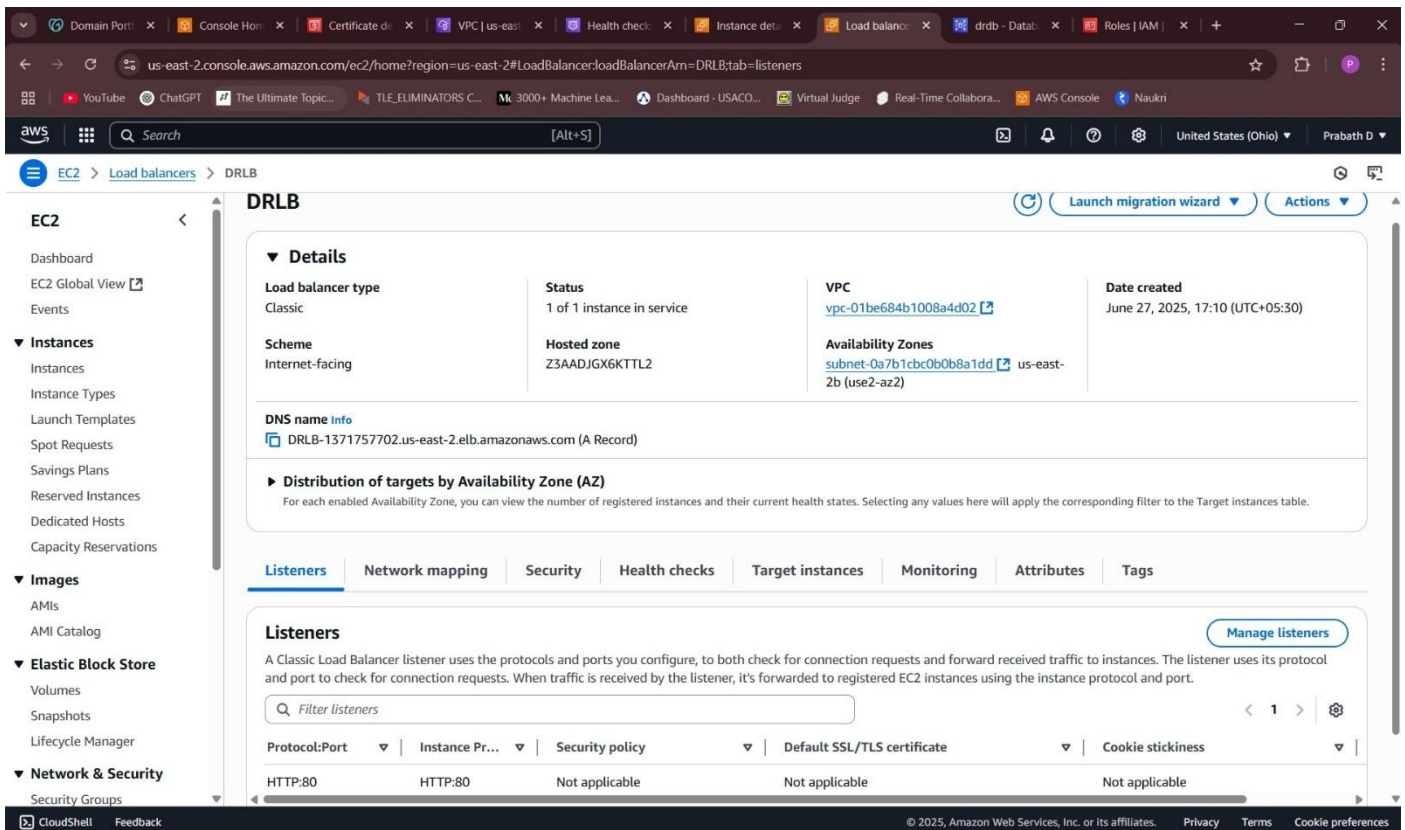
Instance summary for i-0de70d6ce0c7738af (PROD-EC2)		
Instance ID i-0de70d6ce0c7738af	Public IPv4 address 3.141.28.126 open address	Private IPv4 addresses 10.1.1.79
IPv6 address -	Instance state Running	Public DNS -
Hostname type IP name: ip-10-1-1-79.us-east-2.compute.internal	Private IP DNS name (IPv4 only) ip-10-1-1-79.us-east-2.compute.internal	Elastic IP addresses -
Answer private resource DNS name -	Instance type t2.micro	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendations. Learn more
Auto-assigned IP address 3.141.28.126 [Public IP]	VPC ID vpc-01be684b1008a4d02 (vpc-ohio)	Auto Scaling Group name -
IAM Role awsprojectrole27062025	Subnet ID subnet-0517a58cb03974aaa (ohio-pub-subnet-1)	Managed false
IMDSv2 Required	Instance ARN arn:aws:ec2:us-east-2:767828764671:instance/i-0de70d6ce0c7738af	
Operator -		



x. AWS ELB Creation & Setup:

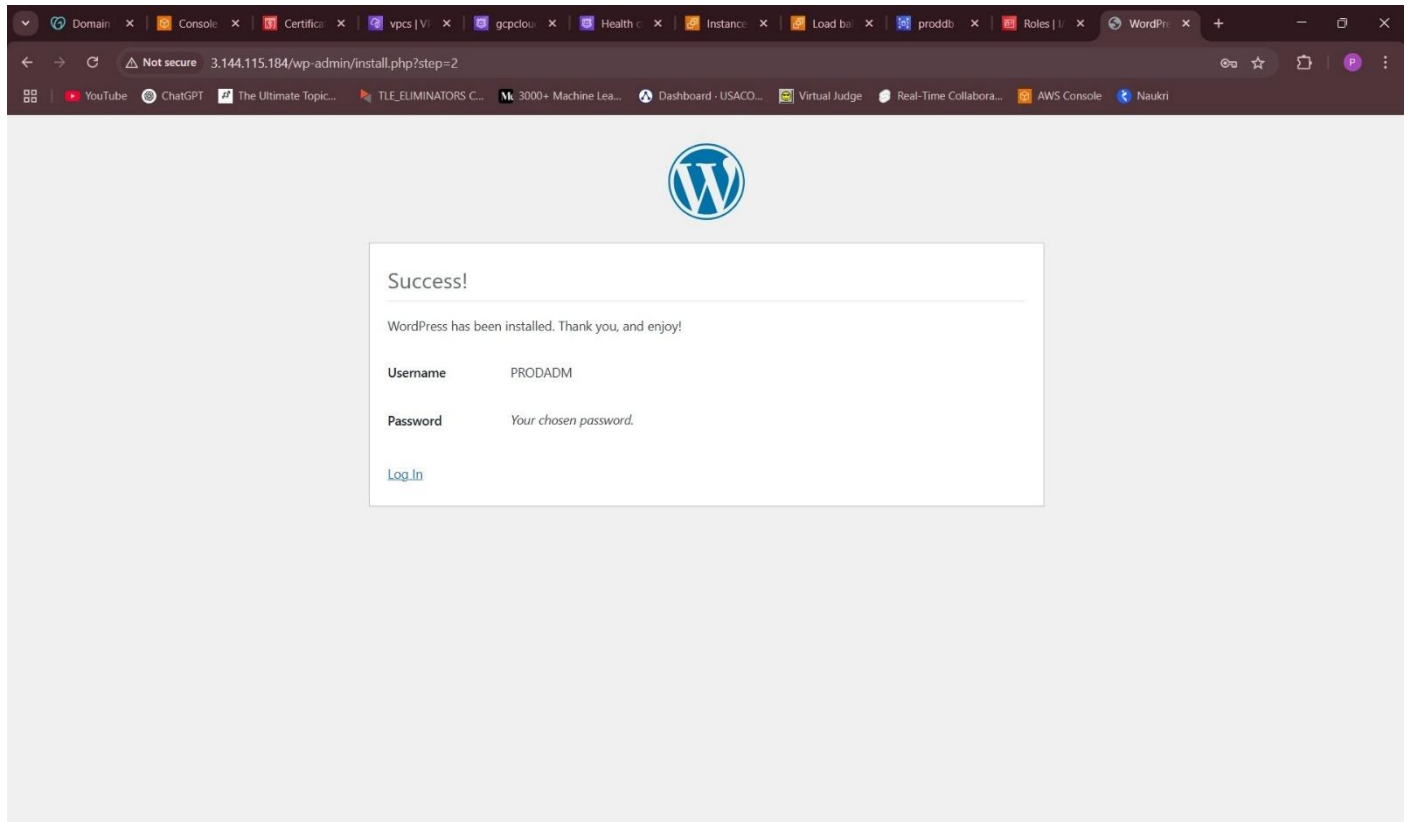
- Create two Classic ELB and map appropriate EC2 to it.

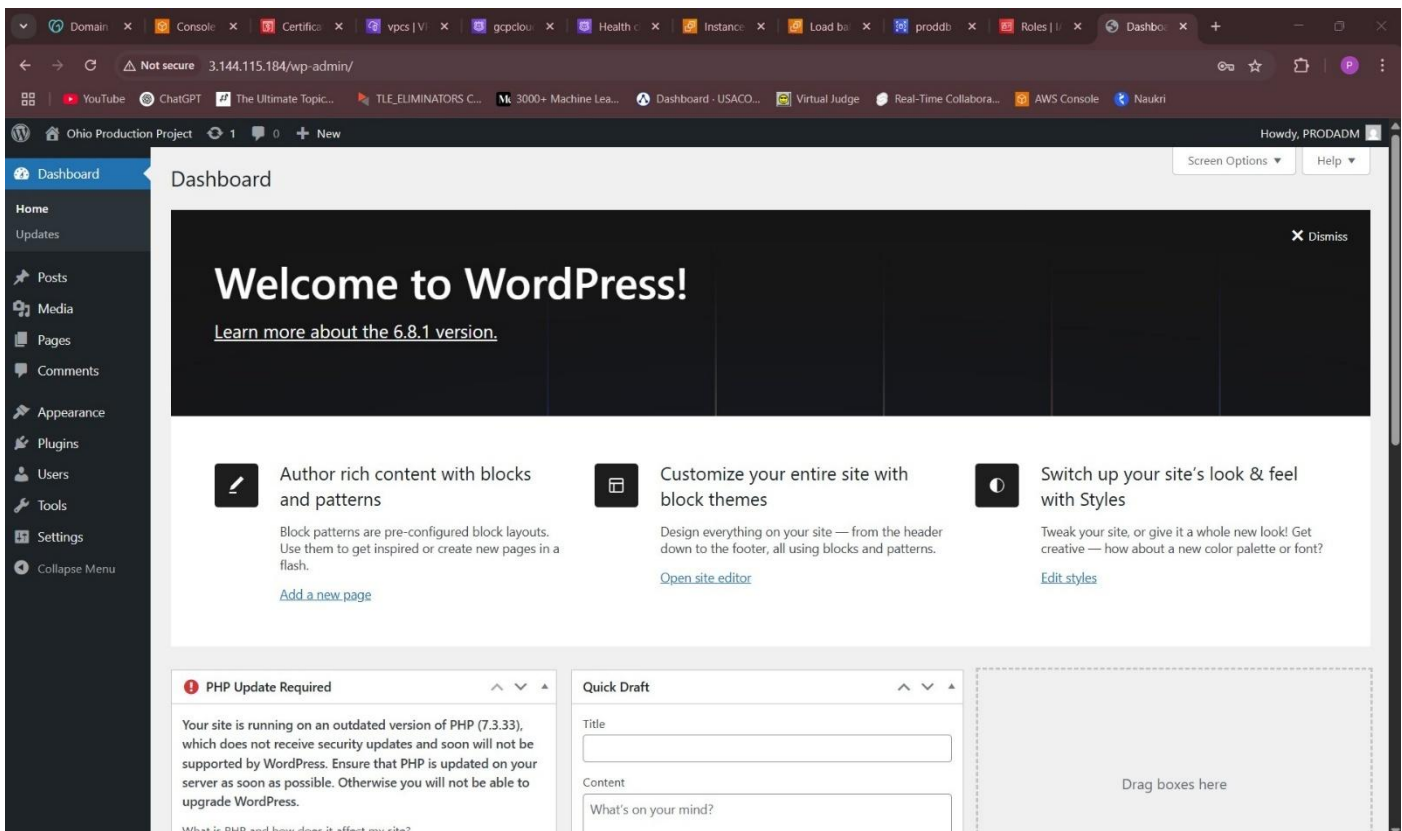
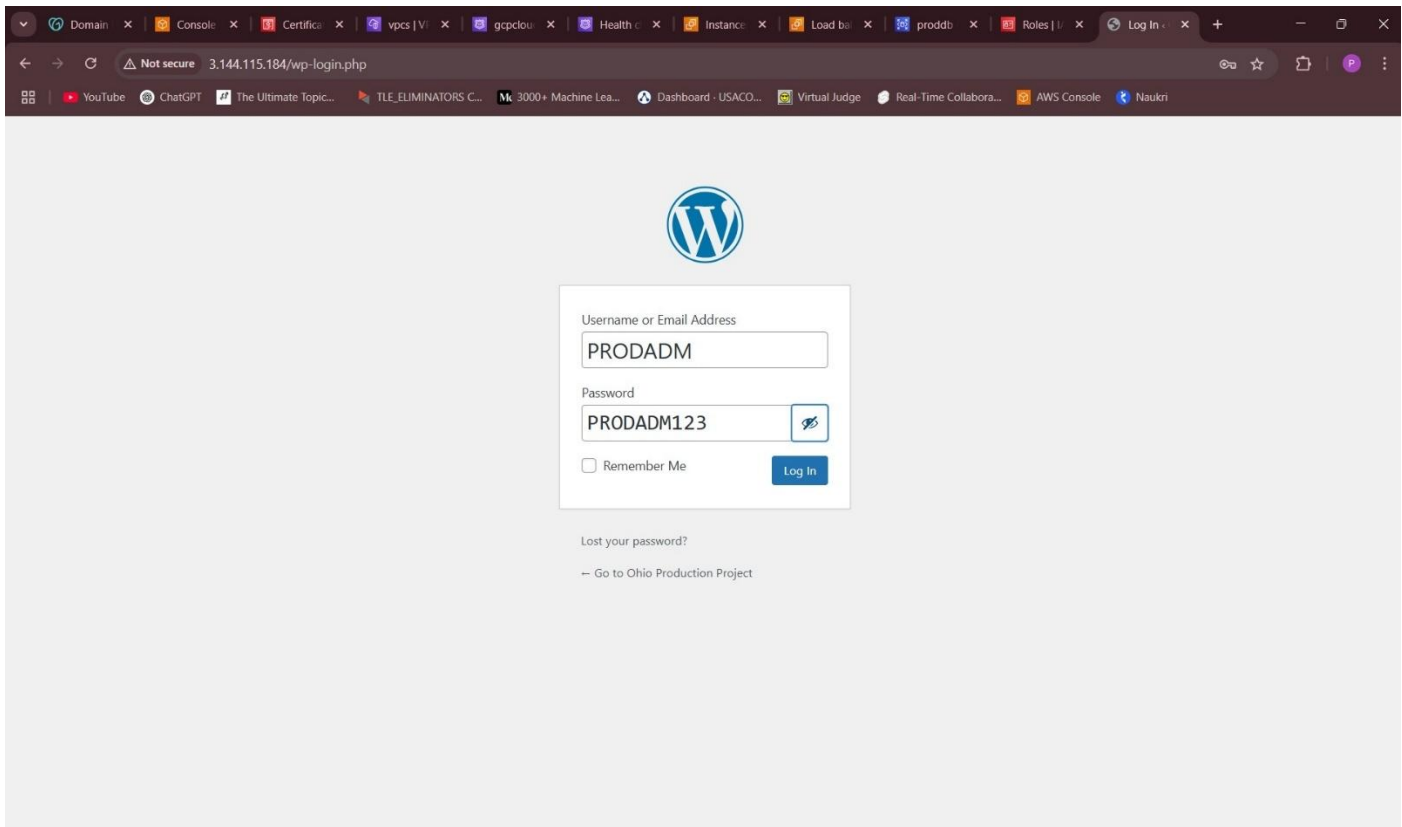


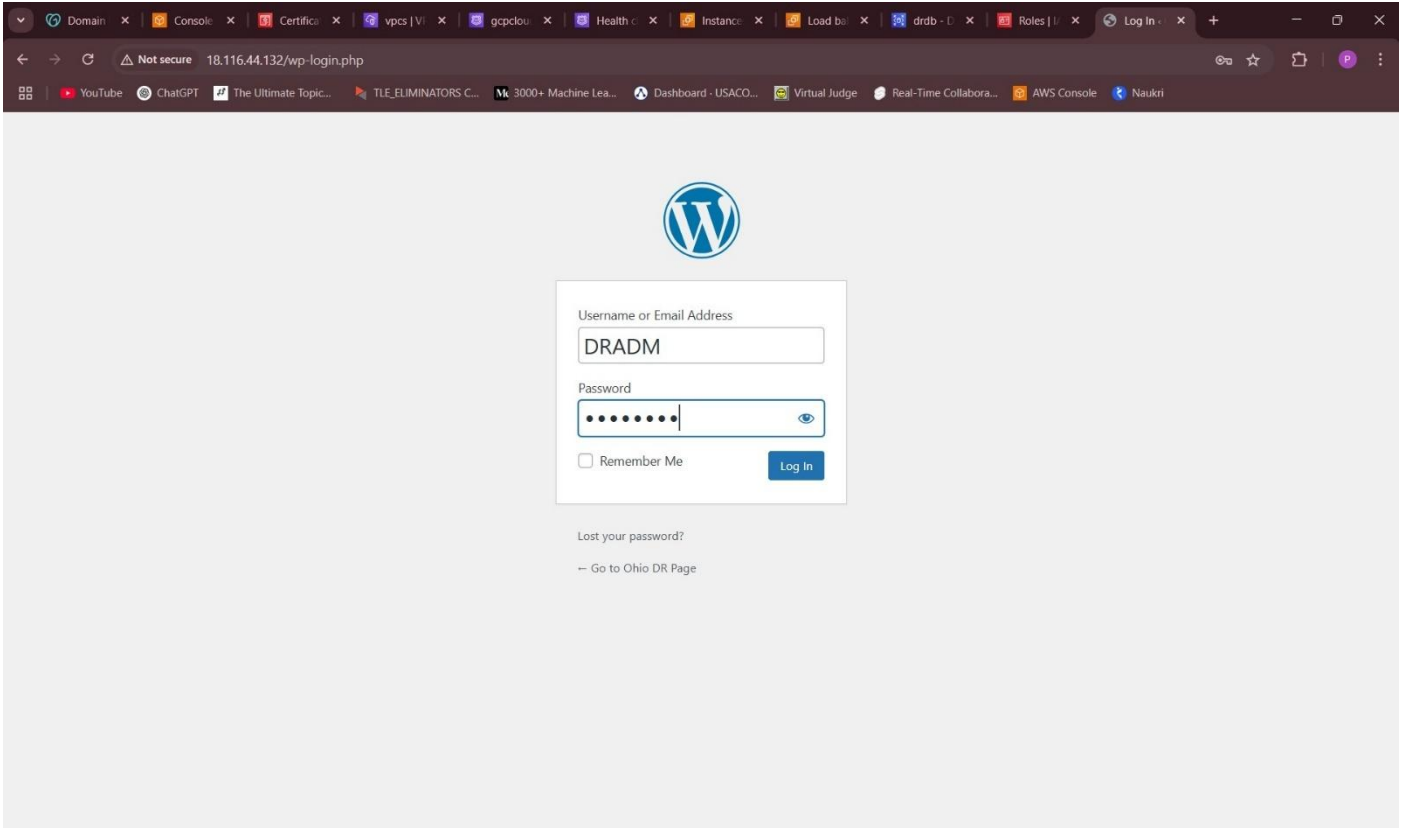
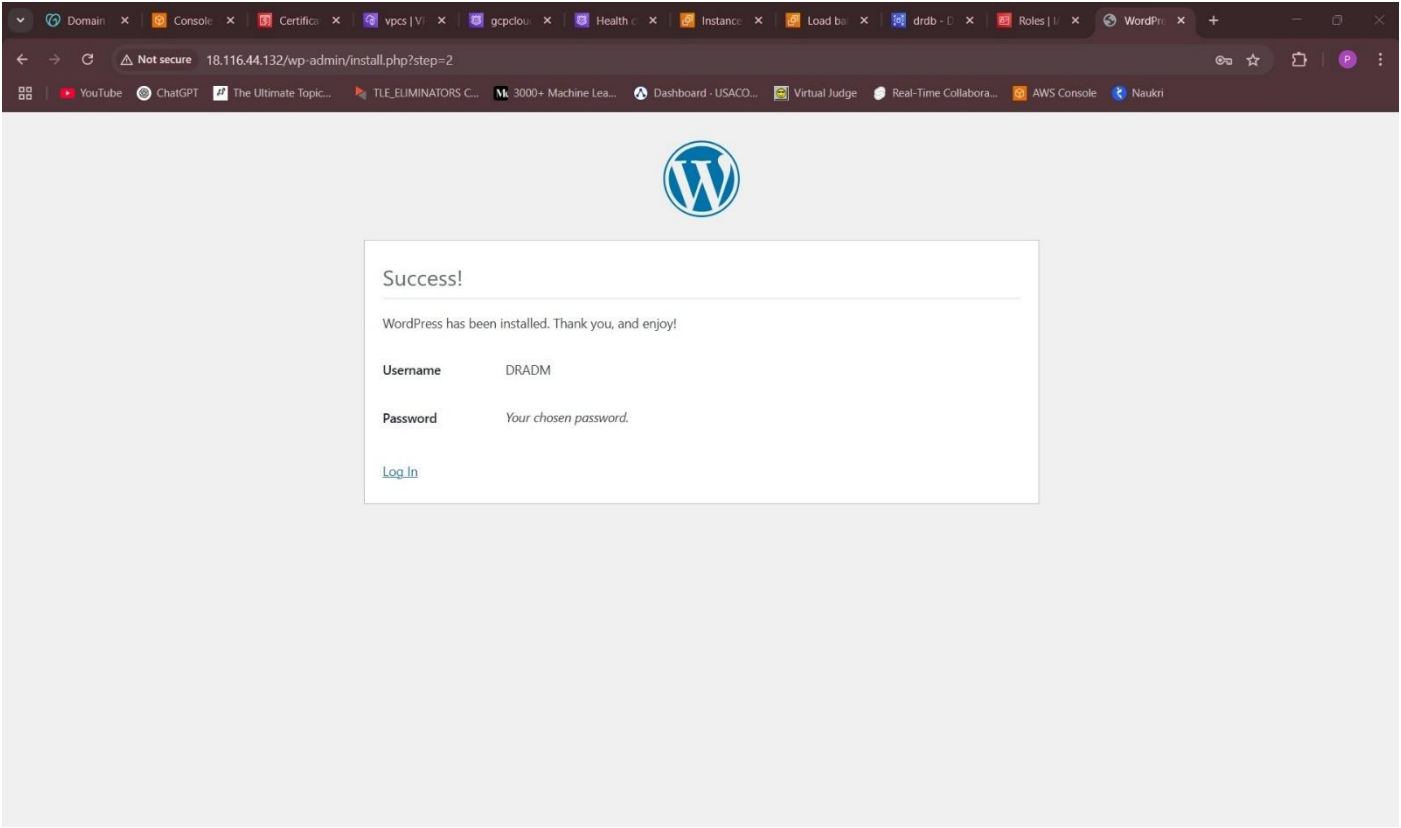


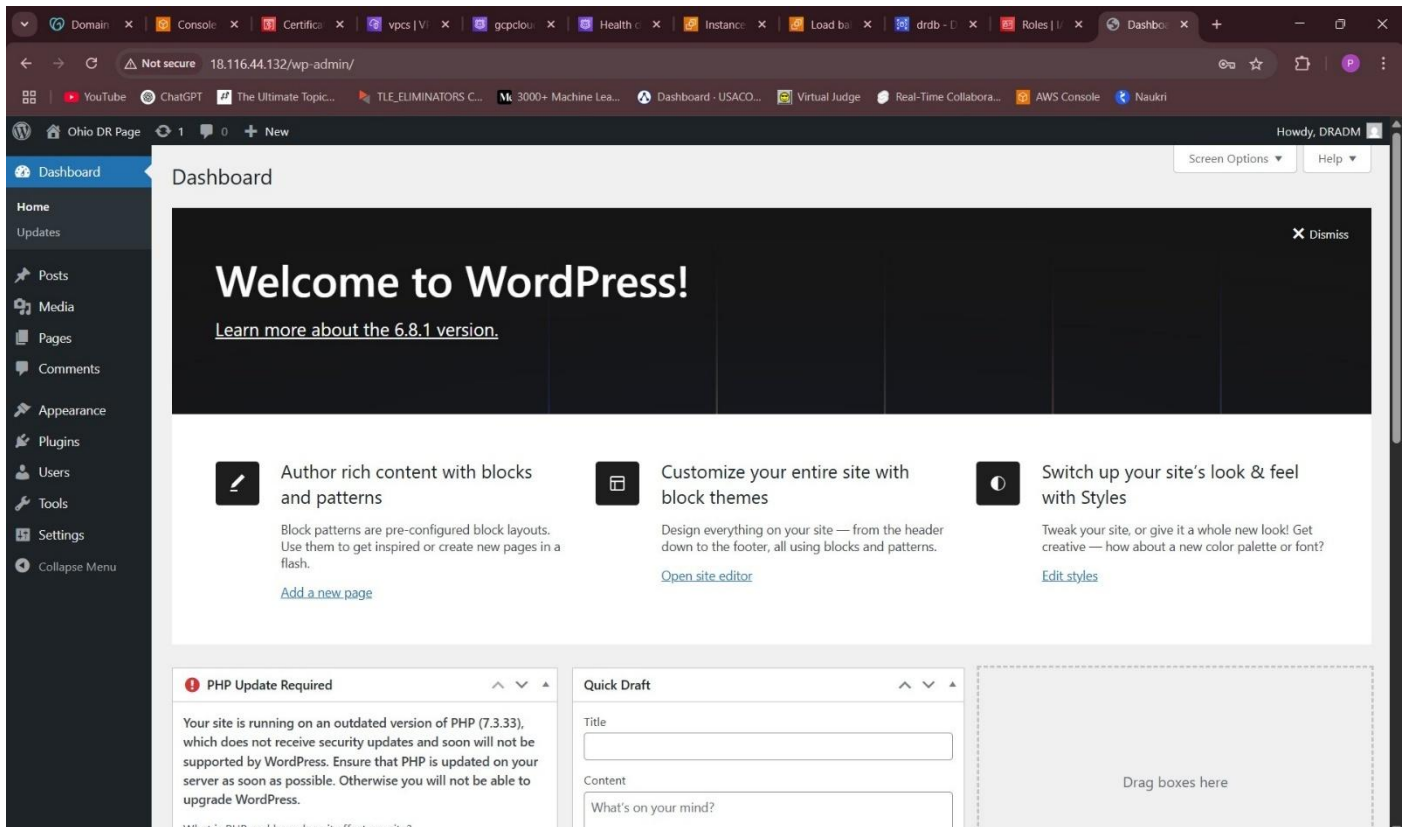
xi. Add the DB details under WordPress Application for both DR & Production:

- Create and edit the wp-config.php file in both the EC2s.



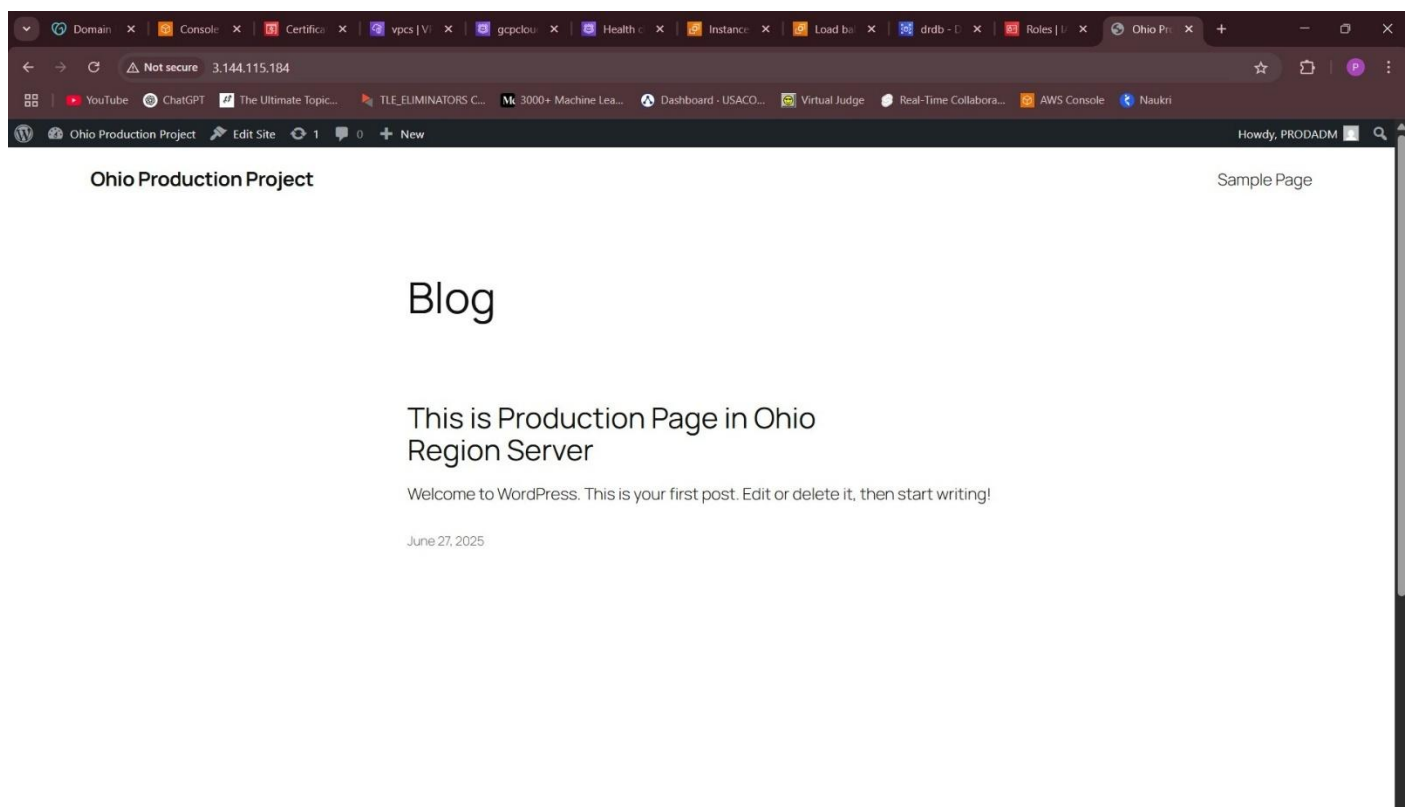


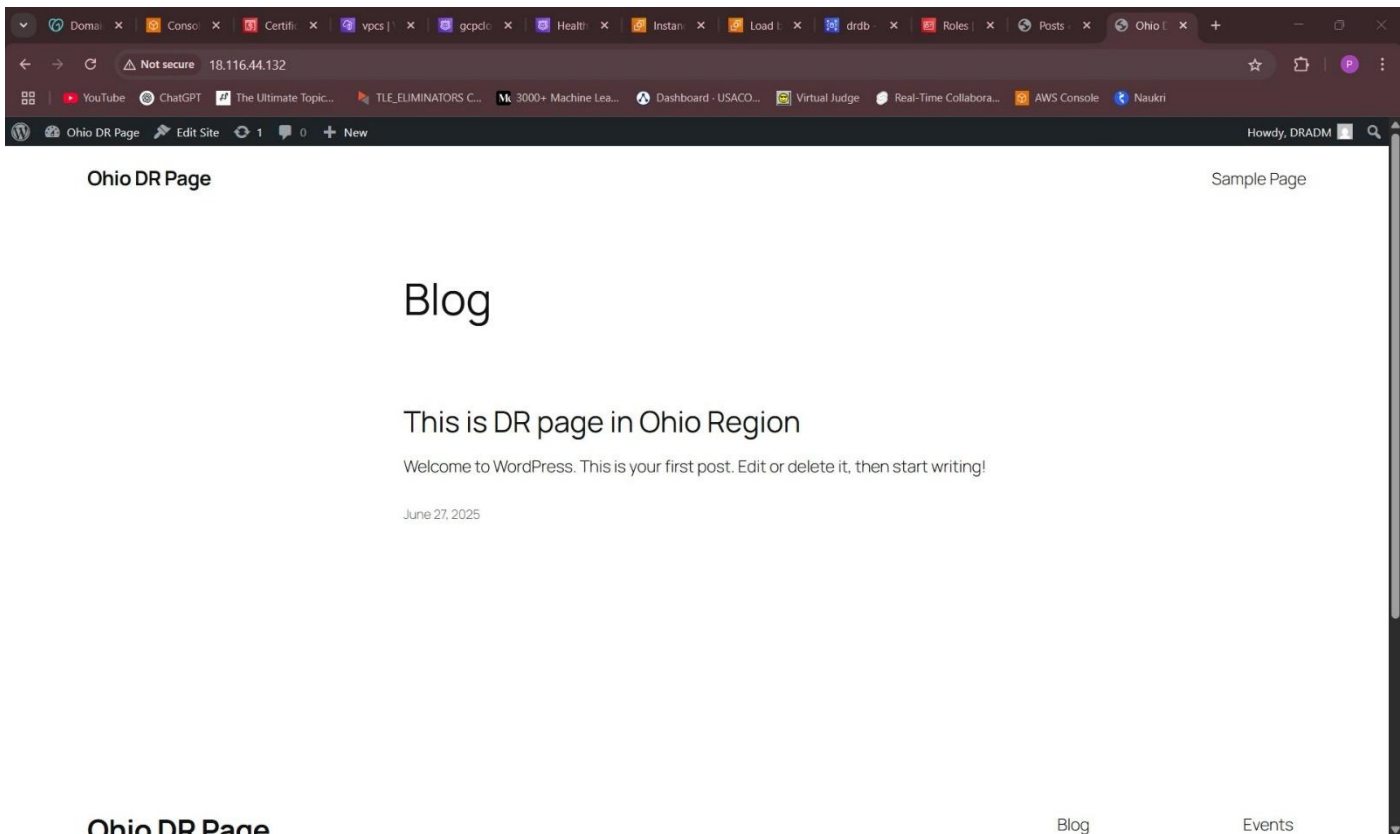




xii. Launch the WordPress page & post it with appropriate DR & Production texts to get difference in the application page:

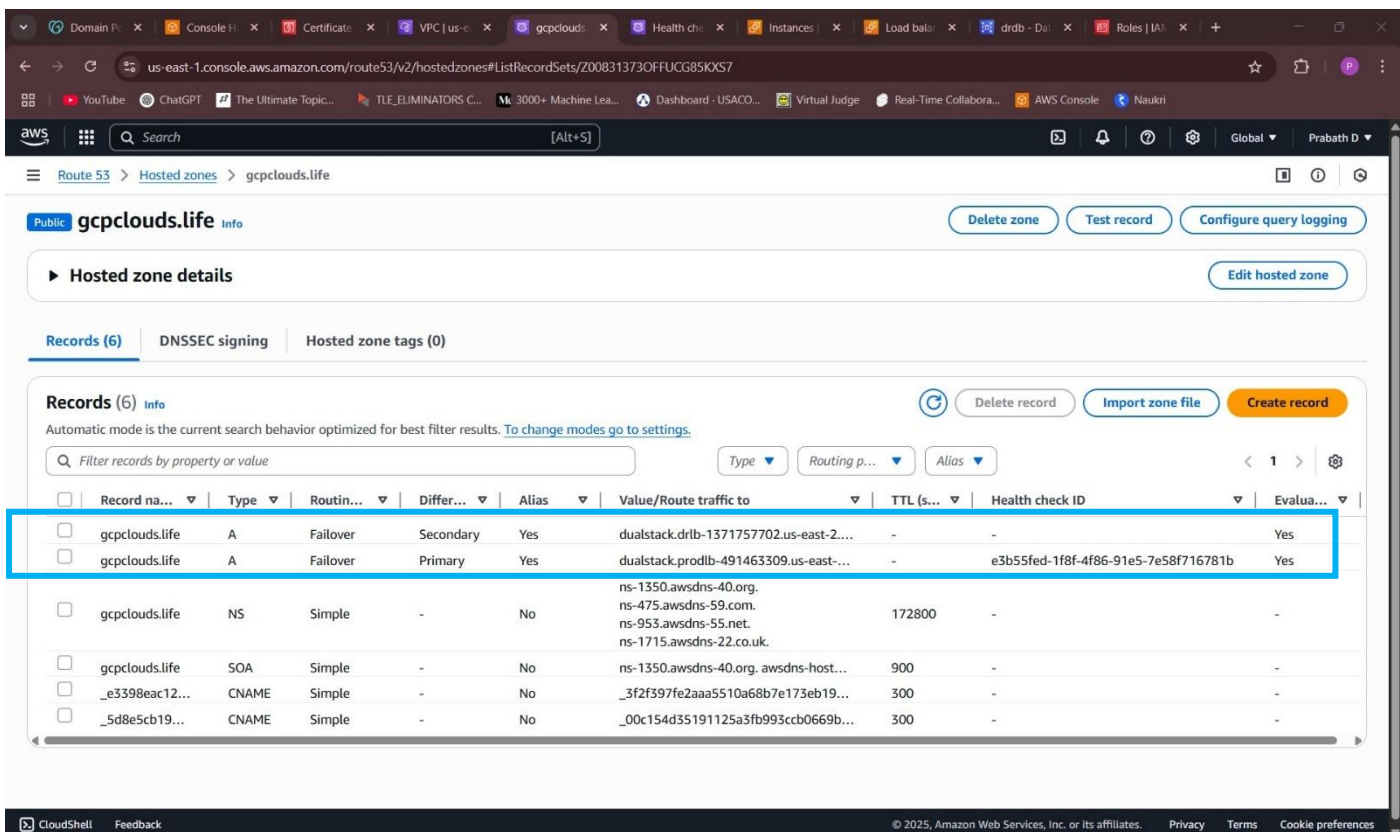
- Now, with the help of DNS name of the prod1b and dr1b or IP Address of any one of EC2, paste it in the browser and load the page.





xiii. AWS R53 & ELB Sync:

- Refresh the R53
- Go to primary health check. Click create health check, then give a name then by mapping the created production ELB by giving the path in '/' as index.html.
- Create Record set and map appropriate ELB under alias by adding health checks in a particular region.



xiv. AWS S3 Bucket Creation:

- Go to S3 click create bucket, then create two S3 buckets for Production and DR.
- Set the crontab job's in the EC2 in small intervals to copy the content from production to S3 & S3 to DR.
- Add the media sets for production and DR.
- The media sets are:
 - wordpress-mediaasset-project
 - wordpress-code-project
- The content that must be given in server instance,

Production:

```
* / 2 * * * * aws s3 sync --delete /var/www/html/wp-content/uploads s3://wordpress-mediaasset-project
```

```
* / 2 * * * * aws s3 sync --delete /var/www/html/ s3://wordpress-code-project
```

DR:

```
* / 2 * * * * aws s3 sync --delete s3://wordpress-mediaasset-project  
/var/www/html/wp-content/uploads
```

```
* / 2 * * * * aws s3 sync --delete s3://wordpress-code-project /var/www/html/
```

The screenshot shows the AWS S3 console interface. At the top, there's a navigation bar with the AWS logo and a search bar. Below it, the 'Buckets' section is active. A banner at the top of the main content area says 'Account snapshot - updated every 24 hours' with a 'View Storage Lens dashboard' button. The 'General purpose buckets' tab is selected, showing a list of two buckets. The table has columns for Name, AWS Region, IAM Access Analyzer, and Creation date. Both buckets are in the 'US East (Ohio) us-east-2' region and were created on June 27, 2025.

Name	AWS Region	IAM Access Analyzer	Creation date
wordpress-code-project27062025	US East (Ohio) us-east-2	View analyzer for us-east-2	June 27, 2025, 19:52:23 (UTC+05:30)
wordpress-mediaasset-project27062025	US East (Ohio) us-east-2	View analyzer for us-east-2	June 27, 2025, 19:51:51 (UTC+05:30)


```
root@ip-10-1-2-117:~  
[root@ip-10-1-2-117 ~]# crontab -e  
no crontab for root - using an empty one  
crontab: installing new crontab  
[root@ip-10-1-2-117 ~]# crontab -l  
*/2 * * * * aws s3 sync --delete s3://wordpress-mediaasset-project27062025 /var  
/www/html/wp-content/uploads  
*/2 * * * * aws s3 sync --delete s3://wordpress-code-project27062025 /var/www/  
html/  
  
[root@ip-10-1-2-117 ~]#
```

Crontab(Prod)

```
root@ip-10-1-1-79:~  
[root@ip-10-1-1-79 ~]# crontab -e  
no crontab for root - using an empty one  
crontab: installing new crontab  
[root@ip-10-1-1-79 ~]# ls  
[root@ip-10-1-1-79 ~]# crontab -ls  
Cannot obtain SELinux process context  
[root@ip-10-1-1-79 ~]# crontab -l  
*/2 * * * * aws s3 sync --delete /var/www/html/wp-content/uploads s3://wordpress  
-mediaasset-project27062025  
*/2 * * * * aws s3 sync --delete /var/www/html/ s3://wordpress-code-project27062  
025  
  
[root@ip-10-1-1-79 ~]#
```

Crontab(DR)

- Now, wait for few minutes and then refresh the page.

- In this task, specifically the failover routing must be applied to view the Production page and DR page, if one availability zone is failed it redirects to the other zone, because of the custom VPC that is used in this.

Hosted zone details

Records (1/6)

Record name	Type	Routing...	Differ...	Alias	Value/Route traffic to
☒ gcpclouds.life	A	Failover	Primary	Yes	dualstack.prod1b-723851817.ap-south-1.elb.amazonaws.com
☐ gcpclouds.life	A	Failover	Secondary	Yes	dualstack.drlb-154749572.ap-south-1.elb.amazonaws.com
☐ gcpclouds.life	NS	Simple	-	No	ns-1211.awsdns-23.org, ns-154.awsdns-19.com, ns-1554.awsdns-02.co.uk, ns-744.awsdns-29.net
☐ gcpclouds.life	SOA	Simple	-	No	ns-1211.awsdns-23.org
☐ _e3398eac1285c7aa2026a98ed618822d.gcpclouds.life	CNAME	Simple	-	No	_3f2f397fe2aaa5510a68
☐ _5d8e5cb192ab0a4d5ecdff8080af79.dr.gcpclouds.life	CNAME	Simple	-	No	_00c154d35191125a3fb

Record details

Record name: gcpclouds.life

Record type: A

Value: dualstack.prod1b-723851817.ap-south-1.elb.amazonaws.com

Alias: Yes

TTL (seconds): -

Routing policy: Failover

Failover record type: Primary

Health check ID: 1192c1b6-c952-48b6-8249-196287627d4d

Record ID: prod-lb

Hosted zone details

Records (1/6)

Record name	Type	Routing...	Differ...	Alias	Value/Route traffic to
☐ gcpclouds.life	A	Failover	Primary	Yes	dualstack.prod1b-723851817.ap-south-1.elb.amazonaws.com
☒ gcpclouds.life	A	Failover	Secondary	Yes	dualstack.drlb-154749572.ap-south-1.elb.amazonaws.com
☐ gcpclouds.life	NS	Simple	-	No	ns-1211.awsdns-23.org, ns-154.awsdns-19.com, ns-1554.awsdns-02.co.uk, ns-744.awsdns-29.net
☐ gcpclouds.life	SOA	Simple	-	No	ns-1211.awsdns-23.org
☐ _e3398eac1285c7aa2026a98ed618822d.gcpclouds.life	CNAME	Simple	-	No	_3f2f397fe2aaa5510a68
☐ _5d8e5cb192ab0a4d5ecdff8080af79.dr.gcpclouds.life	CNAME	Simple	-	No	_00c154d35191125a3fb

Record details

Record name: gcpclouds.life

Record type: A

Value: dualstack.drlb-154749572.ap-south-1.elb.amazonaws.com

Alias: Yes

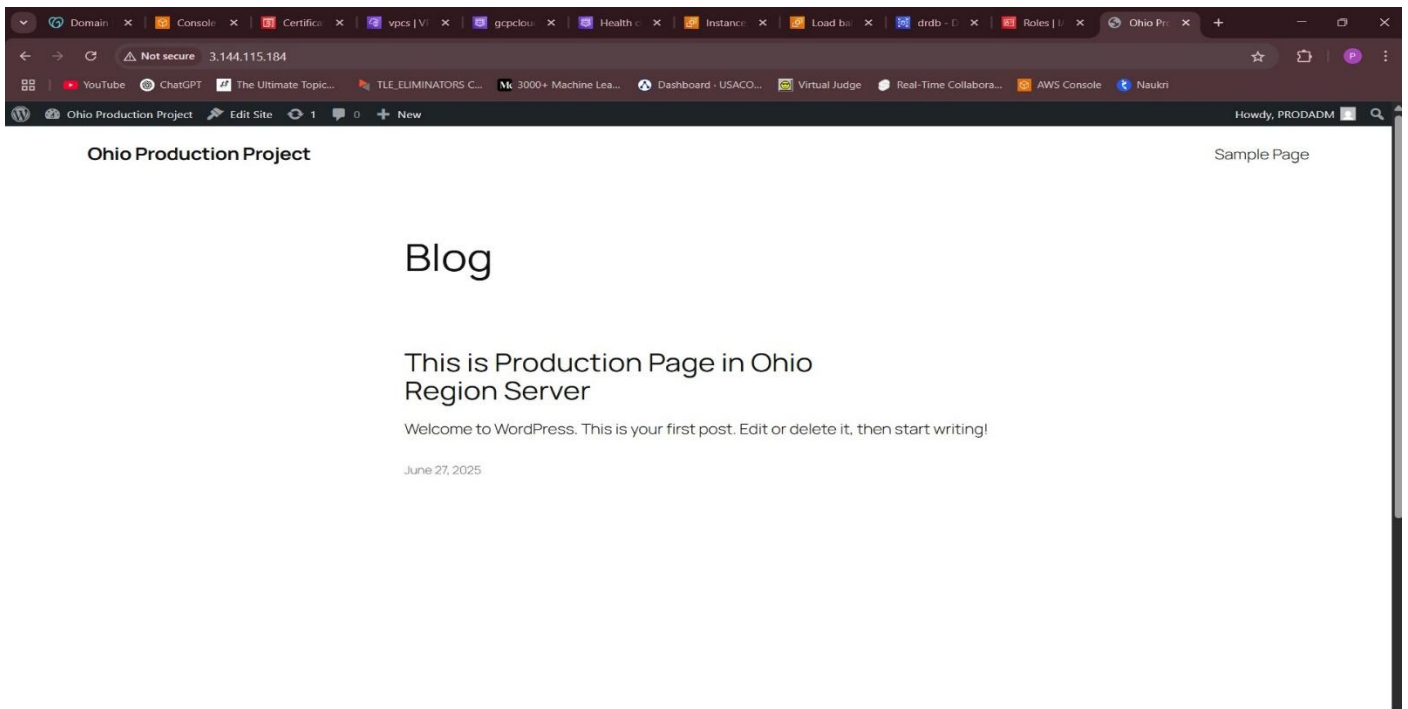
TTL (seconds): -

Routing policy: Failover

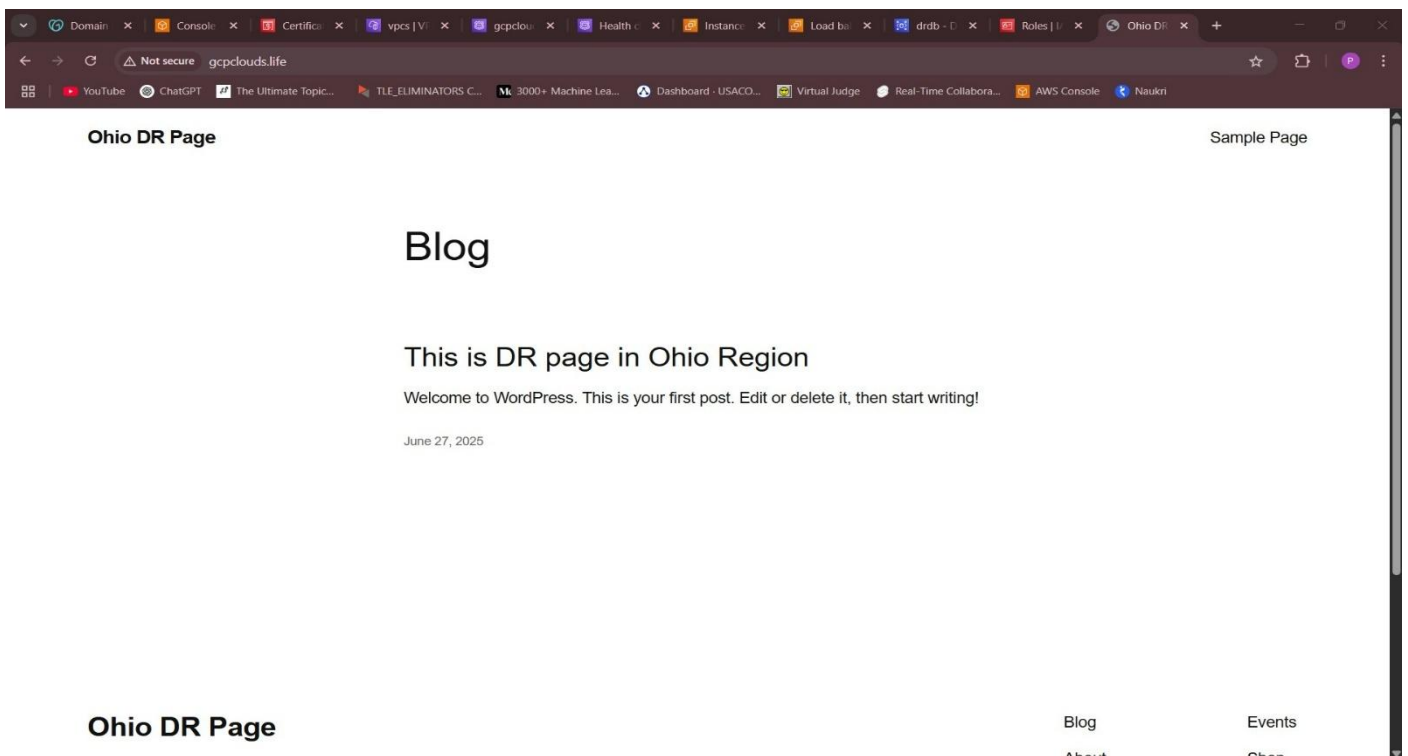
Failover record type: Secondary

Record ID: dr-lb

- Now, by implementing in the same way of already done Mumbai region.



- Now, go to EC2 instances and stop the Production instance for some time, after 2-3 mins, the health check in the route53 becomes unhealthy
- With that, by refreshing the page production DNS name that is hosted in the web browser it will redirect to the DR DNS name.
- By giving the name as gpcclouds.life it will show the DR Blog.
- Once, again if you start the instance of Production instance the web page will get refresh and shows the Production Blog



- **The setup for deploying the WordPress page in this region also completed. With that adding VPC components and using multi – regions is completed.**

Task 2: Implement Failover Routing Policy in Route53

Objective:

Set up a custom VPC in multiple AWS regions (Mumbai) with complete networking (subnets, route tables, NAT, IGW), configure EC2 & RDS, IAM role, set up S3 bucket implement SSL certificate request and DNS linking using ACM + Route 53, connected to GoDaddy domain. It is same as implemented for task 1, but with only one region.

Mumbai Region:

Steps:

i.VPC Creation:

- Click your VPC, then click create VPC then give a name as “vpc-mumbai-25”
- Then set the IP address 10.0.0.0/16, then set both DNS hostname and DNS resolution as enabled, then click create.

The screenshot shows the AWS Management Console for the Mumbai region. The VPC details page for 'vpc-0176d1147d86984e3 / vpc-mumbai-25' is displayed. The details section includes:

- VPC ID:** vpc-0176d1147d86984e3
- State:** Available
- Block Public Access:** Off
- DNS hostnames:** Enabled
- DNS resolution:** Enabled
- Tenancy:** default
- DHCP option set:** dopt-0a69edd3a573d82ac
- Main network ACL:** acl-0589b74eeac3c2e2f
- Default VPC:** No
- Main route table:** rtb-0516b15ccfa14dd81
- IPv6 CIDR (Network border group):** -
- Network Address Usage metrics:** Disabled
- IPv4 CIDR:** 10.0.0.0/16
- Route 53 Resolver DNS Firewall rule groups:** -
- IPv6 pool:** -
- Owner ID:** 767828764671

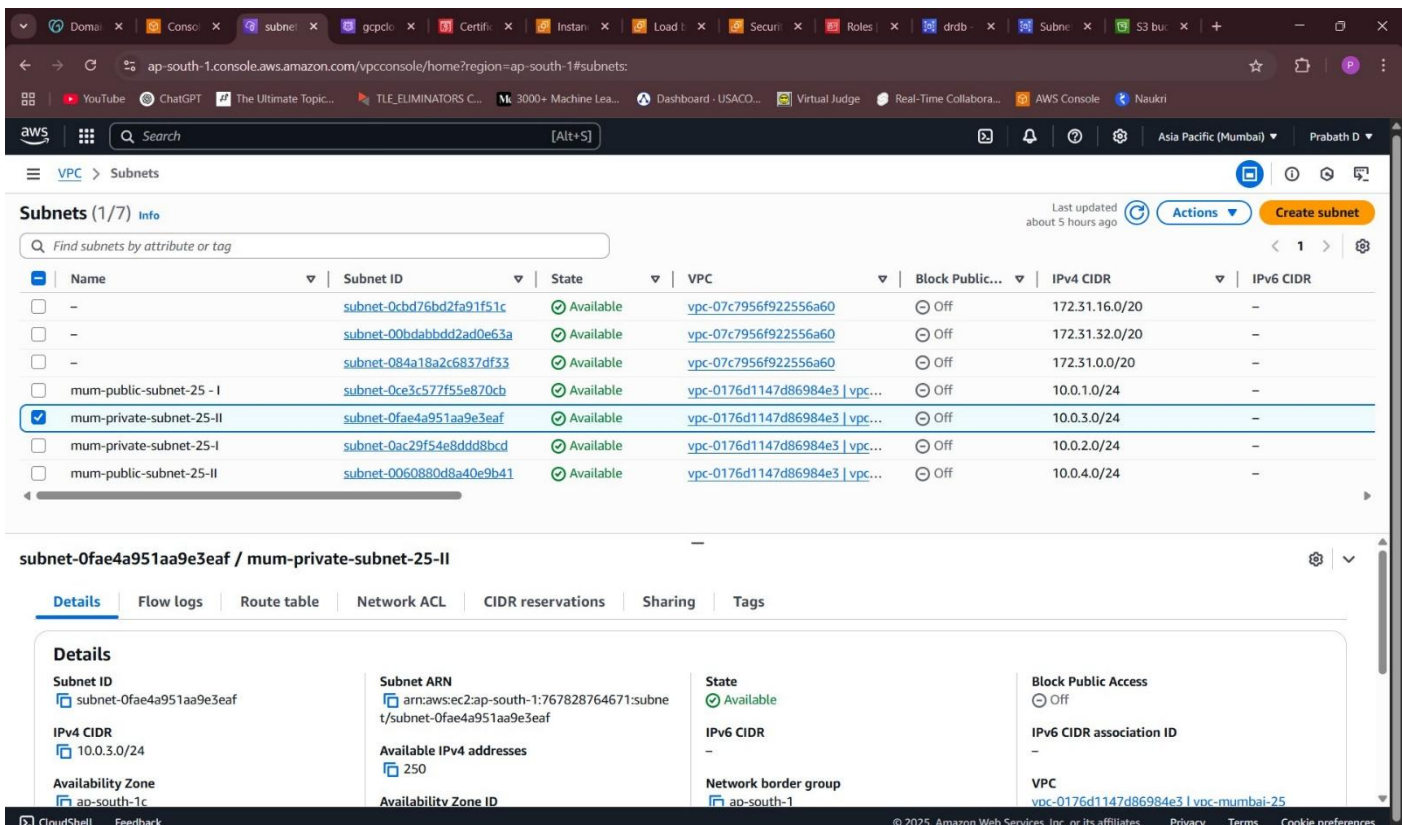
The Resource map section shows the VPC connected to four subnets (ap-south-1a, ap-south-1b, mum-public-subnet-25 - I, and mum-public-subnet-25 - II), three route tables (rtb-private-mum-25, rtb-0516b15ccfa14dd81, and rtb-public-mum-25), and two network connections (igw-mumbai and mum-nat-gw-25).

ii. Subnet Setup:

- Go to subnets, click create subnet after that assign a name for the subnets, in this task we need two subnets for public and private.
- Then enter the below details particularly for public and private subnet, by assigning the IP Addresses, then click create subnet
- Public Subnet 1: For EC2 + NAT
- Public Subnet 2: For EC2
- Private Subnet 1: For RDS Prod
- Private Subnet 2: For RDS DR

CIDR Blocks (Mumbai):

- 10.0.1.0/24 (Public subnet 1)
- 10.0.2.0/24 (Public subnet 2)
- 10.0.3.0/24 (Private subnet 1)
- 10.0.4.0/24 (Private subnet 2)



The screenshot displays the AWS Management Console interface for the 'Subnets' page in the 'Asia Pacific (Mumbai)' region. The page shows a list of subnets under the 'Subnets (1/7) Info' section. The selected subnet, 'mum-private-subnet-25-II', is highlighted in blue. Below the list, the 'Details' tab is active, showing the following information:

Attribute	Value
Subnet ID	subnet-0fae4a951aa9e3eaf
Subnet ARN	arn:aws:ec2:ap-south-1:767828764671:subnet/subnet-0fae4a951aa9e3eaf
State	Available
IP Version	IPv4
Available IPv4 addresses	250
Availability Zone	ap-south-1c
Network border group	ap-south-1
Block Public Access	Off
IPv6 CIDR association ID	-
VPC	vpc-0176d1147d86984e3 vpc-mumbai-25

ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#SubnetDetails:subnetId=subnet-0ce3c577f5e870cb

Search [Alt+S]

Asia Pacific (Mumbai) Prabath D

VPC > Subnets > subnet-0ce3c577f5e870cb

subnet-0ce3c577f5e870cb / mum-public-subnet-25 - I

Actions

Details

Subnet ID

subnet-0ce3c577f5e870cb

IPv4 CIDR

10.0.1.0/24

Availability Zone

ap-south-1a

Route table

rtb-01d7b447432b5b441 | rtb-public-mum-25

Auto-assign IPv6 address

No

IPv4 CIDR reservations

-

Resource name DNS A record

Disabled

Subnet ARN

arn:aws:ec2:ap-south-1:767828764671:subnet/subnet-0ce3c577f5e870cb

Available IPv4 addresses

250

Availability Zone ID

aps1-az1

Network ACL

-

Auto-assign customer-owned IPv4 address

No

IPv6 CIDR reservations

-

Resource name DNS AAAA record

Disabled

State

Available

IPv6 CIDR

-

Network border group

ap-south-1

Default subnet

No

Customer-owned IPv4 pool

-

IPv6-only

No

DNS64

Disabled

Block Public Access

Off

IPv6 CIDR association ID

-

VPC

vpc-0176d1147d86984e3 | vpc-mumbai-25

Auto-assign public IPv4 address

No

Outpost ID

-

Hostname type

IP name

Owner

767828764671

Flow logs

Route table

Network ACL

CIDR reservations

Sharing

Tags

Flow logs

Search

Actions

Create flow log

1

Name

Flow log ID

Filter

Destination type

Destination name

IAM role ARN

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ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#SubnetDetails:subnetId=subnet-0ac29f54e8ddd8bcd

Search [Alt+S]

Asia Pacific (Mumbai) Prabath D

VPC > Subnets > subnet-0ac29f54e8ddd8bcd

subnet-0ac29f54e8ddd8bcd / mum-private-subnet-25-I

Actions

Details

Subnet ID

subnet-0ac29f54e8ddd8bcd

IPv4 CIDR

10.0.2.0/24

Availability Zone

ap-south-1b

Route table

rtb-0aaee2fc7e3c115ad | rtb-private-mum-25

Auto-assign IPv6 address

No

IPv4 CIDR reservations

-

Resource name DNS A record

Disabled

Subnet ARN

arn:aws:ec2:ap-south-1:767828764671:subnet/subnet-0ac29f54e8ddd8bcd

Available IPv4 addresses

250

Availability Zone ID

aps1-az3

Network ACL

-

Auto-assign customer-owned IPv4 address

No

IPv6 CIDR reservations

-

Resource name DNS AAAA record

Disabled

State

Available

IPv6 CIDR

-

Network border group

ap-south-1

Default subnet

No

Customer-owned IPv4 pool

-

IPv6-only

No

DNS64

Disabled

Block Public Access

Off

IPv6 CIDR association ID

-

VPC

vpc-0176d1147d86984e3 | vpc-mumbai-25

Auto-assign public IPv4 address

No

Outpost ID

-

Hostname type

IP name

Owner

767828764671

Flow logs

Route table

Network ACL

CIDR reservations

Sharing

Tags

Flow logs

Search

Actions

Create flow log

1

Name

Flow log ID

Filter

Destination type

Destination name

IAM role ARN

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subnet-0060880d8a40e9b41 / mum-public-subnet-25-II

Details

Subnet ID subnet-0060880d8a40e9b41 IPv4 CIDR 10.0.4.0/24 Availability Zone ap-south-1b Route table rtb-01d7b447432b5b441 rtb-public-mum-25 Auto-assign IPv6 address No IPv4 CIDR reservations - Resource name DNS A record Disabled	Subnet ARN arn:aws:ec2:ap-south-1:767828764671:subnet/subnet-0060880d8a40e9b41 Available IPv4 addresses 251 Availability Zone ID aps1-az3 Network ACL - Auto-assign customer-owned IPv4 address No IPv6 CIDR reservations - Resource name DNS AAAA record Disabled	State ✔ Available IPv6 CIDR - Network border group ap-south-1 Default subnet No Customer-owned IPv4 pool - IPv6-only No DNS64 Disabled	Block Public Access ☹ Off IPv6 CIDR association ID - VPC vpc-0176d1147d86984e3 vpc-mumbai-25 Auto-assign public IPv4 address No Outpost ID - Hostname type IP name Owner 767828764671
---	--	---	---

Flow logs

Route table | Network ACL | CIDR reservations | Sharing | Tags

Search

Name	Flow log ID	Filter	Destination type	Destination name	IAM role ARN

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iii. Internet Gateway + NAT Gateway:

- Create and attach IGW in Mumbai VPC.

igw-0867acd1a59c8cf48 / igw-mumbai

Details

Internet gateway ID igw-0867acd1a59c8cf48	State ✔ Attached	VPC ID vpc-0176d1147d86984e3 vpc-mumbai-25	Owner 767828764671
---	---	--	--

Tags

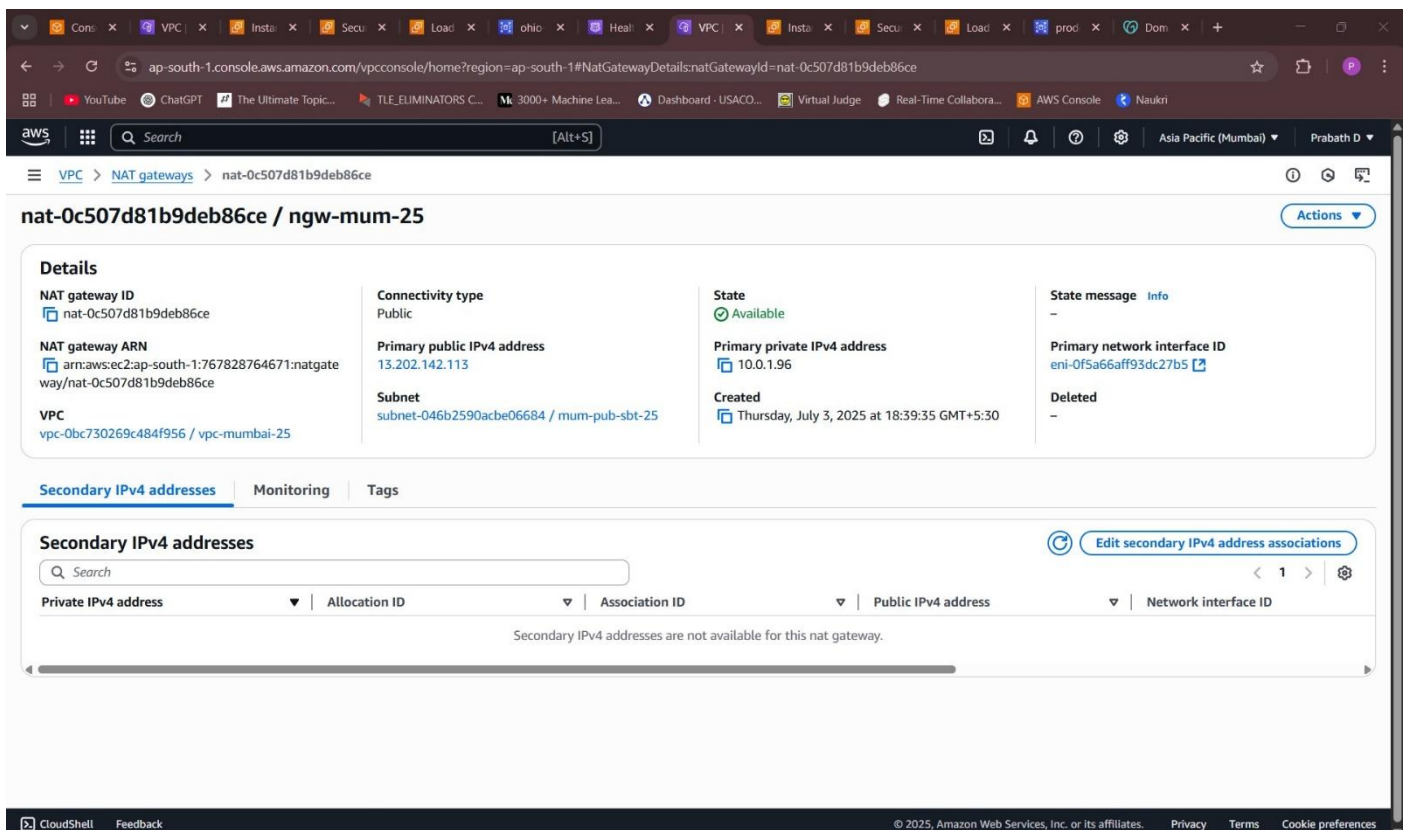
Search tags

Key	Value
Name	igw-mumbai

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- Create NAT Gateway in Public Subnet (with Elastic IP)



iv. Route Tables (Mumbai):

- Then after creating both Internet Gateway and NAT Gateway, go to route tables, click create, then assign names for both public and private route tables, then associate both the tables by clicking routes then attach the network
- Public Route Table → 0.0.0.0/0 → IGW.
- Then click subnet associations in public route table, then click edit association after that click both public subnet 1 and public subnet 2.
- Private Route Table → 0.0.0.0/0 → NAT.
- Then click subnet associations in private route table, then click edit association after that click both private subnet 1 and private subnet 2.

ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#RouteTableDetails:RouteTableId=rtb-01d7b447432b5b441

Search [Alt+S]

Asia Pacific (Mumbai) Prabath D

VPC > Route tables > rtb-01d7b447432b5b441

rtb-01d7b447432b5b441 / rtb-public-mum-25

Details Info

Route table ID

rtb-01d7b447432b5b441

VPC

vpc-0176d1147d86984e3 | vpc-mumbai-25

Main

No

Explicit subnet associations

2 subnets

Edge associations

-

Owner ID

767828764671

Routes Subnet associations Edge associations Route propagation Tags

Explicit subnet associations (2)

Find subnet association

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
mum-public-subnet-25 - I	subnet-0ce3c577f55e870cb	10.0.1.0/24	-
mum-public-subnet-25-II	subnet-0060880d8a40e9b41	10.0.4.0/24	-

Edit subnet associations

Subnets without explicit associations (0)

The following subnets have not been explicitly associated with any route tables and are therefore associated with the main route table:

Find subnet association

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
------	-----------	-----------	-----------

No subnets without explicit associations

All your subnets are associated with a route table.

Edit subnet associations

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ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#RouteTableDetails:RouteTableId=rtb-01d7b447432b5b441

Search [Alt+S]

Asia Pacific (Mumbai) Prabath D

VPC > Route tables > rtb-01d7b447432b5b441

rtb-01d7b447432b5b441 / rtb-public-mum-25

Details Info

Route table ID

rtb-01d7b447432b5b441

VPC

vpc-0176d1147d86984e3 | vpc-mumbai-25

Main

No

Explicit subnet associations

2 subnets

Edge associations

-

Owner ID

767828764671

Routes Subnet associations Edge associations Route propagation Tags

Routes (2)

Filter routes

Both Edit routes

Destination	Target	Status	Propagated
0.0.0.0/0	igw-0867acd1a59c8cf48	Active	No
10.0.0.0/16	local	Active	No

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ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#RouteTableDetails:RouteTableId=rtb-0aaee2fc7e3c115ad

VPC > Route tables > rtb-0aaee2fc7e3c115ad

rtb-0aaee2fc7e3c115ad / rtb-private-mum-25

Actions

Details

Route table ID
rtb-0aaee2fc7e3c115ad

VPC
vpc-0176d1147d86984e3 | vpc-mumbai-25

Main

No

Owner ID
767828764671

Explicit subnet associations

2 subnets

Edge associations

-

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (2)

Filter routes

Both

Edit routes

Destination	Target	Status	Propagated
0.0.0.0/0	nat-0f80ac4cfb0ae2260	Active	No
10.0.0/16	local	Active	No

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ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#RouteTableDetails:RouteTableId=rtb-0aaee2fc7e3c115ad

VPC > Route tables > rtb-0aaee2fc7e3c115ad

rtb-0aaee2fc7e3c115ad / rtb-private-mum-25

Actions

Details

Route table ID
rtb-0aaee2fc7e3c115ad

VPC
vpc-0176d1147d86984e3 | vpc-mumbai-25

Main

No

Owner ID
767828764671

Explicit subnet associations

2 subnets

Edge associations

-

Routes

Subnet associations

Edge associations

Route propagation

Tags

Explicit subnet associations (2)

Find subnet association

Edit subnet associations

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
mum-private-subnet-25-II	subnet-0fae4a951aa9e3eaf	10.0.3.0/24	-
mum-private-subnet-25-I	subnet-0ac29f54e8ddd8bcd	10.0.2.0/24	-

Subnets without explicit associations (0)

The following subnets have not been explicitly associated with any route tables and are therefore associated with the main route table:

Find subnet association

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
------	-----------	-----------	-----------

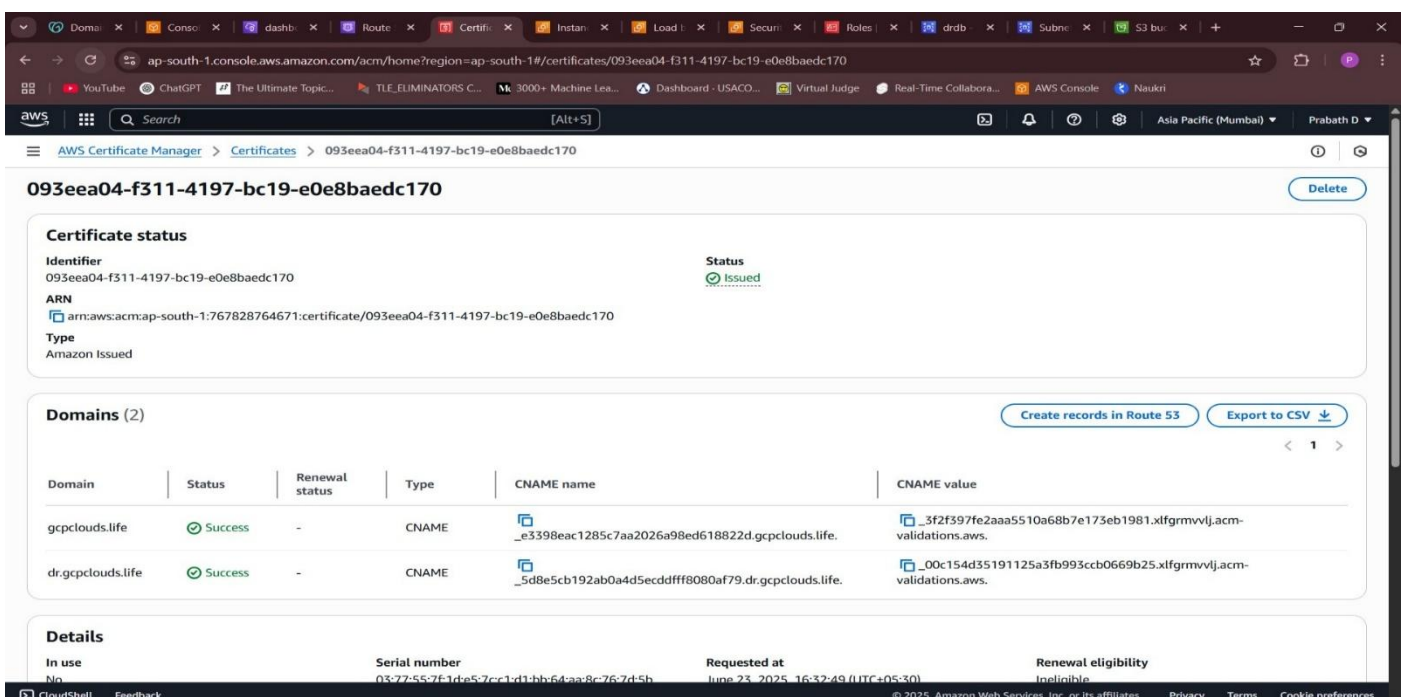
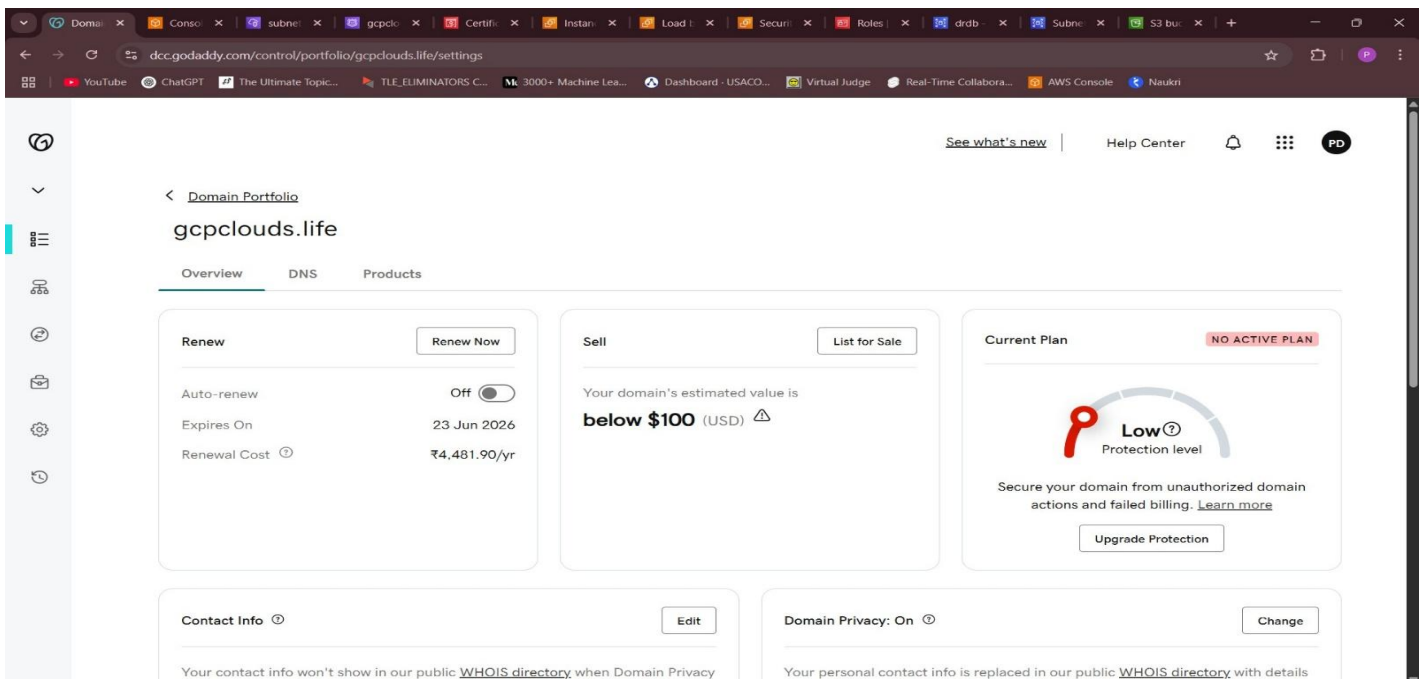
No subnets without explicit associations
All your subnets are associated with a route table.

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v. AWS Certificate Manager Creation:

- Request public certificate as per the registered domain.
- Go to ACM click request, then click public then name it with gcpclouds.life, then click add another certificate in that name it as 'dr.gcpclouds.life'.
- Then click create, after the creation records click create records in Route53, then after selecting both SSL certificate that click create.
- Then refresh the Route53 after the creation of hosted zone, it would be appearing with CNAME and CVALUE.
- Purchased a domain in GoDaddy named with 'gcpclouds.life'.



vi. AWS Route53 setup & link to GoDaddy domain DNS:

- Create hosted zone.
- Map the nameservers under GoDaddy domain.
- Create records set from ACM to R53.

The screenshot shows the AWS Route 53 console for the domain gcpclouds.life. The 'Hosted zone details' section is visible, and the 'Records (6)' tab is selected. The 'NS' record is highlighted, showing the nameservers: ns-1211.awsdns-23.org, ns-154.awsdns-19.com, ns-1554.awsdns-02.co.uk, and ns-744.awsdns-29.net.

Record name	Type	Routing	Difference	Alias	Value/Route traffic to	TTL (seconds)
gcpclouds.life	A	Failover	Primary	Yes	dualstack.prodlib-723851817...	-
gcpclouds.life	A	Failover	Secondary	Yes	dualstack.drlb-154749572.a...	-
gcpclouds.life	NS	Simple	-	No	ns-1211.awsdns-23.org. ns-154.awsdns-19.com. ns-1554.awsdns-02.co.uk. ns-744.awsdns-29.net.	172800
gcpclouds.life	SOA	Simple	-	No	ns-1211.awsdns-23.org. aws...	900
_e3398eac1285c7aa2026a98ed618822d.gcpclouds.life	CNAME	Simple	-	No	_3f2f397fe2aaa5510a68b7e...	300
_5d8e5cb192ab0a4d5ecdff8080af79.dr.gcpclouds.life	CNAME	Simple	-	No	_00c154d35191125a3fb993...	300

The screenshot shows the GoDaddy DNS management console for the domain gcpclouds.life. The 'Nameservers' tab is selected, and the nameservers are listed as ns-1211.awsdns-23.org, ns-154.awsdns-19.com, ns-1554.awsdns-02.co.uk, and ns-744.awsdns-29.net.

Nameservers determine where your DNS is hosted and where you add, edit or delete your DNS records.

Using custom nameservers [Change Nameservers](#)

Nameservers

ns-1211.awsdns-23.org

ns-154.awsdns-19.com

ns-1554.awsdns-02.co.uk

ns-744.awsdns-29.net

vii. AWS RDS Setup:

- Create Two RDS
 - V. New Security Group
 - VI. New Subnet Group
 - VII. PRODUCTION DB (SECURITY GROUP MAKE AS ALL TRAFFIC)
 - VIII. DR DB (SECURITY GROUP MAKE AS ALL TRAFFIC)

The screenshot displays the AWS Management Console for the 'awsproject-subnet' subnet group. A green notification bar at the top indicates 'Successfully created awsproject-subnet. View subnet group'. The main content area shows the following details:

- Subnet group details:**
 - VPC ID: [vpc-0176d1147d86984e3](#)
 - ARN: `arn:aws:rds:ap-south-1:767828764671:subgrp:awsproject-subnet`
 - Supported network types: IPv4
 - Description: awsproject-subnet
- Subnets (2):**

Availability zone	Subnet name	Subnet ID	CIDR block
ap-south-1c	mum-private-subnet-25-II	subnet-0fae4a951aa9e3eaf	10.0.3.0/24
ap-south-1b	mum-private-subnet-25-I	subnet-0ac29f54e8ddd8bcd	10.0.2.0/24
- Tags (0):** [Manage tags](#)

The screenshot displays the AWS Management Console for the 'proddb' database instance. The left sidebar shows the navigation menu with 'Aurora and RDS' selected. The main content area shows the following details:

- DB identifier:** prodadb
- Status:** Available
- Role:** Instance
- Engine:** MySQL Community
- Recommendations:**
- CPU:** 3.76%
- Class:** db.t3.micro
- Current activity:** 0 Connections
- Region & AZ:** ap-south-1c

The 'Connectivity & security' tab is selected, showing the following details:

- Endpoint & port:**
 - Endpoint: [prodadb.cngac84c2in0.ap-south-1.rds.amazonaws.com](#)
 - Port: 3306
- Networking:**
 - Availability Zone: ap-south-1c
 - VPC: [vpc-mumbai-25 \(vpc-0176d1147d86984e3\)](#)
 - Subnet group: awsproject-subnet
 - Subnets: [subnet-0ac29f54e8ddd8bcd](#), [subnet-0fae4a951aa9e3eaf](#)
 - Network type: IPv4
- Security:**
 - VPC security groups: [awsprojectsg \(sg-067a443aeced89167\)](#) (Active)
 - Publicly accessible: No
 - Certificate authority: [rds-ca-rsa2048-g1](#) (Info)
 - Certificate authority date: May 20, 2061, 00:10 (UTC+05:30)
 - DB instance certificate expiration date: June 26, 2026, 16:07 (UTC+05:30)

Aurora and RDS

Dashboard

Databases

Query editor

Performance insights

Snapshots

Exports in Amazon S3

Automated backups

Reserved instances

Proxies

Subnet groups

Parameter groups

Option groups

Custom engine versions

Zero-ETL integrations

New

Events

Event subscriptions

Recommendations

0

Certificate update

No proxies

You don't have any proxies.

Create proxy

Security group rules (3)

Filter by Security group rules

Security group	Type	Rule
awsprojectsg (sg-067a443aecef89167)	CIDR/IP - Inbound	0.0.0.0/0
awsprojectsg (sg-067a443aecef89167)	CIDR/IP - Inbound	0.0.0.0/0
awsprojectsg (sg-067a443aecef89167)	CIDR/IP - Outbound	0.0.0.0/0

Replication (1)

Filter by Replication

DB identifier	Role	Region & AZ	Replication source	Replication state	Lag
proddb	Instance	ap-south-1c	-	-	-

CloudShell

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Terms

Cookie preferences

Aurora and RDS

Dashboard

Databases

Query editor

Performance insights

Snapshots

Exports in Amazon S3

Automated backups

Reserved instances

Proxies

Subnet groups

Parameter groups

Option groups

Custom engine versions

Zero-ETL integrations

New

Events

Event subscriptions

Recommendations

0

Certificate update

DB identifier

drdb

CPU

2.90%

Status

Available

Class

db.t3.micro

Role

Instance

Current activity

0 Connections

Engine

MySQL Community

Region & AZ

ap-south-1b

Recommendations

Connectivity & security

Monitoring

Logs & events

Configuration

Zero-ETL integrations

Maintenance & backups

Data

Connectivity & security

Endpoint & port

Endpoint

[drdb.cngac84c2in0.ap-south-1.rds.amazonaws.com](#)

Port

3306

Networking

Availability Zone

ap-south-1b

VPC

[vpc-mumbai-25 \(vpc-0176d1147d86984e3\)](#)

Subnet group

[awsproject-subnet](#)

Subnets

[subnet-0ac29f54e8ddd8bcd](#)
[subnet-0fae4a951aa9e3eaf](#)

Network type

IPv4

Security

VPC security groups

[awsprojectsg \(sg-067a443aecef89167\)](#)

Active

Publicly accessible

No

Certificate authority

[Info](#)
rds-ca-rsa2048-g1

Certificate authority date

May 20, 2061, 00:10 (UTC+05:30)

DB instance certificate expiration date

June 26, 2026, 15:59 (UTC+05:30)

CloudShell

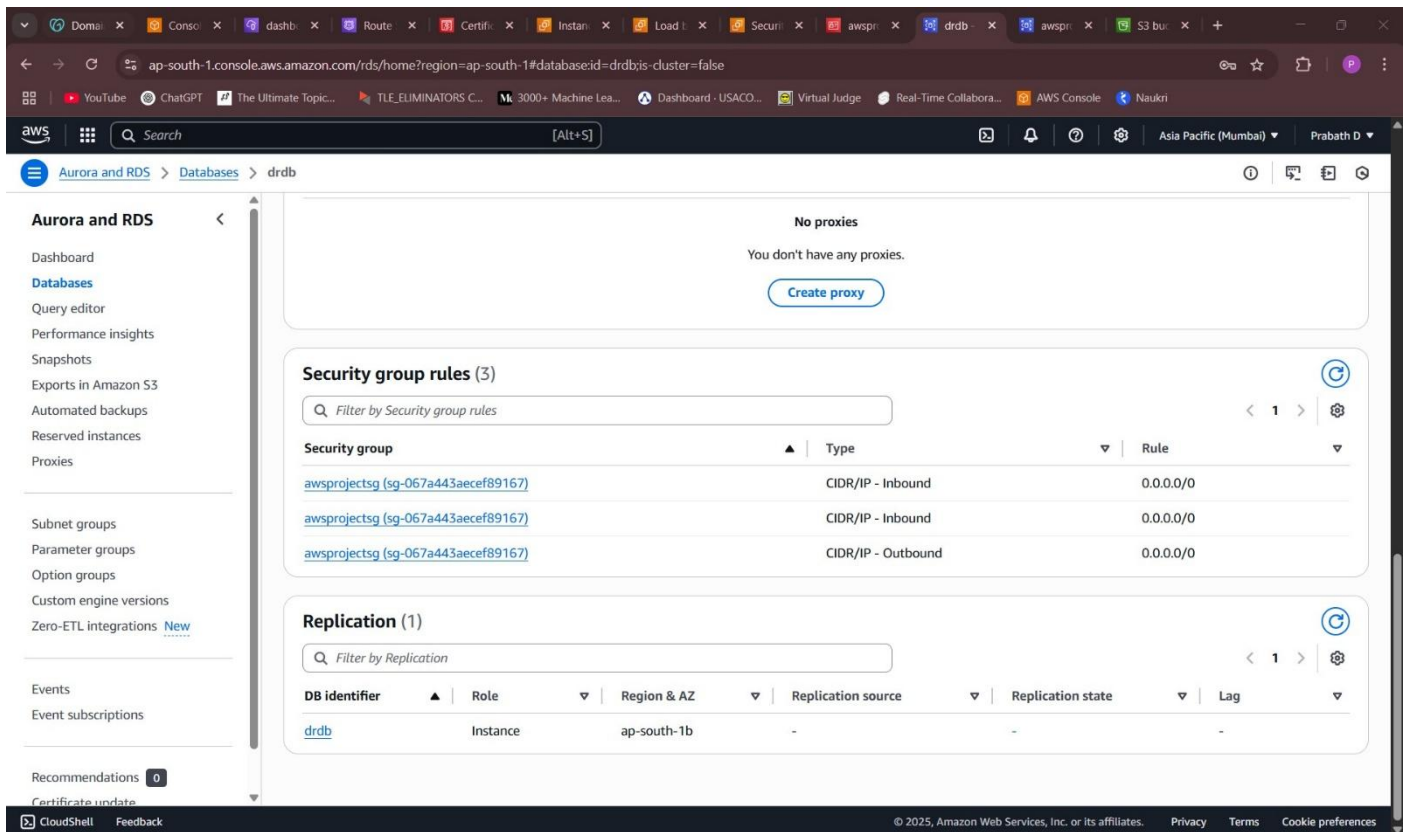
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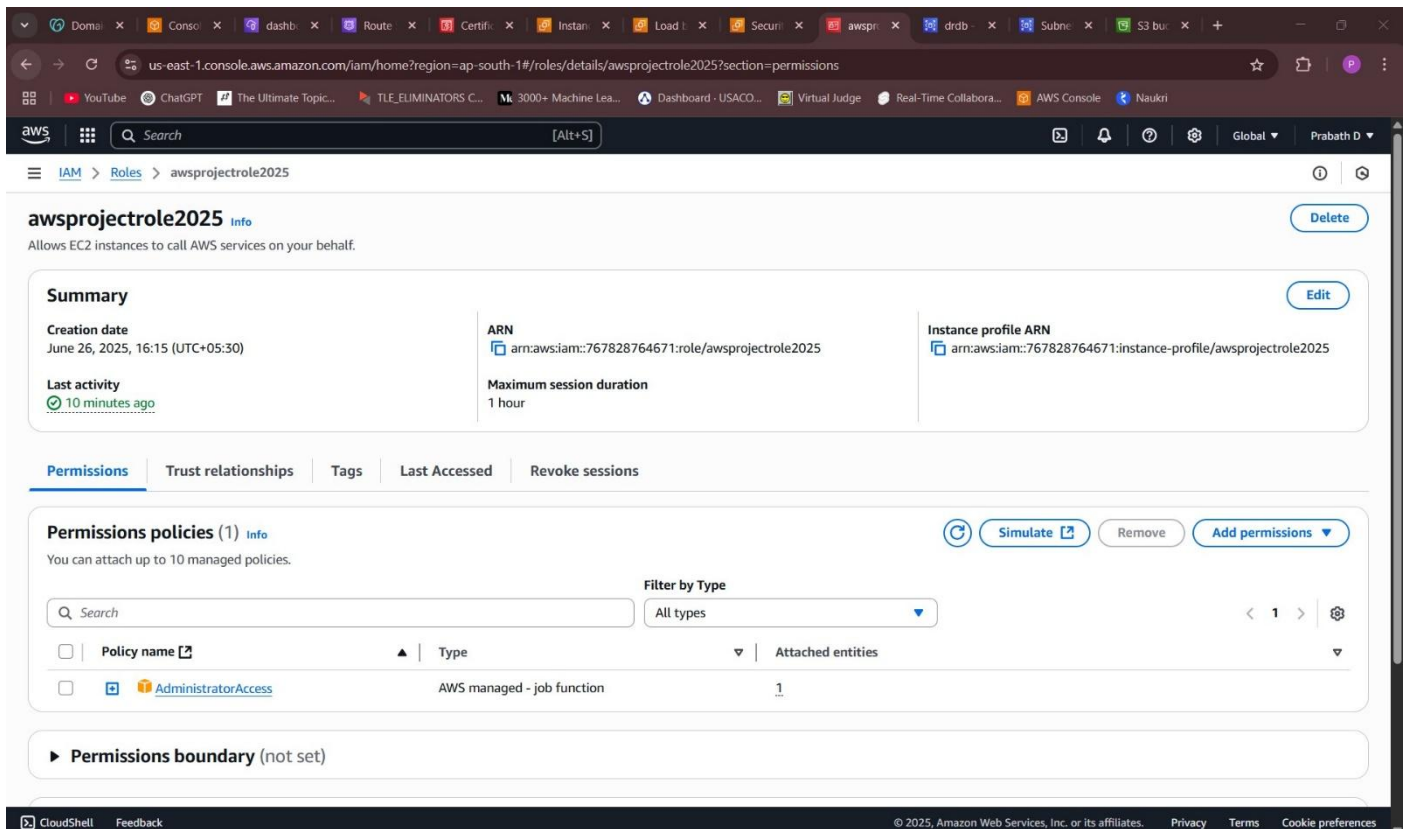
Terms

Cookie preferences



viii. AWS IAM Role:

- Create an IAM role with full access.



ix. AWS EC2 Creation:

- Create two EC2 with IAM roles enables & add user data under advanced section
- Two instances need to setup for Mumbai region and adding the details in advanced section.

User data:

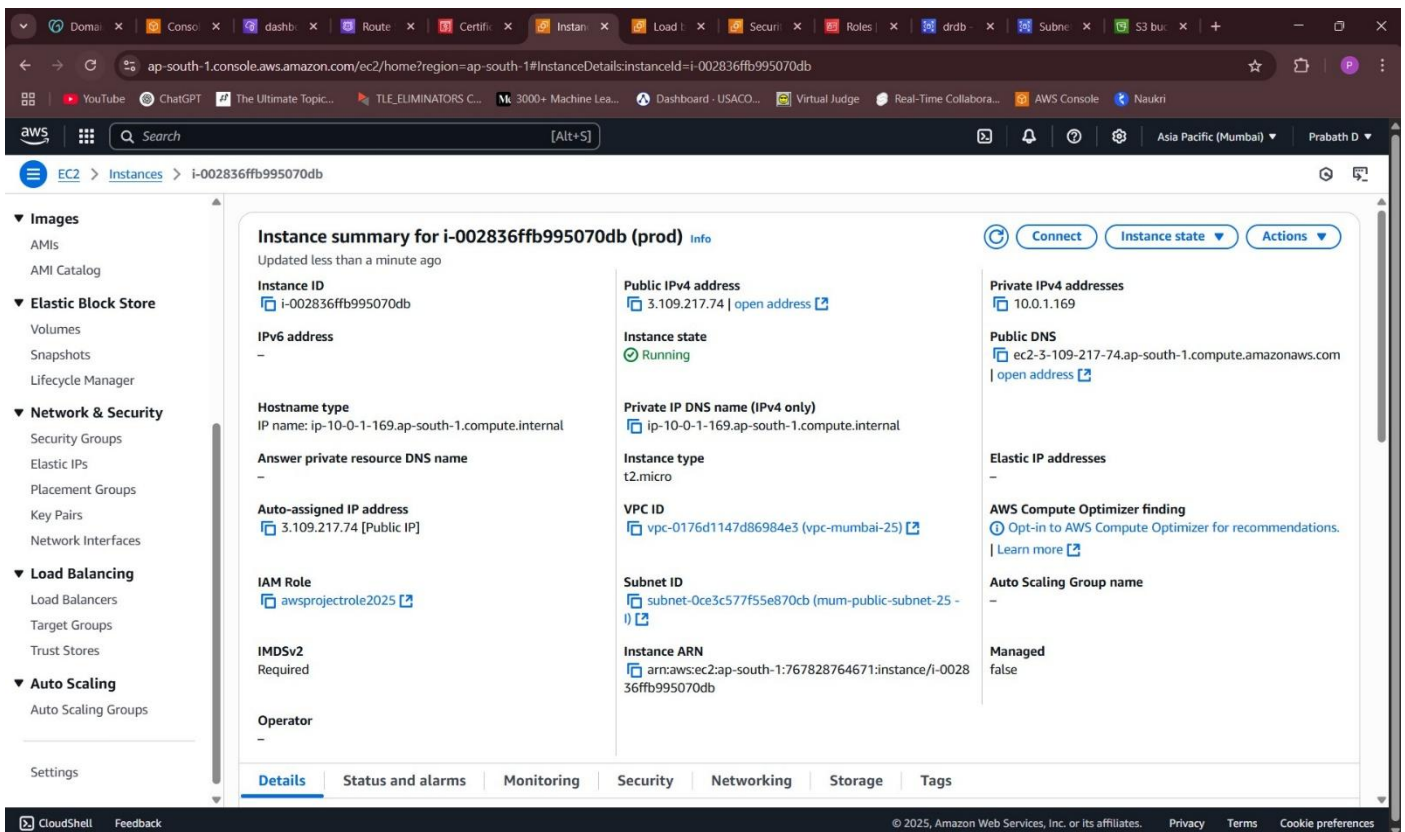
```
#!/bin/bash
yum install httpd php-mysql -y
amazon-linux-extras install -y php7.3
cd /var/www/html
echo "healthy" > healthy.html
wget https://wordpress.org/latest.tar.gz
tar -xzf latest.tar.gz
cp -r wordpress/* /var/www/html/
rm -rf wordpress
rm -rf latest.tar.gz
chmod -R 755 wp-content
chown -R apache:apache wp-content
wget https://s3.amazonaws.com/bucketforwordpresslab-donotdelete/htaccess.txt
mv htaccess.txt .htaccess
chkconfig httpd on
service httpd start
```

- This is the data need to be added in both EC2 instances.

The screenshot displays the AWS Management Console interface for an EC2 instance. The left sidebar shows navigation options like Images, Elastic Block Store, Network & Security, Load Balancing, and Auto Scaling. The main content area shows the 'Instance summary for i-08fc18592dcdbf44a (dr)'. Key details include:

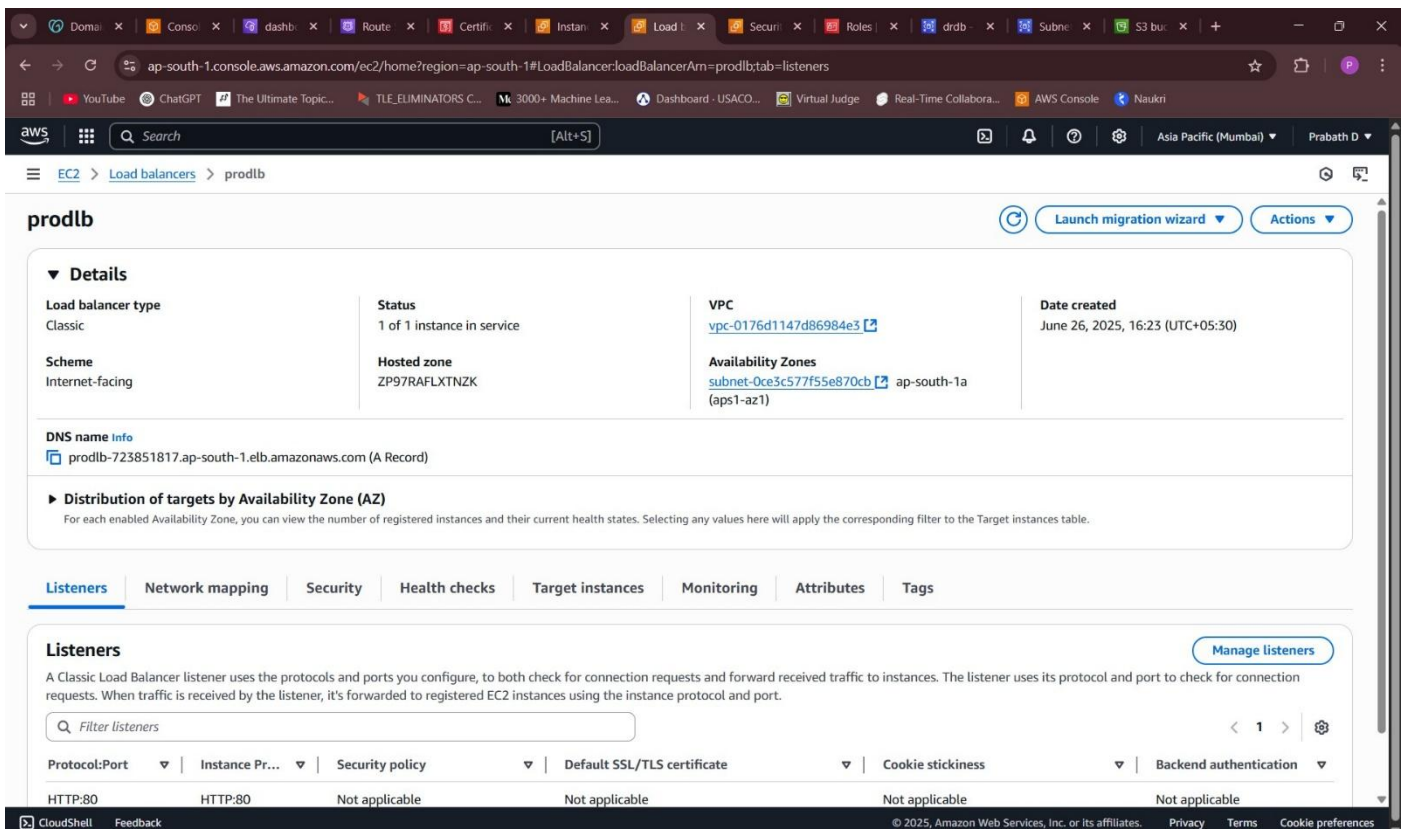
- Instance ID:** i-08fc18592dcdbf44a
- Public IPv4 address:** 3.110.46.203
- Instance state:** Running
- Private IP DNS name (IPv4 only):** ip-10-0-4-163.ap-south-1.compute.internal
- Instance type:** t2.micro
- VPC ID:** vpc-0176d1147d86984e3 (vpc-mumbai-25)
- Subnet ID:** subnet-0060880d8a40e9b41 (mum-public-subnet-25-1)
- IAM Role:** awsprojectrole2025
- IMDSv2:** Required
- Auto Scaling Group name:** -
- Managed:** false

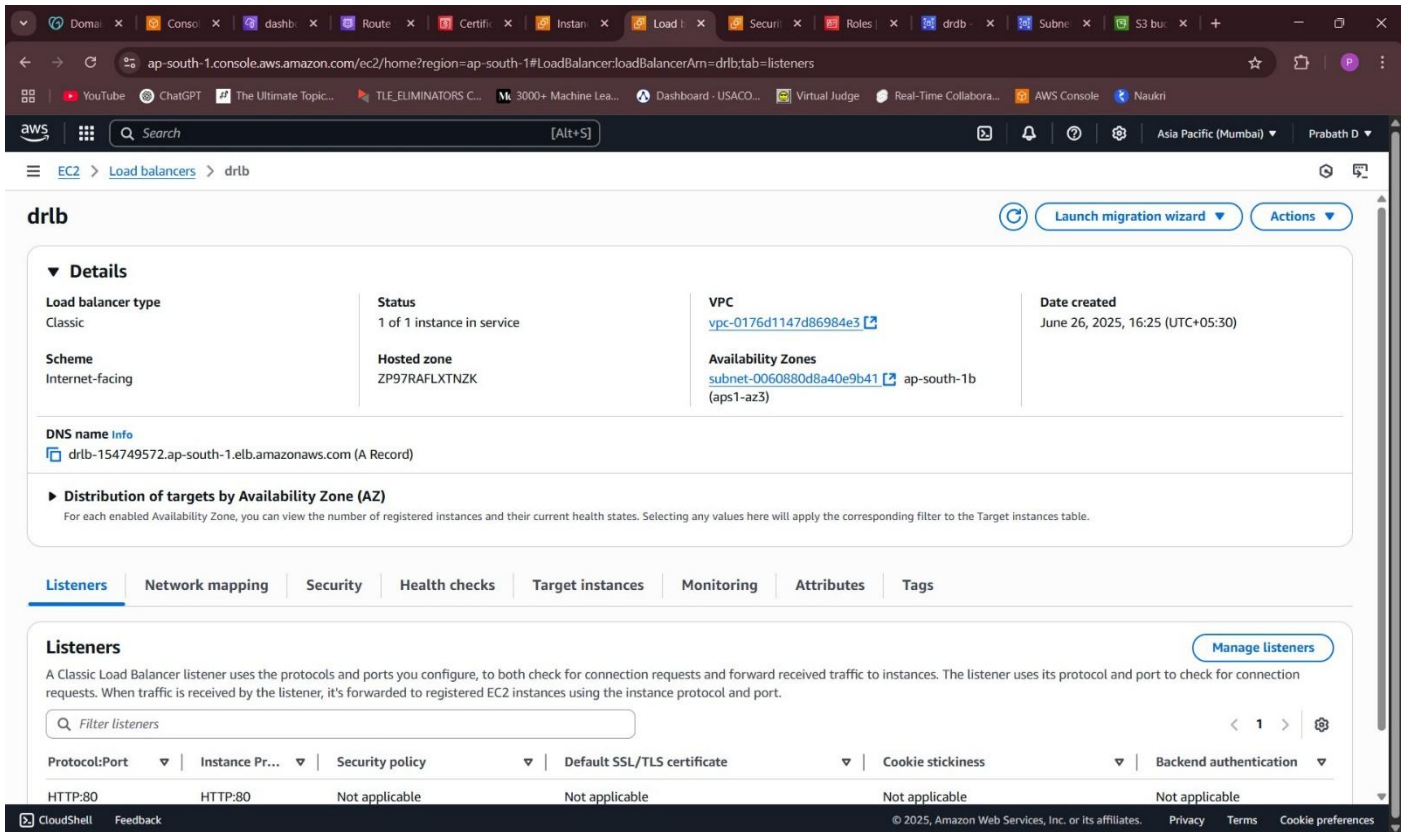
The bottom of the console shows tabs for Details, Status and alarms, Monitoring, Security, Networking, Storage, and Tags.



x. AWS ELB Creation & Setup:

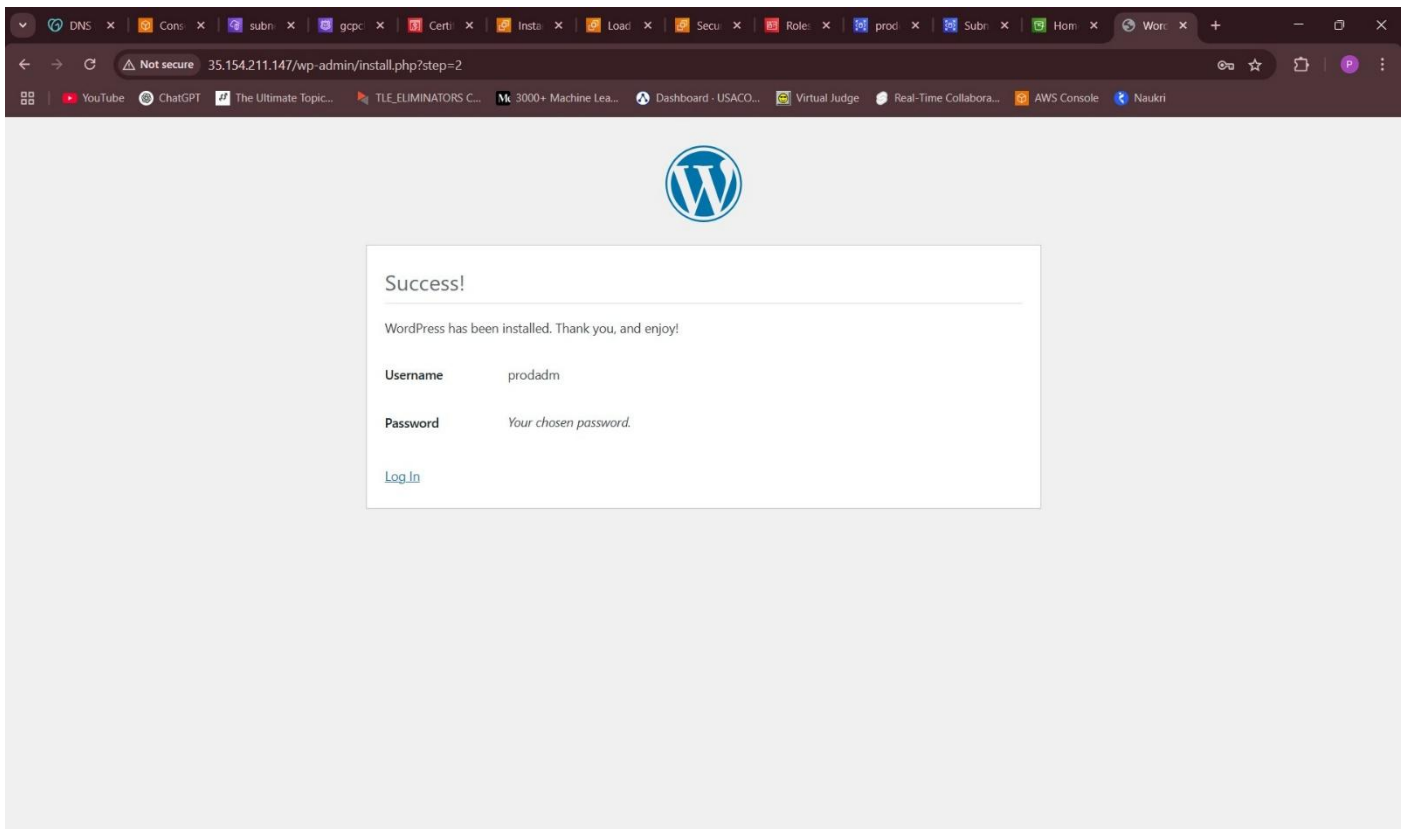
- Create two Classic ELB and map appropriate EC2 to it.

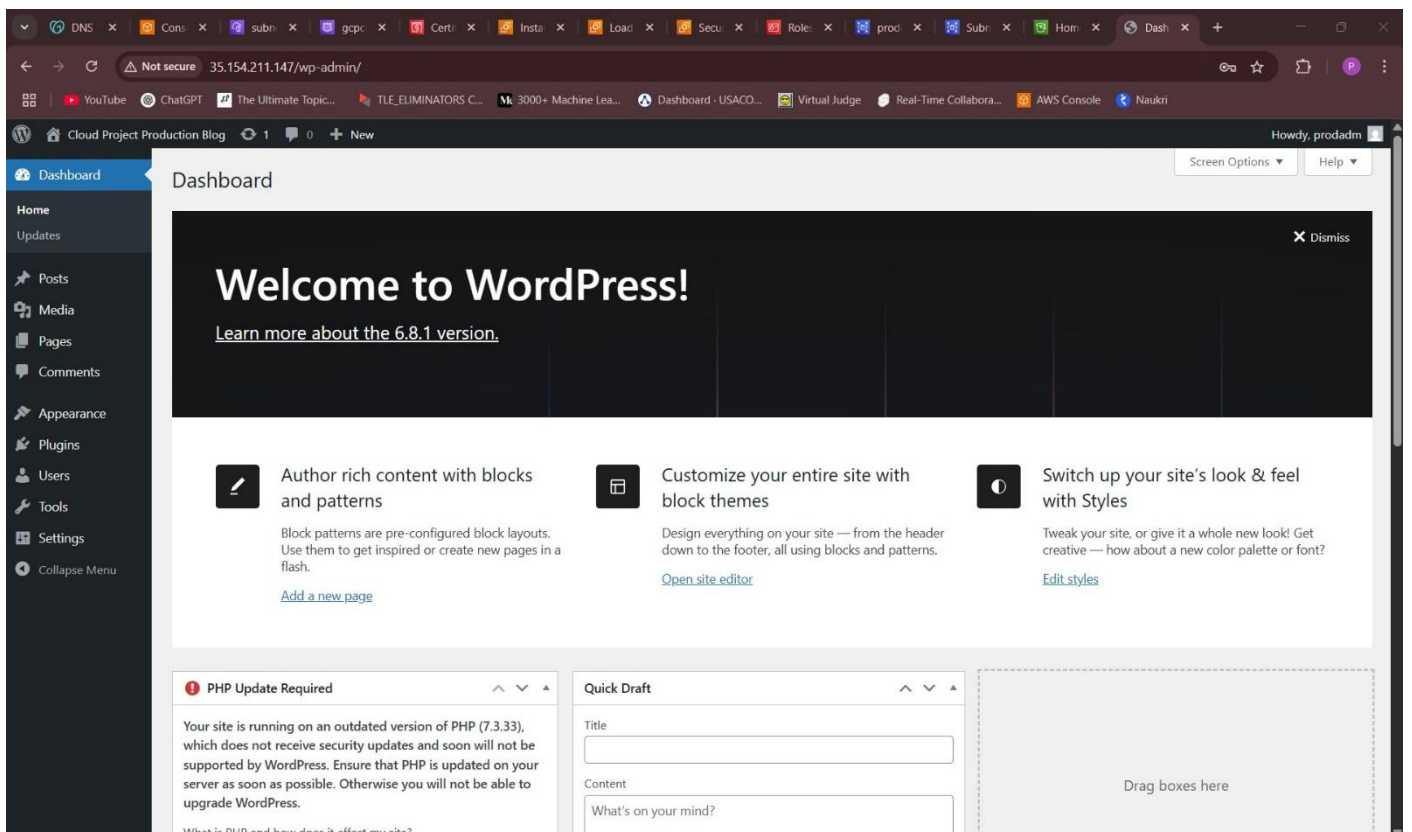
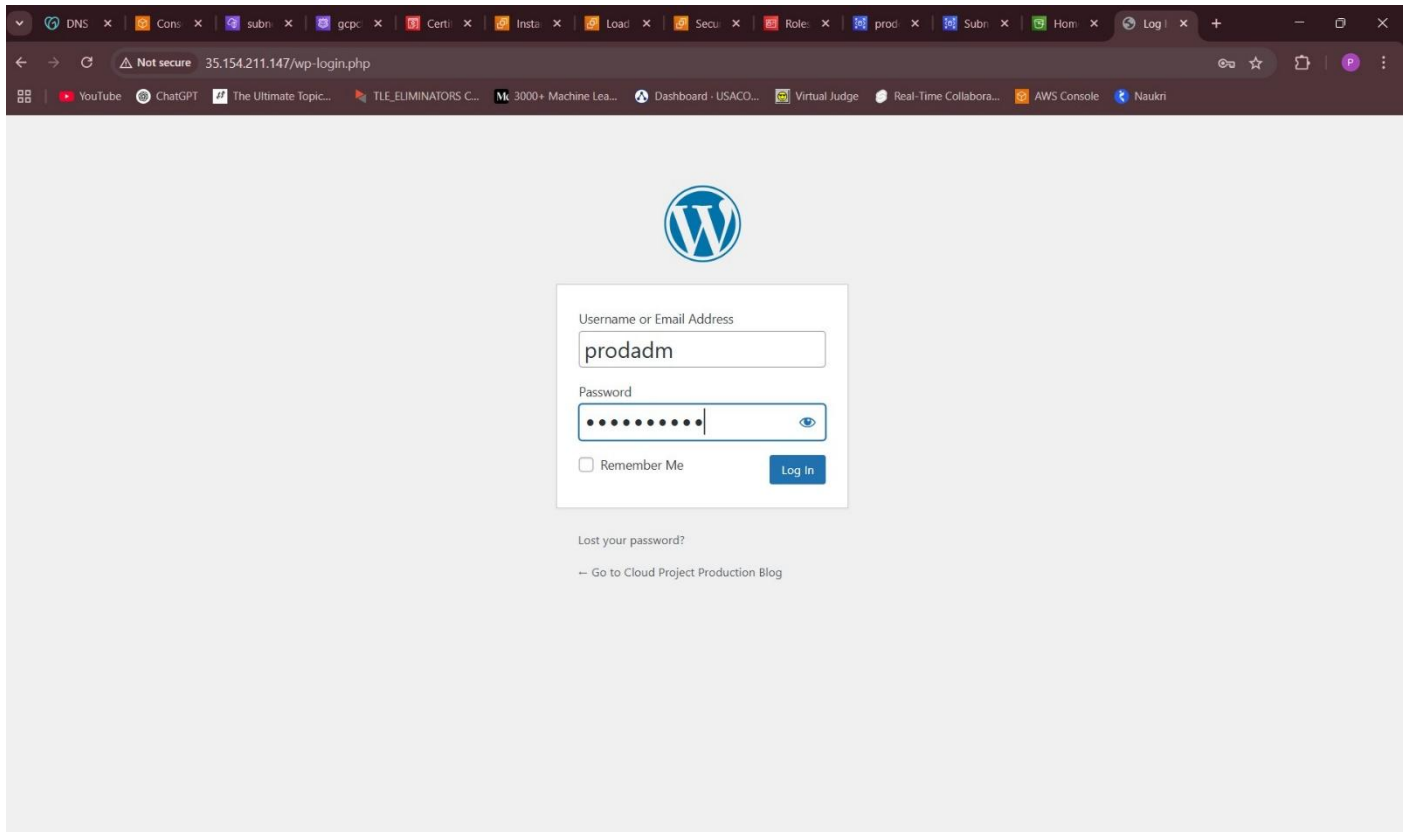


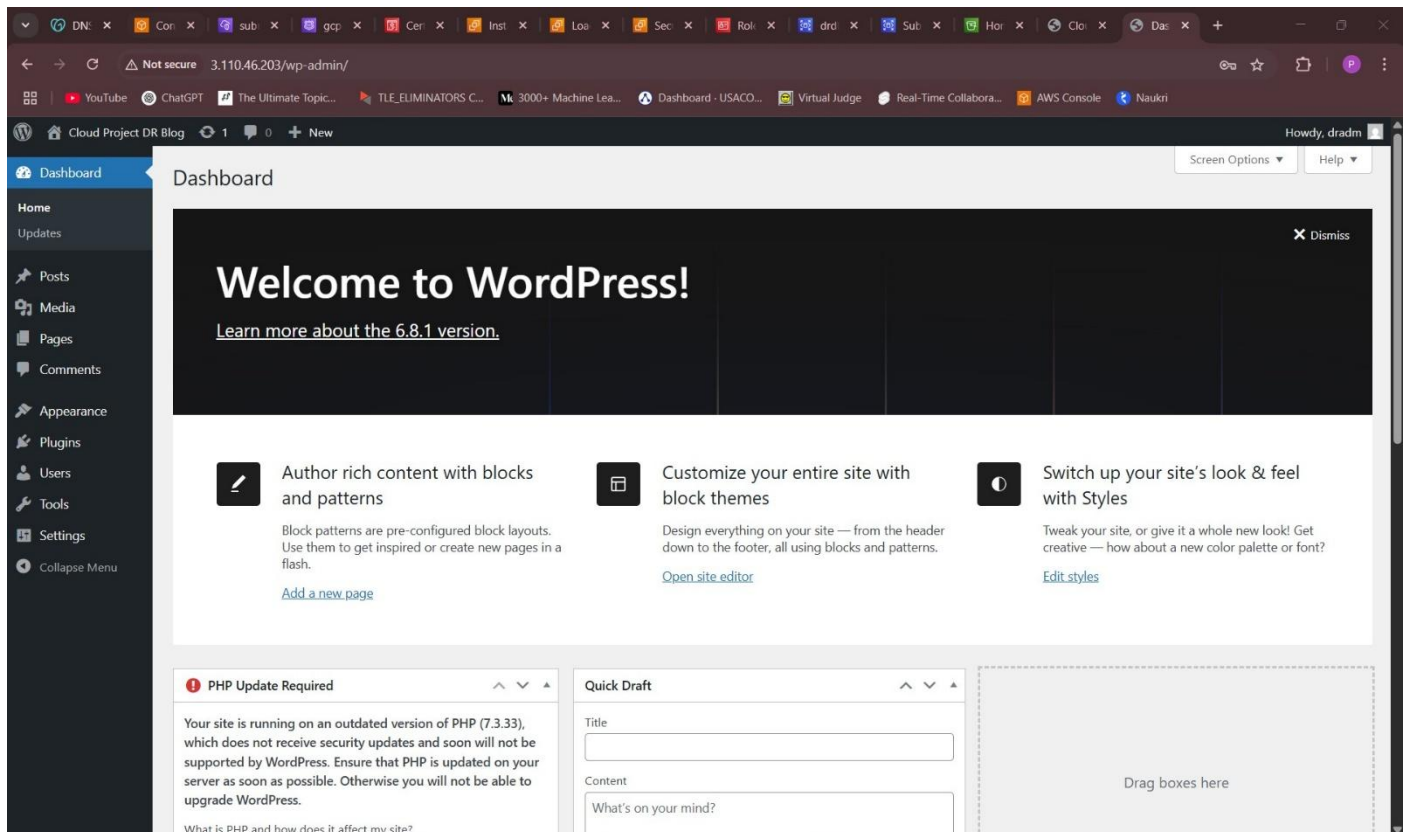
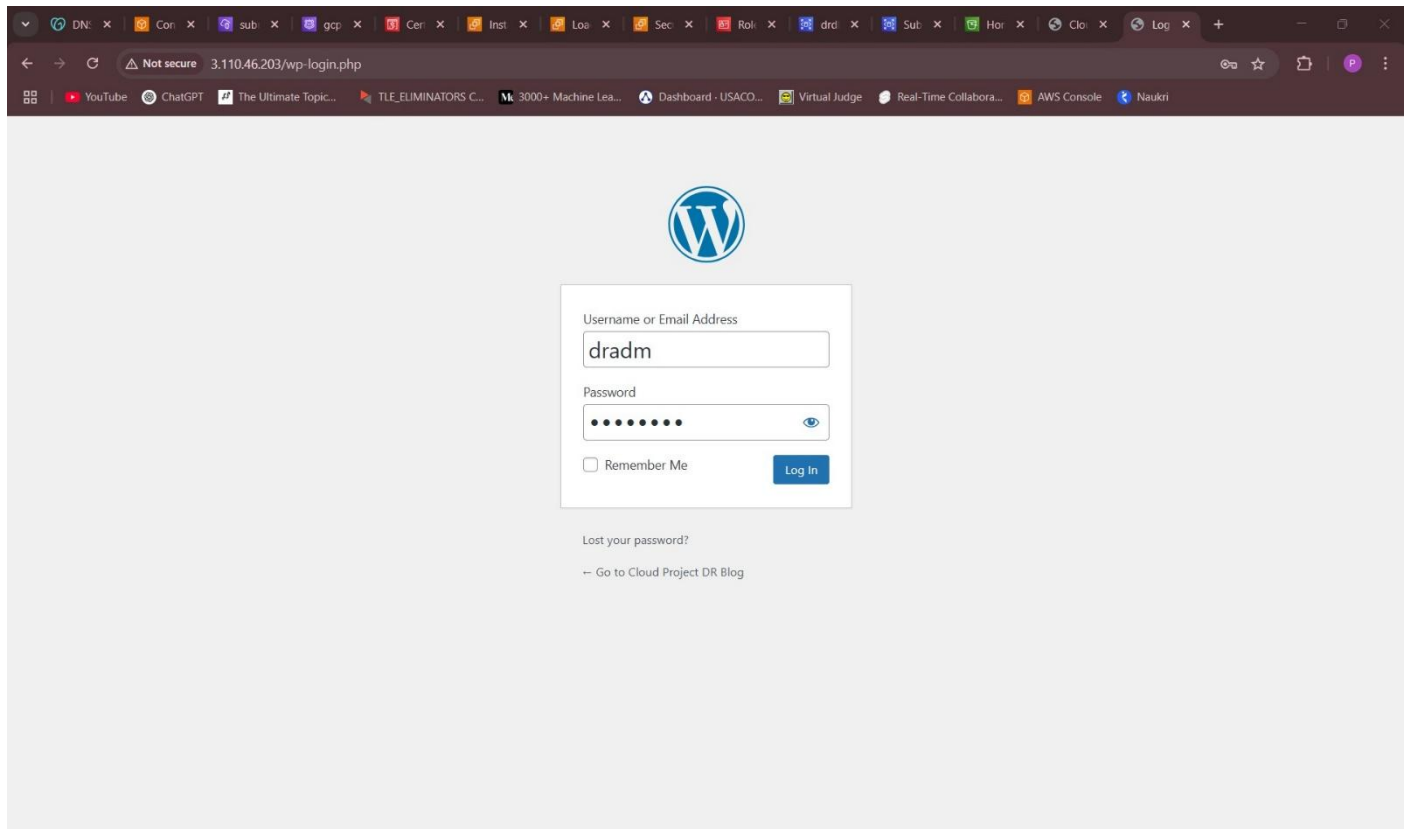


xi. Add the DB details under WordPress Application for both DR & Production:

- Create and edit the wp-config.php file in both the EC2s.

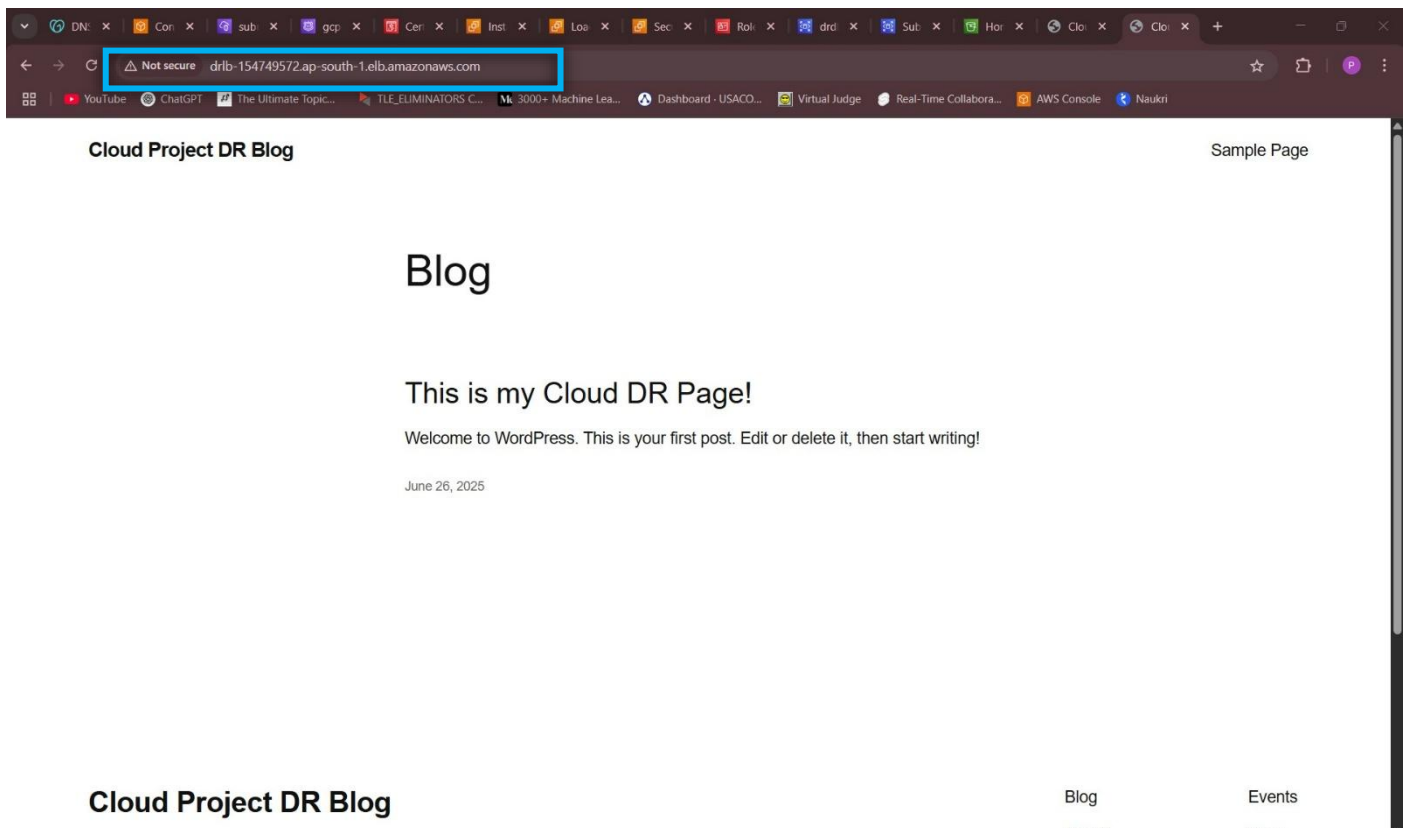
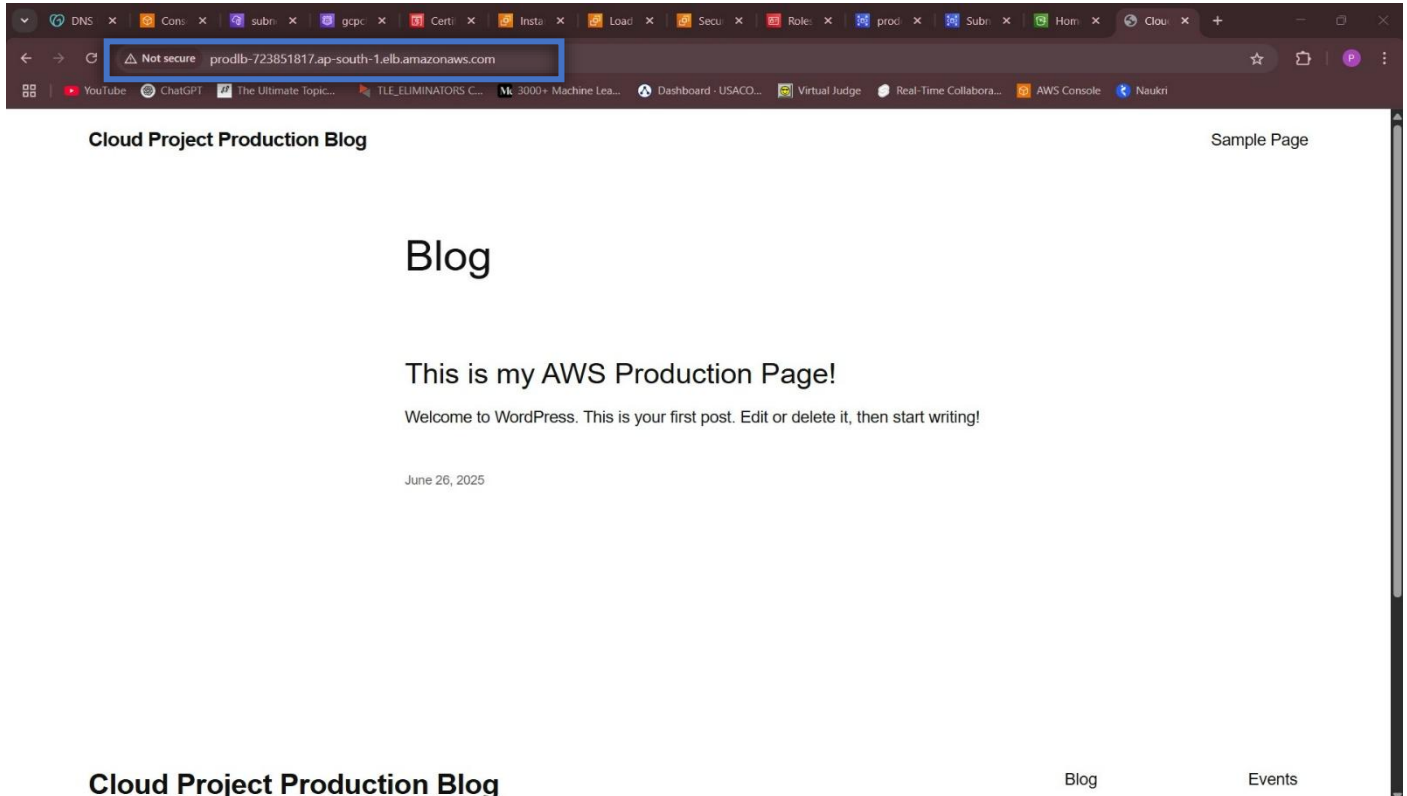






xii. Launch the WordPress page & post it with appropriate DR & Production texts to get difference in the application page:

- Now, with the help of DNS name of the prod1b and dr1b or IP Address of any one of EC2, paste it in the browser and load the page.



xiii. AWS R53 & ELB Sync:

- Refresh the R53
- Go to primary health check. Click create health check, then give a name then by mapping the created production ELB by giving the path in '/' as index.html.
- Create Record set and map appropriate ELB under alias by adding health checks in a particular region.

The screenshot shows the AWS Route 53 console for a health check named 'primary-health-check'. The configuration details are as follows:

Property	Value
ID	1192c1b6-c952-48b6-8249-196287627d4d
URL	http://prod1b-723851817.ap-south-1.elb.amazonaws.com:80/
Specified endpoint by	Domain name
State	Enabled
Status	Healthy
Inverted	No

Below the configuration, there are tabs for Metrics, Alarms, Health checkers, and Tags. The Metrics tab is active, showing a line graph for 'Health check status' and a bar chart for 'Health checkers that report the endpoint healthy (percent)'.

The screenshot shows the AWS Route 53 console for a hosted zone named 'gpcclouds.life'. The 'Records (6)' tab is active, displaying a table of DNS records:

Record name	Type	Routing...	Differ...	Alias	Value/Route traffic to	TTL (seconds)
gpcclouds.life	A	Failover	Primary	Yes	dualstack.prod1b-723851817...	-
gpcclouds.life	A	Failover	Secondary	Yes	dualstack.dr1b-154749572.a...	-
gpcclouds.life	NS	Simple	-	No	ns-1211.awsdns-23.org. ns-154.awsdns-19.com. ns-1554.awsdns-02.co.uk. ns-744.awsdns-29.net.	172800
gpcclouds.life	SOA	Simple	-	No	ns-1211.awsdns-23.org. aws...	900
_e3398eac1285c7aa2026a98ed618822d.gpcclouds.life	CNAME	Simple	-	No	_3f2f397fe2aaa5510a68b7e...	300
_5d8e5cb192ab0a4d5ecddfff8080af79.dr.gpcclouds.life	CNAME	Simple	-	No	_00c154d35191125a3fb993...	300

xiv. AWS S3 Bucket Creation:

- Go to S3 click create bucket, then create two S3 buckets for Production and DR.
- Set the crontab job's in the EC2 in small intervals to copy the content from production to S3 & S3 to DR.
- Add the media sets for production and DR.
- The media sets are:
 - wordpress-mediaasset-project
 - wordpress-code-project
- The content that must be given in server instance,

Production:

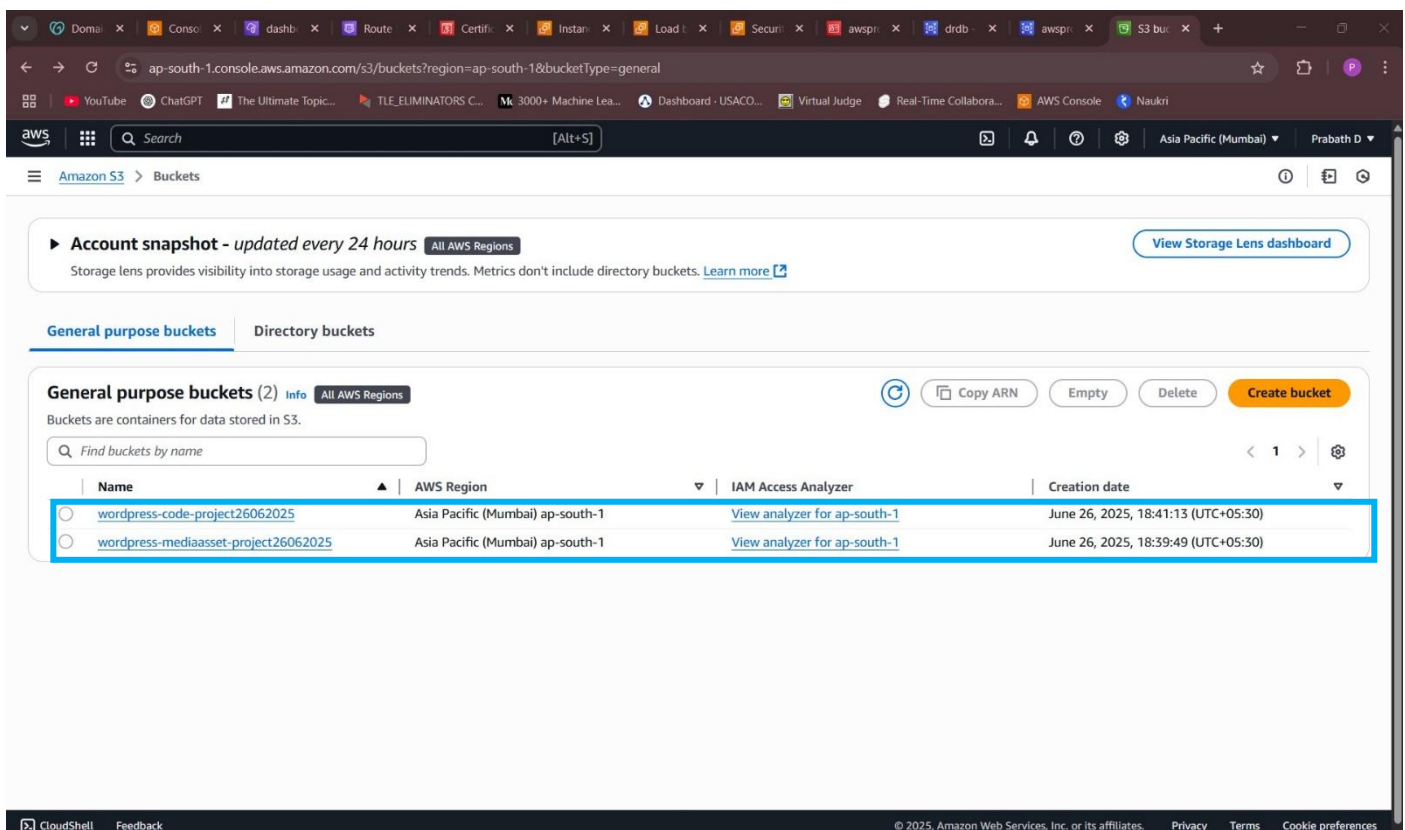
```
* / 2 * * * * aws s3 sync --delete /var/www/html/wp-content/uploads s3://wordpress-mediaasset-project
```

```
* / 2 * * * * aws s3 sync --delete /var/www/html/ s3://wordpress-code-project
```

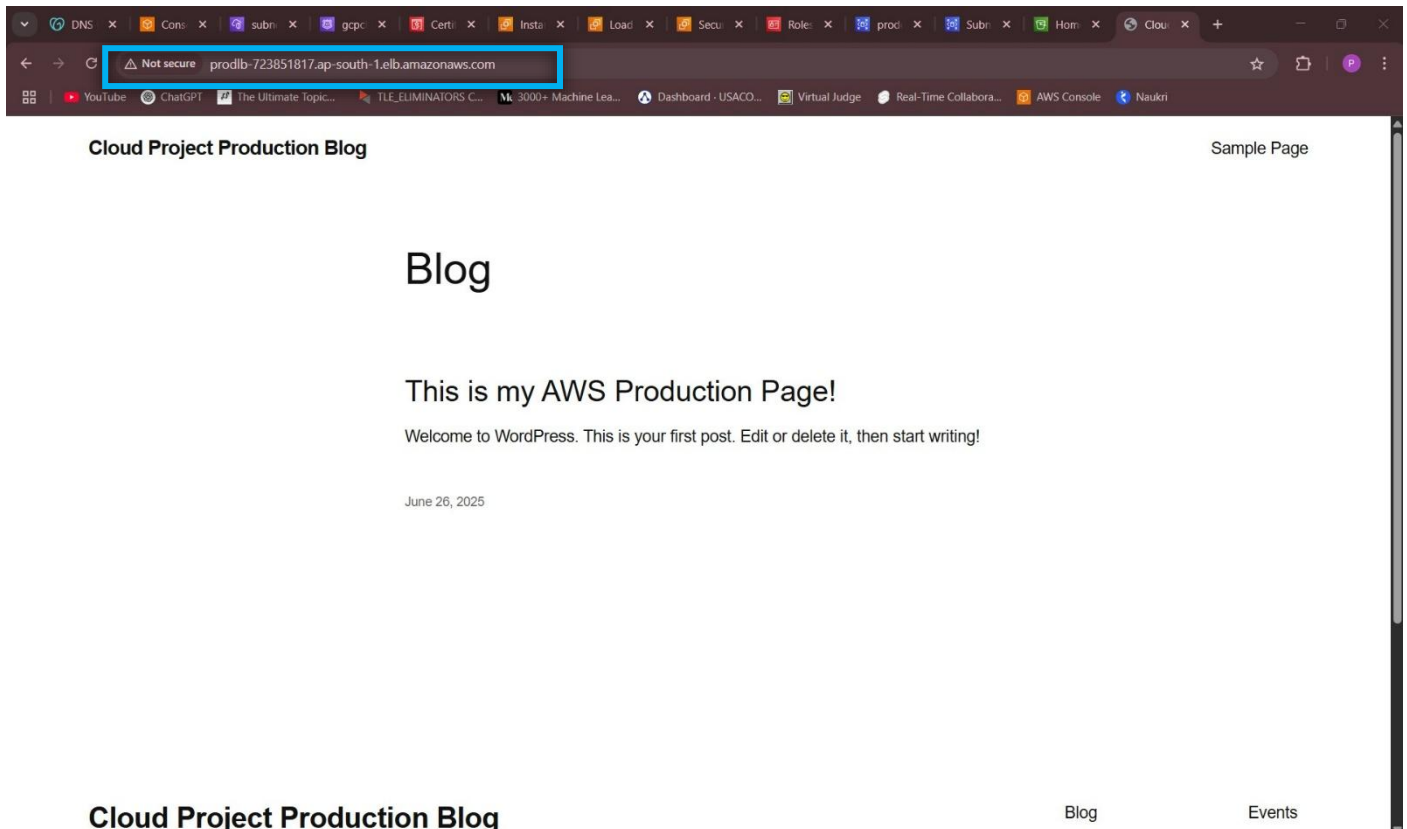
DR:

```
* / 2 * * * * aws s3 sync --delete s3://wordpress-mediaasset-project /var/www/html/wp-content/uploads
```

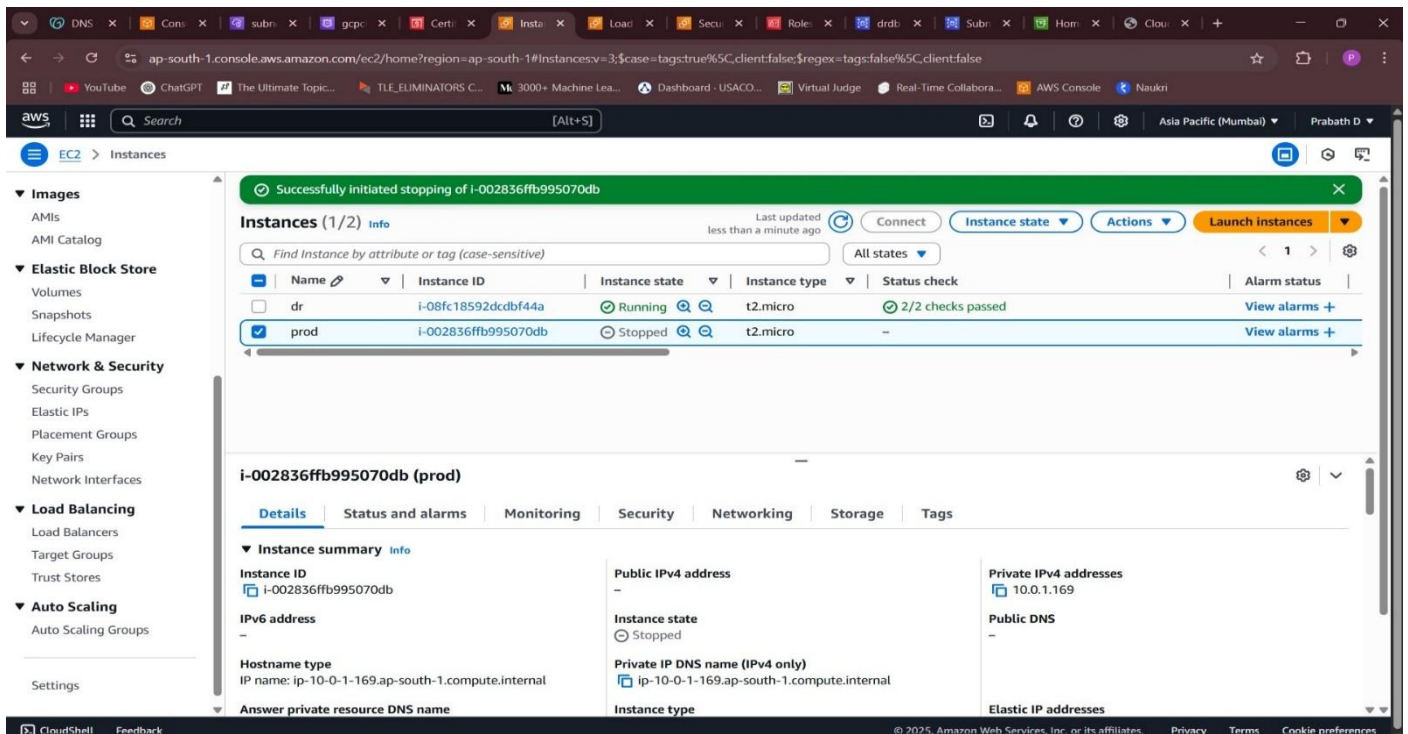
```
* / 2 * * * * aws s3 sync --delete s3://wordpress-code-project /var/www/html/
```



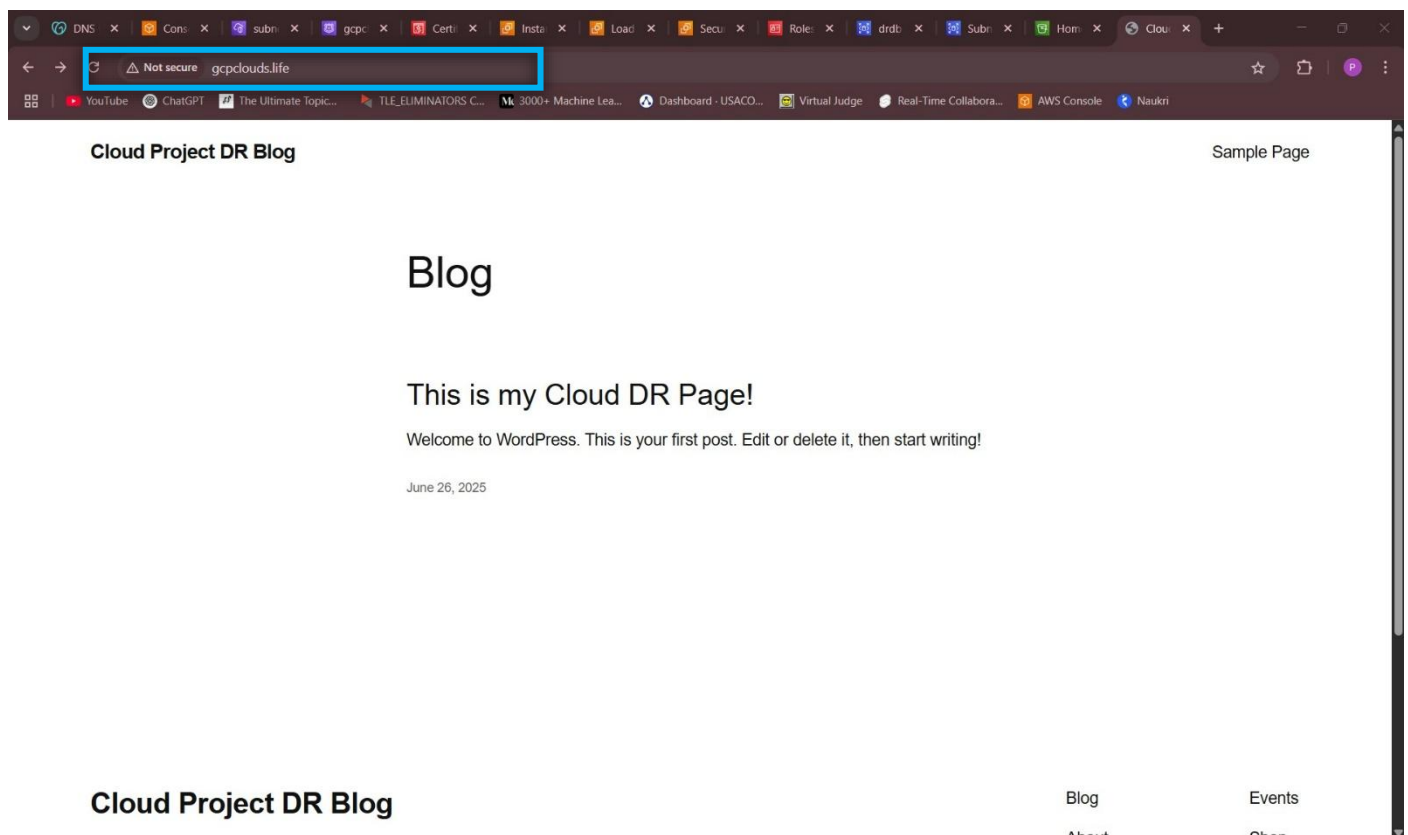
- Now, wait for few minutes and then refresh the page.
- **In this task, specifically the failover routing must be applied to view the Production page and DR page, if one availability zone is failed it redirects to the other zone, because of the custom VPC that is used in this.**
- The production blog that is hosted in the web browser in the form of DNS name.



- Now, go to EC2 instances and stop the Production instance for some time, after 2-3 mins, the health check in the route53 becomes unhealthy
- With that, by refreshing the page production DNS name that is hosted in the web browser it will redirect to the DR DNS name.
- By giving the name as `gpcclouds.life` it will show the DR Blog.



- Once, again if you start the instance of Production instance the web page will get refresh and shows the Production Blog



- The setup for deploying the WordPress page in this region also completed. With that implementing Failover Routing policy through Route53 is completed.

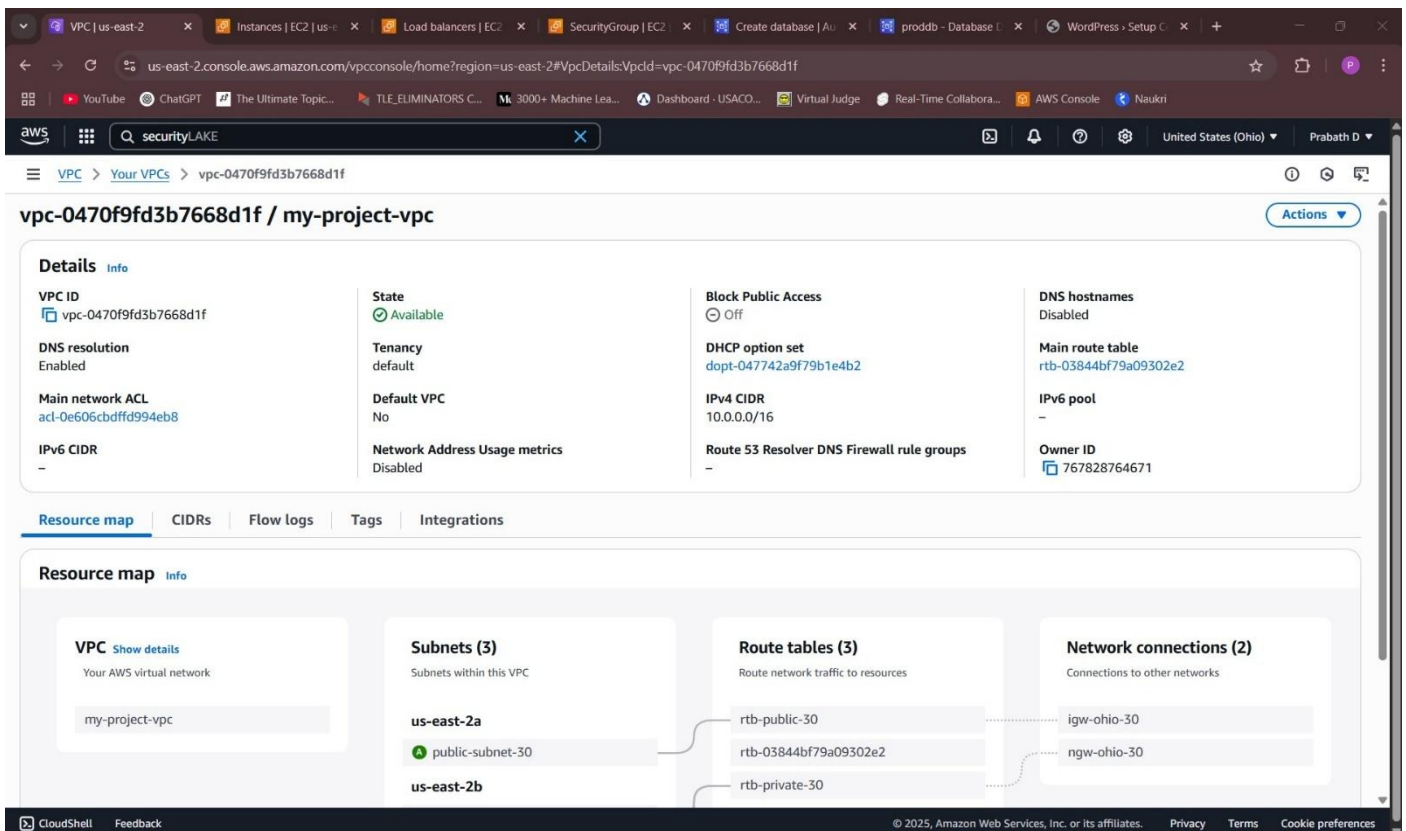
Task 3: Separate security Group for LB,EC2,RDS with restrictions on ports

Mumbai Region:

Steps:

i. VPC Creation:

- Go to your VPC.
- Click your VPC, then click create VPC then give a name as “my-project-vpc”
- Then set the IP address 10.0.0.0/16, then set both DNS hostname and DNS resolution as enabled, then click create.



ii. Subnet Setup:

- Go to subnets, click create subnet after that assign a name for the subnets, in this task we need two subnets for public and private.
- Then enter the below details particularly for public and private subnet, by assigning the IP Addresses, then click create subnet
- Public Subnet 1: For EC2 + NAT
- Public Subnet 2: For EC2
- Private Subnet 1: For RDS Prod

CIDR Blocks (Mumbai):

- 10.1.1.0/24 (Public subnet 1)
- 10.1.2.0/24 (Public subnet 2)
- 10.1.3.0/24 (Private subnet 1)

The screenshot displays the AWS Management Console interface for a subnet named 'subnet-0407b10783f519bb6 / public-subnet-30'. The console is in the 'us-east-2' region. The 'Details' tab is active, showing various configuration parameters for the subnet. The 'Flow logs' tab is also visible at the bottom.

Subnet Details:

- Subnet ID:** subnet-0407b10783f519bb6
- Subnet ARN:** arn:aws:ec2:us-east-2:767828764671:subnet/subnet-0407b10783f519bb6
- State:** Available
- IPv6 CIDR:** -
- Block Public Access:** Off
- IPv6 CIDR association ID:** -
- Route table:** rtb-070780d0e80756952 | rtb-public-30
- Auto-assign IPv6 address:** No
- IPv4 CIDR reservations:** -
- Resource name DNS AAAA record:** Disabled
- Auto-assign customer-owned IPv4 address:** No
- Available IPv4 addresses:** 247
- Availability Zone ID:** use2-az1
- Default subnet:** No
- Customer-owned IPv4 pool:** -
- IPv6-only:** No
- DNS64:** Disabled
- VPC:** vpc-0470f9fd3b7668d1f | my-project-vpc
- Auto-assign public IPv4 address:** No
- Outpost ID:** -
- Hostname type:** IP name
- Owner:** 767828764671

Flow logs:

Search: []

Columns: Name, Flow log ID, Filter, Destination type, Destination name, IAM role ARN

Page: 1

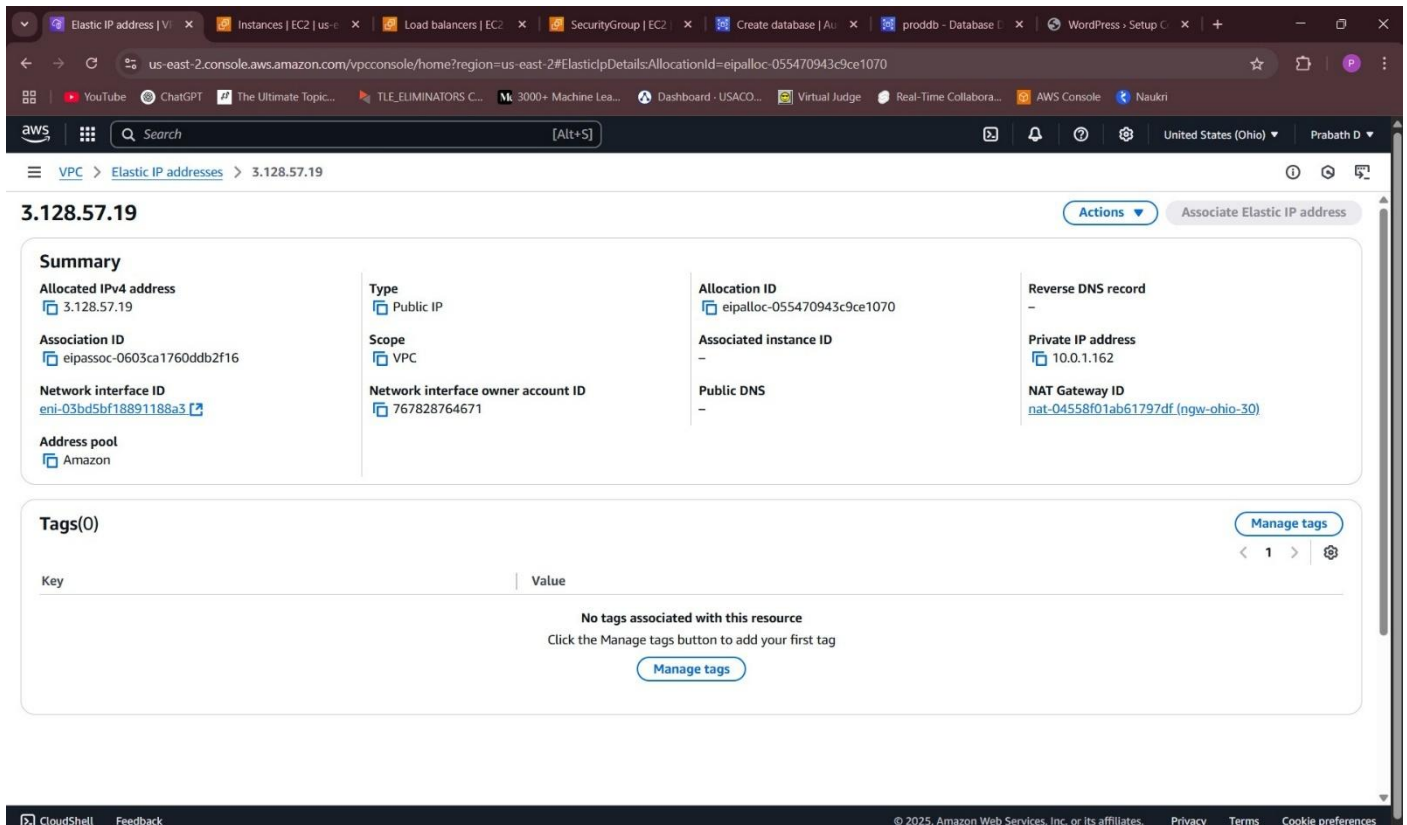
iii. Internet Gateway + NAT Gateway:

- Create and attach IGW in Ohio VPC.

The screenshot shows the AWS Management Console for the us-east-2 region. The breadcrumb navigation is VPC > Internet gateways > igw-039dcd47b9d11ad58. The page title is "igw-039dcd47b9d11ad58 / igw-ohio-30". The "Details" section shows the Internet gateway ID as igw-039dcd47b9d11ad58, the state as "Attached", the VPC ID as vpc-0470f9fd3b7668d1f, and the owner as 767828764671. The "Tags" section shows a single tag with the key "Name" and the value "igw-ohio-30".

- Create NAT Gateway in Public Subnet (with Elastic IP).

The screenshot shows the AWS Management Console for the us-east-2 region. The breadcrumb navigation is VPC > NAT gateways > nat-04558f01ab61797df. The page title is "nat-04558f01ab61797df / ngw-ohio-30". The "Details" section shows the NAT gateway ID as nat-04558f01ab61797df, the connectivity type as "Public", the primary public IPv4 address as 3.128.57.19, the primary private IPv4 address as 10.0.1.162, and the state as "Available". The "Secondary IPv4 addresses" section shows that no secondary addresses are available for this NAT gateway.



iv. Route Tables (Mumbai):

- Then after creating both Internet Gateway and NAT Gateway, go to route tables, click create, then assign names for both public and private route tables, then associate both the tables by clicking routes then attach the network
- Public Route Table → 0.0.0.0/0 → IGW.
- Then click subnet associations in public route table, then click edit association after that click both public subnet 1.
- Private Route Table → 0.0.0.0/0 → NAT.
- Then click subnet associations in private route table, then click edit association after that click both private subnet 1 and private subnet 2.

VPC | us-east-2

Instances | EC2 | us-e

Load balancers | EC2

SecurityGroup | EC2

Create database | Au

proddb - Database

WordPress - Setup C

us-east-2.console.aws.amazon.com/vpcconsole/home?region=us-east-2#RouteTableDetails:RouteTableId=rtb-070780d0e80756952

Search [Alt+S]

United States (Ohio) Prabath D

VPC > Route tables > rtb-070780d0e80756952

rtb-070780d0e80756952 / rtb-public-30

Details Info

Route table ID
rtb-070780d0e80756952

VPC
vpc-0470f9fd3b7668d1f | my-project-vpc

Main
No

Owner ID
767828764671

Explicit subnet associations
subnet-0407b10783f519bb6 / public-subnet-30

Edge associations
-

Routes Subnet associations Edge associations Route propagation Tags

Routes (2)

Filter routes

Both Edit routes

Destination	Target	Status	Propagated
0.0.0.0/0	igw-039dcd47b9d11ad58	Active	No
10.0.0.0/16	local	Active	No

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VPC | us-east-2

Instances | EC2 | us-e

Load balancers | EC2

SecurityGroup | EC2

Create database | Au

proddb - Database

WordPress - Setup C

us-east-2.console.aws.amazon.com/vpcconsole/home?region=us-east-2#RouteTableDetails:RouteTableId=rtb-070780d0e80756952

Search [Alt+S]

United States (Ohio) Prabath D

VPC > Route tables > rtb-070780d0e80756952

rtb-070780d0e80756952 / rtb-public-30

Details Info

Route table ID
rtb-070780d0e80756952

VPC
vpc-0470f9fd3b7668d1f | my-project-vpc

Main
No

Owner ID
767828764671

Explicit subnet associations
subnet-0407b10783f519bb6 / public-subnet-30

Edge associations
-

Routes Subnet associations Edge associations Route propagation Tags

Explicit subnet associations (1)

Find subnet association

Edit subnet associations

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
public-subnet-30	subnet-0407b10783f519bb6	10.0.1.0/24	-

Subnets without explicit associations (0)

Find subnet association

Edit subnet associations

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
------	-----------	-----------	-----------

No subnets without explicit associations
All your subnets are associated with a route table.

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v. Security Groups:

- Go to EC2, Security Groups then create securities for EC2, RDS, LB with restrictions ports on each.

I. EC2 Security Groups:

- In EC2 set the ports in inbound rules,
 - Ports: HTTP - Port Range - 80 - Source - Anywhere/0.0.0.0/0.
- Also allow another port in inbound rules,
 - Ports: HTTP – Port Range – 80 – Source - my-ip.

The screenshot shows the AWS Management Console for the security group **sg-001f1a177720ccda6 - EC2-SG**. The details section shows the security group name, ID, owner, and inbound/outbound rule counts. The inbound rules section shows two rules: one for HTTP (port 80) from anywhere (0.0.0.0/0) and one for SSH (port 22) from a specific IP (106.222.221.143/32).

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
-	sgr-0ed1956265a39a05b	IPv4	HTTP	TCP	80	0.0.0.0/0
-	sgr-0d919b9982ceac7d6	IPv4	SSH	TCP	22	106.222.221.143/32

II. RDS Security Groups:

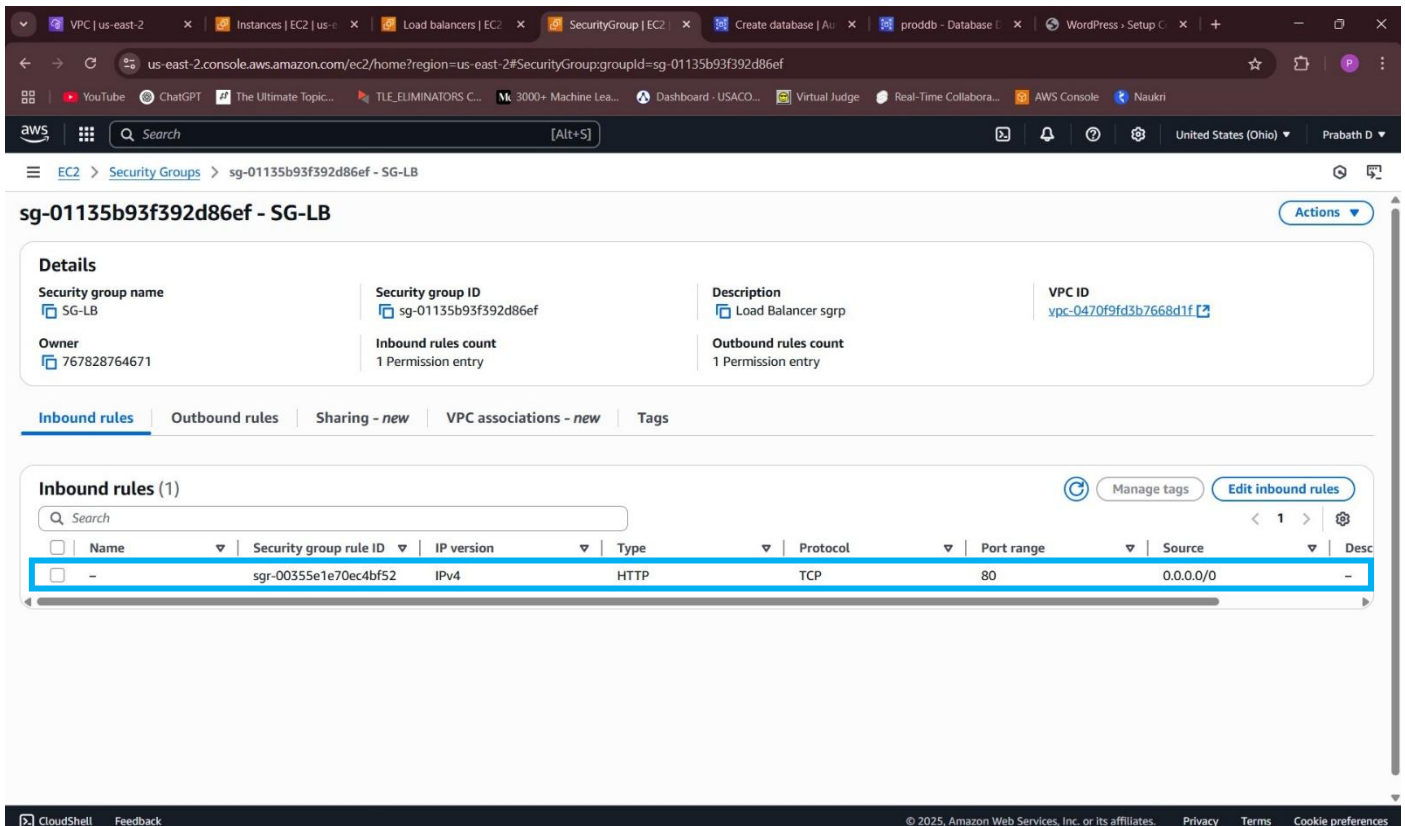
- In RDS set the ports in inbound rules,
 - Ports: MySQL - Port Range - 3306 - Source - EC2-SGRP.

The screenshot shows the AWS Management Console for the security group **sg-0b54cdcd6b365908 - RDS-SG**. The details section shows the security group name, ID, owner, and inbound/outbound rule counts. The inbound rules section shows one rule for MySQL/Aurora (port 3306) from the EC2 security group sg-001f1a177720ccda6.

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
-	sgr-0f2dc65e7917a0445	-	MySQL/Aurora	TCP	3306	sg-001f1a177720ccda6...

III. LB Security Groups:

- In LB set the ports in inbound rules,
 - Ports: HTTP - Port Range - 80 - Source - Anywhere/0.0.0.0/0.

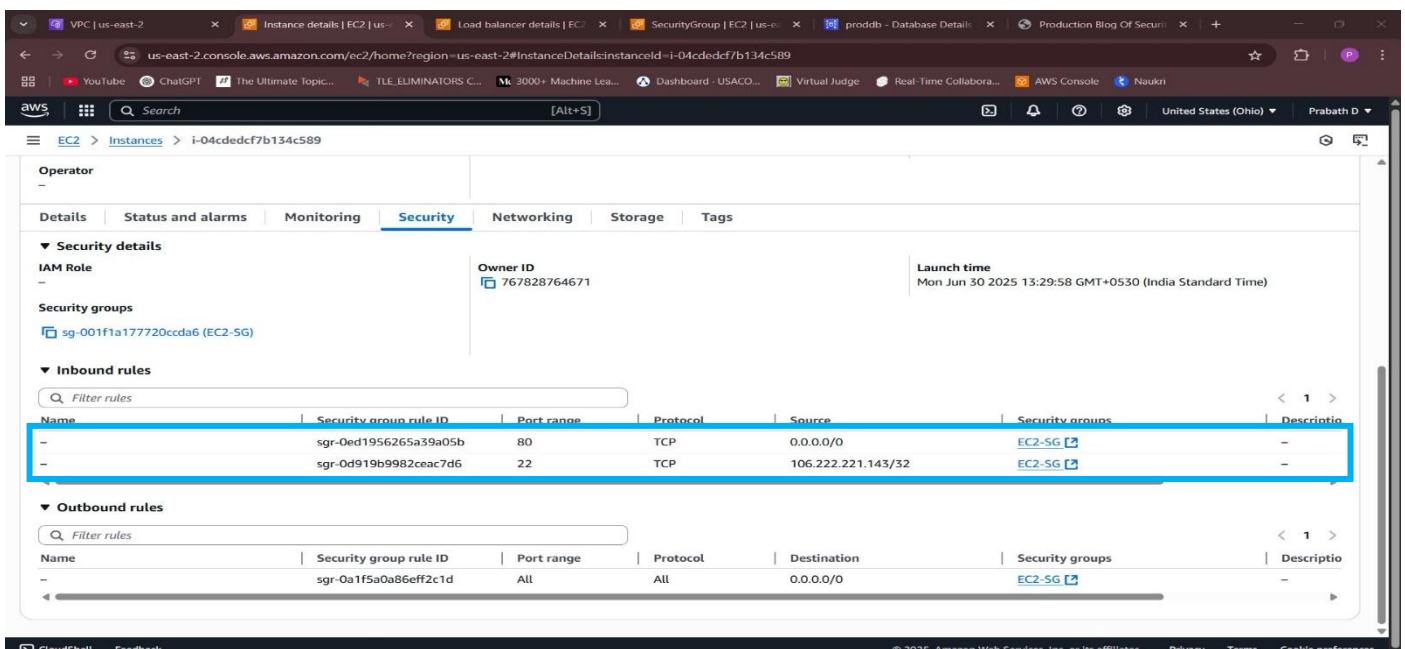


The screenshot displays the AWS Management Console for a Security Group named 'sg-01135b93f392d86ef - SG-LB'. The 'Details' section shows the Security group name, ID, Description ('Load Balancer sgrp'), VPC ID, Owner, Inbound rules count (1), and Outbound rules count (1). The 'Inbound rules' tab is active, showing a table with one rule:

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
-	sgr-00355e1e70ec4bf52	IPv4	HTTP	TCP	80	0.0.0.0/0	-

vi. Attach the Security Groups:

- Attach the security groups with EC2 instance, RDS instance and Load Balancer.
- The respective security allows the web page to load in the specific IP Address.



The screenshot displays the AWS Management Console for an EC2 instance named 'i-04cdedcf7b134c589'. The 'Security' tab is active, showing the 'Security details' section with the IAM Role, Owner ID, Launch time, and Security groups. The 'Inbound rules' section shows a table with two rules:

Name	Security group rule ID	Port range	Protocol	Source	Security groups	Description
-	sgr-0ed1956265a39a05b	80	TCP	0.0.0.0/0	EC2-SG	-
-	sgr-0d919b9982ceac7d6	22	TCP	106.222.221.143/32	EC2-SG	-

VPC | us-east-2Instances | EC2 | us-east-2Load balancer details | EC2SecurityGroup | EC2 | us-east-2aws-subnet-grp-30 - SubnetWordPress - Setup Config

us-east-2.console.aws.amazon.com/rds/home?region=us-east-2#db-subnet-group:id=aws-subnet-grp-30

Search[Alt+S]

United States (Ohio)Prabath D

Aurora and RDSSubnet groupsaws-subnet-grp-30

aws-subnet-grp-30

Subnet group details

VPC ID

vpc-0470f9fd3b7668d1f

ARN

arn:aws:rds:us-east-2:767828764671:subgrp:aws-subnet-grp-30

Supported network types

IPv4

Description

aws-subnet-grp-30

Subnets (2)

Availability zone	Subnet name	Subnet ID	CIDR block
us-east-2c	private-subnet-11-30	subnet-0972d040f89be0a38	10.0.3.0/24
us-east-2b	private-subnet-30	subnet-0b00cc708149d6cae	10.0.2.0/24

Tags (0)

Filter by Tag key

Manage tags

1

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VPC | us-east-2Instances | EC2 | us-east-2Load balancer details | EC2SecurityGroup | EC2 | us-east-2proddb - Database DetailsWordPress - Setup Config

us-east-2.console.aws.amazon.com/rds/home?region=us-east-2#database:id=proddb;is-cluster=false

Search[Alt+S]

United States (Ohio)Prabath D

Aurora and RDS Databasesproddb

proddb

Summary

DB identifier

proddb

CPU

3.63%

Status

Available

Class

db.t3.micro

Role

Instance

Current activity

0 Connections

Engine

MySQL Community

Region & AZ

us-east-2b

Recommendations

Connectivity & security

Monitoring

Logs & events

Configuration

Zero-ETL integrations

Maintenance & backups

Data migrations - new

Tags

Record

Connectivity & security

Endpoint & port

Endpoint

proddb.c7osqm6c4gck.us-east-2.rds.amazonaws.com

Port

3306

Networking

Availability Zone

us-east-2b

VPC

my-project-vpc (vpc-0470f9fd3b7668d1f)

Subnet group

aws-subnet-grp-30

Subnets

subnet-0b00cc708149d6cae
subnet-0972d040f89be0a38

Security

VPC security groups

RDS-SG (sg-0b54cdcd6b365908)

Active

Publicly accessible

No

Certificate authority

rds-ca-rsa2048-g1

Certificate authority date

May 22, 2061, 05:34 (UTC+05:30)

DB instance certificate expiration date

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us-east-2.console.aws.amazon.com/rds/home?region=us-east-2#database-id=proddb;is-cluster=false

Proxy identifier Status Engine family

No proxies

You don't have any proxies.

Create proxy

Security group rules (2)

Filter by Security group rules

Security group	Type	Rule
RDS-SG (sg-0b54cdcd6b365908)	EC2 Security Group - Inbound	sg-001f1a177720ccda6
RDS-SG (sg-0b54cdcd6b365908)	CIDR/IP - Outbound	0.0.0.0/0

Replication (1)

Filter by Replication

DB identifier	Role	Region & AZ	Replication source	Replication state	Lag
proddb	Instance	us-east-2b	-	-	-

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us-east-2.console.aws.amazon.com/ec2/home?region=us-east-2#LoadBalancerloadBalancerArn=prod-LB;tab=security

EC2 > Load balancers > prod-LB

prod-LB Launch migration wizard Actions

▼ Details

Load balancer type Classic	Status 1 of 1 instance in service	VPC vpc-0470f9fd3b7668d1f	Date created June 30, 2025, 13:36 (UTC+05:30)
Scheme Internet-facing	Hosted zone Z3AADJGX6KTTL2	Availability Zones subnet-0407b10783f519bb6 us-east-2a (use2-az1)	

DNS name info
[prod-LB-1763272256.us-east-2.elb.amazonaws.com](#) (A Record)

► Distribution of targets by Availability Zone (AZ)
For each enabled Availability Zone, you can view the number of registered instances and their current health states. Selecting any values here will apply the corresponding filter to the Target instances table.

Listeners Network mapping Security Health checks Target instances Monitoring Attributes Tags

Security groups (1) Edit

A security group is a set of firewall rules that control the traffic to your load balancer.

Security Group ID	Name	Description
sg-01135b93f392d86ef	SG-LB	Load Balancer sggrp

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vi. Installation and Validated Output:

- Update and install the MySQL version in putty configuration through EC2 instance, as the data must be stored via EC2.

```
root@ip-10-0-1-56:/var/www/html# login as: ec2-user
Authenticating with public key "linux25key"
Last login: Mon Jun 30 09:24:18 2025 from 106.222.221.143

#####
/\_   #####      Amazon Linux 2
 ~\ _  #####     AL2 End of Life is 2026-06-30.
 ~\ _  \####|
 ~\ _  \#/
 ~\ _  V~' '->
 ~\ _ /
 ~\ _ /
 ~\ m/' -/-/

A newer version of Amazon Linux is available!

Amazon Linux 2023, GA and supported until 2028-03-15.
https://aws.amazon.com/linux/amazon-linux-2023/

[ec2-user@ip-10-0-1-56 ~]$ sudo -i
[root@ip-10-0-1-56 ~]# sudo yum install mysql -y
Loaded plugins: extras_suggestions, langpacks, priorities, update-motd
amzn2-core                                     | 3.6 kB    00:00
Resolving Dependencies
--> Running transaction check
---> Package mariadb.x86_64 1:5.5.68-1.amzn2.0.1 will be installed
--> Finished Dependency Resolution

Dependencies Resolved

=====
Package             Arch           Version              Repository          Size
=====
Installing:
mariadb            x86_64         1:5.5.68-1.amzn2.0.1 amzn2-core          8.8 M

Transaction Summary
=====
Install 1 Package

Total download size: 8.8 M
Installed size: 49 M
Downloading packages:
mariadb-5.5.68-1.amzn2.0.1.x86_64.rpm        | 8.8 MB    00:00
Running transaction check
Running transaction test
Transaction test succeeded
Running transaction
  Installing : 1:mariadb-5.5.68-1.amzn2.0.1.x86_64                    1/1
```

```
root@ip-10-0-1-56:/var/www/html# login as: ec2-user
Authenticating with public key "linux25key"

#####
~\#####\ Amazon Linux 2
~~~\#####\
~~~\###| AL2 End of Life is 2026-06-30.
~~~\#/
~~~~V~' ->
~~~~
~~~~_..
~~~~/_/' - /

A newer version of Amazon Linux is available!

Amazon Linux 2023, GA and supported until 2028-03-15.
https://aws.amazon.com/linux/amazon-linux-2023/

[ec2-user@ip-10-0-1-56 ~]$ sudo -i
[root@ip-10-0-1-56 ~]# ssh -i linux25key.pem ec2-user@<18.220.92.52>
-bash: syntax error near unexpected token `newline'
[root@ip-10-0-1-56 ~]# ssh -i
option requires an argument -- i
usage: ssh [-l246AaCfGgKkMnNqsTtTvVxXyY] [-b bind_address] [-c cipher_spec]
          [-D [bind_address:]port] [-E log_file] [-e escape_char]
          [-F configfile] [-I pkcs11] [-i identity_file]
          [-J [user@]host[:port]] [-L address] [-l login_name] [-m mac_spec]
          [-O ctl_cmd] [-o option] [-p port] [-Q query_option] [-R address]
          [-S ctl_path] [-W host:port] [-w local_tun[:remote_tun]]
          [user@]hostname [command]
[root@ip-10-0-1-56 ~]# linux25key.pem ec2-user@<18.220.92.52>
-bash: syntax error near unexpected token `newline'
[root@ip-10-0-1-56 ~]# sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; vendor preset
: disabled)
   Drop-In: /usr/lib/systemd/system/httpd.service.d
            └─ php-fpm.conf
   Active: active (running) since Mon 2025-06-30 08:01:30 UTC; 57min ago
     Docs: man:httpd.service(8)
 Main PID: 3544 (httpd)
    Status: "Total requests: 1153; Idle/Busy workers 66/33; Requests/sec: 0.334; B
ytes served/sec: 274 B/sec"
   CGroup: /system.slice/httpd.service
            └─ 508 /usr/sbin/httpd -DFOREGROUND
               515 /usr/sbin/httpd -DFOREGROUND
               586 /usr/sbin/httpd -DFOREGROUND
              3544 /usr/sbin/httpd -DFOREGROUND
              3569 /usr/sbin/httpd -DFOREGROUND
```

```

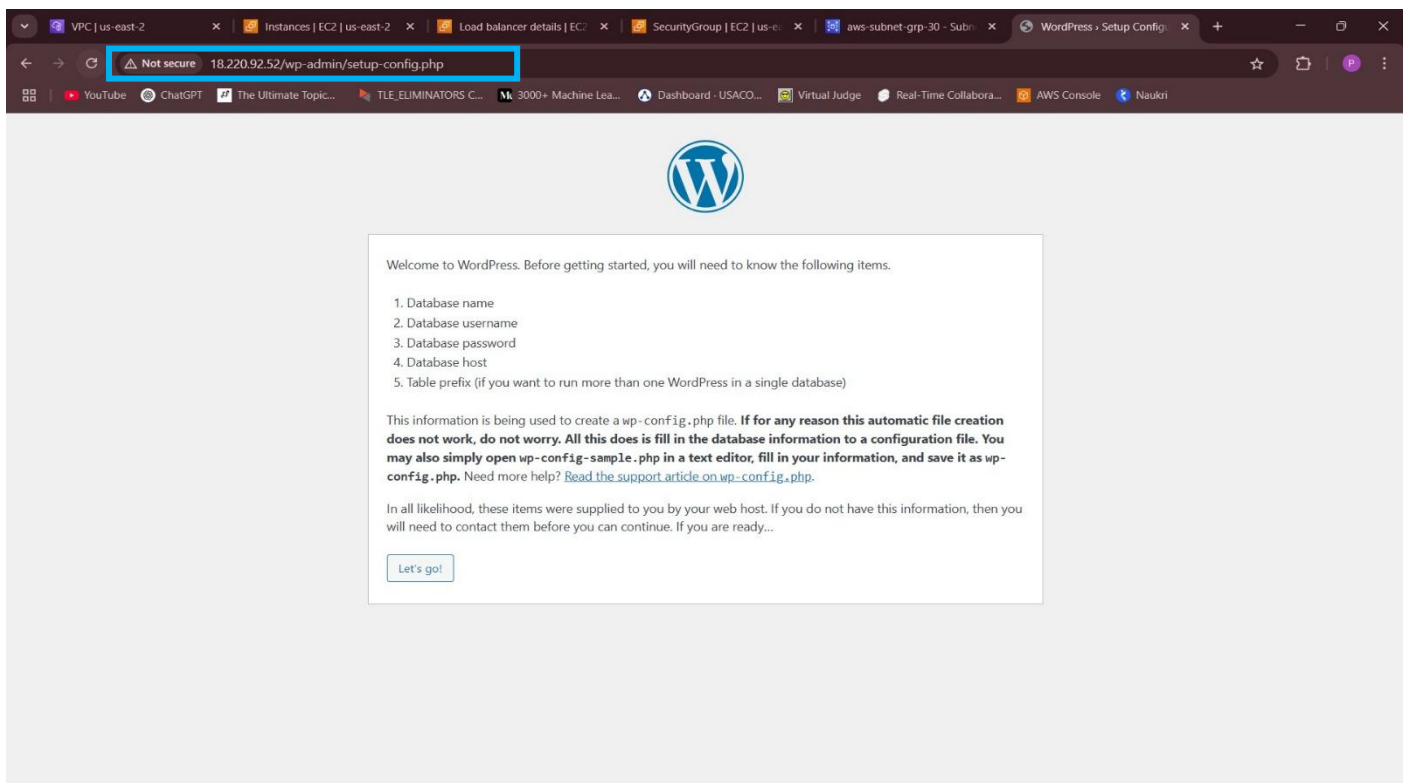
root@ip-10-0-1-56:/var/www/html
[-F configfile] [-I pkcs11] [-i identity_file]
[-J [user@]host[:port]] [-L address] [-l login_name] [-m mac_spec]
[-O ctl_cmd] [-o option] [-p port] [-Q query_option] [-R address]
[-S ctl_path] [-W host:port] [-w local_tun[:remote_tun]]
[user@]hostname [command]
[root@ip-10-0-1-56 ~]# linux25key.pem ec2-user@18.220.92.52>
-bash: syntax error near unexpected token `newline'
[root@ip-10-0-1-56 ~]# sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; vendor preset
: disabled)
   Drop-In: /usr/lib/systemd/system/httpd.service.d
            └─php-fpm.conf
   Active: active (running) since Mon 2025-06-30 08:01:30 UTC; 57min ago
     Docs: man:httpd.service(8)
   Main PID: 3544 (httpd)
  Status: "Total requests: 1153; Idle/Busy workers 66/33;Requests/sec: 0.334; B
ytes served/sec: 274 B/sec"
   CGroup: /system.slice/httpd.service
           └─ 508 /usr/sbin/httpd -DFOREGROUND
              515 /usr/sbin/httpd -DFOREGROUND
              586 /usr/sbin/httpd -DFOREGROUND
             3544 /usr/sbin/httpd -DFOREGROUND
             3569 /usr/sbin/httpd -DFOREGROUND
             3679 /usr/sbin/httpd -DFOREGROUND
             3772 /usr/sbin/httpd -DFOREGROUND
             3773 /usr/sbin/httpd -DFOREGROUND
             3787 /usr/sbin/httpd -DFOREGROUND
             3817 /usr/sbin/httpd -DFOREGROUND
             3879 /usr/sbin/httpd -DFOREGROUND
             3927 /usr/sbin/httpd -DFOREGROUND
             3933 /usr/sbin/httpd -DFOREGROUND

Jun 30 08:01:30 ip-10-0-1-56.us-east-2.compute.internal systemd[1]: Starting ...
Jun 30 08:01:30 ip-10-0-1-56.us-east-2.compute.internal systemd[1]: Started T...
Hint: Some lines were ellipsized, use -l to show in full.
[root@ip-10-0-1-56 ~]# cd /var/www/html
[root@ip-10-0-1-56 html]# ls
healthy.html      wp-admin          wp-cron.php       wp-mail.php
index.php         wp-blog-header.php wp-includes       wp-settings.php
license.txt       wp-comments-post.php wp-links-opml.php wp-signup.php
readme.html      wp-config-sample.php wp-load.php       wp-trackback.php
wp-activate.php  wp-content        wp-login.php      xmlrpc.php
[root@ip-10-0-1-56 html]#

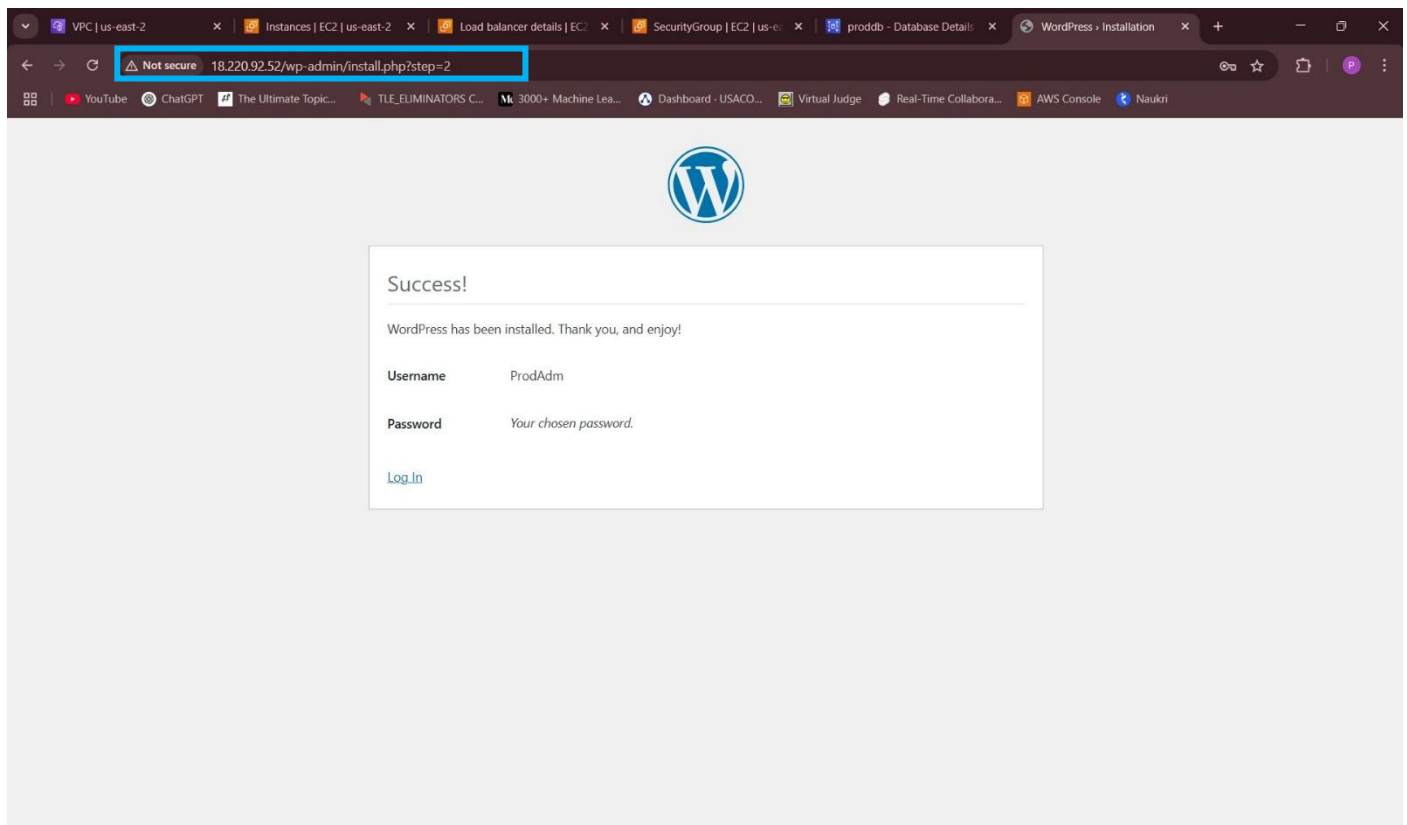
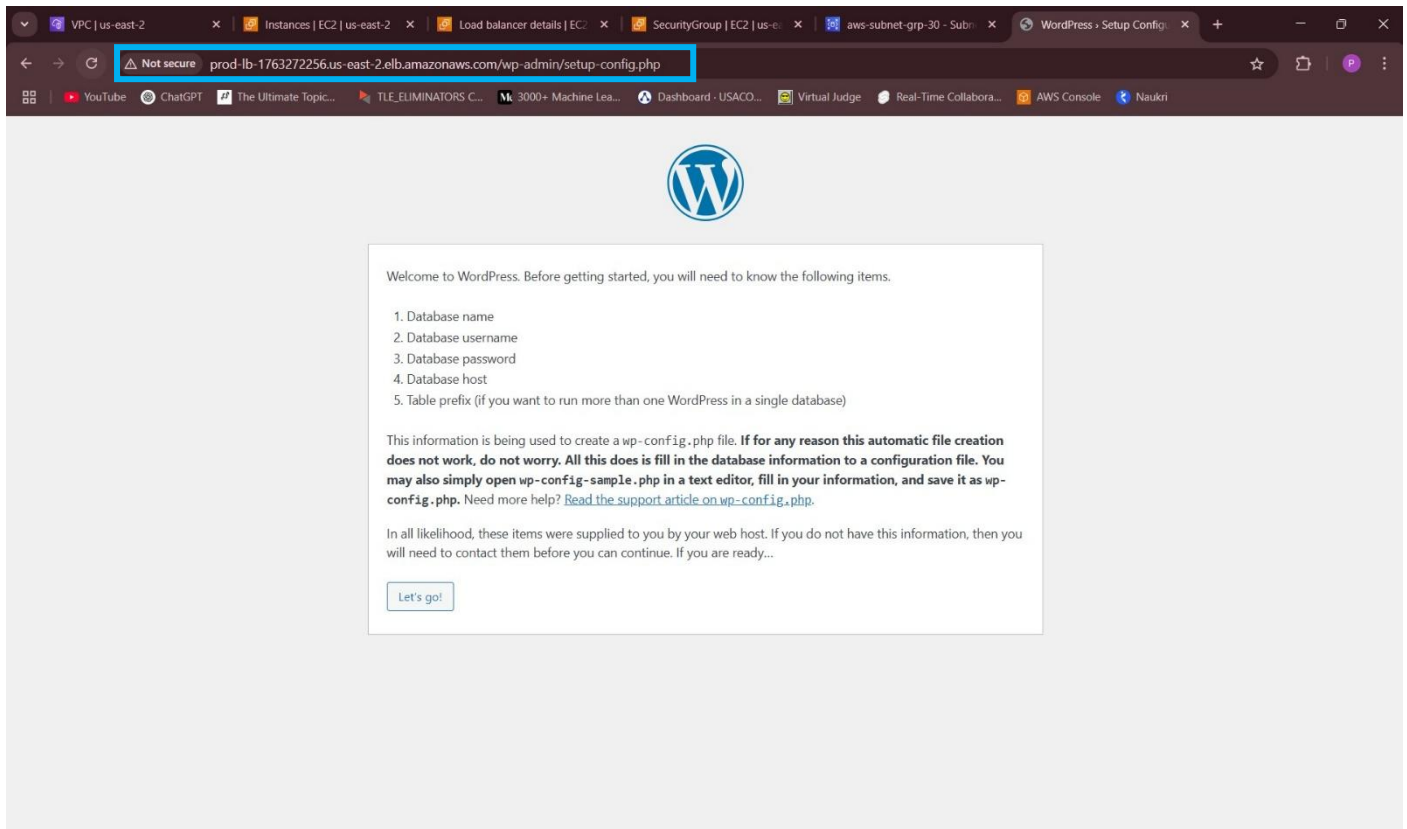
```

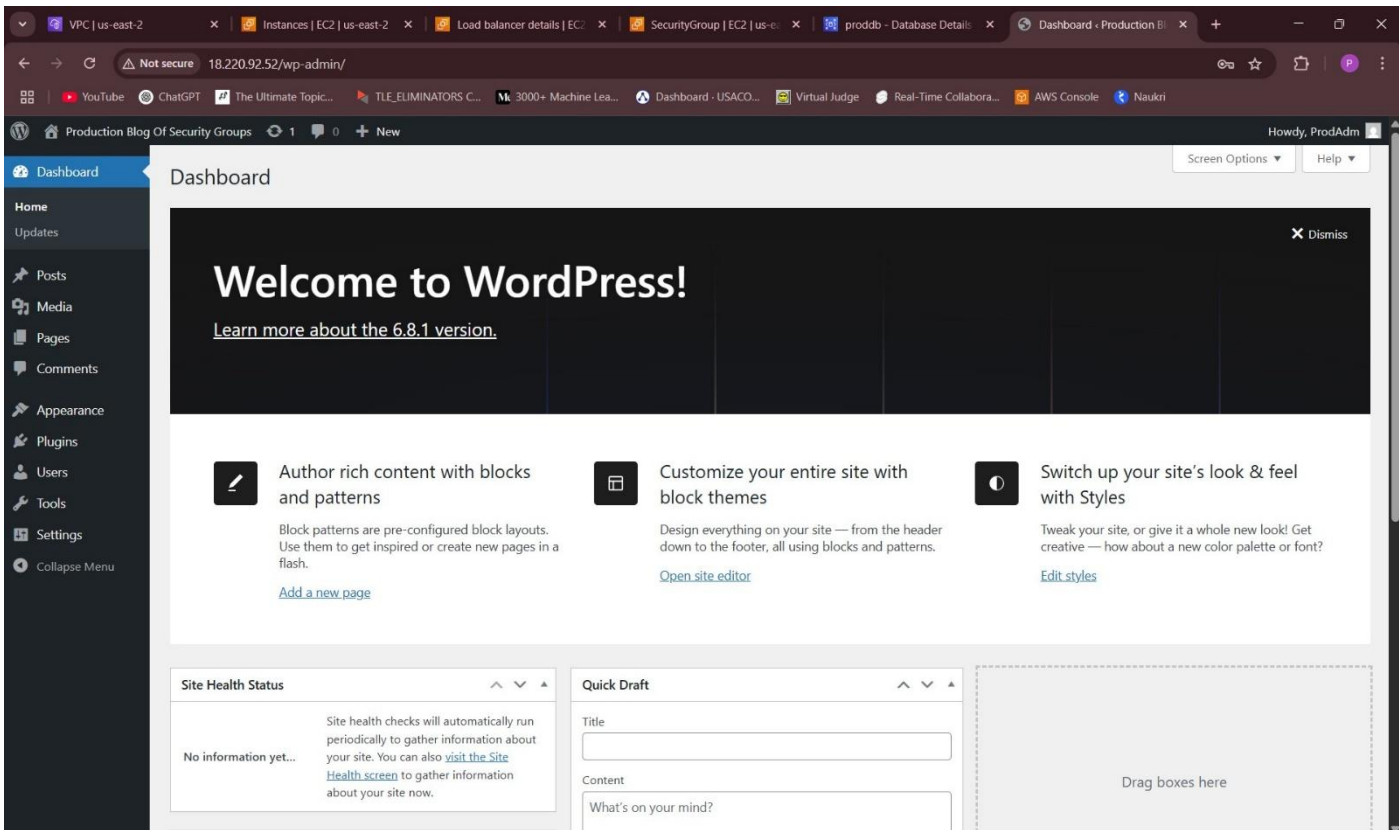
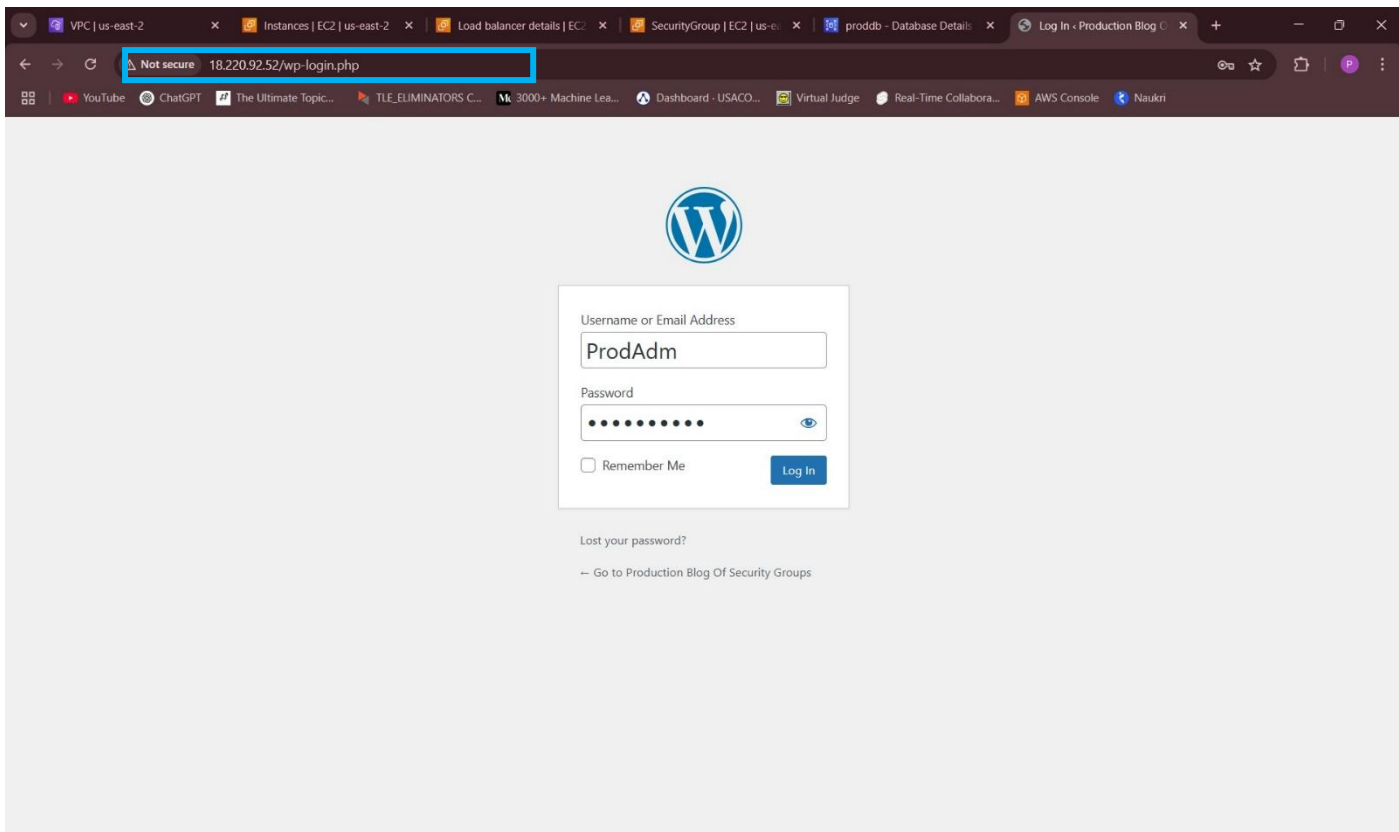
vii. Host WordPress in Browser:

- Copy the IP Address of the created EC2 and paste it in the new tab in the browser.
- After the configuration of wp-config.php in the putty server, then give run installation and setup the blog page.



- The page that is hosted through LB DNS name.





ix. Database Records in Putty:

```
root@ip-10-0-1-56:/var/www/html
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> show databases;
+-----+
| Database |
+-----+
| Proddb   |
| information_schema |
| mysql    |
| performance_schema |
| sys      |
| test_db  |
+-----+
6 rows in set (0.00 sec)

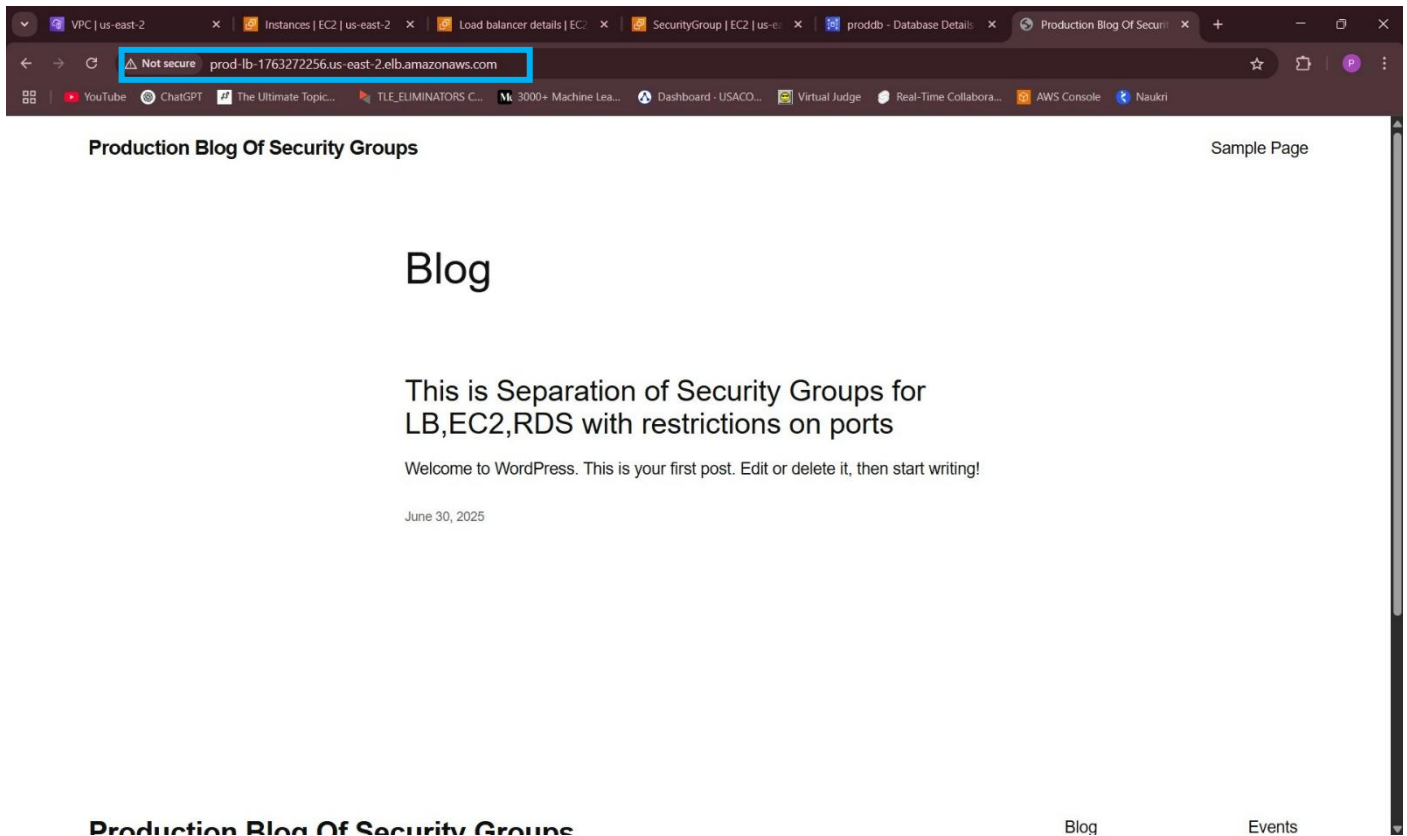
MySQL [(none)]> show tables;
ERROR 1046 (3D000): No database selected
MySQL [(none)]> use Proddb;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MySQL [Proddb]> show tables;
+-----+
| Tables_in_Proddb |
+-----+
| wp_commentmeta    |
| wp_comments       |
| wp_links          |
| wp_options        |
| wp_postmeta       |
| wp_posts          |
| wp_term_relationships |
| wp_term_taxonomy  |
| wp_termmeta       |
| wp_terms          |
| wp_usermeta       |
| wp_users          |
+-----+
12 rows in set (0.00 sec)

MySQL [Proddb]> exit
Bye
[root@ip-10-0-1-56 html]#
```

viii. Blog Page:

The screenshot displays a web browser window with multiple tabs open. The active tab is titled 'Production Blog Of Security Groups'. The browser's address bar shows the URL '18.220.92.52' and a 'Not secure' warning. The page content includes a header with the title 'Production Blog Of Security Groups' and a 'Sample Page' link. The main heading is 'Blog'. Below it, there is a post titled 'This is Separation of Security Groups for LB, EC2, RDS with restrictions on ports'. The post content begins with 'Welcome to WordPress. This is your first post. Edit or delete it, then start writing!' and is dated 'June 30, 2025'.



- **The setup for deploying the WordPress page in this region also completed. With that implementing separate security groups for EC2, RDS and LB with restriction ports is completed.**

Task 4: Implement Multi-Region Architecture using Failover Routing

- In this task, the Production and DR Page that is hosted in WordPress through failover routing the region will be changed host the particular page. And in this task the Production Page is hosted in Mumbai Region and DR Page is hosted in Ohio Region.

Mumbai Region:

Steps:

i. VPC Creation:

- Go to your VPC.
- Click your VPC, then click create VPC then give a name as “vpc-mumbai-25”
- Then set the IP address 10.0.0.0/16, then set both DNS hostname and DNS resolution as enabled, then click create.

The screenshot displays the AWS Management Console for a VPC named vpc-mumbai-25 in the Asia Pacific (Mumbai) region. The VPC ID is vpc-0bc730269c484f956. The state is Available. DNS resolution is Enabled. The main network ACL is acl-01852098dd7d55dc2. The IPv4 CIDR is 10.0.0.0/16. The DHCP option set is dopt-0a69edd3a573d82ac. The Route 53 Resolver DNS Firewall rule groups are none. The DNS hostnames are Disabled. The main route table is rtb-012c7dc3d969fba23. The IPv6 pool is none. The Owner ID is 767828764671. The resource map shows the VPC, subnets (ap-south-1a, ap-south-1b), route tables (mum-pvt-rtb-25, rtb-012c7dc3d969fba23, mum-pub-rtb-25), and network connections (igw-mum-25, ngw-mum-25).

ii. Subnet Setup:

- Go to subnets, click create subnet after that assign a name for the subnets, in this task we need two subnets for public and private.
- Then enter the below details particularly for public and private subnet, by assigning the IP Addresses, then click create subnet
- Public Subnet 1: For EC2 + NAT
- Public Subnet 2: For EC2
- Private Subnet 1: For RDS Prod

CIDR Blocks (Mumbai):

- 10.0.1.0/24 (Public subnet 1)
- 10.0.2.0/24 (Public subnet 2)
- 10.0.3.0/24 (Private subnet 1)

The screenshot displays the AWS Management Console interface for a specific subnet. The browser address bar shows the URL: `ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#SubnetDetails:subnetId=subnet-046b2590acbe06684`. The console header includes the AWS logo, a search bar, and navigation links for VPC, Subnets, and the specific subnet ID.

The main content area is titled **subnet-046b2590acbe06684 / mum-pub-sbt-25**. It contains a 'Details' section with the following information:

- Subnet ID:** subnet-046b2590acbe06684
- IPv4 CIDR:** 10.0.1.0/24
- Availability Zone:** ap-south-1a
- Route table:** rtb-0ad275e1fe011f3a1 | mum-pub-rtb-25
- Auto-assign IPv6 address:** No
- IPv4 CIDR reservations:** -
- Resource name DNS A record:** Disabled
- Subnet ARN:** arn:aws:ec2:ap-south-1:767828764671:subnet/subnet-046b2590acbe06684
- Available IPv4 addresses:** 251
- Availability Zone ID:** aps1-az1
- Network ACL:** -
- Auto-assign customer-owned IPv4 address:** No
- IPv6 CIDR reservations:** -
- Resource name DNS AAAA record:** Disabled
- State:** Available
- IPv6 CIDR:** -
- Network border group:** ap-south-1
- Default subnet:** No
- Customer-owned IPv4 pool:** -
- IPv6-only:** No
- DNS64:** Disabled
- Block Public Access:** Off
- IPv6 CIDR association ID:** -
- VPC:** vpc-0bc730269c484f956 | vpc-mumbai-25
- Auto-assign public IPv4 address:** No
- Outpost ID:** -
- Hostname type:** IP name
- Owner:** 767828764671

Below the details section, there are tabs for 'Flow logs', 'Route table', 'Network ACL', 'CIDR reservations', 'Sharing', and 'Tags'. The 'Flow logs' tab is currently selected, showing a search bar and a table with columns for Name, Flow log ID, Filter, Destination type, Destination name, and IAM role ARN. The table currently displays one entry.

The footer of the console shows the copyright notice: © 2025, Amazon Web Services, Inc. or its affiliates. It also includes links for Privacy, Terms, and Cookie preferences.

Con: xVPC xInsta xSecu xLoad xohio xHeal xVPC xInsta xSecu xLoad xprod xDom x+ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#SubnetDetails:subnetId=subnet-0d2b222eb7533540a

Search[Alt+S]

Asia Pacific (Mumbai)Prabath D

VPCSubnetssubnet-0d2b222eb7533540a

subnet-0d2b222eb7533540a / mum-pvt-sbt-I-25Actions

Details

Subnet ID

subnet-0d2b222eb7533540a

IPv4 CIDR

10.0.2.0/24

Availability Zone

ap-south-1b

Route table

rtb-0f2eb64351bcedeb2 | mum-pvt-rtb-25

Auto-assign IPv6 address

No

IPv4 CIDR reservations

-

Resource name DNS A record

Disabled

Subnet ARN

arn:aws:ec2:ap-south-1:767828764671:subnet/subnet-0d2b222eb7533540a

Available IPv4 addresses

251

Availability Zone ID

aps1-az3

Network ACL

-

Auto-assign customer-owned IPv4 address

No

IPv6 CIDR reservations

-

Resource name DNS AAAA record

Disabled

State

Available

IPv6 CIDR

-

Network border group

ap-south-1

Default subnet

No

Customer-owned IPv4 pool

-

IPv6-only

No

DNS64

Disabled

Block Public Access

Off

IPv6 CIDR association ID

-

VPC

vpc-0bc730269c484f956 | vpc-mumbai-25

Auto-assign public IPv4 address

No

Outpost ID

-

Hostname type

IP name

Owner

767828764671

Flow logsRoute tableNetwork ACLCIDR reservationsSharingTags

Flow logs

Search

ActionsCreate flow log

1

NameFlow log IDFilterDestination typeDestination nameIAM role ARN

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Con: xVPC xInsta xSecu xLoad xohio xHeal xVPC xInsta xSecu xLoad xprod xDom x+ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#SubnetDetails:subnetId=subnet-064d27a213dde0423

Search[Alt+S]

Asia Pacific (Mumbai)Prabath D

VPCSubnetssubnet-064d27a213dde0423

subnet-064d27a213dde0423 / mum-pvt-sbt-II-25Actions

Details

Subnet ID

subnet-064d27a213dde0423

IPv4 CIDR

10.0.3.0/24

Availability Zone

ap-south-1c

Route table

rtb-0f2eb64351bcedeb2 | mum-pvt-rtb-25

Auto-assign IPv6 address

No

IPv4 CIDR reservations

-

Resource name DNS A record

Disabled

Subnet ARN

arn:aws:ec2:ap-south-1:767828764671:subnet/subnet-064d27a213dde0423

Available IPv4 addresses

251

Availability Zone ID

aps1-az2

Network ACL

-

Auto-assign customer-owned IPv4 address

No

IPv6 CIDR reservations

-

Resource name DNS AAAA record

Disabled

State

Available

IPv6 CIDR

-

Network border group

ap-south-1

Default subnet

No

Customer-owned IPv4 pool

-

IPv6-only

No

DNS64

Disabled

Block Public Access

Off

IPv6 CIDR association ID

-

VPC

vpc-0bc730269c484f956 | vpc-mumbai-25

Auto-assign public IPv4 address

No

Outpost ID

-

Hostname type

IP name

Owner

767828764671

Flow logsRoute tableNetwork ACLCIDR reservationsSharingTags

Flow logs

Search

ActionsCreate flow log

1

NameFlow log IDFilterDestination typeDestination nameIAM role ARN

CloudShellFeedback

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iii. Internet Gateway + NAT Gateway:

- Create and attach IGW in Ohio VPC.

The screenshot shows the AWS Management Console for an Internet Gateway. The breadcrumb navigation is VPC > Internet gateways > igw-0867acd1a59c8cf48. The title is igw-0867acd1a59c8cf48 / igw-mum-25. The Details section shows the Internet gateway ID as igw-0867acd1a59c8cf48, the State as Attached, the VPC ID as vpc-0bc730269c484f956 | vpc-mumbai-25, and the Owner as 767828764671. The Tags section shows a single tag with the key Name and value igw-mum-25.

- Create NAT Gateway in Public Subnet (with Elastic IP).

The screenshot shows the AWS Management Console for a NAT Gateway. The breadcrumb navigation is VPC > NAT gateways > nat-0c507d81b9deb86ce. The title is nat-0c507d81b9deb86ce / ngw-mum-25. The Details section shows the NAT gateway ID as nat-0c507d81b9deb86ce, the NAT gateway ARN as arn:aws:ec2:ap-south-1:767828764671:natgateway/nat-0c507d81b9deb86ce, the VPC as vpc-0bc730269c484f956 | vpc-mumbai-25, the Connectivity type as Public, the Primary public IPv4 address as 13.202.142.113, the Subnet as subnet-046b2590acbe06684 | mum-pub-sbt-25, the State as Available, the Primary private IPv4 address as 10.0.1.96, the Created time as Thursday, July 3, 2025 at 18:39:35 GMT+5:30, the State message as -, the Primary network interface ID as eni-0f5a66aff93dc27b5, and the Deleted status as -. The Secondary IPv4 addresses section shows that secondary IPv4 addresses are not available for this NAT gateway.

13.202.142.113

Summary

Allocated IPv4 address 13.202.142.113	Type Public IP	Allocation ID eipalloc-0d29e5137c3812fdc	Reverse DNS record -
Association ID eipassoc-0643c6702545e52e9	Scope VPC	Associated instance ID -	Private IP address 10.0.1.96
Network interface ID eni-0f5a66aff93dc27b5	Network interface owner account ID 767828764671	Public DNS -	NAT Gateway ID nat-0c507d81b9deb86ce (ngw-mum-25)
Address pool Amazon	Network border group ap-south-1		

Tags(0)

No tags associated with this resource
Click the Manage tags button to add your first tag

Manage tags

iv. Route Tables (Mumbai):

- Then after creating both Internet Gateway and NAT Gateway, go to route tables, click create, then assign names for both public and private route tables, then associate both the tables by clicking routes then attach the network
- Public Route Table → 0.0.0.0/0 → IGW.
- Then click subnet associations in public route table, then click edit association after that click both public subnet 1.
- Private Route Table → 0.0.0.0/0 → NAT.
- Then click subnet associations in private route table, then click edit association after that click both private subnet 1 and private subnet 2.

Con: xVPC xInsta xSecu xLoad xohio xHeal xVPC xInsta xSecu xLoad xprod xDom x+ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#RouteTableDetails:RouteTableId=rtb-0ad275e1fe011f3a1

Search[Alt+S]

Asia Pacific (Mumbai)Prabath D

VPCRoute tablesrtb-0ad275e1fe011f3a1

rtb-0ad275e1fe011f3a1 / mum-pub-rtb-25

DetailsInfo

Route table ID
rtb-0ad275e1fe011f3a1

VPC
vpc-0bc730269c484f956 | vpc-mumbai-25

Main
No

Owner ID
767828764671

Explicit subnet associations
subnet-046b2590acbe06684 / mum-pub-sbt-25

Edge associations
-

RoutesSubnet associationsEdge associationsRoute propagationTags

Routes (2)

Filter routes

BothEdit routes

Destination	Target	Status	Propagated
0.0.0.0/0	igw-0867acd1a59c8cf48	Active	No
10.0.0.0/16	local	Active	No

Con: xVPC xInsta xSecu xLoad xohio xHeal xVPC xInsta xSecu xLoad xprod xDom x+ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#RouteTableDetails:RouteTableId=rtb-0ad275e1fe011f3a1

Search[Alt+S]

Asia Pacific (Mumbai)Prabath D

VPCRoute tablesrtb-0ad275e1fe011f3a1

rtb-0ad275e1fe011f3a1 / mum-pub-rtb-25

DetailsInfo

Route table ID
rtb-0ad275e1fe011f3a1

VPC
vpc-0bc730269c484f956 | vpc-mumbai-25

Main
No

Owner ID
767828764671

Explicit subnet associations
subnet-046b2590acbe06684 / mum-pub-sbt-25

Edge associations
-

RoutesSubnet associationsEdge associationsRoute propagationTags

Explicit subnet associations (1)

Find subnet association

Edit subnet associations

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
mum-pub-sbt-25	subnet-046b2590acbe06684	10.0.1.0/24	-

Subnets without explicit associations (0)

The following subnets have not been explicitly associated with any route tables and are therefore associated with the main route table:

Find subnet association

Edit subnet associations

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
------	-----------	-----------	-----------

No subnets without explicit associations
All your subnets are associated with a route table.

Cons xVPC xInsta xSecu xLoad xohio xHeal xVPC xInsta xSecu xLoad xprod xDom x+ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#RouteTableDetails:RouteTableId=rtb-0f2eb64351bcedeb2

Search[Alt+S]

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VPCRoute tablesrtb-0f2eb64351bcedeb2

rtb-0f2eb64351bcedeb2 / mum-pvt-rtb-25Actions

DetailsInfo

Route table ID
rtb-0f2eb64351bcedeb2

VPC
vpc-0bc730269c484f956 | vpc-mumbai-25

Main
No

Owner ID
767828764671

Explicit subnet associations
2 subnets

Edge associations
-

RoutesSubnet associationsEdge associationsRoute propagationTags

Routes (2)

Filter routes

BothEdit routes

Destination	Target	Status	Propagated
0.0.0.0/0	nat-0c507d81b9deb86ce	Active	No
10.0.0.0/16	local	Active	No

Cons xVPC xInsta xSecu xLoad xohio xHeal xVPC xInsta xSecu xLoad xprod xDom x+ap-south-1.console.aws.amazon.com/vpcconsole/home?region=ap-south-1#RouteTableDetails:RouteTableId=rtb-0f2eb64351bcedeb2

Search[Alt+S]

Asia Pacific (Mumbai)Prabath D

VPCRoute tablesrtb-0f2eb64351bcedeb2

rtb-0f2eb64351bcedeb2 / mum-pvt-rtb-25Actions

DetailsInfo

Route table ID
rtb-0f2eb64351bcedeb2

VPC
vpc-0bc730269c484f956 | vpc-mumbai-25

Main
No

Owner ID
767828764671

Explicit subnet associations
2 subnets

Edge associations
-

RoutesSubnet associationsEdge associationsRoute propagationTags

Explicit subnet associations (2)

Find subnet association

Edit subnet associations

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
mum-pvt-sbt-ii-25	subnet-064d27a213dde0423	10.0.3.0/24	-
mum-pvt-sbt-i-25	subnet-0d2b222eb7533540a	10.0.2.0/24	-

Subnets without explicit associations (0)

The following subnets have not been explicitly associated with any route tables and are therefore associated with the main route table:

Find subnet association

Edit subnet associations

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
------	-----------	-----------	-----------

No subnets without explicit associations
All your subnets are associated with a route table.

v. EC2, RDS & LB:

- First create Security Groups separately for EC2, RDS and LB.
- Create Production EC2 by giving the same apache/php code in advanced details.
- After creation of EC2, create RDS and LB also.

The screenshot shows the AWS Management Console for the Security Group **sg-02038d80f2d414b37 - EC2-SGRP**. The details section shows the Security group name as EC2-SGRP, Security group ID as sg-02038d80f2d414b37, and VPC ID as vpc-0bc730269c484f956. The Inbound rules count is 2 and the Outbound rules count is 1. The Inbound rules table is highlighted with a blue box and contains the following data:

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
-	sgr-01f6cc57276e4ad41	IPv4	SSH	TCP	22	106.222.221.218/32
-	sgr-0b307fec90a5a79ba	IPv4	HTTP	TCP	80	0.0.0.0/0

The screenshot shows the AWS Management Console for the Security Group **sg-03baf019ad2bb7e8c - RDS-SGRP**. The details section shows the Security group name as RDS-SGRP, Security group ID as sg-03baf019ad2bb7e8c, and VPC ID as vpc-0bc730269c484f956. The Inbound rules count is 1 and the Outbound rules count is 1. The Inbound rules table is highlighted with a blue box and contains the following data:

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
-	sgr-0fb606b5fca194084	-	MYSQL/Aurora	TCP	3306	sg-02038d80f2d414b3...

Con: xVPC xInsta xSecu xLoad xohio xHeal xVPC xInsta xSecu xLoad xprod xDom x+ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#InstanceDetails:instanceId=i-0a866bd0c8b8ea8b2

Search[Alt+S]

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EC2>Instances>i-0a866bd0c8b8ea8b2

Instance summary for i-0a866bd0c8b8ea8b2 (prod-EC2) Info

Updated less than a minute ago

Instance ID

i-0a866bd0c8b8ea8b2

IPV6 address

-

Hostname type

IP name: ip-10-0-1-172.ap-south-1.compute.internal

Answer private resource DNS name

-

Auto-assigned IP address

43.204.30.89 [Public IP]

IAM Role

-

IMDSv2

Required

Operator

-

Public IPv4 address

43.204.30.89 | open address

Instance state

Running

Private IP DNS name (IPv4 only)

ip-10-0-1-172.ap-south-1.compute.internal

Instance type

t2.micro

VPC ID

vpc-0bc730269c484f956 (vpc-mumbai-25)

Subnet ID

subnet-046b2590acbe06684 (mum-pub-sbt-25)

Instance ARN

arn:aws:ec2:ap-south-1:767828764671:instance/i-0a866bd0c8b8ea8b2

Private IPv4 addresses

10.0.1.172

Public DNS

-

Elastic IP addresses

-

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendations. | Learn more

Auto Scaling Group name

-

Managed

false

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

▼ Instance details Info

AMI ID

Monitoring

Platform details

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Con: xVPC xInsta xSecu xLoad xohio xHeal xVPC xInsta xSecu xLoad xprod xDom x+ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#SecurityGroup:groupId=sg-03baf019ad2bb7e8c

Search[Alt+S]

Asia Pacific (Mumbai)Prabath D

EC2>Security Groups>sg-03baf019ad2bb7e8c - RDS-SGRP

sg-03baf019ad2bb7e8c - RDS-SGRP

Actions

Details

Security group name

RDS-SGRP

Owner

767828764671

Security group ID

sg-03baf019ad2bb7e8c

Inbound rules count

1 Permission entry

Description

Sgrp for DB

Outbound rules count

1 Permission entry

VPC ID

vpc-0bc730269c484f956

Inbound rules

Outbound rules

Sharing - new

VPC associations - new

Tags

Inbound rules (1)

Manage tags

Edit inbound rules

Search

	Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	De
☐	-	sgr-0fb606b5fca194084	-	MySQL/Aurora	TCP	3306	sg-02038d80f2d414b3...	

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Con: xVPC xInsta xSecu xLoad xohio xHeal xVPC xInsta xSecu xLoad xprod xDom x+ap-south-1.console.aws.amazon.com/rds/home?region=ap-south-1#databaseid=proddb&cluster=false

YouTubeChatGPTThe Ultimate Topic...TLE_ELIMINATORS C...3000+ Machine Lea...Dashboard - USACO...Virtual JudgeReal-Time Collabora...AWS ConsoleNaukri

awsSearch[Alt+S]

Asia Pacific (Mumbai)Prabath D

Aurora and RDSDatabasesproddb

proddb

Summary

DB identifier
proddb

CPU
3.76%

Status
Available

Class
db.t3.micro

Role
Instance

Current activity
0 Connections

Engine
MySQL Community

Region & AZ
ap-south-1b

Recommendations

Connectivity & security

Monitoring

Logs & events

Configuration

Zero-ETL integrations

Maintenance & backups

Data migrations - new

Tags

Records

Connectivity & security

Endpoint & port

Endpoint
proddb.cngac84c2in0.ap-south-1.rds.amazonaws.com

Port
3306

Networking

Availability Zone
ap-south-1b

VPC
vpc-mumbai-25 (vpc-0bc730269c484f956)

Subnet group
mum-sbt-25

Subnets
subnet-0d2b222eb7533540a
subnet-064d27a213dde0423

Security

VPC security groups
RDS-SGRP (sg-03baf019ad2bb7e8c)
Active

Publicly accessible
No

Certificate authority
rds-ca-rsa2048-g1

Certificate authority date
May 20, 2061, 00:10 (UTC+05:30)

DB instance certificate expiration date

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Con: xVPC xInsta xSecu xLoad xohio xHeal xVPC xInsta xSecu xLoad xprod xDom x+ap-south-1.console.aws.amazon.com/rds/home?region=ap-south-1#databaseid=proddb&cluster=false

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awsSearch[Alt+S]

Asia Pacific (Mumbai)Prabath D

Aurora and RDSDatabasesproddb

proddb

Proxy identifier

Status

Engine family

No proxies
You don't have any proxies.
Create proxy

Security group rules (2)

Filter by Security group rules

1

Security group	Type	Rule
RDS-SGRP (sg-03baf019ad2bb7e8c)	EC2 Security Group - Inbound	sg-02038d80f2d414b37
RDS-SGRP (sg-03baf019ad2bb7e8c)	CIDR/IP - Outbound	0.0.0.0/0

Replication (1)

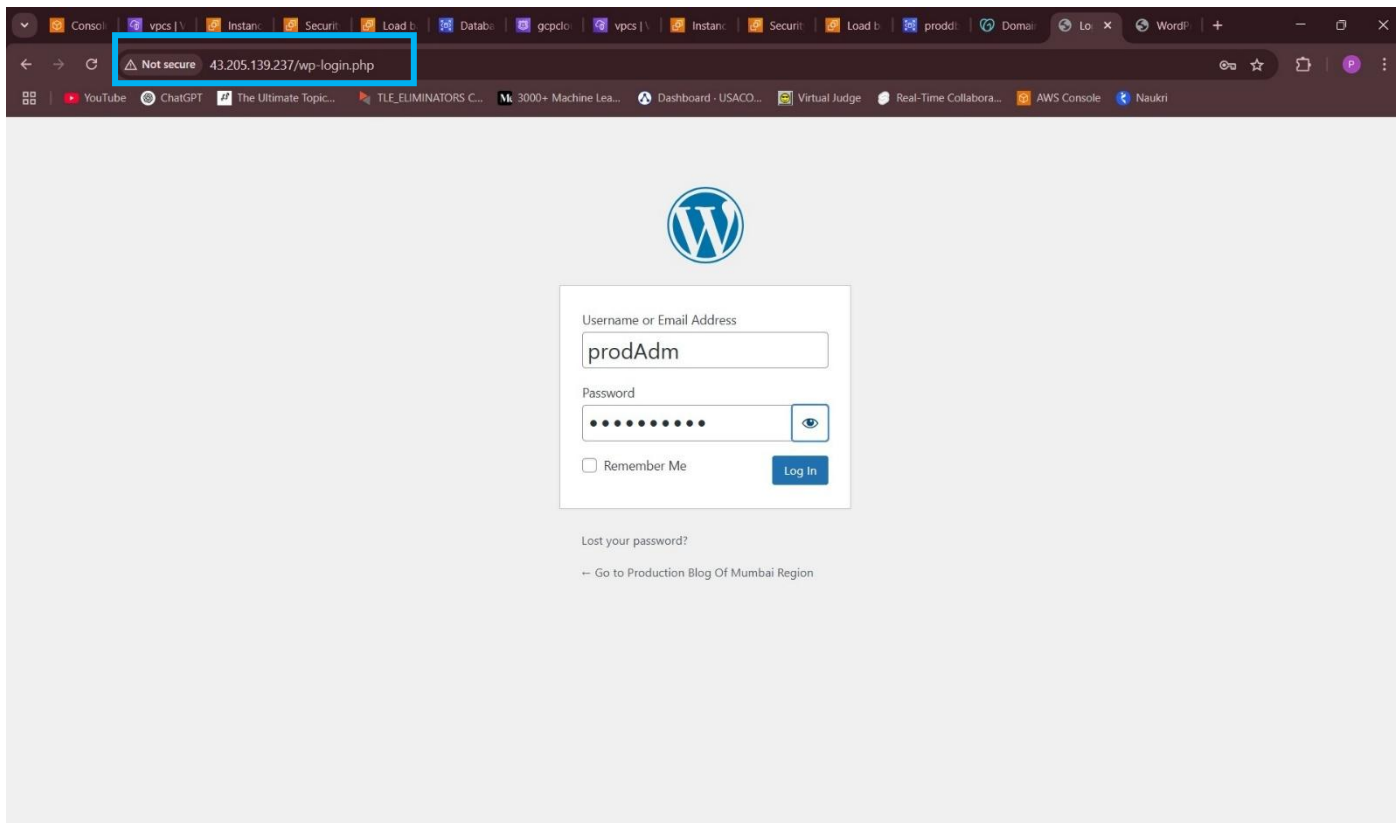
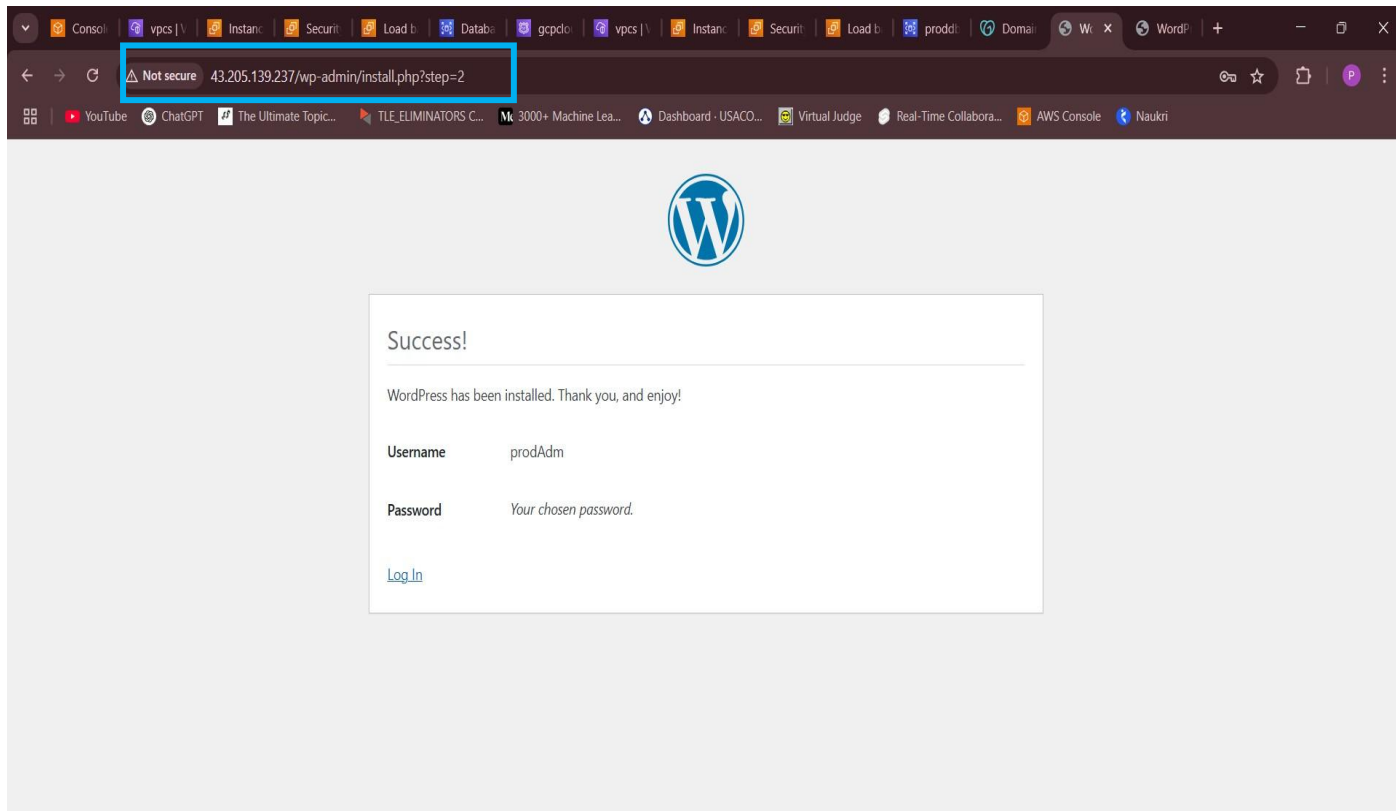
Filter by Replication

1

DB identifier	Role	Region & AZ	Replication source	Replication state	Lag
proddb	Instance	ap-south-1b	-	-	-

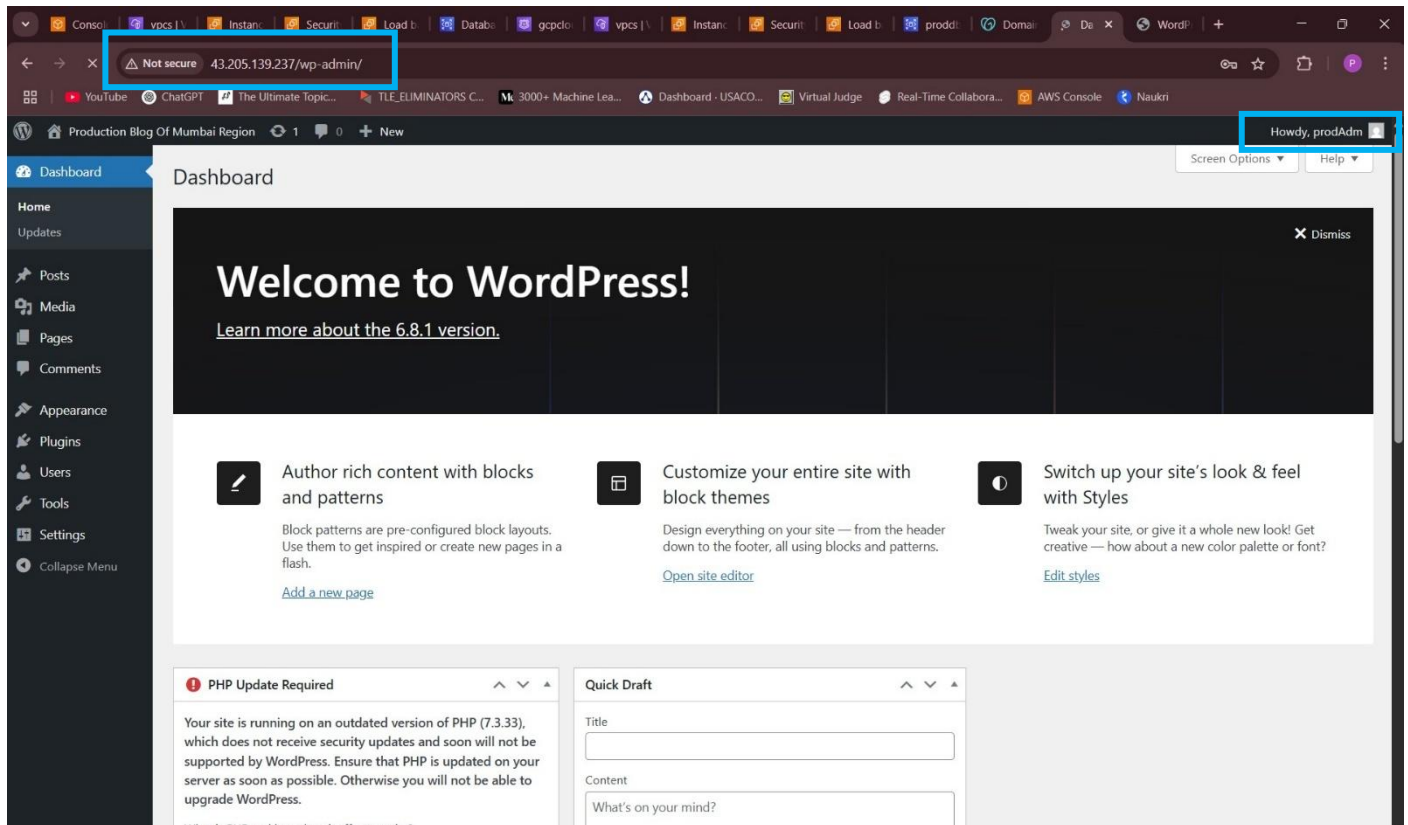
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vii. Edit the Blog Page :

- View the dashboard, then edit the page as per the user's requirement then host the page through the domain name 'gcpcLOUDS.life'.



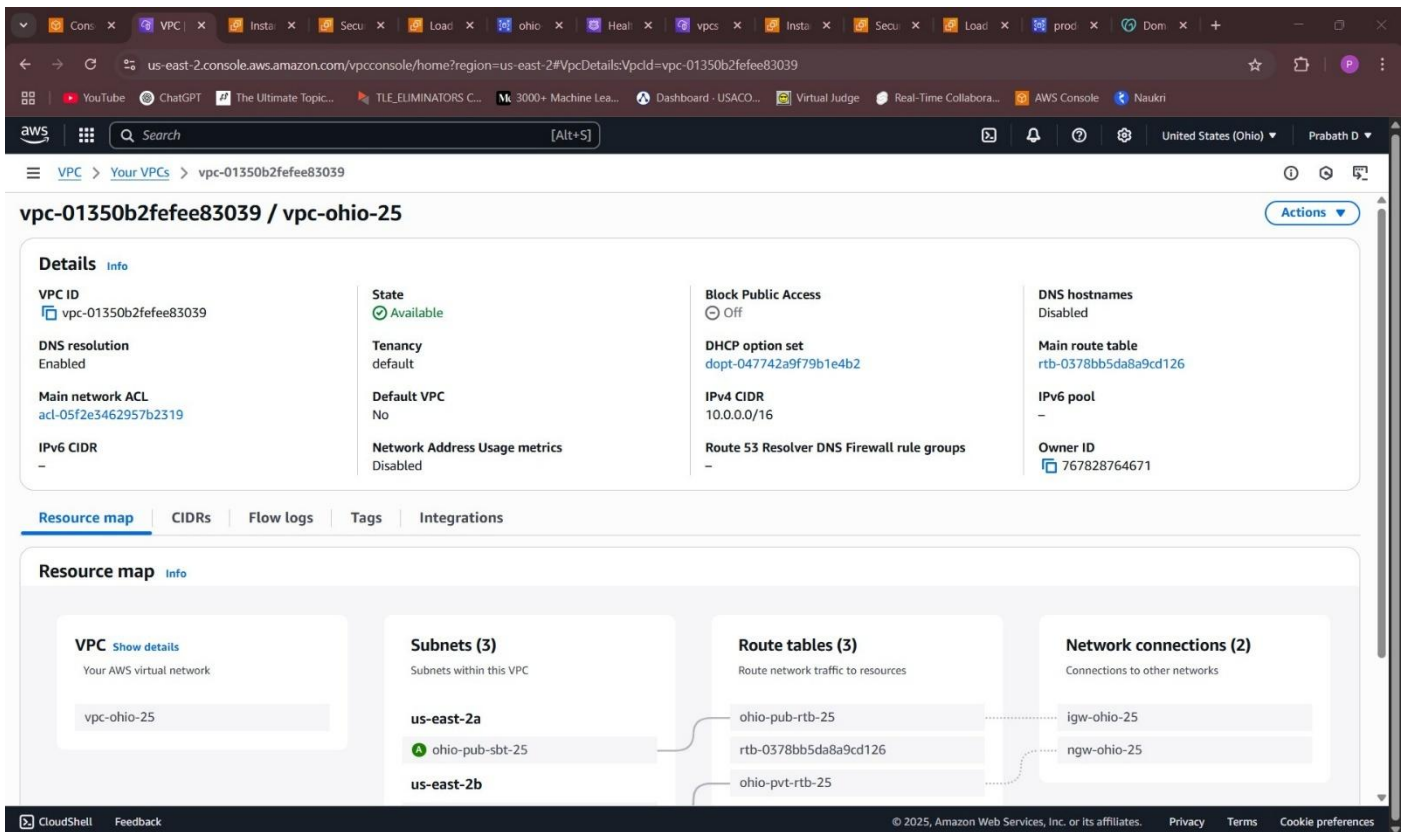
- Now implement the same in the Ohio Region also, and then create a Route53 records then load with the domain name 'gcpcLOUDS.life'.

Ohio Region:

Steps:

i. VPC Creation:

- Go to your VPC.
- Click your VPC, then click create VPC then give a name as “vpc-ohio-25”
- Then set the IP address 10.0.0.0/16, then set both DNS hostname and DNS resolution as enabled, then click create.



ii. Subnet Setup:

- Go to subnets, click create subnet after that assign a name for the subnets, in this task we need two subnets for public and private.
- Then enter the below details particularly for public and private subnet, by assigning the IP Addresses, then click create subnet
- Public Subnet 1: For EC2 + NAT
- Public Subnet 2: For EC2
- Private Subnet 1: For RDS Prod

CIDR Blocks (Mumbai):

- 10.0.1.0/24 (Public subnet 1)
- 10.0.2.0/24 (Public subnet 2)
- 10.0.3.0/24 (Private subnet 1)

Con: xVPC xInsta xSecu xLoad xohio xHeal xvpccs xInsta xSecu xLoad xprod xDom x+

us-east-2.console.aws.amazon.com/vpcconsole/home?region=us-east-2#SubnetDetails:subnetId=subnet-084aadf97e6037c5c

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United States (Ohio)Prabath D

VPCSubnetssubnet-084aadf97e6037c5c

subnetsubnet-084aadf97e6037c5c / ohio-pub-sbt-25

Actions

Details

Subnet ID

subnet-084aadf97e6037c5c

IPv4 CIDR

10.0.1.0/24

Availability Zone

us-east-2a

Network ACL

-

Auto-assign customer-owned IPv4 address

No

IPv6 CIDR reservations

-

Resource name DNS AAAA record

Disabled

Subnet ARN

arn:aws:ec2:us-east-2:767828764671:subnet/subnet-084aadf97e6037c5c

Available IPv4 addresses

250

Availability Zone ID

use2-az1

Default subnet

No

Customer-owned IPv4 pool

-

IPv6-only

No

DNS64

Disabled

State

Available

IPv6 CIDR

-

VPC

vpc-01350b2fefee83039 | vpc-ohio-25

Auto-assign public IPv4 address

No

Outpost ID

-

Hostname type

IP name

Owner

767828764671

Block Public Access

Off

IPv6 CIDR association ID

-

Route table

rtb-0771fe02704157e10 | ohio-pub-rtb-25

Auto-assign IPv6 address

No

IPv4 CIDR reservations

-

Resource name DNS A record

Disabled

Flow logs

Route table

Network ACL

CIDR reservations

Sharing

Tags

Flow logs

Search

Actions

Create flow log

< 1 >

NameFlow log IDFilterDestination typeDestination nameIAM role ARN

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Con: xVPC xInsta xSecu xLoad xohio xHeal xvpccs xInsta xSecu xLoad xprod xDom x+

us-east-2.console.aws.amazon.com/vpcconsole/home?region=us-east-2#SubnetDetails:subnetId=subnet-03bddea7c32655fd4

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VPCSubnetssubnet-03bddea7c32655fd4

subnetsubnet-03bddea7c32655fd4 / ohio-pvt-sbt-l-25

Actions

Details

Subnet ID

subnet-03bddea7c32655fd4

IPv4 CIDR

10.0.2.0/24

Availability Zone

us-east-2b

Network ACL

-

Auto-assign customer-owned IPv4 address

No

IPv6 CIDR reservations

-

Resource name DNS AAAA record

Disabled

Subnet ARN

arn:aws:ec2:us-east-2:767828764671:subnet/subnet-03bddea7c32655fd4

Available IPv4 addresses

251

Availability Zone ID

use2-az2

Default subnet

No

Customer-owned IPv4 pool

-

IPv6-only

No

DNS64

Disabled

State

Available

IPv6 CIDR

-

VPC

vpc-01350b2fefee83039 | vpc-ohio-25

Auto-assign public IPv4 address

No

Outpost ID

-

Hostname type

IP name

Owner

767828764671

Block Public Access

Off

IPv6 CIDR association ID

-

Route table

rtb-0de937a0c8d05d032 | ohio-pvt-rtb-25

Auto-assign IPv6 address

No

IPv4 CIDR reservations

-

Resource name DNS A record

Disabled

Flow logs

Route table

Network ACL

CIDR reservations

Sharing

Tags

Flow logs

Search

Actions

Create flow log

< 1 >

NameFlow log IDFilterDestination typeDestination nameIAM role ARN

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us-east-2.console.aws.amazon.com/vpcconsole/home?region=us-east-2#SubnetDetails:subnetId=subnet-0c632e795553deacb

subnets > subnet-0c632e795553deacb

subnet-0c632e795553deacb / ohio-pvt-sbt-1l-25

Details

Subnet ID
subnet-0c632e795553deacb

IPv4 CIDR
10.0.3.0/24

Availability Zone
us-east-2c

Network ACL
-

Auto-assign customer-owned IPv4 address
No

IPv6 CIDR reservations
-

Resource name DNS AAAA record
Disabled

Subnet ARN
arn:aws:ec2:us-east-2:767828764671:subnet/subnet-0c632e795553deacb

Available IPv4 addresses
251

Availability Zone ID
use2-az3

Default subnet
No

Customer-owned IPv4 pool
-

IPv6-only
No

DNS64
Disabled

State
Available

IPv6 CIDR
-

VPC
vpc-01350b2fefee83039 | vpc-ohio-25

Auto-assign public IPv4 address
No

Outpost ID
-

Hostname type
IP name

Owner
767828764671

Block Public Access
Off

IPv6 CIDR association ID
-

Route table
rtb-0de937a0c8d05d032 | ohio-pvt-rtb-25

Auto-assign IPv6 address
No

IPv4 CIDR reservations
-

Resource name DNS A record
Disabled

Flow logs

Route table

Network ACL

CIDR reservations

Sharing

Tags

Flow logs

Search

Actions

Create flow log

1

Name

Flow log ID

Filter

Destination type

Destination name

IAM role ARN

iii. Internet Gateway + NAT Gateway:

- Create and attach IGW in Ohio VPC.

us-east-2.console.aws.amazon.com/vpcconsole/home?region=us-east-2#InternetGateway:internetGatewayId=igw-0762604099e073a4e

Internet gateways > igw-0762604099e073a4e

igw-0762604099e073a4e / igw-ohio-25

Details

Internet gateway ID
igw-0762604099e073a4e

State
Attached

VPC ID
vpc-01350b2fefee83039 | vpc-ohio-25

Owner
767828764671

Tags

Search tags

Manage tags

1

Key

Value

Name

igw-ohio-25

- Create NAT Gateway in Public Subnet (with Elastic IP).

us-east-2.console.aws.amazon.com/vpcconsole/home?region=us-east-2#NatGatewayDetails:nat-0eac98ae950075581

VPC > NAT gateways > nat-0eac98ae950075581

nat-0eac98ae950075581 / ngw-ohio-25

Details

NAT gateway ID nat-0eac98ae950075581	Connectivity type Public	State Available	State message -
NAT gateway ARN arn:aws:ec2:us-east-2:767828764671:natgateway/nat-0eac98ae950075581	Primary public IPv4 address 3.140.143.223	Primary private IPv4 address 10.0.1.151	Primary network interface ID eni-01cf6ad810120a4d7
VPC vpc-01350b2fefee83039 / vpc-ohio-25	Subnet subnet-084aadf97e6037c5c / ohio-pub-sbt-25	Created Thursday, July 3, 2025 at 19:45:55 GMT+5:30	Deleted -

Secondary IPv4 addresses | Monitoring | Tags

Search

Private IPv4 address	Allocation ID	Association ID	Public IPv4 address	Network interface ID
Secondary IPv4 addresses are not available for this nat gateway.				

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us-east-2.console.aws.amazon.com/vpcconsole/home?region=us-east-2#ElasticIpDetails:AllocationId=eipalloc-0c6bd26a7c1f44c47

VPC > Elastic IP addresses > 3.140.143.223

3.140.143.223

Summary

Allocated IPv4 address 3.140.143.223	Type Public IP	Allocation ID eipalloc-0c6bd26a7c1f44c47	Reverse DNS record -
Association ID eipassoc-07dda36c652ff3e9e	Scope VPC	Associated instance ID -	Private IP address 10.0.1.151
Network interface ID eni-01cf6ad810120a4d7	Network interface owner account ID 767828764671	Public DNS -	NAT Gateway ID nat-0eac98ae950075581 (ngw-ohio-25)
Address pool Amazon			

Tags(0)

Key | Value

No tags associated with this resource
Click the Manage tags button to add your first tag

Manage tags

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iv. Route Tables (Mumbai):

- Then after creating both Internet Gateway and NAT Gateway, go to route tables, click create, then assign names for both public and private route tables, then associate both the tables by clicking routes then attach the network
- Public Route Table → 0.0.0.0/0 → IGW.
- Then click subnet associations in public route table, then click edit association after that click both public subnet 1.
- Private Route Table → 0.0.0.0/0 → NAT.
- Then click subnet associations in private route table, then click edit association after that click both private subnet 1 and private subnet 2.

rtb-0771fe02704157e10 / ohio-pub-rtb-25

Details

Route table ID: rtb-0771fe02704157e10
VPC: vpc-01350b2fefee83039 | vpc-ohio-25
Main: No
Owner ID: 767828764671
Explicit subnet associations: subnet-084aadf97e6037c5c / ohio-pub-sbt-25
Edge associations: -

Routes (2)

Destination	Target	Status	Propagated
0.0.0.0/0	igw-0762604099e073a4e	Active	No
10.0.0.0/16	local	Active	No

rtb-0771fe02704157e10 / ohio-pub-rtb-25

Subnet associations

Explicit subnet associations (1)

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
ohio-pub-sbt-25	subnet-084aadf97e6037c5c	10.0.1.0/24	-

Subnets without explicit associations (0)

The following subnets have not been explicitly associated with any route tables and are therefore associated with the main route table:

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
------	-----------	-----------	-----------

No subnets without explicit associations
All your subnets are associated with a route table.

us-east-2.console.aws.amazon.com/vpcconsole/home?region=us-east-2#RouteTableDetails:RouteTableId=rtb-0de937a0c8d05d032

Search [Alt+S]

United States (Ohio) Prabath D

VPC > Route tables > rtb-0de937a0c8d05d032

rtb-0de937a0c8d05d032 / ohio-pvt-rtb-25

Details Info

Route table ID

rtb-0de937a0c8d05d032

VPC

vpc-01350b2fefee83039 | vpc-ohio-25

Main

No

Explicit subnet associations

2 subnets

Edge associations

-

Routes Subnet associations Edge associations Route propagation Tags

Routes (2)

Both Edit routes

Filter routes

Destination	Target	Status	Propagated
0.0.0.0/0	nat-0eac98ae950075581	Active	No
10.0.0.0/16	local	Active	No

us-east-2.console.aws.amazon.com/vpcconsole/home?region=us-east-2#RouteTableDetails:RouteTableId=rtb-0de937a0c8d05d032

Search [Alt+S]

United States (Ohio) Prabath D

VPC > Route tables > rtb-0de937a0c8d05d032

rtb-0de937a0c8d05d032 / ohio-pvt-rtb-25

Details Info

Route table ID

rtb-0de937a0c8d05d032

VPC

vpc-01350b2fefee83039 | vpc-ohio-25

Main

No

Explicit subnet associations

2 subnets

Edge associations

-

Routes Subnet associations Edge associations Route propagation Tags

Explicit subnet associations (2)

Edit subnet associations

Find subnet association

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
ohio-pvt-sbt-ii-25	subnet-0c632e795553deacb	10.0.3.0/24	-
ohio-pvt-sbt-i-25	subnet-03bddea7c32655fd4	10.0.2.0/24	-

Subnets without explicit associations (0)

Edit subnet associations

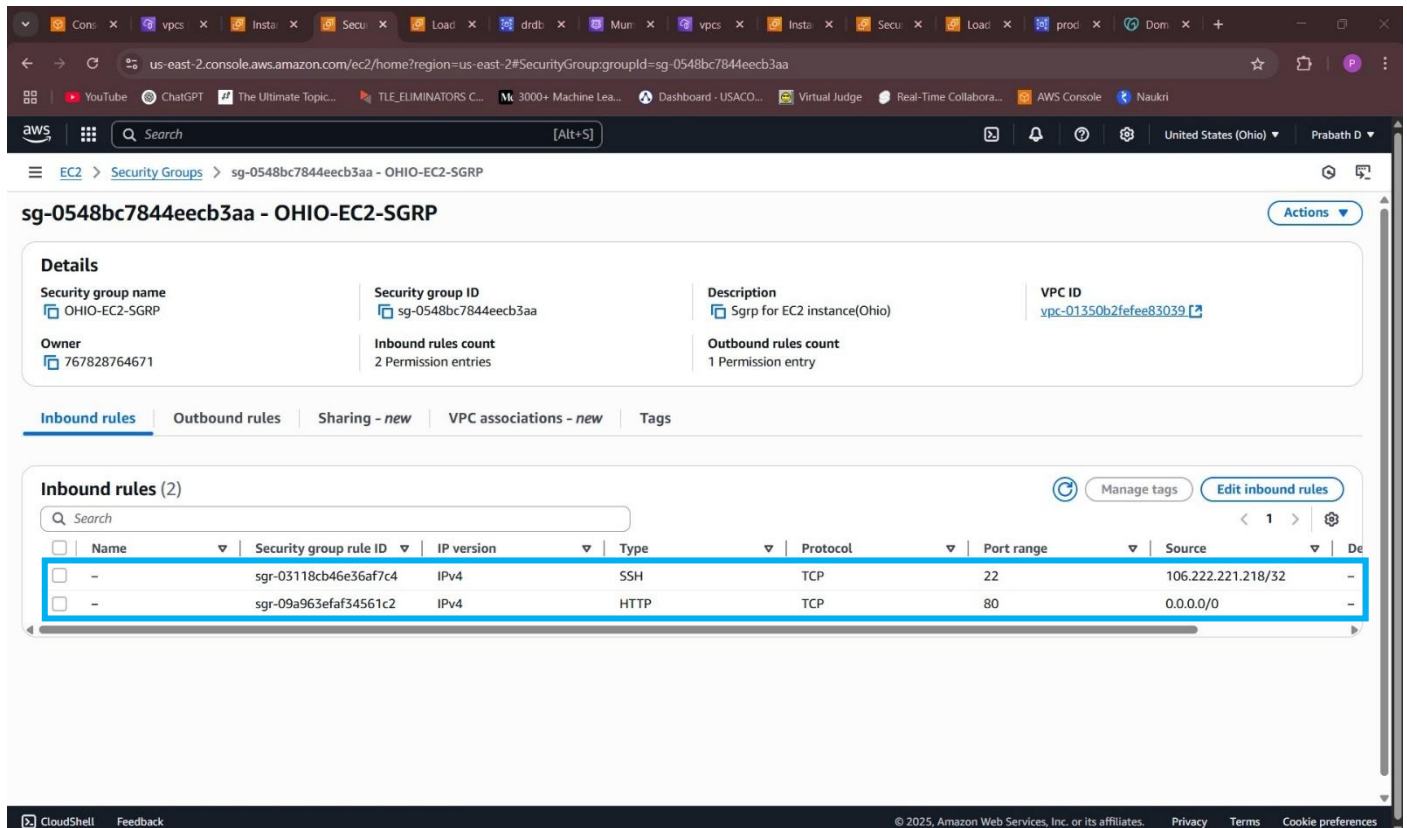
Find subnet association

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
------	-----------	-----------	-----------

No subnets without explicit associations
All your subnets are associated with a route table.

v. EC2, RDS & LB:

- First create Security Groups separately for EC2, RDS and LB.
- Create Production EC2 by giving the same apache/php code in advanced details.
- After creation of EC2, create RDS and LB also.



sg-0548bc7844eecb3aa - OHIO-EC2-SGRP

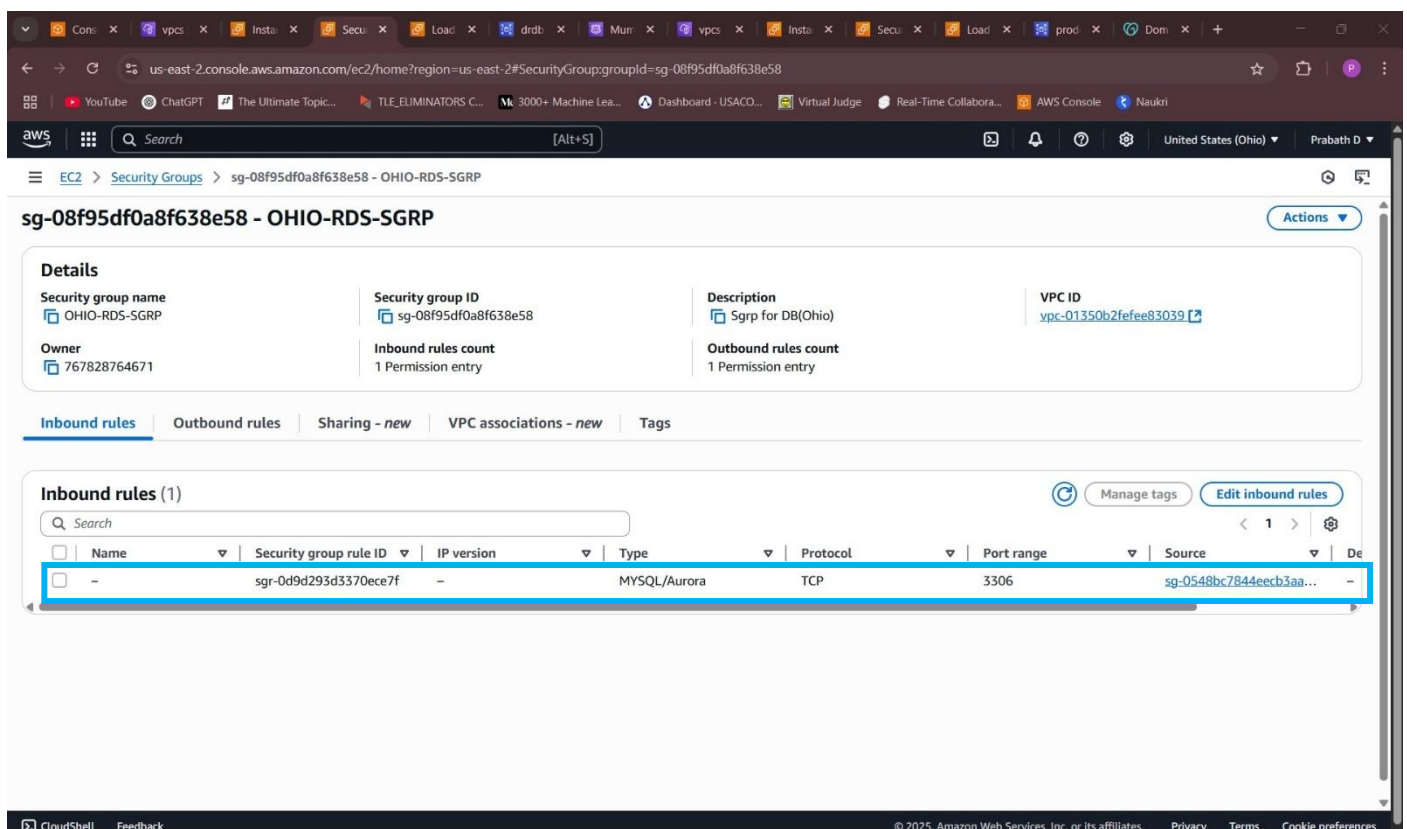
Details

Security group name OHIO-EC2-SGRP	Security group ID sg-0548bc7844eecb3aa	Description Sgrp for EC2 instance(Ohio)	VPC ID vpc-01350b2fefee83039
Owner 767828764671	Inbound rules count 2 Permission entries	Outbound rules count 1 Permission entry	

Inbound rules | Outbound rules | Sharing - new | VPC associations - new | Tags

Inbound rules (2)

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
-	sgr-03118cb46e36af7c4	IPv4	SSH	TCP	22	106.222.221.218/32
-	sgr-09a963efaf34561c2	IPv4	HTTP	TCP	80	0.0.0.0/0



sg-08f95df0a8f638e58 - OHIO-RDS-SGRP

Details

Security group name OHIO-RDS-SGRP	Security group ID sg-08f95df0a8f638e58	Description Sgrp for DB(Ohio)	VPC ID vpc-01350b2fefee83039
Owner 767828764671	Inbound rules count 1 Permission entry	Outbound rules count 1 Permission entry	

Inbound rules | Outbound rules | Sharing - new | VPC associations - new | Tags

Inbound rules (1)

Name	Security group rule ID	IP version	Type	Protocol	Port range	Source
-	sgr-0d9d293d3370ece7f	-	MySQL/Aurora	TCP	3306	sg-0548bc7844eecb3aa...

Cons x vpcs x Insta x Secu x Load x drdb x Mun x vpcs x Insta x Secu x Load x prod x Dom x +

us-east-2.console.aws.amazon.com/rds/home?region=us-east-2#databaseid=drdb;is-cluster=false

Search [Alt+S]

United States (Ohio) ▾ Prabath D ▾

Aurora and RDS > Databases > drdb

Proxy identifier

Status

Engine family

No proxies
You don't have any proxies.
Create proxy

Security group rules (2)

Filter by Security group rules

< 1 > ⚙

Security group	Type	Rule
OHIO-RDS-SGRP (sg-08f95df0a8f638e58)	EC2 Security Group - Inbound	sg-0548bc7844eecb3aa
OHIO-RDS-SGRP (sg-08f95df0a8f638e58)	CIDR/IP - Outbound	0.0.0.0/0

Replication (1)

Filter by Replication

< 1 > ⚙

DB identifier	Role	Region & AZ	Replication source	Replication state	Lag
drdb	Instance	us-east-2b	-	-	-

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Cons x vpcs x Insta x Secu x Load x ohio x Mun x vpcs x Insta x Secu x Load x prod x Dom x +

us-east-2.console.aws.amazon.com/rds/home?region=us-east-2#db-subnet-groupid=ohio-sbt-grp-25

Search [Alt+S]

United States (Ohio) ▾ Prabath D ▾

Aurora and RDS > Subnet groups > ohio-sbt-grp-25

ohio-sbt-grp-25

Subnet group details

VPC ID

vpc-01350b2fefee83039

ARN

arn:aws:rds:us-east-2:767828764671:subgrp:ohio-sbt-grp-25

Supported network types

IPv4

Description

ohio-sbt-grp-25

Subnets (2)

Availability zone	Subnet name	Subnet ID	CIDR block
us-east-2c	ohio-pvt-sbt-ii-25	subnet-0c632e795553deacb	10.0.3.0/24
us-east-2b	ohio-pvt-sbt-i-25	subnet-03bddea7c32655fd4	10.0.2.0/24

Tags (0)

Filter by Tag key

< 1 > ⚙

Manage tags

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Cons x vpcs x Insta x Secu x Load x drdb x Mur x vpcs x Insta x Secu x Load x prod x Dom x +

us-east-2.console.aws.amazon.com/rds/home?region=us-east-2#databaseid=drdbjs-cluster=false

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aws Search [Alt+S]

United States (Ohio) Prabath D

Aurora and RDS > Databases > drdb

drdb

Summary

DB identifier
drdb

CPU
3.71%

Status
Available

Class
db.t3.micro

Role
Instance

Current activity
0 Connections

Engine
MySQL Community

Region & AZ
us-east-2b

Recommendations

Connectivity & security

MonitoringLogs & eventsConfigurationZero-ETL integrationsMaintenance & backupsData migrations - newTagsRecords

Connectivity & security

Endpoint & port

Endpoint
drdb.c7osqm6c4gck.us-east-2.rds.amazonaws.com

Port
3306

Networking

Availability Zone
us-east-2b

VPC
vpc-ohio-25 (vpc-01350b2fefee83039)

Subnet group
ohio-sbt-grp-25

Subnets
subnet-03bdddea7c32655fd4
subnet-0c632e795553deacb

Security

VPC security groups
OHIO-RDS-SGRP (sg-08f95df0a8f638e58)
Active

Publicly accessible
No

Certificate authority
rds-ca-rsa2048-g1

Certificate authority date
May 22, 2061, 05:34 (UTC+05:30)

DB instance certificate expiration date

Cons x vpcs x Insta x Secu x Load x drdb x Mur x vpcs x Insta x Secu x Load x prod x Dom x +

us-east-2.console.aws.amazon.com/ec2/home?region=us-east-2#InstanceDetails:instanceid=i-0e3ab9ec3fabbbc23

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aws Search [Alt+S]

United States (Ohio) Prabath D

EC2 > Instances > i-0e3ab9ec3fabbbc23

Instance summary for i-0e3ab9ec3fabbbc23 (dr-EC2) Info

Connect

Instance state

Actions

Updated less than a minute ago

Instance ID
i-0e3ab9ec3fabbbc23

IPv6 address
-

Hostname type
IP name: ip-10-0-1-93.us-east-2.compute.internal

Answer private resource DNS name
-

Auto-assigned IP address
3.15.25.109 [Public IP]

IAM Role
-

IMDSv2
Required

Operator
-

Public IPv4 address
3.15.25.109 | open address

Instance state
Running

Private IP DNS name (IPv4 only)
ip-10-0-1-93.us-east-2.compute.internal

Instance type
t2.micro

VPC ID
vpc-01350b2fefee83039 (vpc-ohio-25)

Subnet ID
subnet-084aadf97e6037c5c (ohio-pub-sbt-25)

Instance ARN
arn:aws:ec2:us-east-2:767828764671:instance/i-0e3ab9ec3fabbbc23

Private IPv4 addresses
10.0.1.93

Public DNS
-

Elastic IP addresses
-

AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations. | Learn more

Auto Scaling Group name
-

Managed
false

Details

Status and alarmsMonitoringSecurityNetworkingStorageTags

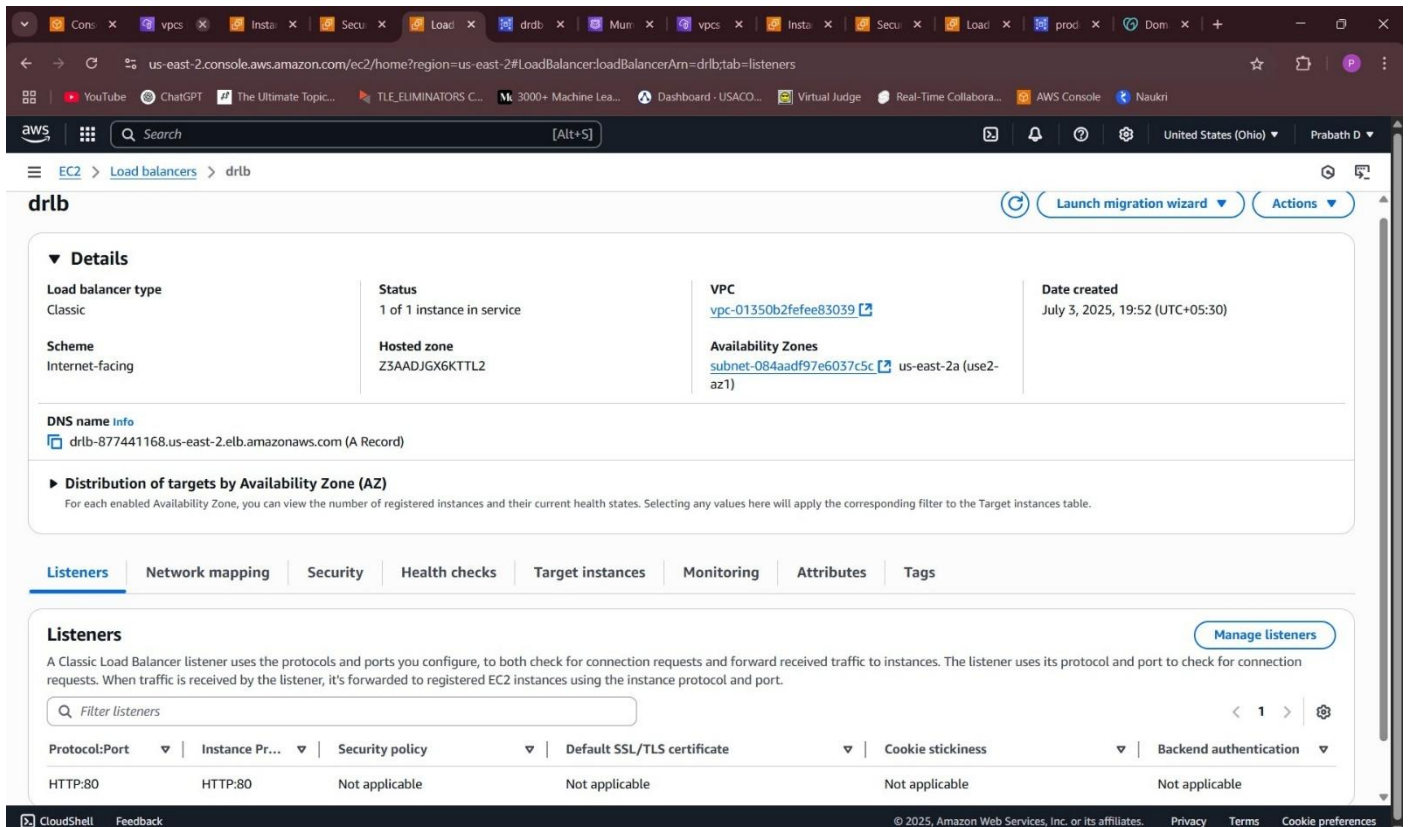
Instance details Info

AMI ID

Monitoring

Platform details

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vi. Configuration :

- Copy and paste the IP address or LB DNS in the browser and then fill the details, then copy the php and paste it in the prod EC2 putty, save it and then run the installation in the wordpress.
- Then setup by giving the details to show the login page should show the message 'success'.
- Then login with the created username and password.

```

root@ip-10-0-1-93:/var/www/html
login as: ec2-user
Authenticating with public key "linux25key"

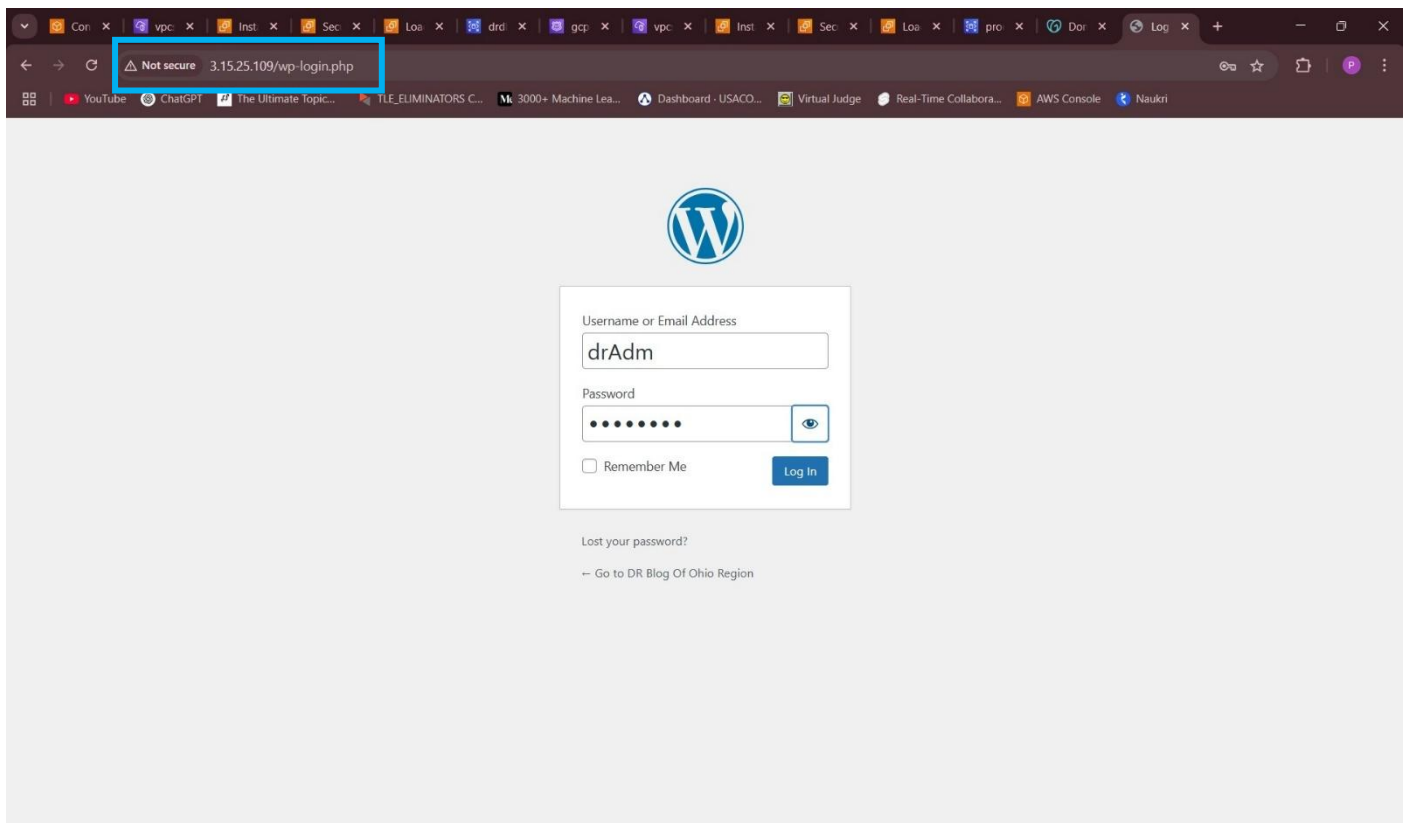
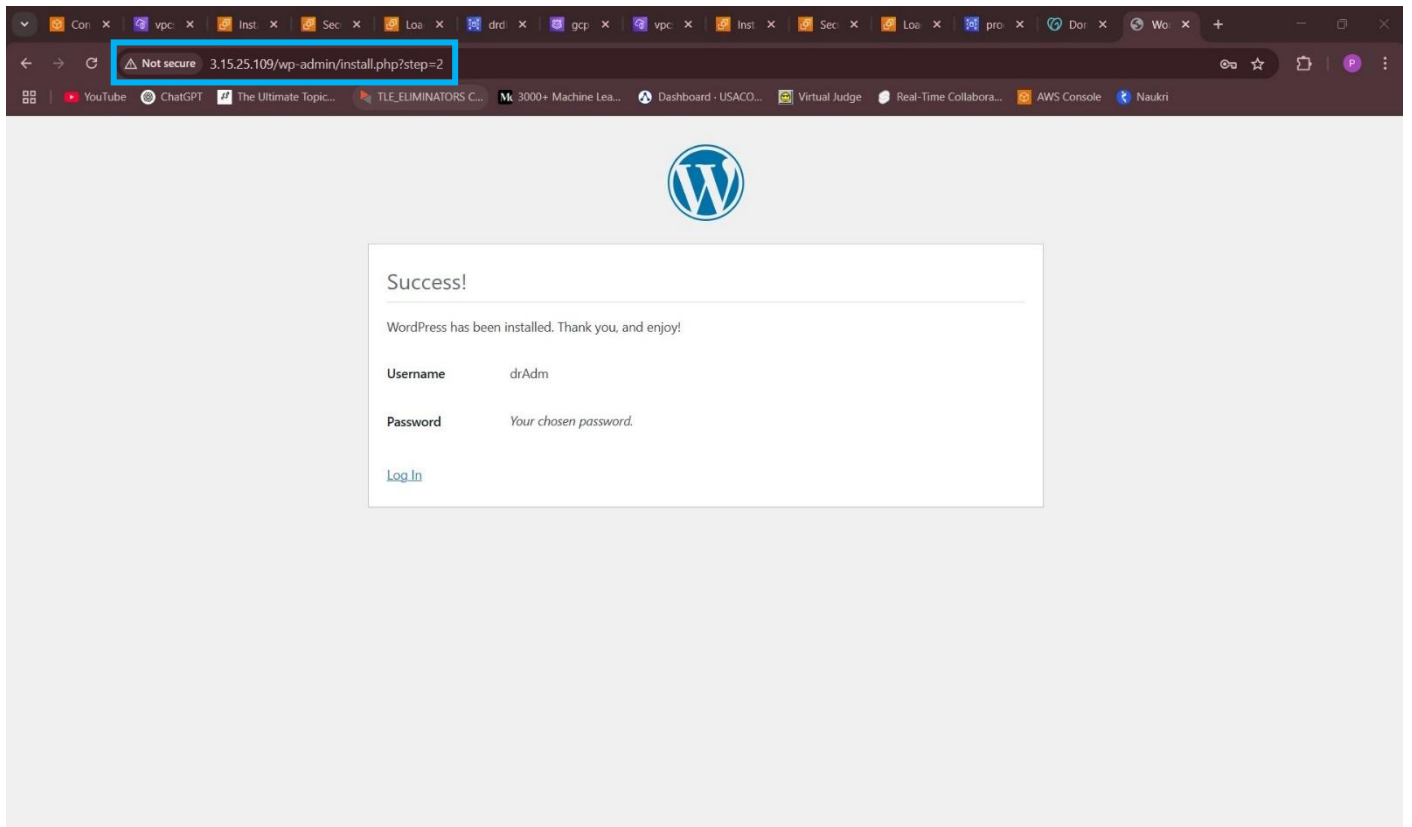
#####
# Amazon Linux 2
#####
AL2 End of Life is 2026-06-30.

A newer version of Amazon Linux is available!

Amazon Linux 2023, GA and supported until 2028-03-15.
https://aws.amazon.com/linux/amazon-linux-2023/

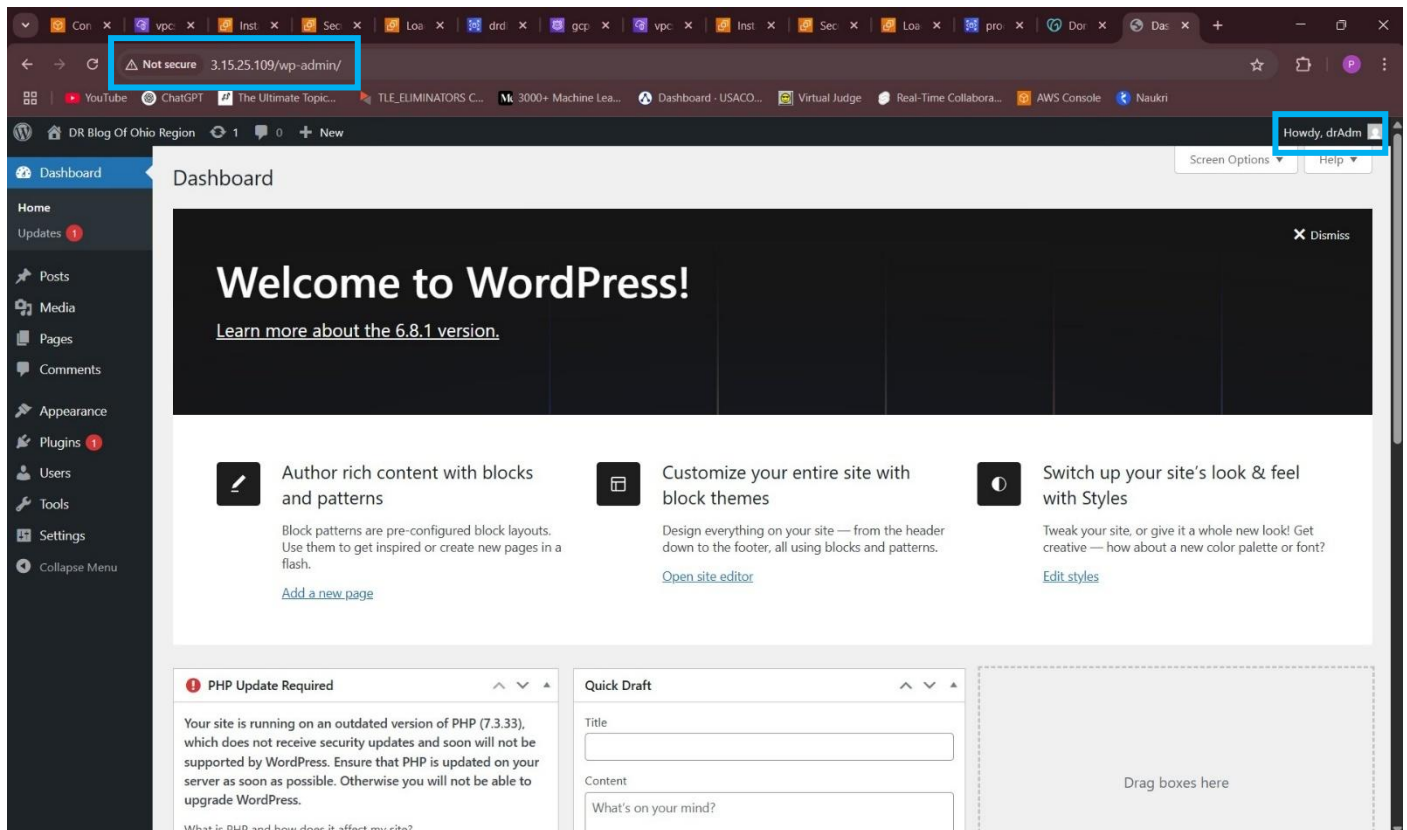
[ec2-user@ip-10-0-1-93 ~]$ sudo -i
[root@ip-10-0-1-93 ~]# clear
[root@ip-10-0-1-93 ~]# cd /var/www/html
[root@ip-10-0-1-93 html]# ls
healthy.html      wp-admin          wp-cron.php       wp-mail.php
index.php         wp-blog-header.php wp-includes        wp-settings.php
license.txt       wp-comments-post.php wp-links-opml.php  wp-signup.php
readme.html      wp-config-sample.php wp-load.php        wp-trackback.php
wp-activate.php  wp-content        wp-login.php       xmlrpc.php
[root@ip-10-0-1-93 html]# vi wp-config.php
[root@ip-10-0-1-93 html]#

```



vii. Edit the Blog Page :

- View the dashboard, then edit the page as per the user's requirement then host the page through the domain name 'gcpclouds.life'.



- Now, sync the Route53 with Load Balancer. And hit the browser with the domain name.

I. Sync R53 with LB:

- Refresh the R53
- Go to primary health check. Click create health check, then give a name then by mapping the created production ELB by giving the path in '/' as index.html.
- Create Record set and map appropriate ELB under alias by adding health checks in a primary for Mumbai region.
- Actually, in this we need to create secondary for Ohio Region.
- Then only when the Production EC2 in Mumbai region becomes unhealthy it redirects to Ohio region and loads the DR page.

us-east-1.console.aws.amazon.com/route53/v2/healthchecks/home#/details/b355b1bc-30c9-4233-b224-8249e4421b85

Route 53 > Health checks > Mumbai-Health-Check

Mumbai-Health-Check Info

Delete Invert Disable Edit

Configuration

ID
b355b1bc-30c9-4233-b224-8249e4421b85

State
Enabled

URL
http://43.205.139.237:80/healthy.html

Status
Healthy

Specified endpoint by
IP address

Inverted
No

Advanced configuration

Metrics Alarms Health checkers Tags

Metrics Info

Alarm recommendations Investigate with AI - new 1h 3h 12h 1d 3d 1w Custom UTC timezone View metrics in CloudWatch Explore related

Health check status

1

0.8

0.6

Health checkers that report the endpoint healthy (percent)

100

80

60

us-east-1.console.aws.amazon.com/route53/v2/hostedzones/ListRecordSets/Z0354002LL5TE1ZJJEM7

Route 53 > Hosted zones > gpcclouds.life

Public gpcclouds.life Info

Delete zone Test record Configure query logging

Hosted zone details Edit hosted zone

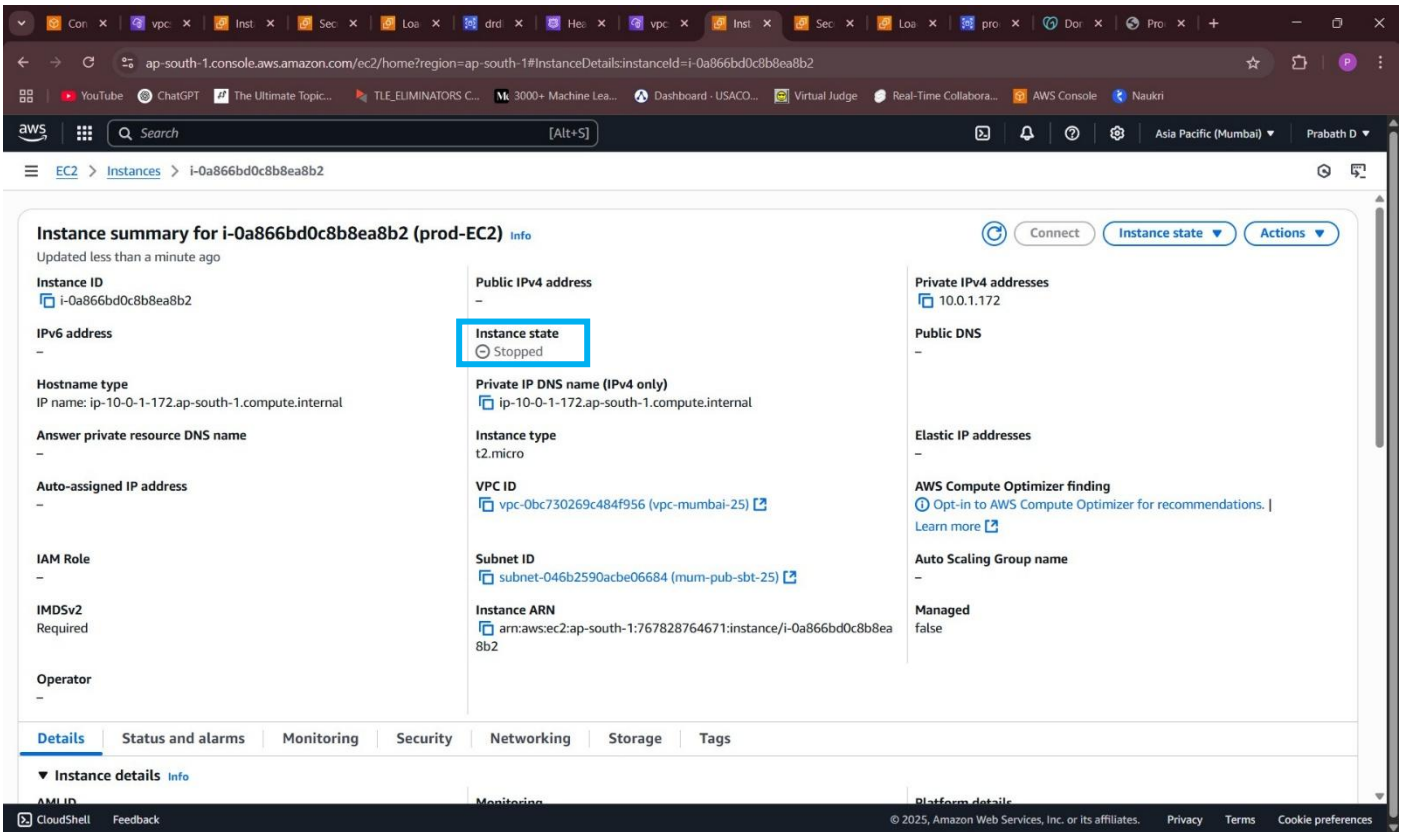
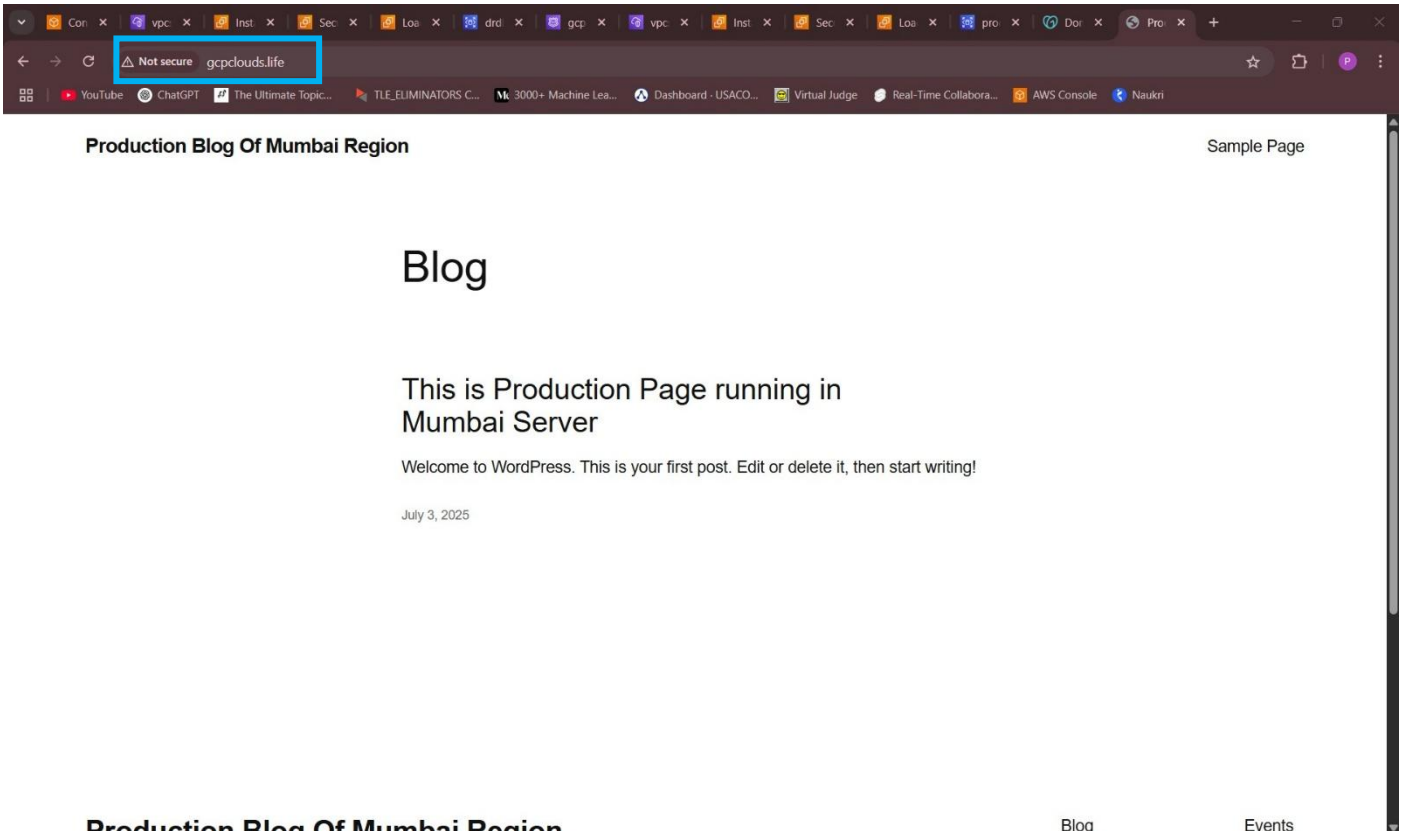
Records (4) DNSSEC signing Hosted zone tags (0)

Records (4) Info

Automatic mode is the current search behavior optimized for best filter results. To change modes go to settings.

Filter records by property or value Type Routing p... Alias

☐	Record ...	Type	Routin...	Differ...	Alias	Value/Route traffic to	TTL (s...	Health ...	Evalua...	Record ID
☐	gpcclouds...	A	Failover	Primary	Yes	dualstack.prodlib-119218341...	-	b355b1bc...	Yes	primary-mumbai
☐	gpcclouds...	A	Failover	Secondary	Yes	dualstack.drlb-877441168.u...	-	-	Yes	secondary-ohio
☐	gpcclouds...	NS	Simple	-	No	ns-1318.awsdns-36.org. ns-308.awsdns-38.com. ns-1940.awsdns-50.co.uk. ns-867.awsdns-44.net.	172800	-	-	-
☐	gpcclouds...	SOA	Simple	-	No	ns-1318.awsdns-36.org. aws...	900	-	-	-



us-east-1.console.aws.amazon.com/route53/v2/healthchecks/home#/details/b355b1bc-30c9-4233-b224-8249e4421b85

Mumbai-Health-Check

Configuration

ID: b355b1bc-30c9-4233-b224-8249e4421b85

URL: http://43.205.139.237:80/healthy.html

Status: Unhealthy

Specified endpoint by IP address

Inverted: No

Advanced configuration

Metrics

Alarm recommendations

Investigate with AI - new

1h 3h 12h 1d 3d 1w Custom UTC timezone

View metrics in CloudWatch

Explore related

Health check status

Health checkers that report the endpoint healthy (percent)

us-east-2.console.aws.amazon.com/ec2/home?region=us-east-2#InstanceDetails:instanceId=i-0e3ab9ec3fabbbc23

EC2 > Instances > i-0e3ab9ec3fabbbc23

Instance summary for i-0e3ab9ec3fabbbc23 (dr-EC2)

Updated less than a minute ago

Instance ID: i-0e3ab9ec3fabbbc23

Public IPv4 address: 3.15.25.109

Instance state: Running

Private IPv4 addresses: 10.0.1.93

Public DNS: -

Private IP DNS name (IPv4 only): ip-10-0-1-93.us-east-2.compute.internal

Instance type: t2.micro

VPC ID: vpc-01350b2fefe83039

Subnet ID: subnet-084aadf97e6037c5c

Instance ARN: arn:aws:ec2:us-east-2:767828764671:instance/i-0e3ab9ec3fabbbc23

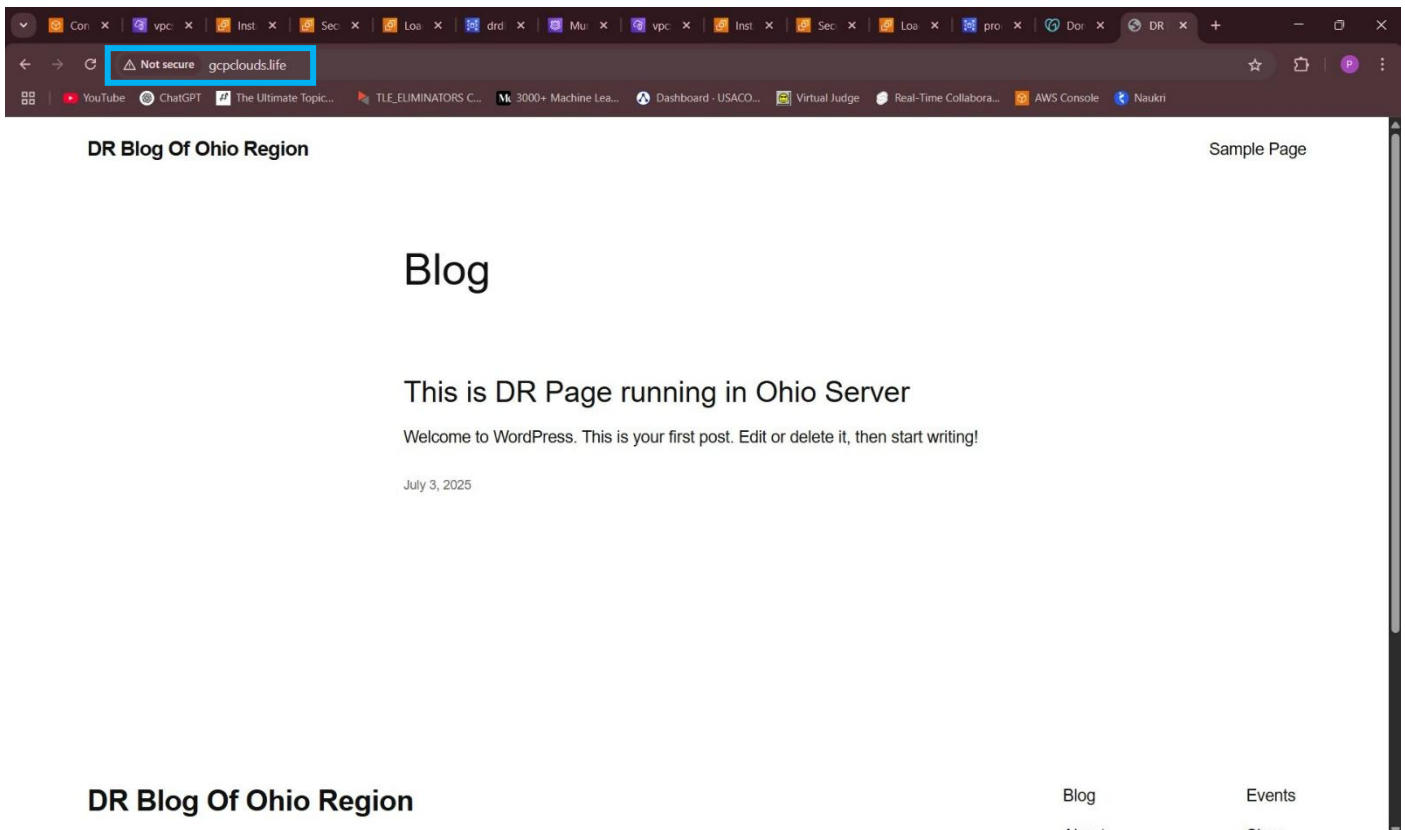
Auto-assigned IP address: 3.15.25.109 [Public IP]

Auto Scaling Group name: -

Managed: false

Details Status and alarms Monitoring Security Networking Storage Tags

Instance details



- The setup for deploying the WordPress page in this task also completed. With that implementing multi-region architecture using failover routing is completed.
- Now, let us see the difference between task 1 and task 4 even though in both the task the failover routing is implemented

Failover Routing Differences:

Feature	Use Multiple Regions	Multi-Region Architecture (Failover)
Purpose	Deploy app in two separate regions	Connect both regions for failover
Connectivity	Independent deployments	Unified via Route 53 + Health Checks
Traffic Handling	Accessed manually via IP/ELB	Automatically routed via domain
High Availability / Disaster Recovery	Not implemented	Implemented
Example Output	Two different blog pages manually	One domain with automated failover

Task 5: Draw our project in Cloud Craft as 3D diagram

Objective:

To create a comprehensive and visually clear 3D architecture diagram of our AWS project setup using CloudCraft, helping stakeholders understand the cloud infrastructure at a glance.

Tools Used:

- CloudCraft (<https://www.cloudcraft.co/>)
- AWS Services: EC2, RDS, ELB, S3, VPC, Route53, IAM

Steps to Create the Diagram:

1. Login to CloudCraft using AWS credentials or manually select components.
2. Drag and Drop Components:
 - EC2 Instances (Prod and DR)
 - RDS Databases (ProdDB and DRDB)
 - Classic Load Balancers for each region
 - Public and Private Subnets within a VPC
 - NAT Gateway and Internet Gateway
 - S3 Buckets for sync backups
 - Route 53 DNS component
 - IAM Role symbol (optional)
3. Group Resources by Region:
 - Separate components into Mumbai (Primary) and Ohio (Secondary) sections to indicate multi-region architecture.
4. Add Labels and Notes:
 - Name each service (e.g., Prod-EC2, DR-RDS, S3-Bucket-Prod)
 - Use icons and network flow lines to show:
 - ELB → EC2
 - EC2 ↔ RDS
 - EC2 ↔ S3
 - Failover logic via Route53 Health Checks

5. Highlight Routing Flow:

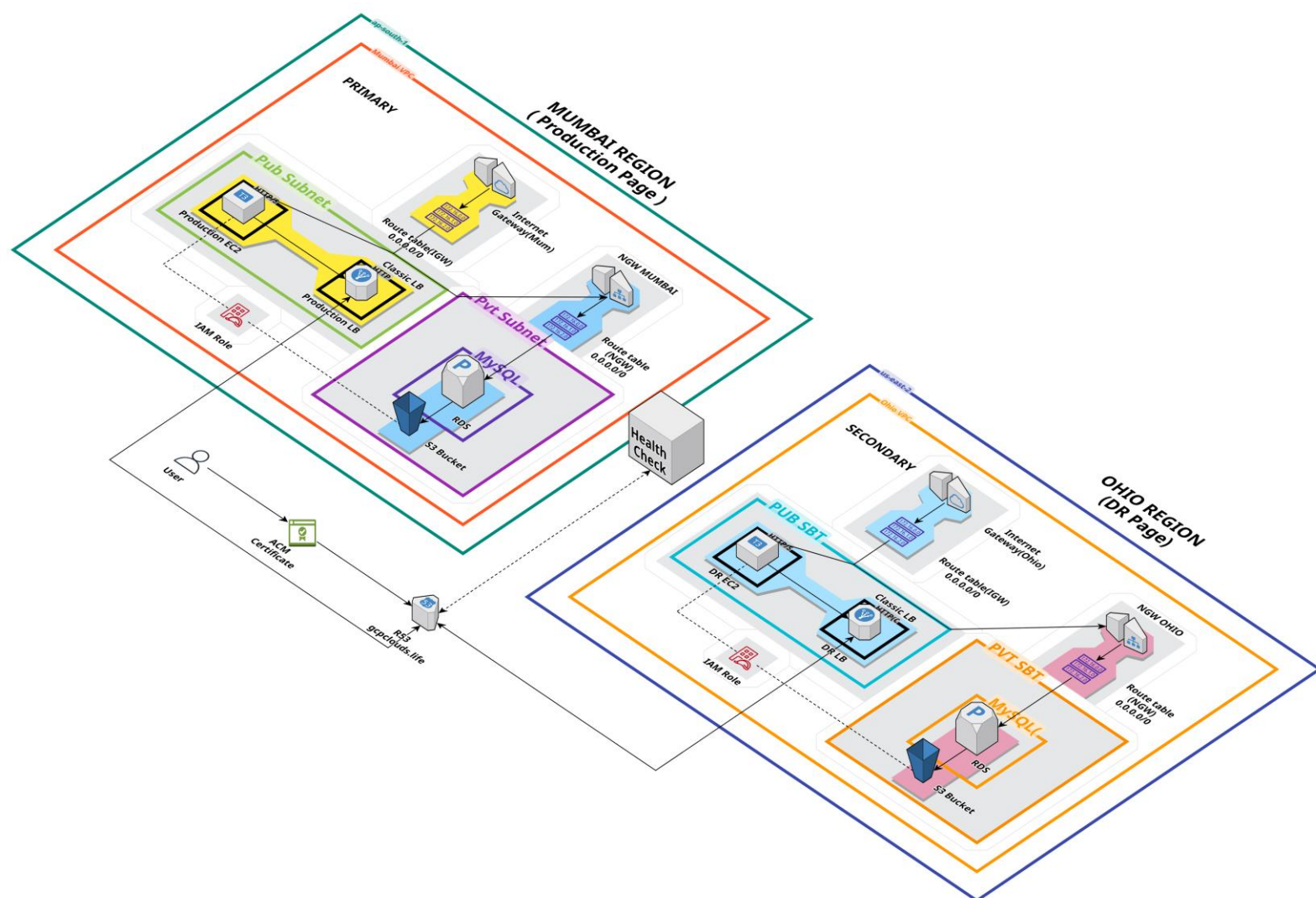
- Show Route53 pointing to ELBs with health check logic for failover routing.

6. Export the Diagram:

- Download the image from CloudCraft in PNG or PDF format.

Conclusion:

Using CloudCraft made it easier to visualize the complex architecture, highlight relationships between AWS services, and demonstrate DR readiness for failover routing. The 3D view helps non-technical stakeholders understand the flow and redundancy built into the system.



3D Multi-Region AWS Architecture – Primary (Mumbai) and DR (Ohio)